

FINAL TRIM AND STABILITY BOOKLET OF KB-33 (YARD NO. CHISTI 019)



AS AMENDED
SEE PAGE NO.9,52,53,54

SEE LETTER E-140898-189263

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 www.cybermarine.net info@cybermarine.net 10, Bukit Batok Crescent, #12-01, The Spire Building, Singapore-658079. 501, Technocity IT Premises Co-op. Soc. Ltd, Plot No. X-5/3, Mahape, Maharashtra- 400710, India. 22503, Katy Freeway #7, Katy, Texas, USA 77450.				

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GENERAL PARTICULARS

Name of the Vessel	KB-33
Client	K.B. SHIPPING & CO. (JAMNAGAR)
Builder	CHISTI MARINE SERVICES
Yard No.	CHISTI-019
Type of Vessel	PONTOON
Official No.	MAR-2394-D
Date of Keel Laid	27-08-2021
Port of Registry	MORMUGAO
Class Notation	㊦SU, PONTOON, INDIAN COASTAL SERVICE DESCRIPTION"FOR CRANE OPERATION" "DECK STRENGTHENED FOR 10T/M ² "

Main Particulars

LOA	50.000 M
LBP	50.000 M
Breadth (Mld.)	18.000 M
Depth (Mld.)	03.000 M
Draft (Mld.)	2.200 M
Draft (Ext.)	2.212 M
Displacement(Ext.)	1893.06 T
Lightship	449.57 T
Deadweight	1443.49 T

Stability Criteria

REGULATION

Intact Stability Criteria in accordance **IS CODE 2008**

- 1) The area under a righting lever curve up to the angle of maximum righting lever should not be less than 0.080 meter-radian.
- 2) The static angle of heel due to a uniformly distributed wind load of 0.54 kPa (wind speed 30 m/s) should not exceed an angle corresponding to half freeboard for the relevant loading condition, where the lever of wind heeling moment is measured from the centroid of the windage area to half the draught.
- 3) The minimum range of stability should be:
For $L \leq 100\text{m}$: 20°
For $L \geq 150\text{m}$: 15°
For intermediate length by interpolation

Intact stability criteria during cargo lifting

The following intact stability criteria are to be complied with:

- $\theta_C \leq 15^\circ$
- $GZC \leq 0,6 GZ_{MAX}$
- $A1 \geq 0.4 ATOT$

where:

θ_C : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 1)

GZC , GZ_{MAX} : Defined in Fig 1

$A1$: Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_C to the heeling angle equal to the lesser of:

- heeling angle θ_R of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 1)
- heeling angle θ_F , corresponding to flooding of unprotected openings

$ATOT$: Total area, in m.rad, below the righting lever curve.

In the above formula, the heeling arm, corresponding to the cargo lifting, is to be obtained, in m, from the following formula:

$$b = (Pd - Zz) / \Delta$$

where:

P : Cargo lifting mass, in t

d : Transversal distance, in m, of lifting cargo to the longitudinal plane (see Fig 1)

Z : Mass, in t, of ballast used for righting the pontoon, if applicable (see Fig 1)

z : Transversal distance, in m, of the centre of gravity of Z to the longitudinal plane (see Fig 1)

Δ : Displacement, in t, at the loading condition

The above check is to be carried out considering the most unfavourable situations of cargo lifting combined with the lesser initial metacentric height GM . The vertical position of the centre of gravity of cargo lifting is to be assumed in correspondence of the suspension point.

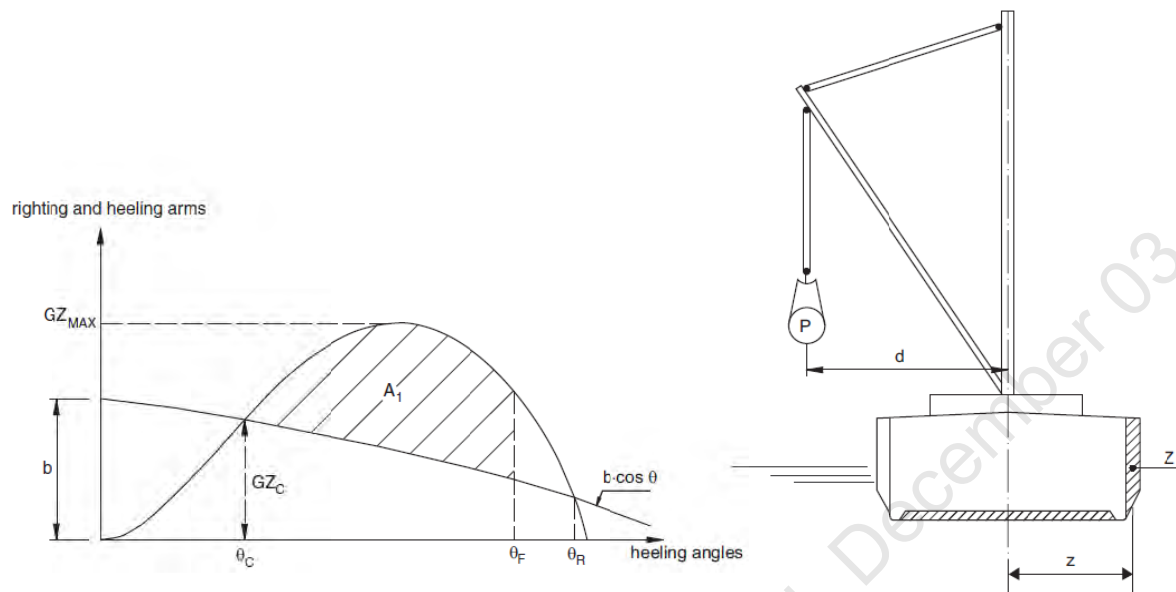


FIGURE 1: CARGO LIFTING

Intact stability criteria in the event of sudden loss of cargo during lifting

This additional requirement is compulsory when counterweights or ballasting of the ship are necessary or when deemed necessary by the Society taking into account the ship dimensions and the weights lifted.

The case of a hypothetical loss of cargo during lifting due to a break of the lifting cable is to be considered.

In this case, the following intact stability criteria are to be compiled with

- $A_2 / A_1 \geq 1$
- $\theta_2 - \theta_3 \geq 20^\circ$

where:

A_1 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_1 to the heeling angle θ_C (see Fig 2)

A_2 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_C to the heeling angle θ_2 (see Fig 2)

A_3 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_C to the heeling angle θ_3 (see Fig 2)

θ_1 : Heeling angle of equilibrium during lifting (see Fig 2)

θ_2 : Heeling angle corresponding to the lesser of θ_R and θ_F

θ_C : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 2)

θ_3 : Maximum heeling angle due to roll, at which $A_3 = A_1$, to be taken not greater than 30° (angle in correspondence of which the loaded cargo on deck is assumed to shift (see Fig 2)

θ_R : Heeling angle of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 2).

θ_F : Heeling angle at which progressive flooding may occur (see Fig 2)

In the above formulae, the heeling arm, induced on the ship by the cargo loss, is to be obtained, in m, from the following formula

$$b = Zz \cos\theta/\Delta$$

HEELING AND RIGHTING ARMS

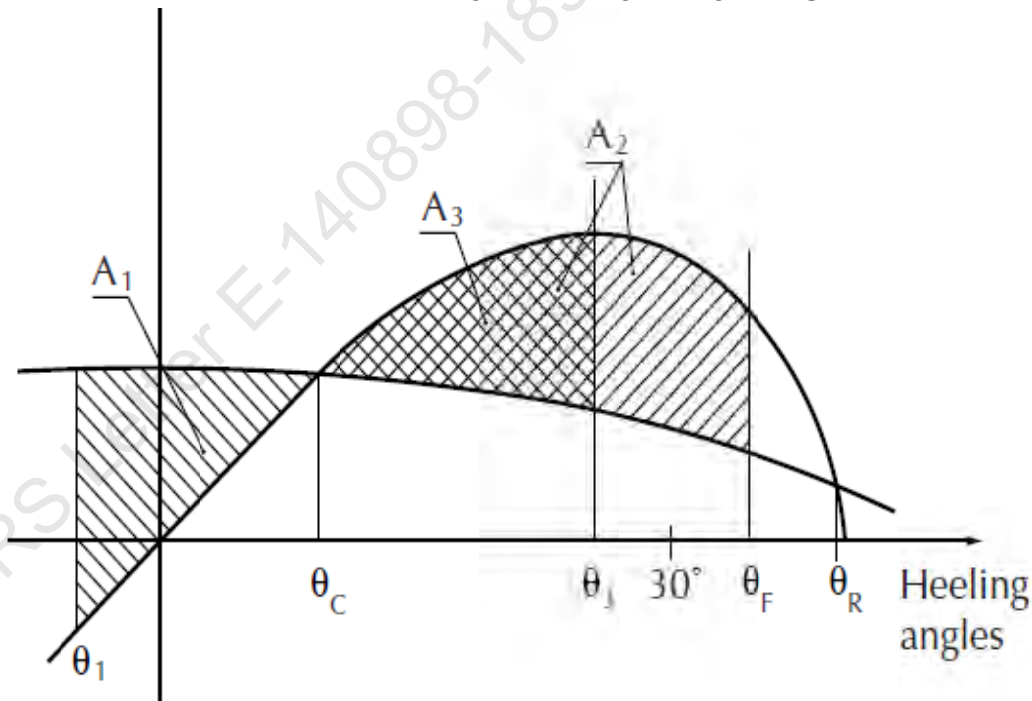


FIGURE 2: CARGO LOSS

Notes to the Master

General precautions against capsizing

1. Compliance with the required minimum stability criteria does not insure immunity against capsizing, regardless of the circumstances, or absolve the Master from his responsibilities. Masters should therefore exercise prudence and good seamanship, having regard to the season of the year, weather forecasts, the navigational zone, and should take appropriate action as to speed and course warranted by the prevailing circumstances.
2. Loading conditions shown in this booklet are typical of those which may be encountered in service and are worked out to be used for guidance only. Further the booklet contains sufficient data to enable the ship's staff to compute the ship's stability in various loading conditions.
3. Excessive trim by forward is to be avoided in all cases as this may lead to difficulties of maneuverability and course keeping as also infringe the minimum Bow-height requirements.
4. All longitudinal levers, namely LCB, LCG, and LCF are computed with reference to the A.P of Ship.
5. In this booklet hydrostatic particulars and cross curves are given at following trims.

-1.00m, -0.50m, 0.00m, 0.50m, 1.00m

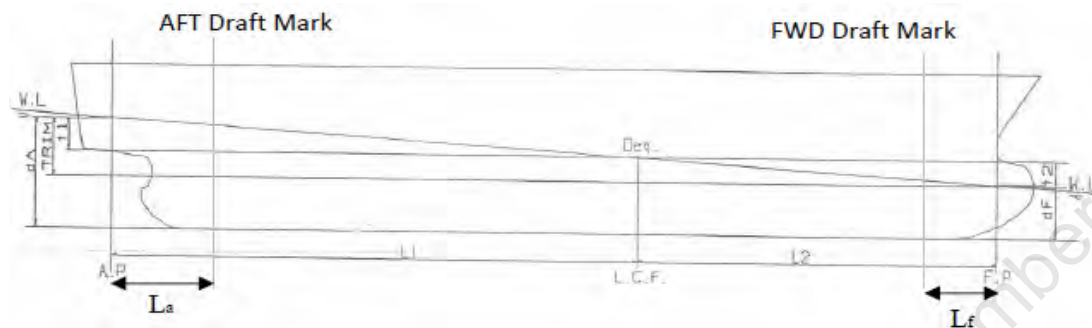
These trims are the expected operational trims.

MAX DECK CARGO HEIGHT - 2 M ABOVE DECK.

Notes on Use of Cross Curves of Stability

The purpose of the Cross-Curves of Stability is to enable Statical Stability curves to be drawn for the ship in any loading condition. Intact stability is then determined on the basis of this Curve. WORKED OUT EXAMPLE demonstrates the use of Cross Curves.

Draft and Trim Calculation



The following data are required for ascertaining trim and draft:

- a) Longitudinal Centre of gravity of the ship calculated as follows:

$$LCG = \frac{\text{Summation of longitudinal weight moments}}{\text{Displacement}}$$

- b) Longitudinal centre of buoyancy, LCB
 c) Displacement, Δ
 d) Moment to change trim, MCT1cm
 e) Longitudinal centre of floatation, LCF

LCB, LCF and MCT1cm are taken from the hydrostatic tables (trim = 0 m) corresponding to the actual displacement. The trim is calculated according to the following formula:

$$\text{Trim} = \frac{\Delta * (LCB - LCG)}{MCT1cm * 100}$$

The Extreme drafts at AP (TAP) and at FP (TFP) are calculated according to the following formulae:

$$T_{AP} = \frac{LCF * \text{Trim}}{LBP} + T (\text{extreme})$$

$$T_{FP} = T_{AP} - \text{Trim}$$

Where T is the Extreme draft taken from the hydrostatic tables (at trim = 0) corresponding to the actual displacement.

The draft at the Aft and Forward reading marks are as follows:

$$T_a = T_{AP} - \frac{\text{Trim} * L_a}{LBP}$$

$$T_f = T_{FP} + \frac{\text{Trim} * L_f}{LBP}$$

Where,

T_a = Draft at Aft Draft Mark

T_f = Draft at Fwd Draft Mark

L_a = Distance of Aft Draft Mark from AP and it is consider positive when Aft Draft Mark is Fwd of AP

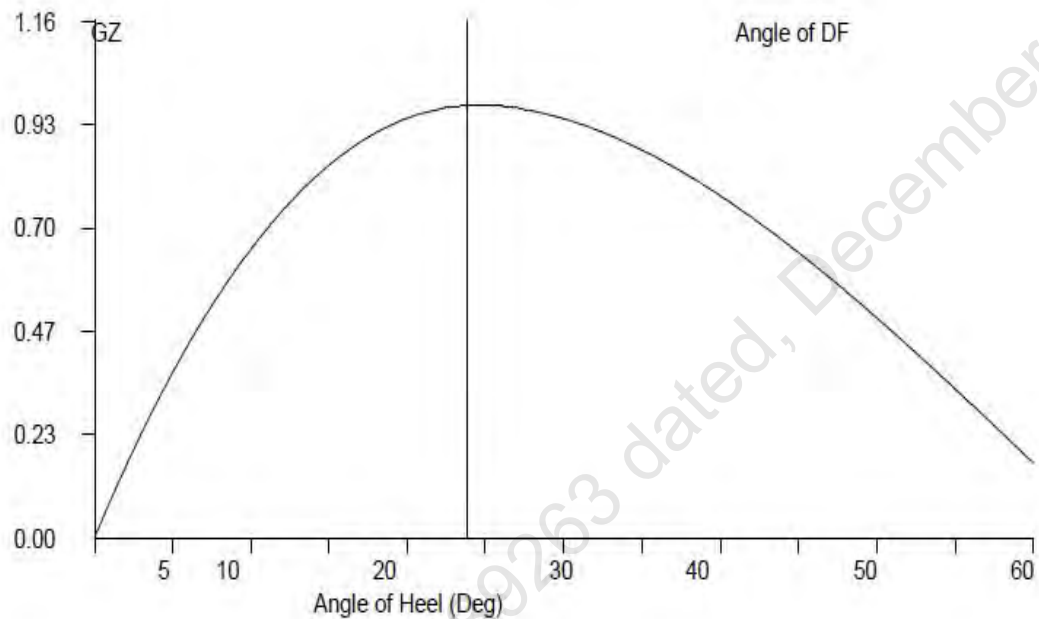
L_f = Distance of Fwd Draft Mark from FP and it is consider positive when Fwd Draft Mark is Aft of FP

GZ Calculation:

- 1) $GZ = KN - KGo \sin (\theta)$
- 2) Area under 0-10 = $10 * 0.01745 * [GZ0 + GZ10] / 2$
- 3) Area under 10-20 = $10 * 0.01745 * [GZ10 + GZ20] / 2$
- 4) Area under 20-30 = $10 * 0.01745 * [GZ20 + GZ30] / 2$
- 5) Area under 30-40 = $10 * 0.01745 * [GZ30 + GZ40] / 2$
- 6) Area under 40-50 = $10 * 0.01745 * [GZ40 + GZ50] / 2$
- 7) Area under 50-60 = $10 * 0.01745 * [GZ50 + GZ60] / 2$
- 8) Area upto 40 deg / Angle of DF = sum of 2+3+4+5
- 9) Max Gz = Read from Gz Curve

Plot GZ values against the angles of heel as above. At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates. Join the point to the origin. This is the Tangent to the GZ Curve at the Origin. Find the

Maxima of GZ curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding & between 30 degrees to 40 degrees/angle of flooding. If maximum GZ occurs at an angle less than 30 degrees then area upto this angle must be calculated.



Worked Out Examples**1 Loading the Vessel**

List out the weights, LCG, VCG, TCG and work out longitudinal moment and vertical moment for each load and finally sum up the entire loads to get Displacement, LCG and VCG of the vessel.

For this worked out Example,

Condition 1: Lightship + Crane + Ballast

Displacement	1084.44	Tonnes
LCG	24.264	M Fwd of AP
VCG (KG)	3.936	M above Base Line
TCG	0.000	M Off centerline

2 Free Surface Correction

A tank which is partially filled, adversely affects the stability of the vessel. This is known as Free Surface Effect and is expressed as 'Loss of GM' or 'Virtual Rise in KG' and is calculated as follows:

$$\text{Free Surface Correction (M)} = \frac{\text{Sum of Free Surface Moments of all Tanks}}{\text{Displacement of Vessel}}$$

$$\text{Free Surface moment for a Tank} = \text{Moment of Inertia (M}^4\text{) for a Tank} \times \text{Density of Liquid in Tank}$$

$$\text{KGo (Fluid)} = \text{KG (Solid)} + \text{Free Surface Correction.}$$

Hydrostatic Particulars

The trim of a particular loading condition is obtained by

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{displacement} / (\text{MCT} \times 100) + \text{Trim}^*$$

Where,

Trim* = Trim at which hydrostatic particulars have been computed.

Condition 1 : Lightship + Crane + Ballast

TANK NAME	VOL(Cu.M.)	SpGr	Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	FSM
				METER	T-M	METER	T-M	METER	T-M	T-M
06_WBTK.C	132.36	1.025	135.67	42.251	5732.19	0.000	0.00	1.499	203.37	135.61
07_WBTK.P	48.39	1.025	49.60	47.660	2363.94	-5.971	-296.16	1.880	93.25	72.32
07_WBTK.S	48.39	1.025	49.60	47.660	2363.94	5.971	296.16	1.880	93.25	72.32
TOTAL TANK			234.87	44.536	10460.07	0.000	0.00	1.660	389.87	280.25

FIXED WEIGHT			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
LIGHT SHIP			449.57	25.030	11,252.74	0.000	0.00	3.000	1,348.71
CRANE			400.00	11.500	4,600.00	0.000	0.00	6.325	2,530.00
TOTAL FIXED WEIGHT			849.57	18.660	15,852.74	0.000	0.00	4.565	3,878.71

ITEM			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
TOTAL TANK			234.870	44.536	10460.065	0.000	0.000	1.660	389.865
TOTAL FIXED WEIGHT			849.570	18.660	15852.737	0.000	0.000	4.565	3878.710
TOTAL WEIGHT			1084.440	24.264	26312.802	0.000	0.000	3.936	4268.575

Condition 1 : Lightship + Crane + Ballast

Free Surface Correction for VCG = FSM/Displacement

	0.258	m
VCGc	4.195	m

From the MaxVCG table,

Displacement	1084.44	T
Allowable Max VCG	14.600	m

VCGc < Maxvcg

PASS

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement =	1084.440	T
Draft ,Tm @ 25.0f =	1.319	m
LCB =	25.000	m
LCF =	25.000	m
MCT =	33.129	T-m/cm
KMT =	22.524	m

The vessels trim is to be determined as follow,

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{Displacement} / (\text{MCT} \times 100)$$

0.241 m, (+) is aft trim, (-) is fwd trim

Determining the Forward perpendicular, Midship and aft Perpendicular Draft

LBP = 50m	
Draft @ 25.00 (MS), Tmid	1.319 m
Draft @ 50.0f (FP), Tfwd	=Tm - Trim/2 1.198 m
Draft@0(AP), Taft	=Tm + Trim/2 1.439 m

Condition 1 : Lightship + Crane + Ballast**From HYDROSTATICS**

Hydrostatic Particulars			Drafts & Trims	
MCT	33.129	T-m	Draft (Mean)	1.319 m
LCB	25.000	m	Draft AP (USK)	1.439 m
			Draft FP (USK)	1.198 m
Metacentric Height				
KMt	22.524	m		
KG	3.936	m		
KG0	4.195	m		
GMT	18.329	m	Trim	0.241 m

The condition shown above is first worked using Even Keel hydrostatic particulars Program then interpolates the hydrostatic particulars from the values available for the displacement

From HYDROSTATICS

Hydrostatic Particulars			Drafts & Trims	
MCT	32.300	T-m	Draft (Mean)	1.318 m
LCB	24.248	m	Draft AP (USK)	1.442 m
			Draft FP (USK)	1.195 m
Metacentric Height				
KMt	22.531	m		
KG	3.936	m		
KG0	4.195	m		
GMT	18.336	m	Trim	0.247 m

Trim is calculated as follows:

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{displacement} / (\text{MCT} \times 100) + \text{Trim}^*$$

Where Trim* is trim at which hydrostatics were calculated.

Condition 1 : Lightship + Crane + Ballast**Intact Stability (Primary Criterion)****Minimum Required**

Displacement	1084.44	T
KG0	4.195	M
GMt	18.336	M

0.150 M

Residual Arm Curve

Angle (Deg)	0	10	20	30	40	50
KN (M)	-0.020	3.784	5.012	5.101	4.423	4.464
GZ (M)	-0.020	3.052	3.574	3.000	1.724	1.248

	Computed values	Minimum Required
Residual Area from abs 0 deg to MAXRA	0.6893 m-rad	0.080 m-rad
Angle from Equilibrium to RAZero or flood	LARGE m-rad	20.00 Deg.
Angle from Equilibrium to 50% DK Imm Angle	5.05 m-rad	0.00 Deg.

Plot GZ values against the angles of heel as above.

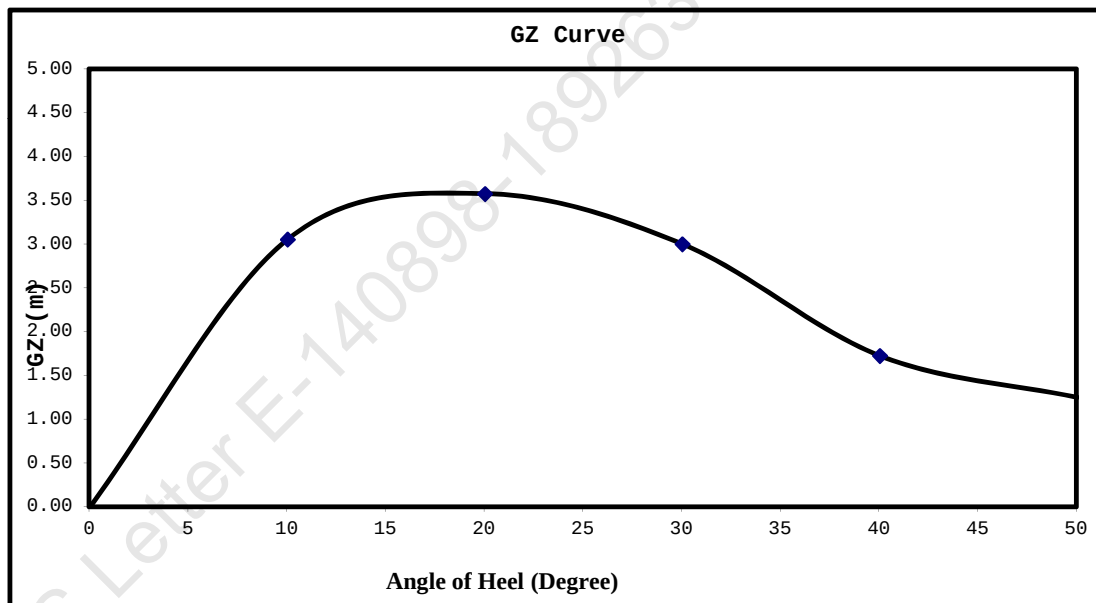
At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates. Join the point to the origin.

This is the Tangent to the GZ Curve at the Origin. Find the Maxima of GZ

curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding

If maximum GZ occurs at an angle less than 30 degrees

then area upto this angle must be calculated.



Worked Out Example Contd.

Mean Draft	1.318	M
LCF from AP	24.769	M
LCB from AP	24.248	M
MCT	32.300	T-M/Cm
Trim	0.247	M
Draft (AP)	1.442	M
Draft (FP)	1.195	M
KG	3.936	M
KGo	4.195	M
GoMt	18.336	M

LOADLINE DRAFT: Check that Mean Draft (Midship) does not exceed the Summer Draft (Ext) for the corresponding season, as per the assigned loadline.

[illegible][illegible][illegible]

	m
VCGc	m

Displacement	T
Allowable Max VCG	m

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement =	T
Draft ,Tm @ 25.0f =	m
LCB =	m
LCF =	m
MCT =	T-m/cm
KMT =	m

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{Displacement} / (\text{MCTX} \times 100)$$

m, (+) is aft trim, (-) is fwd trim

LBP = 50m

Draft @ 25.00 (MS), T_{mid} m

Draft @ 50.00 (FP), T_{fwd} = $T_m - \frac{Trim}{2}$ m

Draft @ 0 (AP), T_{aft} = $T_m + \frac{Trim}{2}$ m

Definitions, Symbols and Sign Conventions:

AP	Fr.0
FP	Fr.100
Midship	Fr.50
Frame Spacing	500mm throughout

List of Symbols

Symbol	Unit	Description
LOA	M	Length Overall
LBP	M	Length between perpendiculars
B (MLD)	M	Moulded Breadth
D (MLD)	M	Moulded Depth
TPCm	T/Cm	Tonnes per 1 Cm immersion
MCTCm	T-M/Cm	Moment to change Trim by 1 Cm
Draft AP (USK)	M	Draft at AP wrt Keel line
Draft FP (USK)	M	Draft at FP wrt Keel line
Draft (Mean)	M	Draft wrt Keel line at Midship
Draft (Mean)	M	[Draft AP(USK) + Draft FP(USK)] x 0.5
VCG, KG	M	Vertical Center of Gravity above Base Line before correcting for free surface
KGo	M	Vertical Center of Gravity above Base Line after correcting for free surface
KB	M	Vertical Center of Buoyancy from Base Line
KMt	M	Transverse Metacenter above Base Line
GoMt	M	Transverse Metacenter above KGo
GZ	M	Righting Arm
FSM	T-M	Free Surface Moment
sp.gr.		Specific Gravity
LCG	M	Longitudinal Center of Gravity
LCB	M	Longitudinal Center of Buoyancy
LCF	M	Longitudinal Center of Floatation

Sign Convention

LCB, LCG, LCF	Fwd of AP is denoted by f and aft of AP is denoted by a
TCG	port is negative and stbd is positive
Trim by Aft	Positive
Trim by Ford	Negative

Metric Conversion Table

<u>MULTIPLY BY</u>	<u>TO CONVERT FROM</u>	<u>TO OBTAIN</u>	
0.03937	millimeters	Inches	25.400
0.39370	Centimeters	Inches	2.5400
3.28080	Meters	Feet	0.3050
2.20460	kilograms	Pounds	0.4540
0.00098	kilogram's	tons(2240 lb.)	1016.1
0.98420	Metric tonnes (i.e. tonnes of 1000 kg.)	tons(2240 lb.)	1.0160
2.49980	Metric tonnes per centimeter of immersion	tons per inch (immersion)	0.4000
8.20140	moment to change trim one centimeter one (tonne meter units)	moment to change trim inch(foot ton units)	0.1220
187.976	Meter Radians	Feet degrees	0.0053
	<u>TO OBTAIN</u>	<u>TO CONVERT FROM</u>	<u>MULTIPLY</u>

Relation between Weight and volume

1000mm cubed	1 Cubic Centimeters
1 Cubic Centimeter of Fresh Water (S.G.1.0)	1 Gram
1000 cubic centimeter of fresh Water (S.G.1.0)	1 Kilogram (1000 Gm.)
1 Cubic Meter of fresh water (S.G.1.0)	1 Tonne (1000 Kg.)
1 Cubic Meter of fresh water (S.G.1.0)	1 tonne
1 Cubic Meter of Salt water (S.G. 1.025)	1.025 Tonnes
1 Tonne of Salt Water (S.G.1.025)	0.975 Cubic Meters

Angle of Downflooding

1. The angle of flooding is the angle of immersion of the sill or coaming of the opening on or above the freeboard deck through which progressive downflooding can occur as the vessel heels.
2. If this opening has a weathertight closing device which cannot be kept closed and secured at all times while the vessel is underway (e.g. engine room air supply trunk), or has no weathertight closing device, progressive downflooding will occur when the sill of the opening immerses. This angle is to be marked on the GZ curve if it comes within the apparent positive range of the GZ curve, and the GZ curve terminates at this angle: i.e., the actual angle of flooding becomes the range of the GZ curve.
3. If an opening is fitted with a weathertight closing device which can be kept closed and secured at all times while the vessel is underway, progressive downflooding can only occur through the opening if its closing device is not closed and secured. The angle of immersion of the lower sill of this opening is a 'potential' angle of flooding. If the closing appliance is not closed and secured, downflooding will occur when the lower sill of this opening reaches the water level.
If the lower sill of the opening through which downflooding can occur immerses at 40 degree or smaller angles of heel, its closing appliances must be kept closed and secured at all times when the vessel is underway or at sea where this opening is used for access in the normal working of the vessel, an alternate access should be used in heavy weather. In moderate weather, the opening may be used for access, but the closing appliances must be closed and secured again immediately after use.

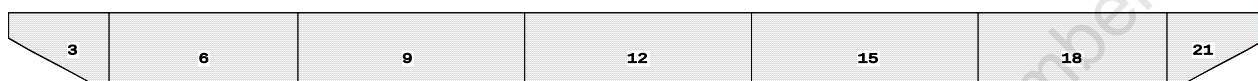
As there is no any unprotected opening exists above freeboard deck, hence the downflooding angle is considered at 50 deg.

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GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONT00N

CAPACITY PLAN AND TABLE

Profile View



Plan View

1	4	7	10	13	16	19
2	5	8	11	14	17	20
3	6	9	12	15	18	21

Tanks

1 01_WBTK.P	4 02_WBTK.P	8 03_WBTK.C	12 04_WBTK.S	16 06_WBTK.P	20 07_WBTK.C
2 01_WBTK.C	5 02_WBTK.C	9 03_WBTK.S	13 05_WBTK.P	17 06_WBTK.C	21 07_WBTK.S
3 01_WBTK.S	6 02_WBTK.S	10 04_WBTK.P	14 05_WBTK.C	18 06_WBTK.S	
	7 03_WBTK.P	11 04_WBTK.C	15 05_WBTK.S	19 07_WBTK.P	

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GHS 17.22 PONT00N

TANK STATUS							
Trim: zero, Heel: zero							
Part-----	Cu.M.----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
01_WBTK.P	48.4	1.025	49.60	2.340f	5.971p	1.880	72.32*
01_WBTK.C	48.9	1.025	50.11	2.336f	0.000	1.872	72.33*
01_WBTK.S	48.4	1.025	49.60	2.340f	5.971s	1.880	72.32*
02_WBTK.P	131.4	1.025	134.72	7.749f	5.980p	1.509	135.61*
02_WBTK.C	132.4	1.025	135.67	7.749f	0.000	1.499	135.61*
02_WBTK.S	131.4	1.025	134.72	7.749f	5.980s	1.509	135.61*
03_WBTK.P	157.7	1.025	161.60	16.000f	5.980p	1.509	162.73*
03_WBTK.C	158.8	1.025	162.73	16.000f	0.000	1.500	162.73*
03_WBTK.S	157.7	1.025	161.60	16.000f	5.980s	1.509	162.73*
04_WBTK.P	157.7	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.C	158.8	1.025	162.73	25.000f	0.000	1.500	162.73*
04_WBTK.S	157.7	1.025	161.60	25.000f	5.980s	1.509	162.73*
05_WBTK.P	157.7	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.C	158.8	1.025	162.73	34.000f	0.000	1.500	162.73*
05_WBTK.S	157.7	1.025	161.60	34.000f	5.980s	1.509	162.73*
06_WBTK.P	131.4	1.025	134.72	42.251f	5.980p	1.509	135.61*
06_WBTK.C	132.4	1.025	135.67	42.251f	0.000	1.499	135.61*
06_WBTK.S	131.4	1.025	134.72	42.251f	5.980s	1.509	135.61*
07_WBTK.P	48.4	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	48.9	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	48.4	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			2,566.63	25.000f	0.000	1.549	2712.14
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

HYDROSTATIC PARTICULARS

31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

HYDROSTATIC PROPERTIES

Trim: Fwd 1.000/50.000, No Heel, Fixed VCG = 0.000

Draft@	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
25.000f	---Weight(MT)---	LCB----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT	
0.112	133.29	37.689f	0.175	5.02	33.490f	6.52	244.36	96.881	
0.212	188.30	36.107f	0.211	5.98	31.124f	10.90	289.46	82.456	
0.312	252.92	34.523f	0.246	6.94	28.751f	16.92	334.36	71.772	
0.412	327.16	32.938f	0.281	7.91	26.375f	24.82	379.30	63.564	
0.512	407.76	31.583f	0.319	8.13	26.034f	26.65	326.70	53.012	
0.612	489.54	30.651f	0.361	8.23	25.975f	27.51	280.91	45.226	
0.712	572.35	29.971f	0.406	8.33	25.923f	28.38	247.92	39.491	
0.812	656.10	29.452f	0.452	8.42	25.886f	29.26	222.97	35.068	
0.912	740.70	29.044f	0.499	8.50	25.870f	30.12	203.29	31.535	
1.012	826.02	28.717f	0.548	8.57	25.880f	30.81	186.46	28.540	
1.112	912.01	28.450f	0.597	8.63	25.878f	31.57	173.02	26.147	
1.212	998.69	28.226f	0.646	8.70	25.875f	32.35	161.94	24.171	
1.312	1,086.07	28.037f	0.696	8.78	25.872f	33.17	152.69	22.519	
1.412	1,174.15	27.875f	0.746	8.85	25.869f	34.03	144.90	21.124	
1.512	1,262.91	27.735f	0.797	8.91	25.870f	34.71	137.41	19.834	
1.612	1,352.12	27.609f	0.848	8.94	25.783f	35.14	129.91	18.700	
1.712	1,441.67	27.493f	0.899	8.97	25.694f	35.58	123.36	17.712	
1.812	1,531.58	27.384f	0.950	9.01	25.603f	36.03	117.61	16.848	
1.912	1,621.83	27.283f	1.000	9.05	25.508f	36.51	112.52	16.088	
2.012	1,712.42	27.187f	1.051	9.08	25.436f	36.88	107.68	15.374	
2.112	1,803.32	27.097f	1.102	9.11	25.351f	37.30	103.41	14.750	
2.212	1,894.54	27.011f	1.153	9.14	25.263f	37.74	99.58	14.190	
2.312	1,986.08	26.928f	1.203	9.17	25.174f	38.19	96.11	13.689	
2.412	2,077.95	26.849f	1.254	9.21	25.081f	38.65	92.98	13.238	
2.512	2,170.15	26.772f	1.305	9.23	25.013f	39.01	89.86	12.803	
2.612	2,258.72	26.664f	1.353	8.40	22.760f	29.59	65.48	11.408	
2.712	2,338.10	26.490f	1.395	7.48	20.257f	21.09	45.09	10.043	
2.812	2,408.27	26.272f	1.431	6.56	17.755f	14.46	30.01	8.792	
2.912	2,469.20	26.032f	1.462	5.63	15.252f	9.46	19.15	7.632	
3.012	2,520.92	25.786f	1.488	4.71	12.750f	5.86	11.62	6.543	

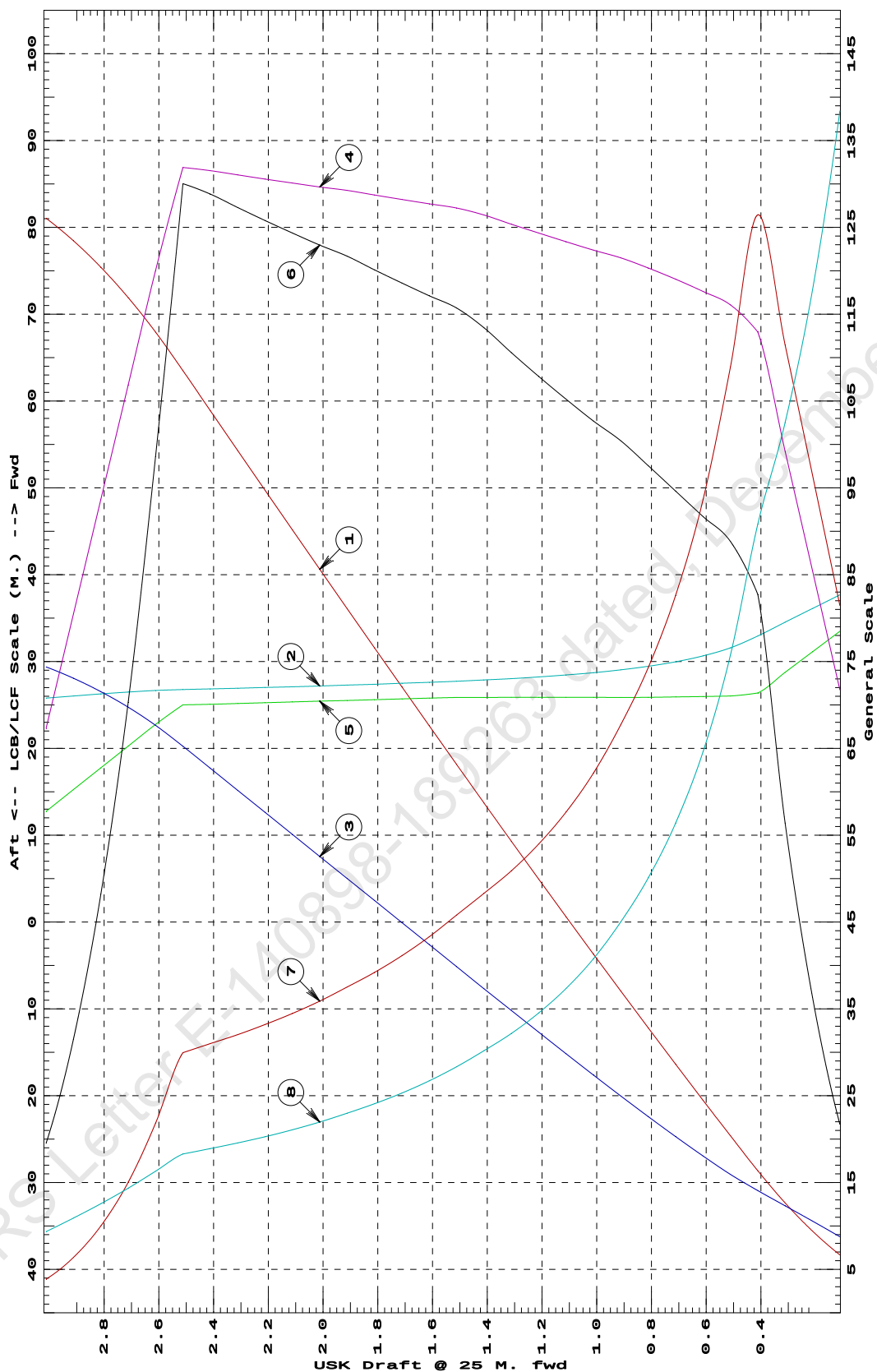
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

Trim is per 50.00m.

Draft is from USK.

31-07-20 12:18:46
GHS 17.22Cybermarine Technologies PTE. Ltd.
PONTON

HYDROSTATIC PROPERTIES at 1 M. FWD TRIM



- ① Displacement 1=20 MT
 ② LCB (use top scale)
 ③ VCB (KB) 1=.02 M.
 ④ Immersion 1=.07 MT/cm
 ④ WPA 1=6.83 Sq.M.
 ⑤ LCF (use top scale)
 ⑥ Moment/Trim 1=.3 M.-MT/cm
 ⑦ KML 1=3 M.
 ⑧ KMT 1=.7 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

HYDROSTATIC PROPERTIES

Trim: Fwd 0.500/50.000, No Heel, Fixed VCG = 0.000

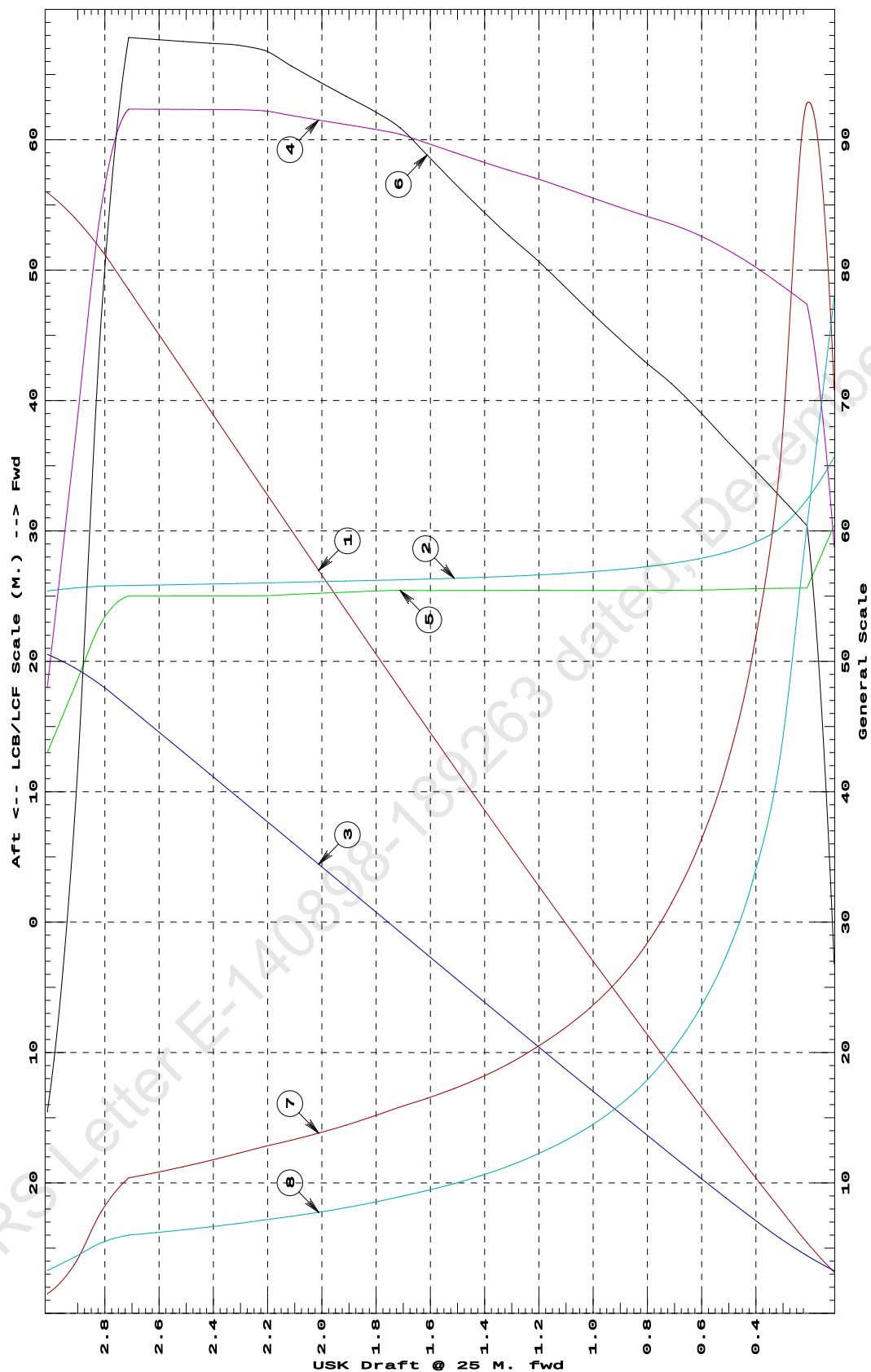
Draft@	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
25.000f	---Weight(MT)---	LCB----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT	
0.112	94.61	35.657f	0.098	5.88	30.439f	10.72	566.32	155.843	
0.212	162.65	32.417f	0.132	7.74	25.617f	24.15	742.47	120.816	
0.312	240.79	30.208f	0.173	7.89	25.602f	25.06	520.34	85.097	
0.412	320.44	29.059f	0.218	8.04	25.566f	25.94	404.70	66.389	
0.512	401.45	28.349f	0.266	8.16	25.509f	26.81	333.87	54.610	
0.612	483.61	27.862f	0.315	8.27	25.457f	27.70	286.38	46.395	
0.712	566.72	27.508f	0.365	8.35	25.449f	28.52	251.60	40.187	
0.812	650.57	27.242f	0.415	8.42	25.440f	29.21	224.49	35.343	
0.912	735.11	27.034f	0.466	8.49	25.439f	29.96	203.76	31.632	
1.012	820.35	26.868f	0.517	8.56	25.438f	30.74	187.34	28.689	
1.112	906.31	26.733f	0.567	8.63	25.436f	31.56	174.08	26.309	
1.212	992.99	26.620f	0.619	8.71	25.451f	32.35	162.89	24.309	
1.312	1,080.33	26.524f	0.670	8.77	25.439f	33.07	153.05	22.576	
1.412	1,168.34	26.443f	0.722	8.83	25.438f	33.86	144.89	21.132	
1.512	1,257.02	26.372f	0.773	8.90	25.437f	34.68	137.92	19.898	
1.612	1,346.39	26.310f	0.825	8.97	25.435f	35.53	131.95	18.836	
1.712	1,436.46	26.255f	0.877	9.04	25.448f	36.37	126.58	17.889	
1.812	1,527.11	26.206f	0.929	9.08	25.389f	36.90	120.81	16.991	
1.912	1,618.11	26.157f	0.981	9.12	25.304f	37.34	115.37	16.198	
2.012	1,709.44	26.109f	1.033	9.15	25.217f	37.79	110.53	15.496	
2.112	1,801.11	26.062f	1.085	9.19	25.127f	38.26	106.21	14.872	
2.212	1,893.13	26.014f	1.137	9.22	25.034f	38.75	102.33	14.317	
2.312	1,985.42	25.968f	1.189	9.23	25.012f	38.91	97.97	13.759	
2.412	2,077.73	25.925f	1.240	9.23	25.009f	38.96	93.75	13.255	
2.512	2,170.06	25.886f	1.292	9.23	25.006f	39.02	89.90	12.799	
2.612	2,262.39	25.850f	1.343	9.23	25.004f	39.08	86.36	12.384	
2.712	2,354.73	25.817f	1.394	9.24	25.001f	39.14	83.10	12.006	
2.812	2,445.61	25.772f	1.445	8.50	23.000f	30.67	62.71	10.846	
2.912	2,521.33	25.617f	1.485	6.65	18.000f	15.11	29.97	8.622	
3.012	2,578.59	25.395f	1.516	4.80	13.000f	6.19	12.01	6.555	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

31-07-20 12:18:46
GHS 17.22Cybermarine Technologies PTE. Ltd.
PONTON

HYDROSTATIC PROPERTIES at 0.5 M. FWD TRIM



- ① Displacement 1=30 MT
 ② LCB (use top scale)
 ③ VCB (KB) 1=.03 M.
 ④ Immersion 1=.1 MT/cm
 ⑤ WPA 1=9.76 Sq.M.
 ⑥ LCF (use top scale)
 ⑦ Moment/Trim 1=.4 M.-MT/cm
 ⑧ KML 1=8 M.
 ⑨ KMT 1=2 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

HYDROSTATIC PROPERTIES
No Trim, No Heel, Fixed VCG = 0.000

Draft@	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
25.000f	---Weight(MT)---	LCB----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT	
0.112	83.79	25.000f	0.044	7.57	25.000f	23.17	1382.58	223.643	
0.212	160.29	25.000f	0.095	7.73	25.000f	24.11	752.15	122.246	
0.312	238.38	25.000f	0.145	7.89	25.000f	25.08	526.13	85.934	
0.412	318.09	25.000f	0.196	8.06	25.000f	26.09	410.04	67.317	
0.512	399.40	25.000f	0.248	8.20	25.000f	26.98	337.80	55.685	
0.612	481.78	25.000f	0.300	8.27	25.000f	27.70	287.43	46.641	
0.712	564.87	25.000f	0.351	8.35	25.000f	28.44	251.70	40.223	
0.812	648.69	25.000f	0.403	8.42	25.000f	29.21	225.14	35.449	
0.912	733.24	25.000f	0.454	8.50	25.000f	30.02	204.72	31.774	
1.012	818.51	25.000f	0.506	8.56	25.000f	30.72	187.65	28.736	
1.112	904.44	25.000f	0.558	8.63	25.000f	31.47	173.99	26.304	
1.212	991.06	25.000f	0.610	8.70	25.000f	32.26	162.74	24.297	
1.312	1,078.37	25.000f	0.661	8.77	25.000f	33.07	153.35	22.621	
1.412	1,166.40	25.000f	0.713	8.84	25.000f	33.93	145.46	21.208	
1.512	1,255.11	25.000f	0.765	8.90	25.000f	34.66	138.07	19.915	
1.612	1,344.46	25.000f	0.818	8.97	25.000f	35.45	131.83	18.822	
1.712	1,434.46	25.000f	0.870	9.03	25.000f	36.27	126.41	17.872	
1.812	1,525.13	25.000f	0.922	9.10	25.000f	37.12	121.69	17.042	
1.912	1,616.48	25.000f	0.974	9.17	25.000f	38.01	117.57	16.315	
2.012	1,708.51	25.000f	1.027	9.23	25.000f	38.71	113.30	15.610	
2.112	1,800.77	25.000f	1.079	9.23	25.000f	38.78	107.66	14.924	
2.212	1,893.06	25.000f	1.132	9.23	25.000f	38.84	102.58	14.309	
2.312	1,985.36	25.000f	1.184	9.23	25.000f	38.90	97.98	13.756	
2.412	2,077.68	25.000f	1.236	9.23	25.000f	38.97	93.78	13.257	
2.512	2,170.03	25.000f	1.287	9.24	25.000f	39.04	89.95	12.804	
2.612	2,262.38	25.000f	1.339	9.24	25.000f	39.09	86.38	12.385	
2.712	2,354.73	25.000f	1.390	9.24	25.000f	39.14	83.10	12.003	
2.812	2,447.09	25.000f	1.442	9.24	25.000f	39.19	80.07	11.654	
2.912	2,539.44	25.000f	1.493	9.24	25.000f	39.24	77.26	11.334	
3.012	2,631.79	25.000f	1.544	9.24	25.000f	39.29	74.65	11.040	

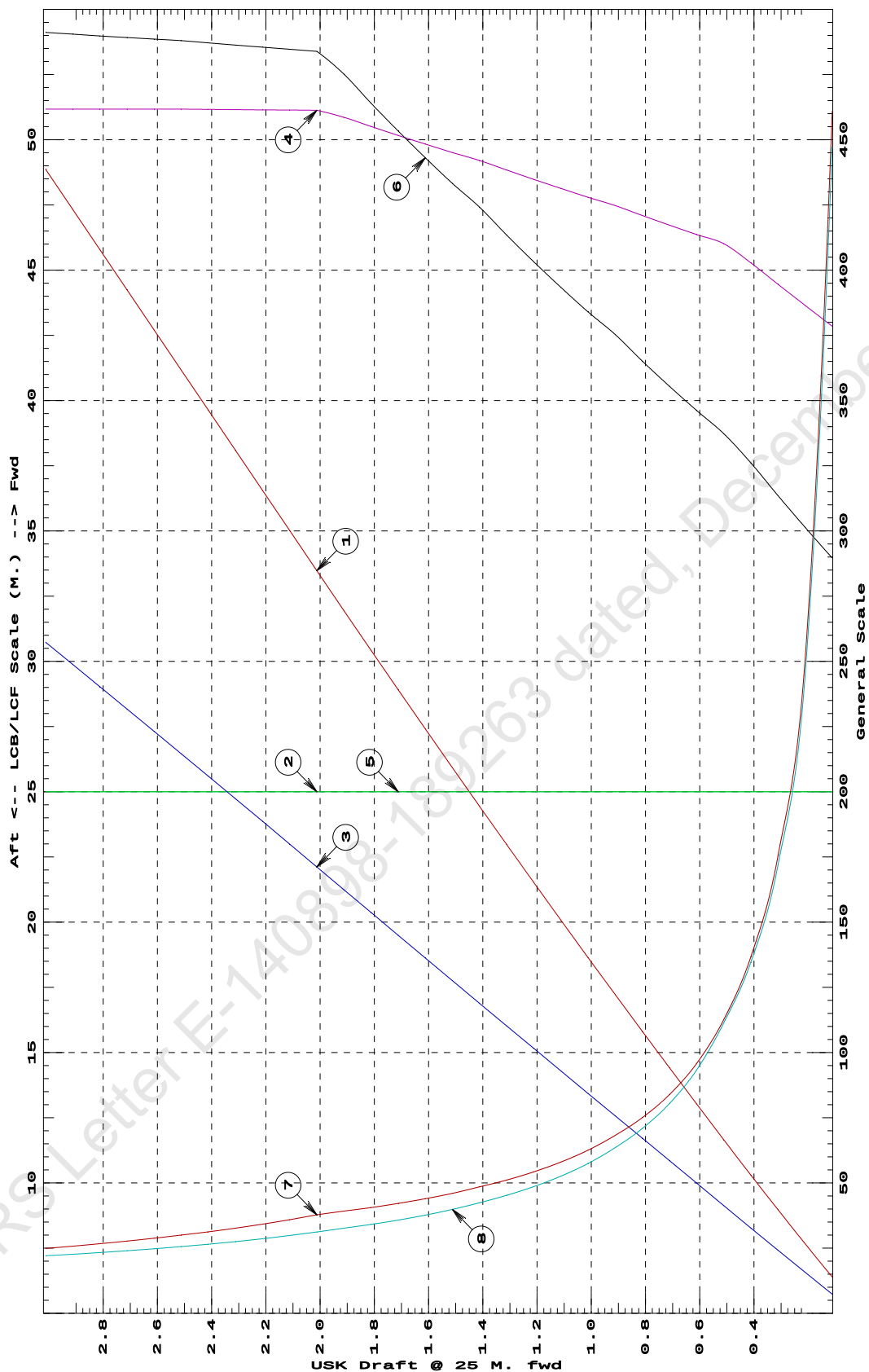
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

31-07-20 12:18:46
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTON

HYDROSTATIC PROPERTIES at LEVEL TRIM



- ① Displacement 1=6 MT
- ② LCB (use top scale)
- ③ VCB (KB) 1=.006 M.
- ④ Immersion 1=.02 MT/cm
- ④ WPA 1=1.95 Sq.M.
- ⑤ LCF (use top scale)
- ⑥ Moment/Trim 1=.08 M.-MT/cm
- ⑦ KML 1=3 M.
- ⑧ KMT 1=.5 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

HYDROSTATIC PROPERTIES

Trim: Aft 0.500/50.000, No Heel, Fixed VCG = 0.000

Draft@	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
25.000f	---Weight(MT)---	LCB----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT	
0.112	94.61	14.343f	0.098	5.88	19.561f	10.72	566.32	155.836	
0.212	162.65	17.583f	0.132	7.74	24.383f	24.15	742.47	120.811	
0.312	240.79	19.792f	0.173	7.89	24.398f	25.06	520.34	85.098	
0.412	320.44	20.941f	0.218	8.04	24.434f	25.94	404.70	66.390	
0.512	401.45	21.651f	0.266	8.16	24.491f	26.81	333.87	54.611	
0.612	483.61	22.138f	0.315	8.27	24.543f	27.70	286.38	46.395	
0.712	566.72	22.492f	0.365	8.35	24.551f	28.52	251.60	40.186	
0.812	650.57	22.758f	0.415	8.42	24.560f	29.21	224.49	35.343	
0.912	735.11	22.966f	0.466	8.49	24.561f	29.96	203.76	31.632	
1.012	820.35	23.132f	0.517	8.56	24.562f	30.74	187.34	28.689	
1.112	906.31	23.267f	0.567	8.63	24.564f	31.56	174.08	26.309	
1.212	992.99	23.380f	0.619	8.71	24.549f	32.35	162.89	24.308	
1.312	1,080.33	23.476f	0.670	8.77	24.561f	33.07	153.05	22.576	
1.412	1,168.34	23.557f	0.722	8.83	24.562f	33.86	144.89	21.132	
1.512	1,257.02	23.628f	0.773	8.90	24.563f	34.68	137.92	19.898	
1.612	1,346.39	23.690f	0.825	8.97	24.565f	35.53	131.95	18.836	
1.712	1,436.46	23.745f	0.877	9.04	24.552f	36.37	126.58	17.889	
1.812	1,527.11	23.794f	0.929	9.08	24.611f	36.90	120.81	16.991	
1.912	1,618.10	23.843f	0.981	9.12	24.696f	37.34	115.37	16.198	
2.012	1,709.44	23.891f	1.033	9.15	24.783f	37.79	110.53	15.496	
2.112	1,801.11	23.938f	1.085	9.19	24.873f	38.26	106.21	14.872	
2.212	1,893.13	23.986f	1.137	9.22	24.966f	38.75	102.33	14.317	
2.312	1,985.42	24.032f	1.189	9.23	24.988f	38.91	97.97	13.758	
2.412	2,077.73	24.075f	1.240	9.23	24.991f	38.96	93.75	13.255	
2.512	2,170.06	24.114f	1.292	9.23	24.994f	39.02	89.90	12.799	
2.612	2,262.39	24.150f	1.343	9.23	24.996f	39.08	86.36	12.384	
2.712	2,354.73	24.183f	1.394	9.24	24.999f	39.14	83.10	12.006	
2.812	2,445.61	24.228f	1.445	8.50	27.000f	30.67	62.71	10.846	
2.912	2,521.33	24.383f	1.485	6.65	32.000f	15.11	29.97	8.622	
3.012	2,578.59	24.605f	1.516	4.80	37.000f	6.19	12.01	6.555	

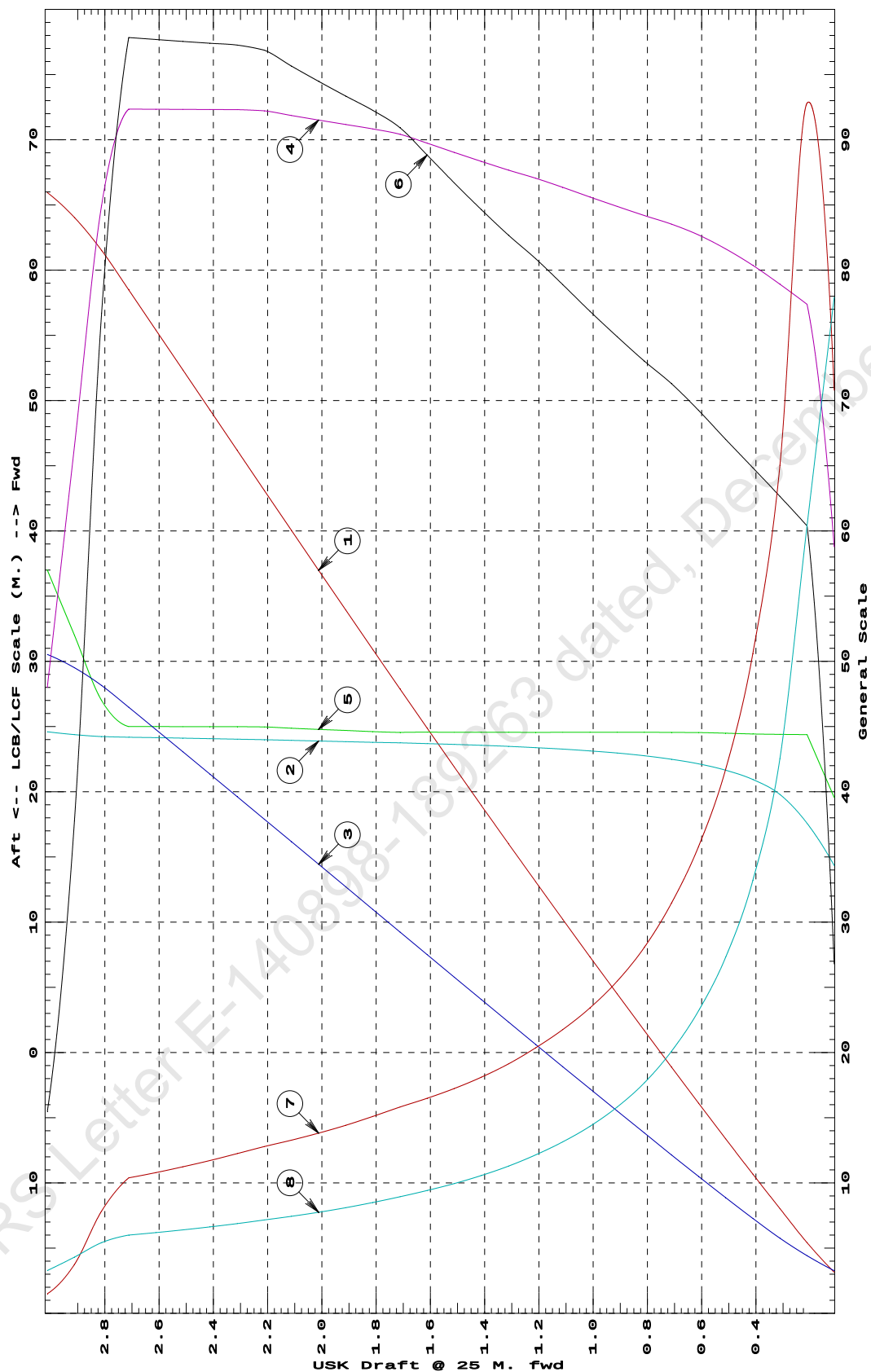
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

Trim is per 50.00m.

Draft is from USK.

31-07-20 12:18:46
GHS 17.22Cybermarine Technologies PTE. Ltd.
PONTON

HYDROSTATIC PROPERTIES at 0.5 M. AFT TRIM



31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

HYDROSTATIC PROPERTIES

Trim: Aft 1.000/50.000, No Heel, Fixed VCG = 0.000

Draft@ 25.000f	Displacement ---Weight(MT)---	Buoyancy-Ctr. ----LCB-----VCB-----	Weight/ -----cm-----	Moment/ cm trim-----KML-----KMT
0.112	133.29	12.311f 0.175	5.02 16.510f	6.52 244.36 96.880
0.212	188.30	13.893f 0.211	5.98 18.876f	10.90 289.46 82.454
0.312	252.92	15.477f 0.246	6.94 21.249f	16.92 334.36 71.771
0.412	327.16	17.062f 0.281	7.91 23.625f	24.82 379.31 63.562
0.512	407.76	18.417f 0.319	8.13 23.966f	26.65 326.70 53.012
0.612	489.54	19.349f 0.361	8.23 24.026f	27.51 280.91 45.226
0.712	572.35	20.029f 0.406	8.33 24.077f	28.38 247.92 39.491
0.812	656.10	20.548f 0.452	8.42 24.114f	29.26 222.97 35.069
0.912	740.70	20.956f 0.499	8.50 24.130f	30.12 203.29 31.535
1.012	826.02	21.283f 0.548	8.57 24.120f	30.81 186.46 28.540
1.112	912.01	21.550f 0.597	8.63 24.122f	31.57 173.02 26.147
1.212	998.69	21.774f 0.646	8.70 24.125f	32.35 161.94 24.171
1.312	1,086.07	21.963f 0.696	8.78 24.128f	33.17 152.69 22.519
1.412	1,174.15	22.125f 0.746	8.85 24.131f	34.03 144.90 21.125
1.512	1,262.91	22.265f 0.797	8.91 24.130f	34.71 137.41 19.834
1.612	1,352.12	22.391f 0.848	8.94 24.217f	35.14 129.91 18.700
1.712	1,441.67	22.507f 0.899	8.97 24.306f	35.58 123.36 17.712
1.812	1,531.58	22.616f 0.950	9.01 24.397f	36.03 117.61 16.848
1.912	1,621.83	22.717f 1.000	9.05 24.492f	36.51 112.52 16.088
2.012	1,712.42	22.813f 1.051	9.08 24.564f	36.88 107.68 15.374
2.112	1,803.32	22.903f 1.102	9.11 24.649f	37.30 103.41 14.749
2.212	1,894.54	22.989f 1.153	9.14 24.737f	37.74 99.58 14.190
2.312	1,986.08	23.072f 1.203	9.17 24.826f	38.19 96.11 13.689
2.412	2,077.95	23.151f 1.254	9.21 24.919f	38.65 92.98 13.238
2.512	2,170.15	23.228f 1.305	9.23 24.987f	39.01 89.86 12.803
2.612	2,258.72	23.336f 1.353	8.40 27.240f	29.59 65.48 11.408
2.712	2,338.10	23.510f 1.395	7.48 29.743f	21.09 45.09 10.042
2.812	2,408.27	23.728f 1.431	6.56 32.245f	14.46 30.01 8.792
2.912	2,469.21	23.968f 1.462	5.63 34.748f	9.46 19.15 7.632
3.012	2,520.92	24.214f 1.488	4.71 37.250f	5.86 11.62 6.543

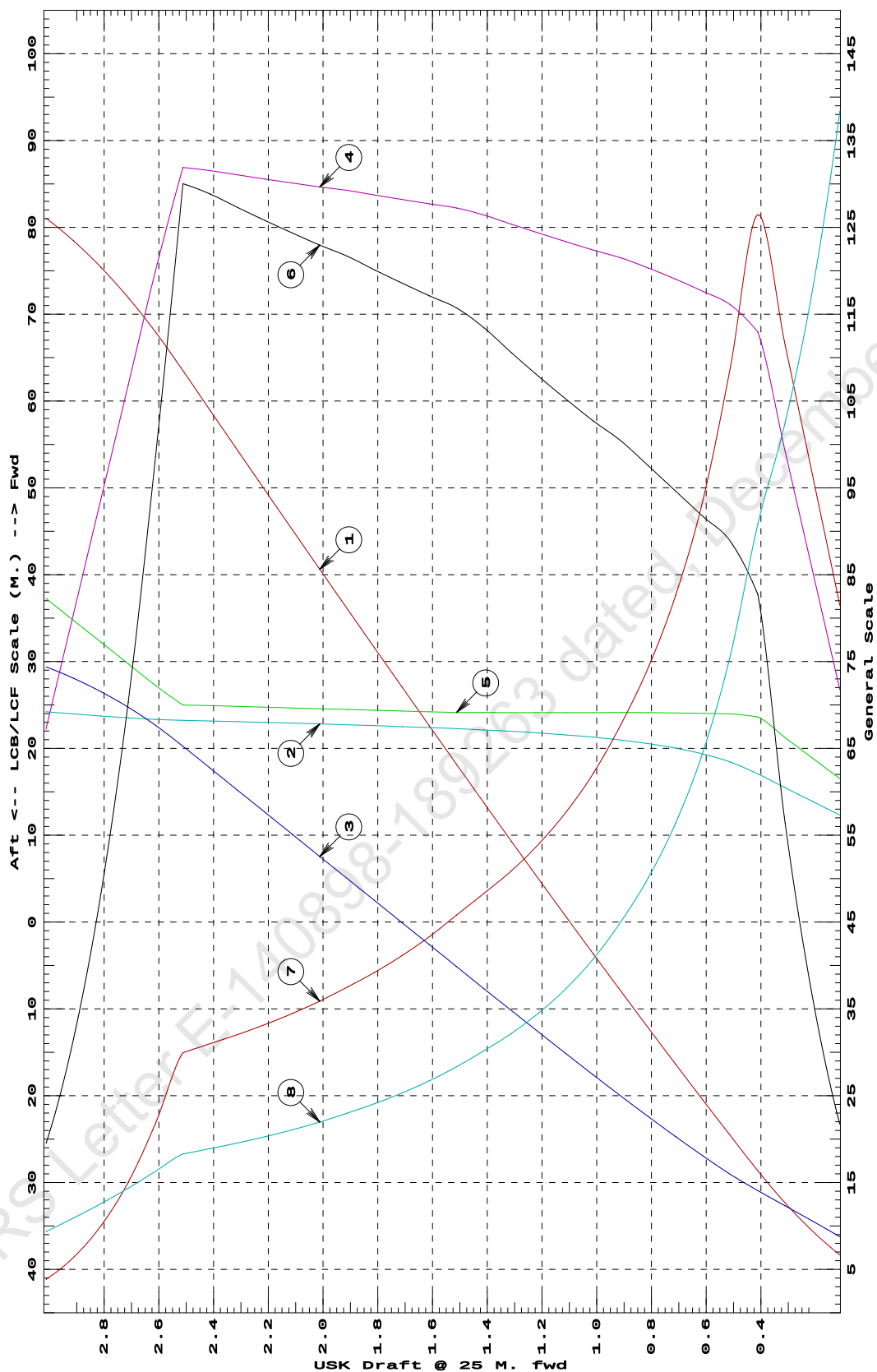
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

Trim is per 50.00m.

Draft is from USK.

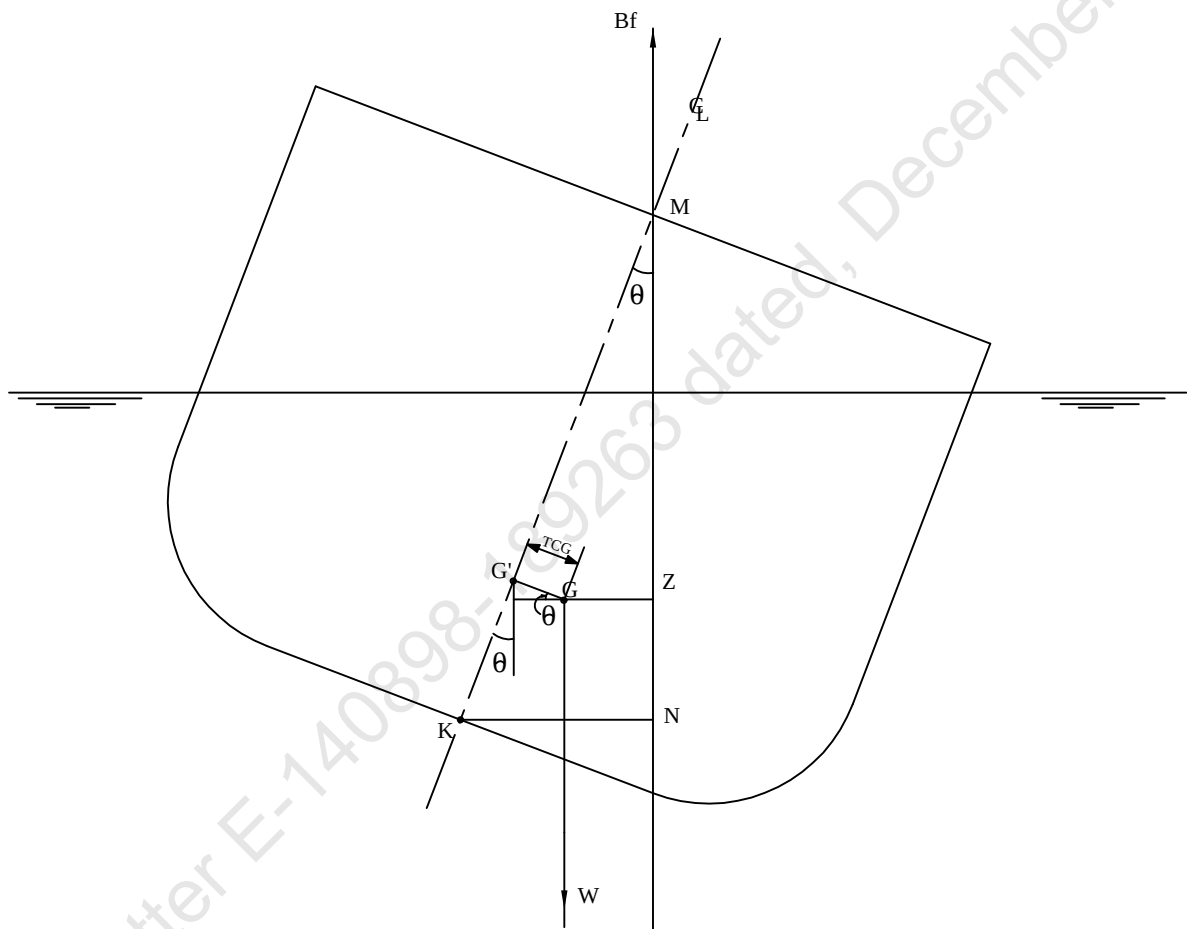
31-07-20 12:18:46
GHS 17.22Cybermarine Technologies PTE. Ltd.
PONTON

HYDROSTATIC PROPERTIES at 1 M. AFT TRIM



Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

KN CONCEPT



$$\underline{GZ = KN - KG' \sin \theta - TCG \cos \theta}$$

01-08-20 21:16:24
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONT00N

VOLUME USED FOR KN COMPUTATION

Profile View



Plan View



CROSS CURVE TABLES

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

CROSS CURVES OF STABILITY

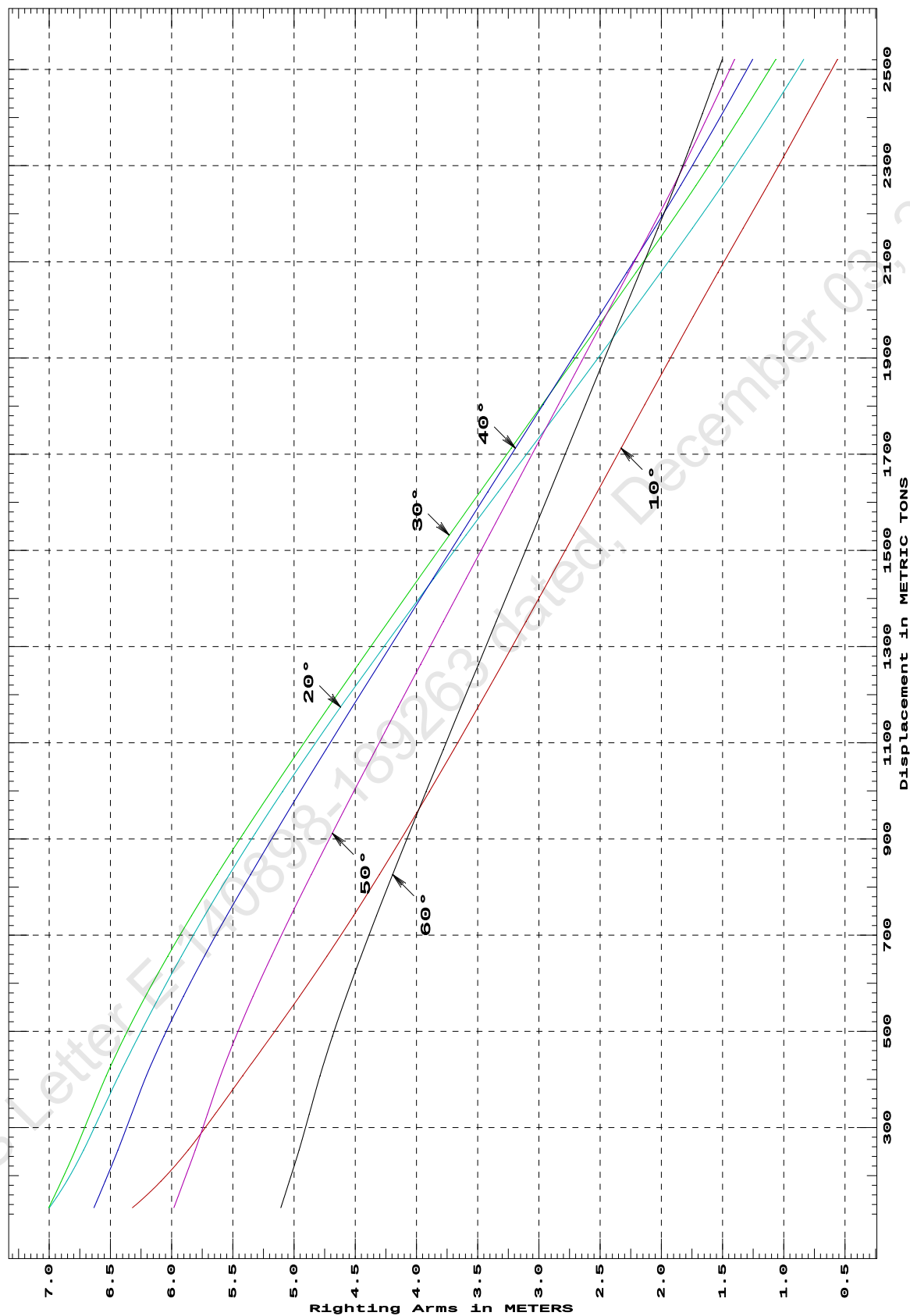
Showing righting arms in heel at VCG = 0.00

Trim: Fwd 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
133.29	6.320s	6.999s	7.003s	6.634s	5.981s	5.107s
188.30	6.087s	6.857s	6.900s	6.542s	5.901s	5.038s
252.92	5.865s	6.719s	6.786s	6.438s	5.806s	4.959s
327.16	5.646s	6.581s	6.666s	6.327s	5.707s	4.876s
407.76	5.419s	6.431s	6.534s	6.205s	5.600s	4.788s
489.54	5.186s	6.270s	6.382s	6.063s	5.474s	4.686s
572.35	4.956s	6.100s	6.214s	5.904s	5.335s	4.574s
656.10	4.731s	5.920s	6.031s	5.732s	5.184s	4.453s
740.70	4.513s	5.729s	5.836s	5.548s	5.025s	4.326s
826.02	4.303s	5.528s	5.629s	5.355s	4.859s	4.194s
912.01	4.097s	5.316s	5.412s	5.155s	4.687s	4.059s
998.69	3.895s	5.094s	5.186s	4.949s	4.512s	3.921s
1,086.07	3.695s	4.861s	4.955s	4.739s	4.333s	3.781s
1,174.15	3.497s	4.620s	4.717s	4.524s	4.151s	3.638s
1,262.91	3.300s	4.371s	4.475s	4.306s	3.966s	3.494s
1,352.12	3.105s	4.116s	4.229s	4.086s	3.781s	3.349s
1,441.67	2.910s	3.858s	3.982s	3.864s	3.594s	3.203s
1,531.58	2.716s	3.596s	3.732s	3.641s	3.406s	3.057s
1,621.83	2.523s	3.331s	3.480s	3.417s	3.218s	2.911s
1,712.42	2.329s	3.065s	3.227s	3.192s	3.029s	2.765s
1,803.32	2.134s	2.798s	2.972s	2.965s	2.840s	2.618s
1,894.54	1.938s	2.532s	2.717s	2.738s	2.649s	2.471s
1,986.08	1.740s	2.268s	2.462s	2.511s	2.459s	2.323s
2,077.95	1.538s	2.006s	2.207s	2.283s	2.268s	2.175s
2,170.15	1.332s	1.746s	1.954s	2.056s	2.077s	2.027s
2,258.72	1.132s	1.503s	1.718s	1.843s	1.897s	1.888s
2,338.10	0.958s	1.294s	1.514s	1.660s	1.743s	1.767s
2,408.27	0.806s	1.115s	1.338s	1.502s	1.610s	1.664s
2,469.20	0.674s	0.963s	1.190s	1.368s	1.497s	1.576s
2,520.92	0.561s	0.836s	1.066s	1.255s	1.402s	1.502s
Distances in METERS.---Specific Gravity = 1.025.-----						

31-07-20 12:20:24
GHS 17.22Cybermarine Technologies PTE. Ltd.
PONTON

CROSS CURVES OF STABILITY - Stbd Heel
at 1 M. FWD TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

CROSS CURVES OF STABILITY

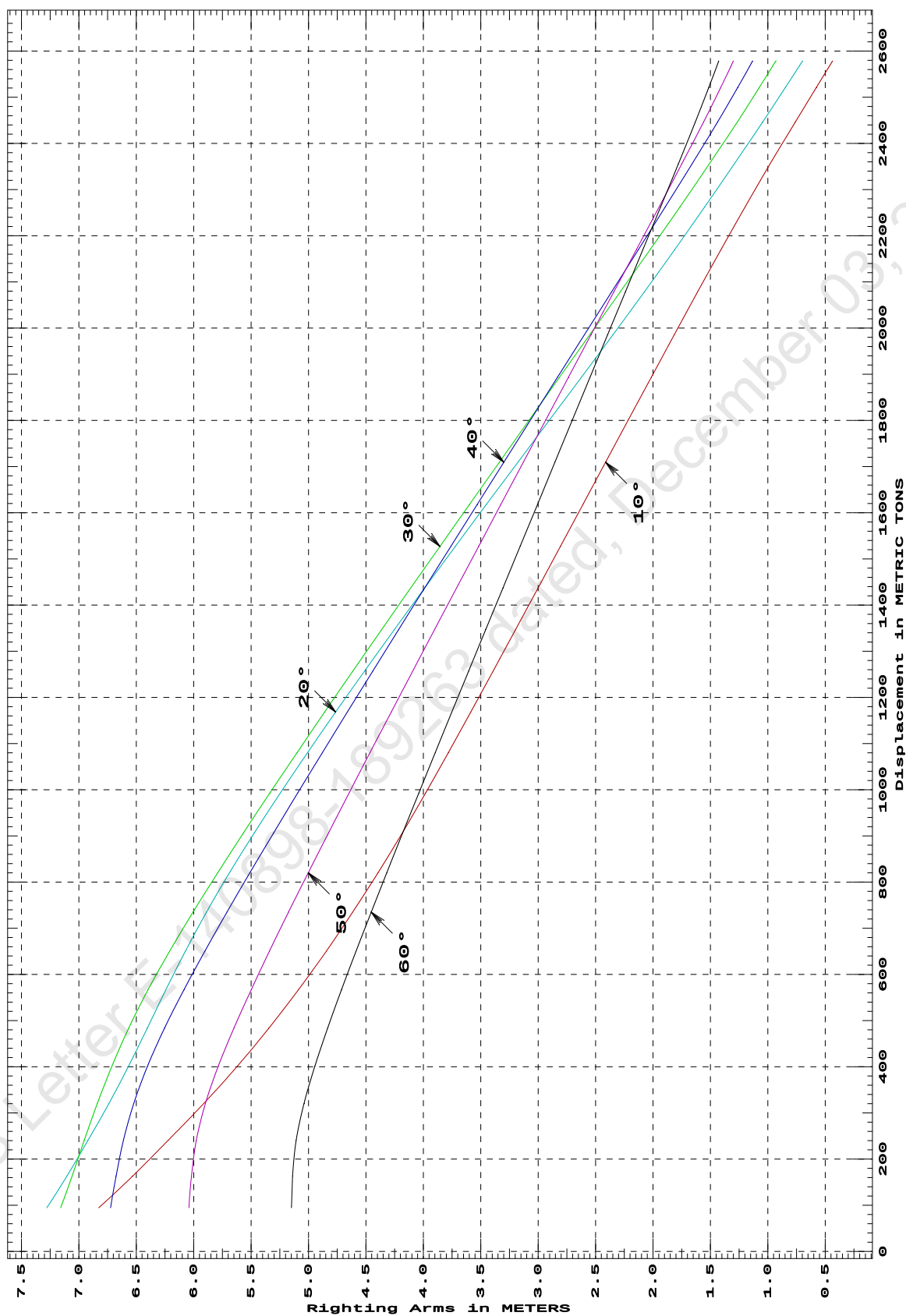
Showing righting arms in heel at VCG = 0.00

Trim: Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
94.61	6.827s	7.278s	7.158s	6.723s	6.041s	5.149s
162.65	6.533s	7.105s	7.059s	6.676s	6.020s	5.139s
240.79	6.221s	6.920s	6.953s	6.612s	5.977s	5.107s
320.44	5.912s	6.738s	6.839s	6.522s	5.897s	5.039s
401.45	5.619s	6.567s	6.712s	6.401s	5.785s	4.944s
483.61	5.346s	6.404s	6.566s	6.254s	5.649s	4.833s
566.72	5.088s	6.242s	6.397s	6.085s	5.499s	4.712s
650.57	4.845s	6.071s	6.209s	5.902s	5.340s	4.585s
735.11	4.615s	5.887s	6.005s	5.710s	5.173s	4.454s
820.35	4.395s	5.688s	5.790s	5.510s	5.002s	4.318s
906.31	4.184s	5.473s	5.567s	5.305s	4.826s	4.180s
992.99	3.980s	5.246s	5.336s	5.094s	4.647s	4.039s
1,080.33	3.782s	5.009s	5.100s	4.880s	4.465s	3.896s
1,168.34	3.587s	4.764s	4.859s	4.662s	4.280s	3.752s
1,257.02	3.393s	4.512s	4.614s	4.441s	4.093s	3.605s
1,346.39	3.198s	4.253s	4.364s	4.217s	3.903s	3.457s
1,436.46	3.003s	3.990s	4.111s	3.990s	3.712s	3.308s
1,527.11	2.807s	3.722s	3.856s	3.761s	3.519s	3.157s
1,618.11	2.610s	3.451s	3.598s	3.530s	3.325s	3.006s
1,709.44	2.413s	3.177s	3.338s	3.299s	3.130s	2.854s
1,801.11	2.215s	2.901s	3.077s	3.066s	2.934s	2.702s
1,893.13	2.016s	2.624s	2.815s	2.833s	2.738s	2.549s
1,985.42	1.816s	2.349s	2.551s	2.598s	2.540s	2.396s
2,077.73	1.613s	2.077s	2.287s	2.363s	2.343s	2.242s
2,170.06	1.408s	1.811s	2.025s	2.128s	2.145s	2.088s
2,262.39	1.199s	1.551s	1.766s	1.894s	1.947s	1.934s
2,354.73	0.983s	1.295s	1.512s	1.662s	1.751s	1.781s
2,445.61	0.763s	1.048s	1.268s	1.440s	1.562s	1.633s
2,521.33	0.578s	0.848s	1.074s	1.263s	1.411s	1.515s
2,578.59	0.436s	0.699s	0.931s	1.133s	1.301s	1.428s
Distances in METERS.---Specific Gravity = 1.025.-----						

31-07-20 12:20:24
GHS 17.22Cybermarine Technologies PTE. Ltd.
PONTON

CROSS CURVES OF STABILITY - Stbd Heel
at 0.5 M. FWD TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

CROSS CURVES OF STABILITY

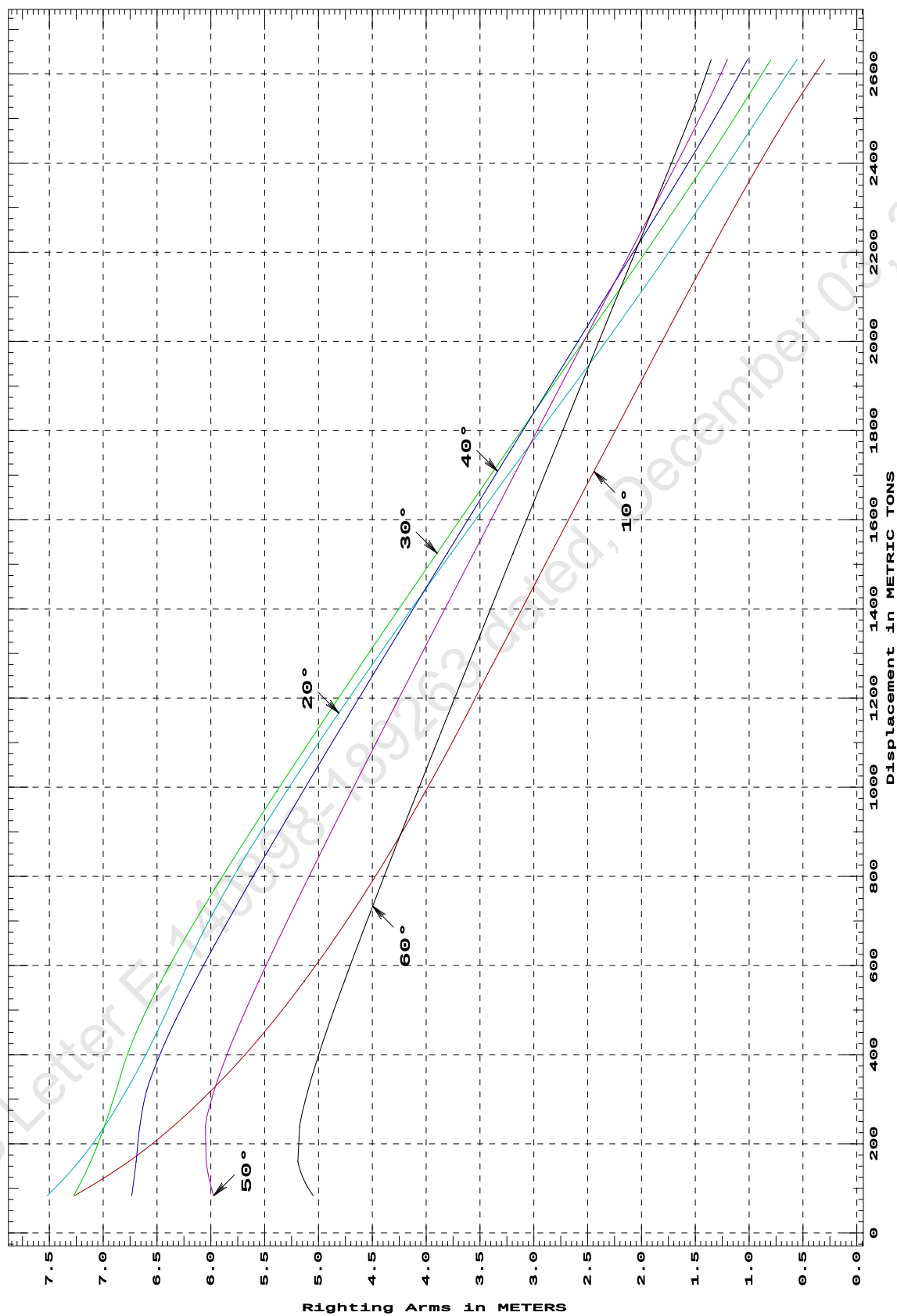
Showing righting arms in heel at VCG = 0.00

Trim: zero at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
83.79	7.268s	7.519s	7.274s	6.735s	5.973s	5.049s
160.29	6.763s	7.226s	7.116s	6.694s	6.043s	5.192s
238.38	6.354s	6.990s	6.990s	6.660s	6.051s	5.176s
318.09	6.003s	6.788s	6.881s	6.597s	5.971s	5.100s
399.40	5.685s	6.606s	6.776s	6.474s	5.851s	4.999s
481.78	5.397s	6.438s	6.634s	6.319s	5.709s	4.883s
564.87	5.132s	6.283s	6.462s	6.146s	5.555s	4.760s
648.69	4.884s	6.127s	6.269s	5.960s	5.392s	4.630s
733.24	4.648s	5.945s	6.061s	5.764s	5.224s	4.497s
818.51	4.425s	5.743s	5.844s	5.562s	5.050s	4.360s
904.44	4.211s	5.525s	5.619s	5.355s	4.873s	4.221s
991.06	4.006s	5.297s	5.387s	5.143s	4.693s	4.080s
1,078.37	3.807s	5.058s	5.149s	4.927s	4.509s	3.936s
1,166.40	3.615s	4.812s	4.907s	4.708s	4.324s	3.791s
1,255.11	3.427s	4.559s	4.660s	4.486s	4.135s	3.643s
1,344.46	3.231s	4.300s	4.410s	4.261s	3.945s	3.495s
1,434.46	3.036s	4.035s	4.156s	4.033s	3.753s	3.345s
1,525.13	2.840s	3.767s	3.899s	3.803s	3.559s	3.193s
1,616.48	2.642s	3.493s	3.640s	3.571s	3.363s	3.040s
1,708.51	2.442s	3.216s	3.377s	3.336s	3.165s	2.886s
1,800.77	2.241s	2.937s	3.113s	3.100s	2.966s	2.731s
1,893.06	2.041s	2.656s	2.848s	2.864s	2.767s	2.576s
1,985.36	1.840s	2.374s	2.582s	2.627s	2.568s	2.421s
2,077.68	1.638s	2.100s	2.316s	2.390s	2.369s	2.265s
2,170.03	1.433s	1.832s	2.049s	2.153s	2.169s	2.110s
2,262.38	1.225s	1.570s	1.786s	1.916s	1.969s	1.954s
2,354.73	1.010s	1.314s	1.529s	1.679s	1.769s	1.798s
2,447.09	0.789s	1.063s	1.280s	1.449s	1.571s	1.642s
2,539.44	0.554s	0.813s	1.038s	1.228s	1.380s	1.491s
2,631.79	0.298s	0.558s	0.799s	1.016s	1.204s	1.354s
Distances in METERS.---Specific Gravity = 1.025.-----						

31-07-20 12:20:24
GHS 17.22Cybermarine Technologies PTE. Ltd.
PONTON

CROSS CURVES OF STABILITY - Stbd Heel
at LEVEL TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

CROSS CURVES OF STABILITY

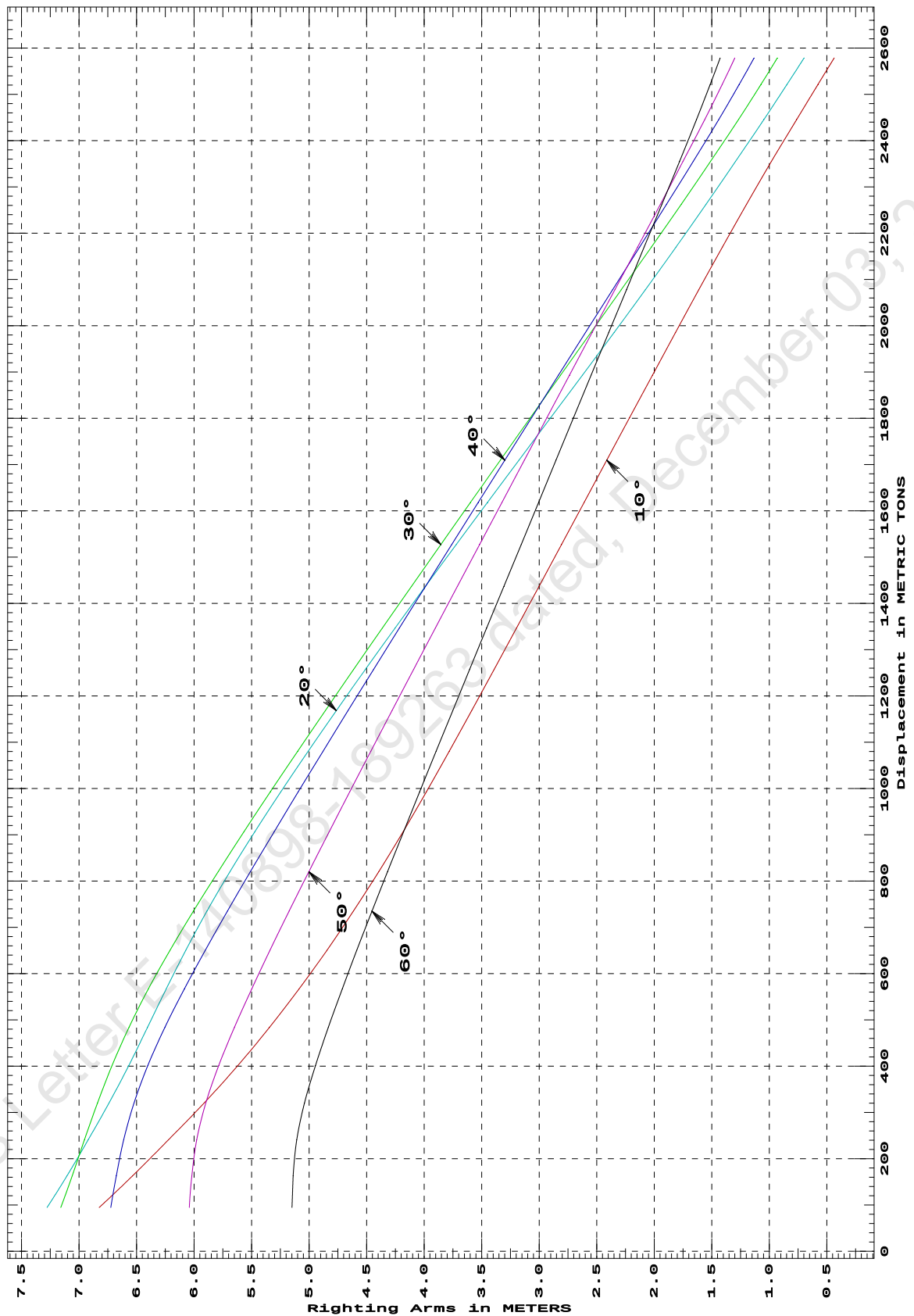
Showing righting arms in heel at VCG = 0.00

Trim: Aft 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
94.61	6.824s	7.277s	7.158s	6.723s	6.041s	5.149s
162.65	6.535s	7.105s	7.059s	6.676s	6.020s	5.139s
240.79	6.222s	6.920s	6.953s	6.612s	5.977s	5.107s
320.44	5.912s	6.738s	6.839s	6.522s	5.897s	5.040s
401.45	5.619s	6.567s	6.712s	6.401s	5.785s	4.944s
483.61	5.346s	6.405s	6.566s	6.254s	5.649s	4.833s
566.72	5.088s	6.242s	6.397s	6.085s	5.499s	4.712s
650.57	4.845s	6.071s	6.209s	5.902s	5.340s	4.585s
735.11	4.615s	5.887s	6.005s	5.710s	5.173s	4.454s
820.35	4.395s	5.688s	5.790s	5.510s	5.002s	4.318s
906.31	4.184s	5.473s	5.567s	5.305s	4.826s	4.180s
992.99	3.980s	5.246s	5.336s	5.094s	4.647s	4.039s
1,080.33	3.782s	5.009s	5.100s	4.880s	4.465s	3.896s
1,168.34	3.587s	4.764s	4.859s	4.662s	4.280s	3.752s
1,257.02	3.393s	4.511s	4.614s	4.441s	4.093s	3.605s
1,346.39	3.198s	4.253s	4.364s	4.216s	3.903s	3.457s
1,436.46	3.003s	3.990s	4.111s	3.990s	3.712s	3.308s
1,527.11	2.807s	3.722s	3.855s	3.761s	3.519s	3.157s
1,618.10	2.610s	3.451s	3.598s	3.530s	3.325s	3.006s
1,709.44	2.413s	3.177s	3.338s	3.299s	3.130s	2.854s
1,801.11	2.215s	2.901s	3.077s	3.066s	2.934s	2.702s
1,893.13	2.016s	2.624s	2.815s	2.833s	2.738s	2.549s
1,985.42	1.816s	2.349s	2.551s	2.598s	2.540s	2.396s
2,077.73	1.613s	2.077s	2.287s	2.363s	2.343s	2.242s
2,170.06	1.408s	1.811s	2.025s	2.128s	2.145s	2.088s
2,262.39	1.198s	1.551s	1.766s	1.894s	1.947s	1.934s
2,354.73	0.983s	1.295s	1.512s	1.662s	1.751s	1.781s
2,445.61	0.763s	1.048s	1.268s	1.440s	1.562s	1.633s
2,521.33	0.578s	0.848s	1.074s	1.263s	1.411s	1.515s
2,578.59	0.436s	0.698s	0.931s	1.133s	1.301s	1.428s
Distances in METERS.---Specific Gravity = 1.025.-----						

31-07-20 12:20:24
GHS 17.22Cybermarine Technologies PTE. Ltd.
PONTON

CROSS CURVES OF STABILITY - Stbd Heel
at 0.5 M. AFT TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

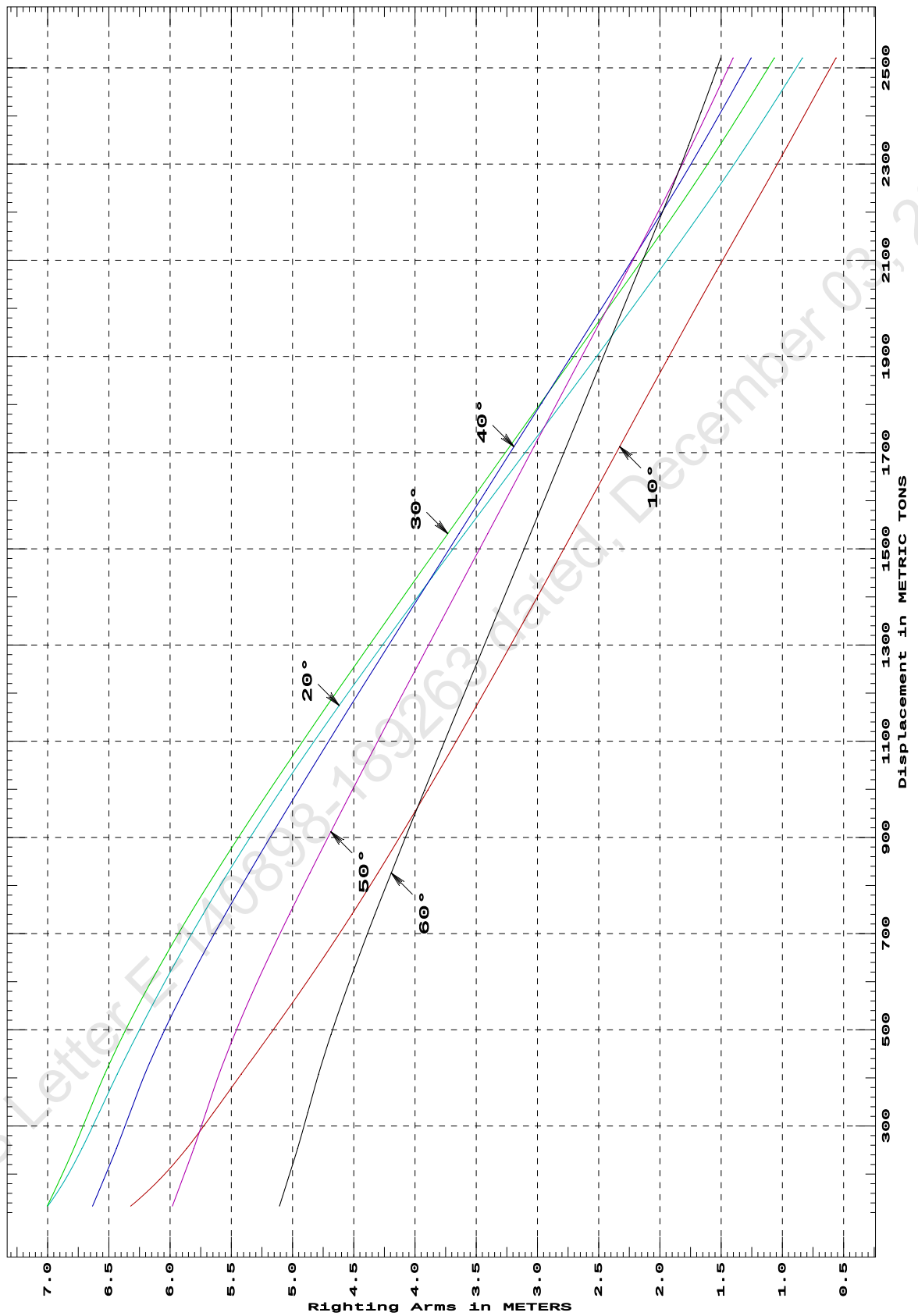
Trim: Aft 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
133.29	6.322s	6.999s	7.004s	6.633s	5.981s	5.107s
188.30	6.087s	6.857s	6.900s	6.542s	5.901s	5.038s
252.92	5.864s	6.719s	6.786s	6.438s	5.807s	4.959s
327.16	5.646s	6.580s	6.666s	6.327s	5.708s	4.876s
407.76	5.419s	6.431s	6.533s	6.205s	5.600s	4.787s
489.54	5.186s	6.270s	6.382s	6.063s	5.474s	4.686s
572.35	4.956s	6.100s	6.214s	5.904s	5.335s	4.574s
656.10	4.731s	5.920s	6.031s	5.732s	5.184s	4.453s
740.70	4.513s	5.729s	5.836s	5.548s	5.025s	4.326s
826.02	4.303s	5.528s	5.629s	5.355s	4.858s	4.194s
912.01	4.097s	5.316s	5.411s	5.155s	4.687s	4.059s
998.69	3.895s	5.093s	5.186s	4.949s	4.511s	3.921s
1,086.07	3.695s	4.861s	4.954s	4.738s	4.333s	3.780s
1,174.15	3.497s	4.619s	4.717s	4.524s	4.151s	3.638s
1,262.91	3.300s	4.370s	4.474s	4.306s	3.966s	3.494s
1,352.12	3.105s	4.116s	4.229s	4.085s	3.780s	3.348s
1,441.67	2.910s	3.858s	3.981s	3.863s	3.593s	3.203s
1,531.58	2.716s	3.596s	3.732s	3.640s	3.406s	3.057s
1,621.83	2.523s	3.330s	3.480s	3.416s	3.218s	2.911s
1,712.42	2.329s	3.065s	3.227s	3.192s	3.029s	2.764s
1,803.32	2.134s	2.798s	2.973s	2.966s	2.840s	2.618s
1,894.54	1.938s	2.532s	2.717s	2.738s	2.650s	2.471s
1,986.08	1.740s	2.268s	2.462s	2.511s	2.458s	2.323s
2,077.95	1.538s	2.006s	2.207s	2.283s	2.268s	2.175s
2,170.15	1.332s	1.746s	1.954s	2.056s	2.077s	2.027s
2,258.72	1.132s	1.503s	1.718s	1.843s	1.898s	1.888s
2,338.10	0.958s	1.294s	1.513s	1.660s	1.743s	1.768s
2,408.27	0.806s	1.115s	1.339s	1.502s	1.610s	1.664s
2,469.21	0.674s	0.963s	1.189s	1.367s	1.497s	1.576s
2,520.92	0.561s	0.836s	1.065s	1.254s	1.402s	1.503s
Distances in METERS.---Specific Gravity = 1.025.-----						

31-07-20 12:20:24
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTON

CROSS CURVES OF STABILITY - Stbd Heel
at 1 M. AFT TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

DEADWEIGHT TABLE

Draft	Disp	Dwt	
M	T	T	
0.612	481.78	75.81	
0.712	564.87	158.90	
0.812	648.69	242.72	
0.912	733.24	327.27	
1.012	818.51	412.54	
1.112	904.44	498.47	
1.212	991.06	585.09	
1.312	1078.37	672.40	
1.412	1166.40	760.43	
1.512	1255.11	849.14	
1.612	1344.46	938.49	
1.712	1434.46	1028.49	
1.812	1525.13	1119.16	
1.912	1616.48	1210.51	
2.012	1708.51	1302.54	
2.112	1800.77	1394.80	
2.212	1893.06	1443.49	Full Load Displacement

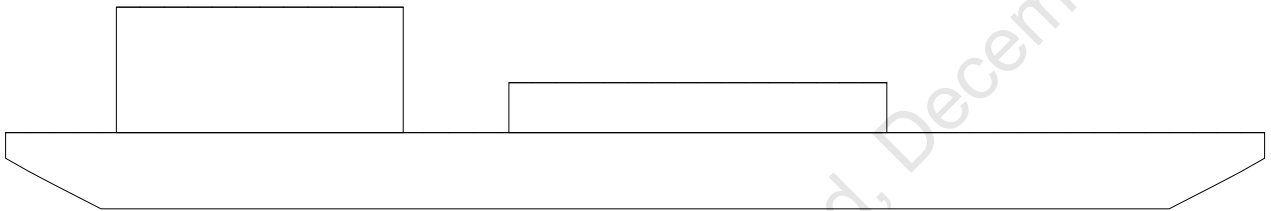
01-08-20 22:26:06 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

USK draft refers to the line:

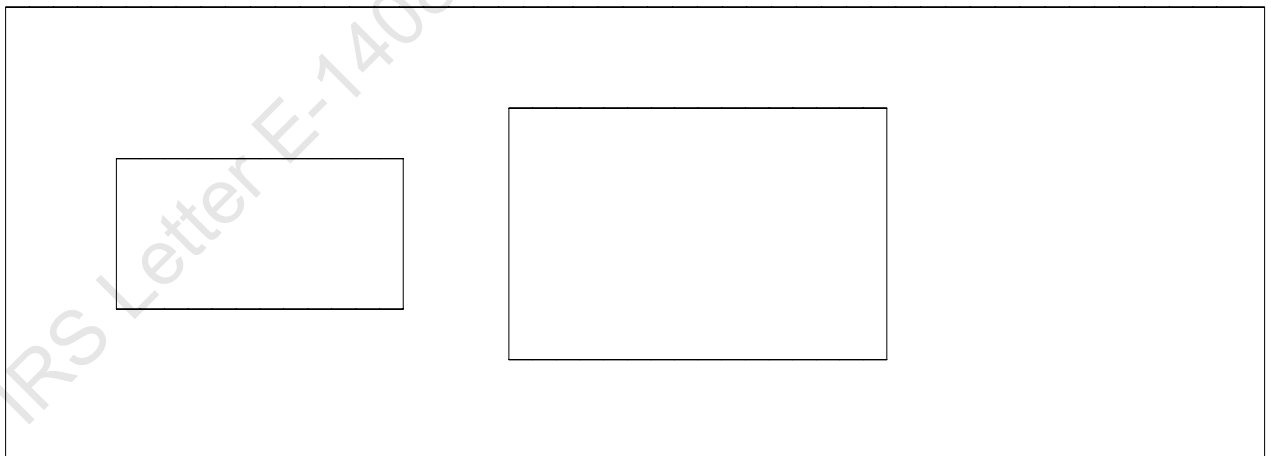
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

Condition Graphic

Profile View



Plan View



01-08-20 22:26:06 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

MAXIMUM VCG vs. DISPLACEMENT

Heeling moment is present from: wind

Trim = Fwd 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement		--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3	
400.00	18.489	122%	0d	7d	
600.00	17.349	115%	0d	6d	
800.00	16.087	112%	0d	5d	
1,000.00	14.631	73%	0d	4d	
1,200.00	13.079	51%	0d	3d	
1,400.00	11.429	30%	0d	3d	
1,600.00	9.697 9.770	12%	0d	2d	
1,800.00	7.619 7.778	0%	1d	1d	
2,000.00	5.293 5.433	0%	5d	1d	

Heeling moment is present from: wind

Trim = Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement		--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3	
400.00	18.792	138%	0d	10d	
600.00	17.834	103%	0d	7d	
800.00	16.573	83%	0d	6d	
1,000.00	15.128	70%	0d	5d	
1,200.00	13.510	57%	0d	4d	
1,400.00	11.824	35%	0d	3d	
1,600.00	10.137	7%	0d	3d	
1,800.00	7.959	0%	2d	2d	
2,000.00	5.373 5.618	3%	6d	1d	

Heeling moment is present from: wind

Trim = zero at zero heel (trim righting arm held at zero)

Displacement		--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3	
400.00	18.899	137%	0d	14d	
600.00	17.990	117%	0d	10d	
800.00	16.745	80%	0d	7d	
1,000.00	15.270	69%	0d	6d	
1,200.00	13.683	59%	0d	5d	
1,400.00	11.943	37%	0d	4d	
1,600.00	10.176	8%	0d	3d	
1,800.00	8.007	0%	2d	3d	
2,000.00	5.351 5.806	0%	5d	2d	

Heeling moment is present from: wind

Trim = Aft 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement		--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3	
400.00	18.793	138%	0d	10d	
600.00	17.835	103%	0d	7d	
800.00	16.573	83%	0d	6d	
1,000.00	15.129	70%	0d	5d	
1,200.00	13.510	57%	0d	4d	

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GHS 17.22 PONT00N

1,400.00	11.825	35%	0d	3d
1,600.00	10.136	7%	0d	3d
1,800.00	7.958	0%	2d	2d
2,000.00	5.363	4%	6d	1d

Heeling moment is present from: wind
Trim = Aft 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement --- Margins ---
METRIC TONS Max VCG LIM1 LIM2 LIM3

400.00	18.491	119%	0d	7d
600.00	17.392	114%	0d	6d
800.00	16.095	112%	0d	5d
1,000.00	14.632	73%	0d	4d
1,200.00	13.079	51%	0d	3d
1,400.00	11.430	31%	0d	3d
1,600.00	9.699	12%	0d	2d
1,800.00	7.661	0%	1d	1d
2,000.00	5.312	0%	5d	1d

Distances in METERS.---Specific Gravity = 1.025.---d = degrees.

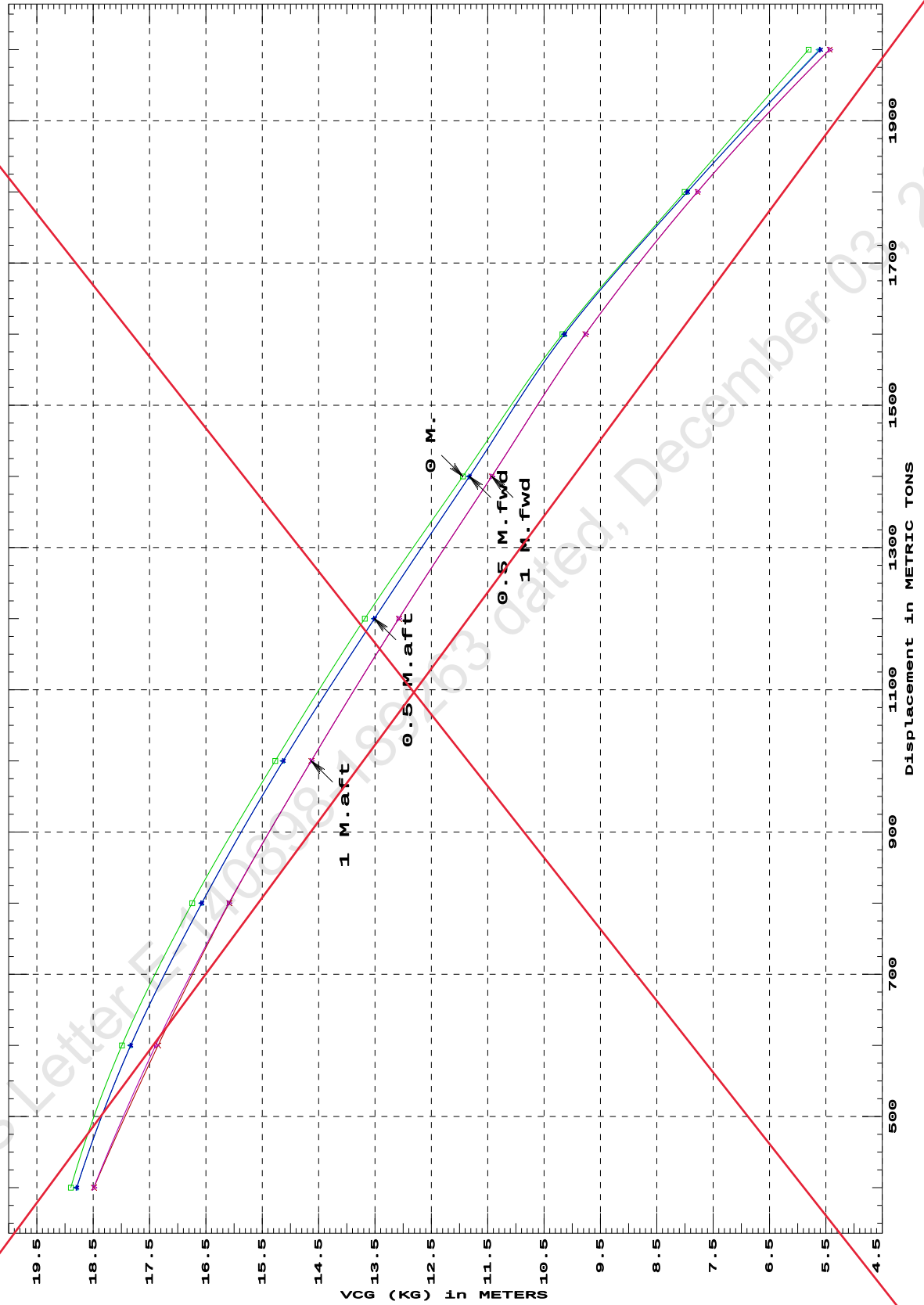
LIM-----IS CODE 2008 CRITERION-----Min/Max
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg

MAX VCG DOES NOT COVER CRANE OPERATION/LIFTING OF WEIGHTS AND VALID ONLY
FOR UNMANNED TOW, WITH CRANE IN STOWED POSITION.

01-08-20 22:26:06
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTON

IS CODE 2008 criterion MAXIMUM VCG (KG)
at various trims (initial)



Specific Gravity = 1.025
Trim is per 50 M. "K" = BPL

LOADING CONDITIONS

SUMMARY OF CONDITIONS				
CONDITION NO.	1	2	3	4
CONDITION	Condition 00 : Lightship Condition	Condition 1 : Lightship + Crane + Ballast	Condition 2 : Lightship + Crane + Deck Cargo + Ballast / Maximum Draft Condition	Condition 3 : Lightship + Deck Cargo / Maximum Deck Cargo Condition
TOTAL TANK LOAD (T)	0.000	234.870	234.870	0.000
DWT CONST.(T)	0.000	400.000	1208.620	1443.490
DEADWEIGHT (T)	0.000	634.870	1443.490	1443.490
LIGHT WEIGHT (T)	449.570	449.570	449.570	449.570
DISPLACEMENT (T)	449.570	1084.440	1893.060	1893.060
FWD DRAFT (M)	0.575	1.195	2.207	2.214
AFT DRAFT (M)	0.571	1.442	2.217	2.210
MEAN DRAFT (M)	0.573	1.319	2.212	2.212
TRIM (M)	0.005f	0.247a	0.010a	0.000
L.C.G. (FROM AP) (M)	25.030f	24.264f	24.980f	25.007f
L.C.B. (FROM AP) (M)	25.030f	24.248f	24.979f	25.007f
M.C.T. (T-CM)	27.140	32.300	37.340	37.410
KG (M)	3.000	3.936	3.963	3.763
GM _T (M)	46.779	18.336	10.197	10.546
MAX. GZ (M)	5.454	3.634	1.362	1.452
Note:For details of deadweight constant refer respective loading condition				

SUMMARY OF CONDITIONS				
CONDITION NO.	5	6	7	8
CONDITION	Condition 4 : Operating Condition of 400T Crane Crane Lifting 40T @ 30m Radius aft	Condition 5 : Operating Condition of 400T Crane Crane Lifting 40T @ 30m Radius stbd	Condition 6 : Operating Condition of 400T Crane Crane Lifting 40T @ 30m Radius fwd	Condition 7 : Operating Condition of 350T Crane Crane Lifting 40T @ 35m Radius aft
TOTAL TANK LOAD (T)	284.980	284.980	284.980	422.400
DWT CONST.(T)	440.000	440.000	440.000	390.000
DEADWEIGHT (T)	724.980	724.980	724.980	812.400
LIGHT WEIGHT (T)	449.570	449.570	449.570	449.570
DISPLACEMENT (T)	1174.550	1174.550	1174.550	1261.970
FWD DRAFT (M)	1.199	1.390	1.578	1.296
AFT DRAFT (M)	1.639	1.452	1.262	1.740
MEAN DRAFT (M)	1.419	1.421	1.420	1.518
TRIM (M)	0.440a	0.062a	0.316f	0.443a
L.C.G. (FROM AP) (M)	23.772f	24.828f	25.883f	23.815f
L.C.B. (FROM AP) (M)	23.735f	24.823f	25.910f	23.787f
M.C.T. (T-CM)	32.760	32.870	32.770	33.700
KG (M)	4.893	4.893	4.893	4.038
GM _T (M)	15.848	15.898	15.850	15.426
MAX. GZ (M)	3.095	3.128	3.110	3.069

SUMMARY OF CONDITIONS				
CONDITION NO.	9	10	11	12
CONDITION	Condition 8 : Operating Condition of 350T Crane Crane Lifting 40T @ 35m Radius stbd	Condition 9 : Operating Condition of 350T Crane Crane Lifting 40T @ 35m Radius fwd	Condition 10 : Operating Condition of 250T Crane Crane Lifting 36.6T @ 18m Radius aft	Condition 11 : Operating Condition of 250T Crane Crane Lifting 36.6T @ 18m Radius stbd
TOTAL TANK LOAD (T)	422.400	422.400	422.400	422.400
DWT CONST.(T)	390.000	390.000	286.600	286.600
DEADWEIGHT (T)	812.400	812.400	709.000	709.000
LIGHT WEIGHT (T)	449.570	449.570	449.570	449.570
DISPLACEMENT (T)	1261.970	1261.970	1158.570	1158.570
FWD DRAFT (M)	1.506	1.712	1.326	1.427
AFT DRAFT (M)	1.534	1.325	1.480	1.379
MEAN DRAFT (M)	1.520	1.519	1.403	1.403
TRIM (M)	0.028a	0.387f	0.155a	0.048f
L.C.G. (FROM AP) (M)	24.925f	26.034f	24.565f	25.134f
L.C.B. (FROM AP) (M)	24.923f	26.060f	24.551f	25.139f
M.C.T. (T-CM)	33.700	33.710	32.570	32.630
KG (M)	4.038	4.038	5.300	5.300
GM _T (M)	15.415	15.427	15.589	15.620
MAX. GZ (M)	3.102	3.076	3.033	3.037

SUMMARY OF CONDITIONS				
CONDITION NO.	13	14	15	16
CONDITION	Condition 12 : Operating Condition of 250T Crane Crane Lifting 36.6T @ 18m Radius fwd	Condition 13 : Operating Condition of 150T Crane Crane Lifting 51.3T @ 12m Radius aft	Condition 14 : Operating Condition of 150T Crane Crane Lifting 51.3T @ 12m Radius stbd	Condition 15 : Operating Condition of 150T Crane Crane Lifting 51.3T @ 12m Radius fwd
TOTAL TANK LOAD (T)	422.400	323.200	323.200	323.200
DWT CONST.(T)	286.600	201.300	201.300	201.300
DEADWEIGHT (T)	709.000	524.500	524.500	524.500
LIGHT WEIGHT (T)	449.570	449.570	449.570	449.570
DISPLACEMENT (T)	1158.570	974.070	974.070	974.070
FWD DRAFT (M)	1.527	0.997	1.096	1.195
AFT DRAFT (M)	1.278	1.386	1.288	1.190
MEAN DRAFT (M)	1.403	1.192	1.192	1.193
TRIM (M)	0.250f	0.389a	0.192a	0.005f
L.C.G. (FROM AP) (M)	25.702f	23.752f	24.384f	25.016f
L.C.B. (FROM AP) (M)	25.725f	23.722f	24.369f	25.017f
M.C.T. (T-CM)	32.560	31.280	31.250	31.240
KG (M)	5.300	4.436	4.436	4.436
GM _T (M)	15.581	19.922	19.899	19.891
MAX. GZ (M)	3.026	3.735	3.757	3.763

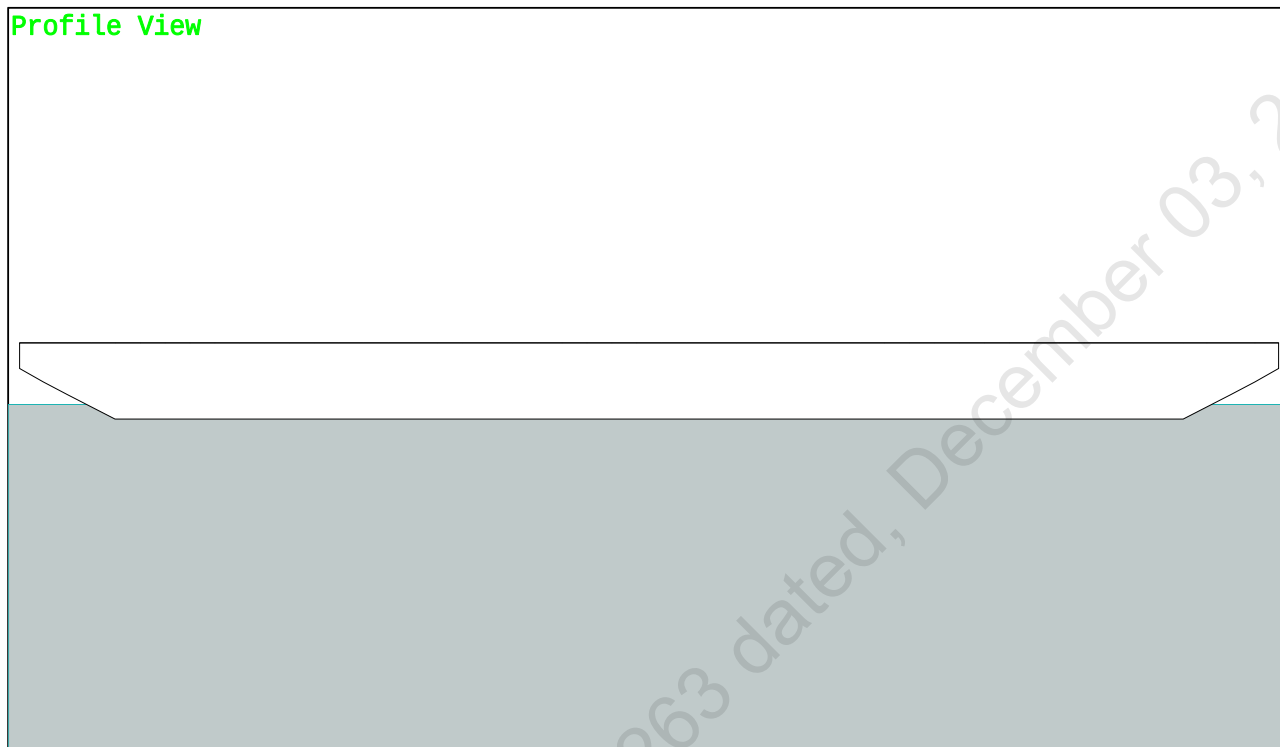
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GHS 17.90

PONTOON

Condition 00 : Lightship
Lightship condition

Condition Graphic - Draft: 0.575 @ 50.000f, 0.571 @ 0.000 Heel: zero

Profile View



Plan View



01-12-21 09:40:20 Cybermarine Technologies PTE. Ltd.
GHS 17.90

PONTOON

Condition 00 : Lightship
Lightship condition

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 0.575 @ 50.00f, 0.573 @ 25.00f, 0.571 @ 0.00

Trim: Fwd 0.005/50.000, Heel: zero

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
WEIGHT	449.57	25.030f	0.000	3.000	
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
Total Tanks----->		0.00			0.00
	Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL	1.025	449.57	25.030f	0.000	0.280

Righting Arms:		0.000	0.000		
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----BMT
Total Waterplane---->	1.025	804.4	25.004f	0.000	304.62 49.499
		MT/cm-----	m.-MT/cm-----	GML-----	GMT
		8.25	27.14	301.90	46.779

Distances in METERS.-----Moments in m.-MT.

SUMMARY OF LOADING

0.0 Cu.M. (0%) SALT WATER

DRAFT AT FP (M) = 0.575
DRAFT AT MS (M) = 0.573
DRAFT AT AP (M) = 0.571
DRAFT AT FWD MARK (M) = 0.575
DRAFT AT AFT MARK (M) = 0.571

HYDROSTATIC PROPERTIES

Trim: Fwd 0.005/50.000, No Heel, VCG = 3.000

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----LCF-----cm trim-----GML-----GMT
0.573	449.57	25.030f	0.280	8.25 25.004f 27.14 301.90 46.779

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 0.575 @ 50.00f, 0.573 @ 25.00f, 0.571 @ 0.00

Trim: Fwd 0.005/50.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm-----	Pressure-----	Moment-----
HULL	118.7	1.249	1.532	0.05500	10.00

Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

LCG = 25.030f TCG = 0.000 VCG = 3.000

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---	Trim---	Heel---	Weight(MT)---	Area---Angle
			in Trim--in Heel-->	
0.559	0.01f	0.00	449.56 0.000 -0.022	0.0000 12.637
0.558	0.01f	0.03s	449.56 0.000 0.000	-0.0000 12.610
0.520	0.01f	5.03s	449.58 0.000 3.698	0.1613 7.610
0.258	0.01f	10.03s	449.57 0.000 4.967	0.5508 2.610
0.062	0.01f	12.64s	449.57 0.000 5.229	0.7840 0.000
-0.138	0.01f	15.03s	449.58 0.000 5.364	1.0050 -2.390
-0.603	0.01f	20.03s	449.57 0.000 5.454	1.4806 -7.390
-1.110	0.01f	25.03s	449.60 0.000 5.380	1.9545 -12.390

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GHS 17.90

PONTOON

Condition 00 : Lightship
Lightship condition

-1.631	0.01f	30.03s	449.57	0.000	5.171	2.4159	-17.390
-2.139	0.02f	35.03s	449.57	0.000	4.841	2.8536	-22.390
-2.631	0.02f	40.03s	449.57	0.000	4.430	3.2587	-27.390
-3.103	0.02f	45.03s	449.57	0.000	3.960	3.6252	-32.390
-3.551	0.02f	50.03s	449.57	0.000	3.443	3.9485	-37.390
-3.973	0.03f	55.03s	449.57	0.000	2.889	4.2250	-42.390
-4.364	0.03f	60.03s	449.57	0.000	2.306	4.4519	-47.390

Distances in METERS.----Specific Gravity = 1.025.-----Area in m.-Rad.

Note: No tank loads are present.

Note: The Residual Righting Arms shown above are in excess of the
wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 10.00 (constant)

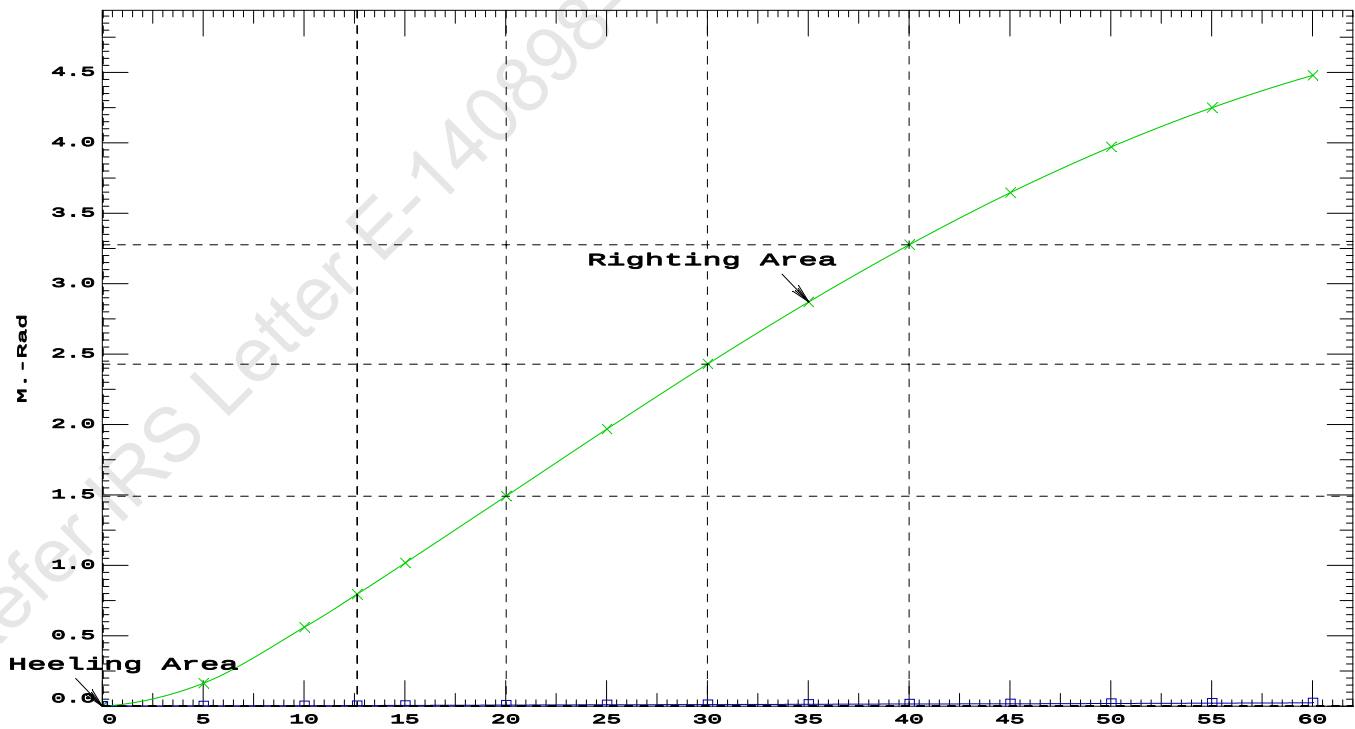
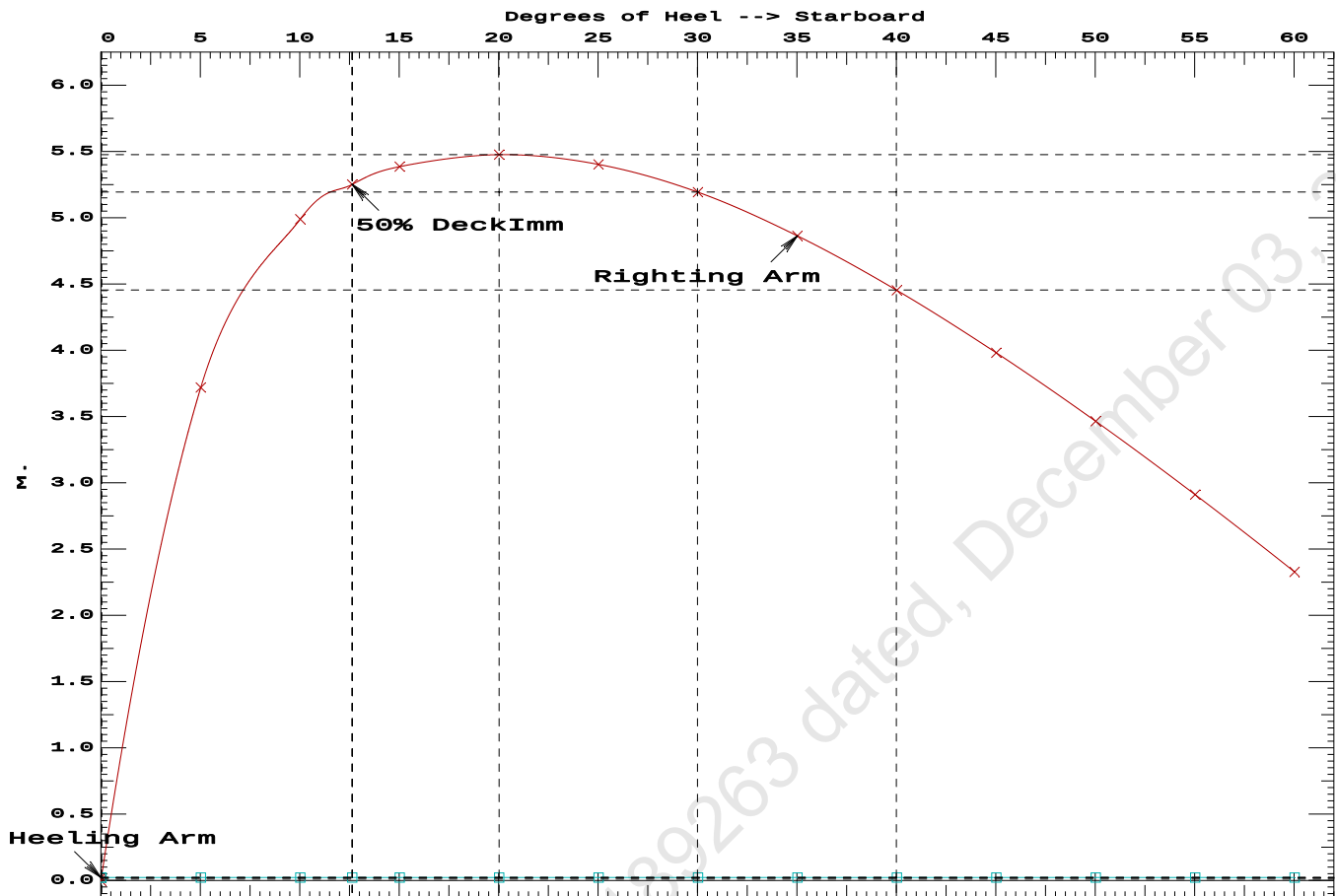
LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 1.4806 P
(2) Angle from Equilibrium to RZero or Flood > 20.00 deg LARGE
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 12.61 P

01-12-21 09:40:20
GHS 17.90

Cybermarine Technologies PTE. Ltd.

PONT00N

Condition 00 : Lightship
Lightship condition



GHS 17.90

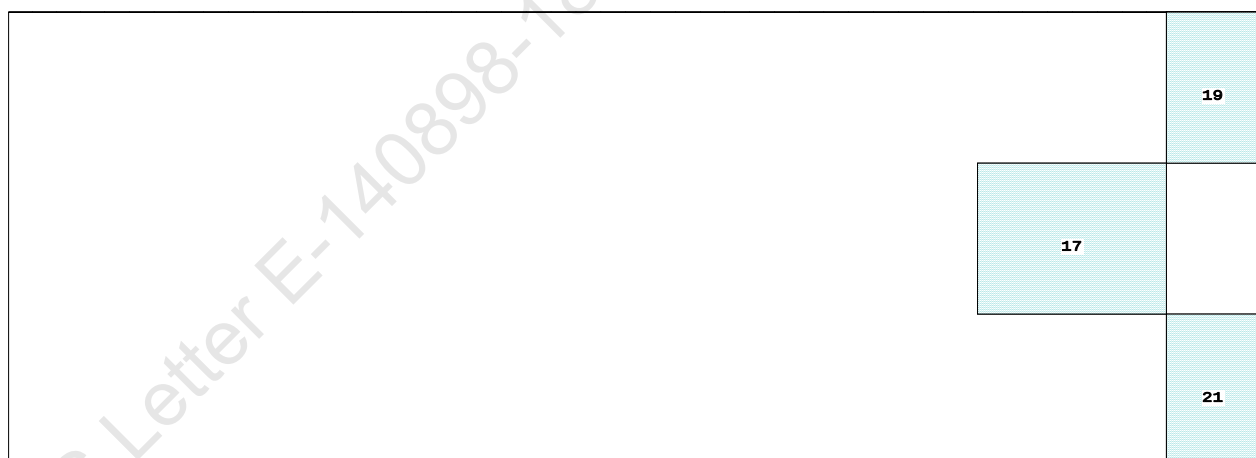
PONTOON

Condition 01 : Lightship + Crane + Ballast

Profile View



Plan View



17 06_WBTK.C.100% SALT WATER Intact

19 07_WBTK.P.100% SALT WATER Intact

21 07_WBTK.S.100% SALT WATER Intact

01-12-21 09:37:57 Cybermarine Technologies PTE. Ltd.
GHS 17.90

PONTOON

Condition 01 : Lightship + Crane + Ballast

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.195 @ 50.00f, 1.318 @ 25.00f, 1.442 @ 0.00

Trim: Aft 0.247/50.000, Heel: zero

Part-----	Weight(MT)	LCG	TCG	VCG	
LIGHT SHIP	449.57	25.030f	0.000	3.000	
Crane	400.00	11.500f	0.000	6.325	
Total Fixed----->	849.57	18.660f	0.000	4.565	
Load-----	SpGr	Weight(MT)	LCG	TCG	VCG
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s
Total Tanks----->			234.87	44.536f	0.000
Total Weight----->			1,084.44	24.264f	0.000
			Displ(MT)	LCB	TCB
HULL	1.025		1,084.39	24.248f	0.000
					VCB
					0.667

Righting Arms:			0.000	0.000	
Part-----	SpGr	WPA	LCF	TCF	BML
Total Waterplane---->	1.025	856.1	24.769f	0.000	152.20
		MT/cm	m	MT/cm	GML
		8.77		32.30	148.93
					18.336
Distances in METERS.-----			Moments in m.-MT.		

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

229.1 Cu.M. (9%) SALT WATER

400.00 MT Crane

DRAFT AT FP (M) = 1.195
DRAFT AT MS (M) = 1.318
DRAFT AT AP (M) = 1.442
DRAFT AT FWD MARK (M) = 1.214
DRAFT AT AFT MARK (M) = 1.422

HYDROSTATIC PROPERTIES

Trim: Aft 0.247/50.000, No Heel, VCG = 3.936

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)	LCB	VCB	cm
1.319	1,084.39	24.248f	0.667	8.77
				24.769f
				32.30
				148.93
				18.336
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.				
Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 280.2 m.-MT

01-12-21 09:37:57 Cybermarine Technologies PTE. Ltd.
GHS 17.90

PONTOON

Condition 01 : Lightship + Crane + Ballast

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.195 @ 50.00f, 1.318 @ 25.00f, 1.442 @ 0.00

Trim: Aft 0.247/50.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm----	Pressure-----	Moment
HULL	84.2	0.861	1.508	0.05500	6.99
CRANE	57.0	4.120	4.767	0.05500	14.95
Total wind heeling moment to starboard----->					21.93
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.264f TCG = 0.000 VCG = 3.936

Free Surface Adjustment: 0.258

Adjusted CG: LCG = 24.265f TCG = 0.000 VCG = 4.195

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm	
Depth---	Trim----	Heel----	Weight(MT)----	Area-----	
			in Trim--in Heel---	Angle	
1.430	0.28a	0.00	1,084.4	0.0000	5.103
1.430	0.28a	0.05s	1,084.4	0.0000	5.053
1.415	0.28a	5.05s	1,084.4	0.0700	0.053
1.415	0.28a	5.10s	1,084.4	0.0715	0.000
1.369	0.31a	10.05s	1,084.4	0.2745	-4.947
1.266	0.44a	15.05s	1,084.4	0.5716	-9.947
1.227	0.50a	16.91s	1,084.4	0.6893	-11.805
1.158	0.59a	20.05s	1,084.4	0.8879	-14.947
1.045	0.75a	25.05s	1,084.4	1.1918	-19.947
0.927	0.91a	30.05s	1,084.4	1.4689	-24.947
0.805	1.06a	35.05s	1,084.4	1.7140	-29.947
0.679	1.22a	40.05s	1,084.4	1.9233	-34.947
0.551	1.37a	45.05s	1,084.4	2.0939	-39.947
0.420	1.51a	50.05s	1,084.4	2.2237	-44.947
0.287	1.64a	55.05s	1,084.4	2.3113	-49.947
0.151	1.76a	60.05s	1,084.4	2.3558	-54.947
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.					

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 280.2 m.-MT was applied to artificially modify the CG.

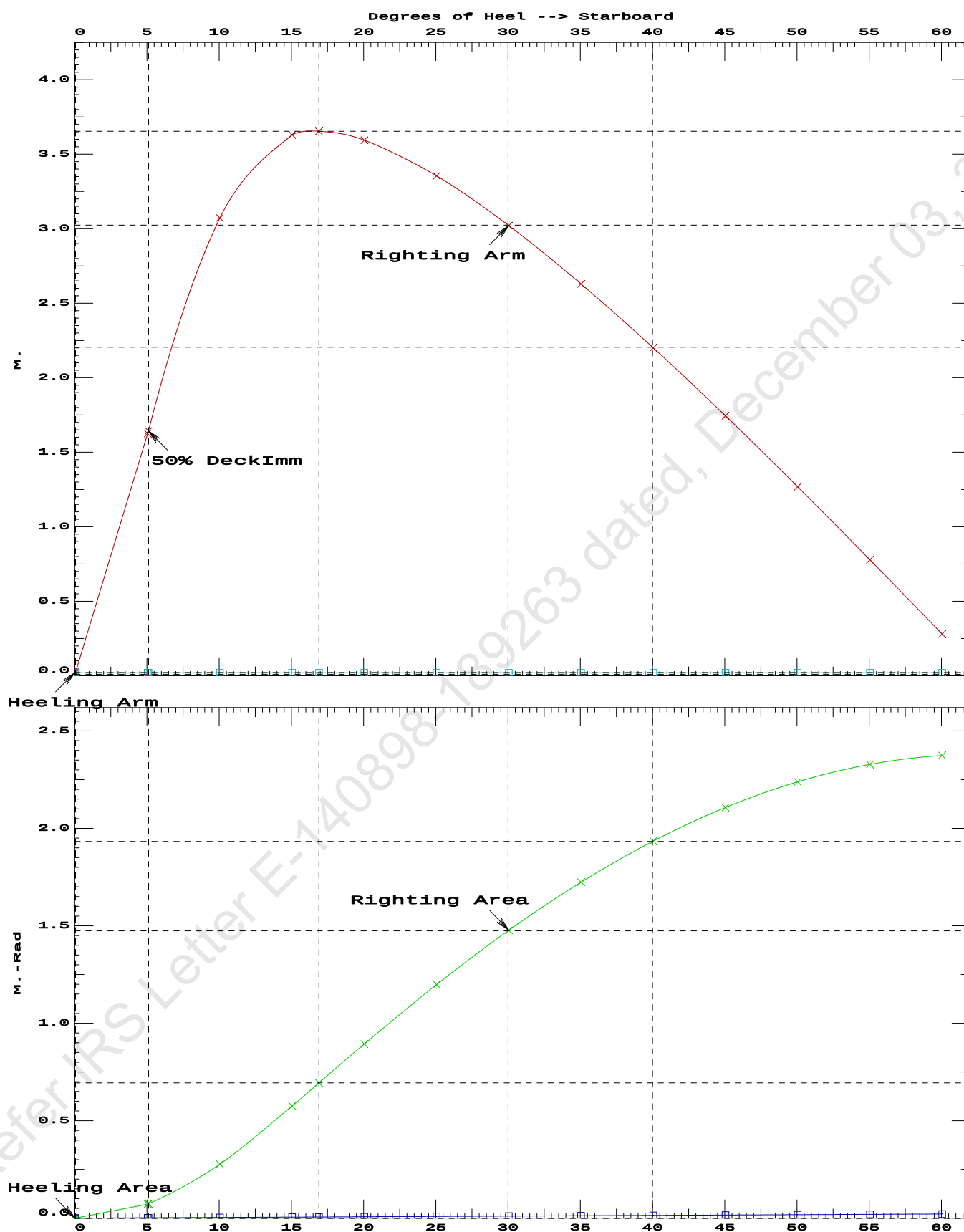
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.93 (constant)

LIM-----	IS CODE 2008 PART B CHAPTER-2 SEC.2----	Min/Max-----	Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.6893 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	LARGE
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	5.05 P

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GHS 17.90

PONT00N

Condition 01 : Lightship + Crane + Ballast

01-12-21 09:39:33 Cybermarine Technologies PTE. Ltd.

GHS 17.90

PONT00N

Condition 02 : Lightship + Crane + Deck Cargo + BallastMaximum Draft Condition

Condition Graphic - Draft: 2.207 @ 50.000f, 2.217 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

17 06_WBTK.C.100% SALT WATER Intact

19 07_WBTK.P.100% SALT WATER Intact

21 07_WBTK.S.100% SALT WATER Intact

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GHS 17.90

PONTOON

Condition 02 : Lightship + Crane + Deck Cargo + Ballast
Maximum Draft Condition

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.207 @ 50.00f, 2.212 @ 25.00f, 2.217 @ 0.00

Trim: Aft 0.010/50.000, Heel: zero

Part-----	Weight(MT)	LCG	TCG	VCG			
LIGHT SHIP	449.57	25.030f	0.000	3.000			
Crane	400.00	11.500f	0.000	6.325			
Deck Cargo	808.62	25.940f	0.000	4.000			
Total Fixed----->	1,658.19	22.210f	0.000	4.290			
Load-----	SpGr-----	Weight(MT)	LCG	TCG	VCG	FSM	
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			234.87	44.536f	0.000	1.660	280.25
Total Weight----->			1,893.06	24.980f	0.000	3.963	
			Displ(MT)	LCB	TCB	VCB	
HULL	1.025		1,893.06	24.979f	0.000	1.132	

Righting Arms:			0.000	0.000			
Part-----	SpGr-----	WPA	LCF	TCF	BML	BMT	
Total Waterplane---->	1.025	900.4	25.000f	0.000	101.45	13.029*	
		MT/cm	m	MT/cm	GML	GMT	
		9.23		37.34	98.62	10.197	
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

229.1 Cu.M. (9%) SALT WATER

400.00 MT Crane

808.62 MT Deck Cargo

DRAFT AT FP (M) = 2.207
DRAFT AT MS (M) = 2.212
DRAFT AT AP (M) = 2.217
DRAFT AT FWD MARK (M) = 2.208
DRAFT AT AFT MARK (M) = 2.216

HYDROSTATIC PROPERTIES

Trim: Aft 0.010/50.000, No Heel, VCG = 3.963

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/	
Draft----	Weight(MT)	LCB	VCB	cm	LCF
2.212	1,893.06	24.979f	1.132	9.23	25.000f
					37.34
					98.62
					10.197
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.					
Trim is per 50.00m.					

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 280.2 m.-MT

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PONTOON

Condition 02 : Lightship + Crane + Deck Cargo + Ballast
Maximum Draft Condition

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.207 @ 50.00f, 2.212 @ 25.00f, 2.217 @ 0.00

Trim: Aft 0.010/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	40.5	0.405	1.478	0.05500	3.29
CRANE	57.0	3.297	4.370	0.05500	13.70
DECKCARGO.C	29.4	1.801	2.874	0.05500	4.65
Total wind heeling moment to starboard-->					21.64
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.980f TCG = 0.000 VCG = 3.963

Free Surface Adjustment: 0.148

Adjusted CG: LCG = 24.980f TCG = 0.000 VCG = 4.112

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. 50% DeckImm Area	Angle
2.205	0.01a	0.00	1,893.1	0.000	-0.011	0.0000 2.559
2.205	0.01a	0.06s	1,893.1	0.000	0.000	-0.0000 2.497
2.202	0.01a	2.56s	1,893.1	0.000	0.449	0.0098 0.000
2.194	0.01a	5.06s	1,893.1	0.000	0.892	0.0391 -2.503
2.292	0.02a	10.06s	1,893.1	0.000	1.318	0.1406 -7.503
2.437	0.02a	13.38s	1,893.1	0.000	1.362	0.2191 -10.822
2.521	0.02a	15.06s	1,893.1	0.000	1.351	0.2589 -12.503
2.794	0.03a	20.06s	1,893.1	0.000	1.236	0.3735 -17.503
3.059	0.04a	25.06s	1,893.1	0.000	1.030	0.4730 -22.503
3.301	0.04a	30.06s	1,893.1	0.000	0.777	0.5522 -27.503
3.518	0.05a	35.06s	1,893.1	0.000	0.499	0.6081 -32.503
3.708	0.06a	40.06s	1,893.1	0.000	0.206	0.6389 -37.503
3.823	0.06a	43.51s	1,893.1	0.000	0.000	0.6452 -40.946
3.870	0.07a	45.06s	1,893.1	0.000	-0.094	0.6439 -42.503
4.002	0.07a	50.06s	1,893.1	0.000	-0.397	0.6225 -47.503
4.104	0.08a	55.06s	1,893.1	0.000	-0.700	0.5746 -52.503
4.177	0.09a	60.06s	1,893.1	0.000	-1.000	0.5003 -57.503
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.						

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 280.2 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.64 (constant)

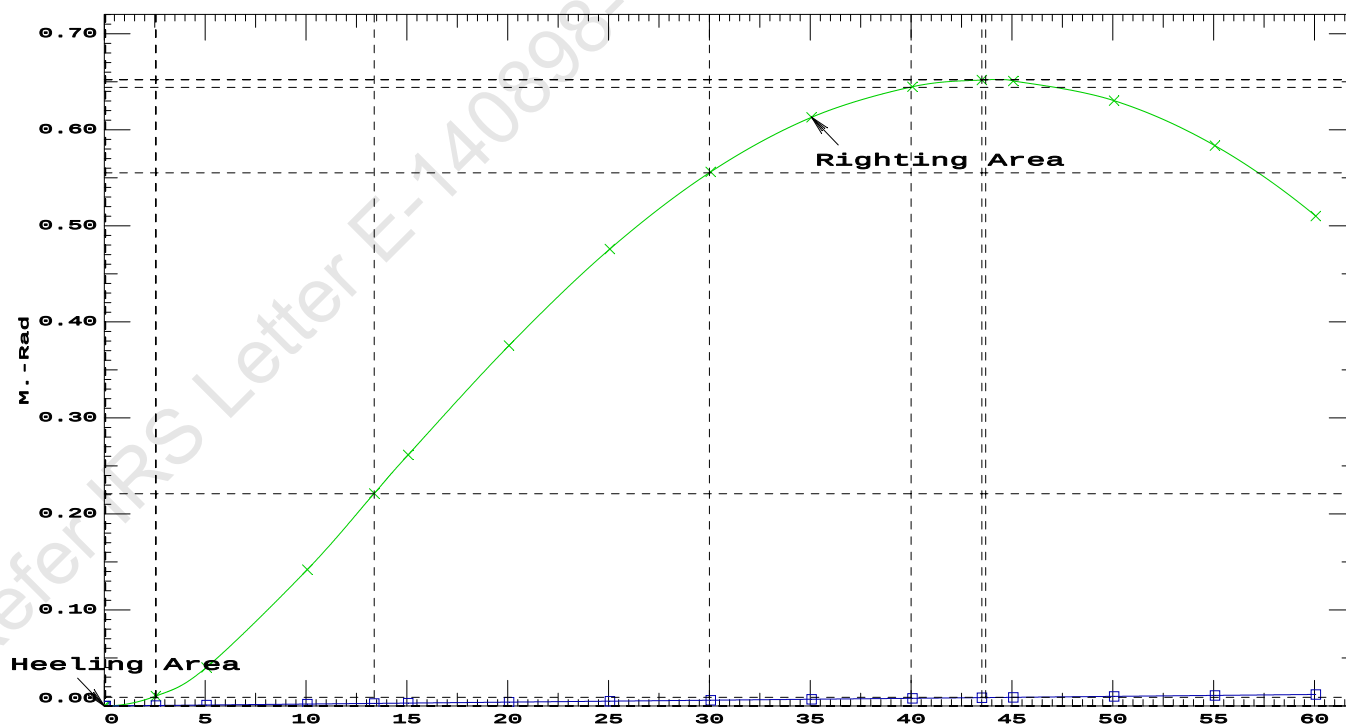
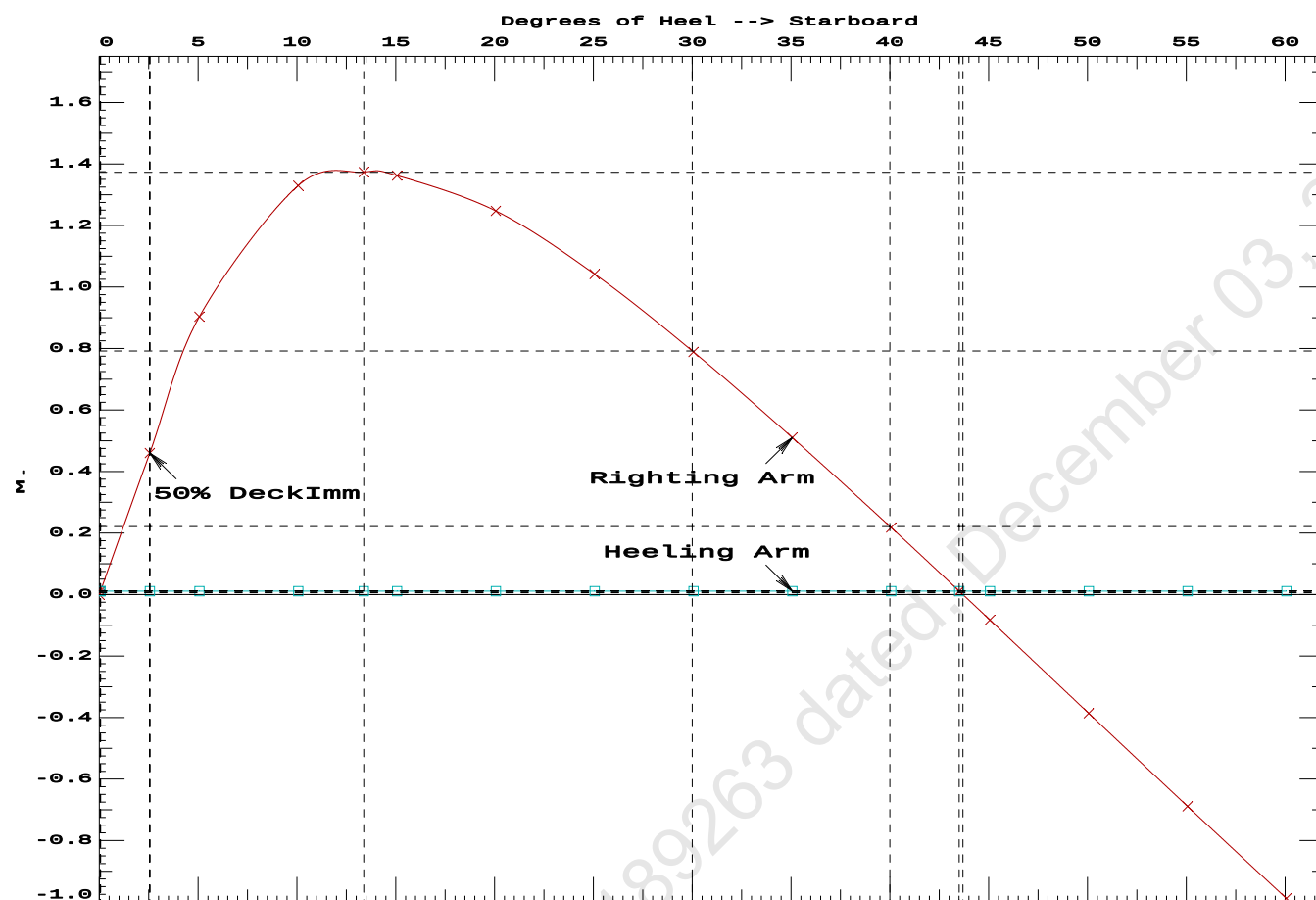
LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2-----Min/Max-----Attained

(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.2191 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	43.44 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	2.50 P

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GHS 17.90

PONT00N

Condition 02 : Lightship + Crane + Deck Cargo + BallastMaximum Draft Condition

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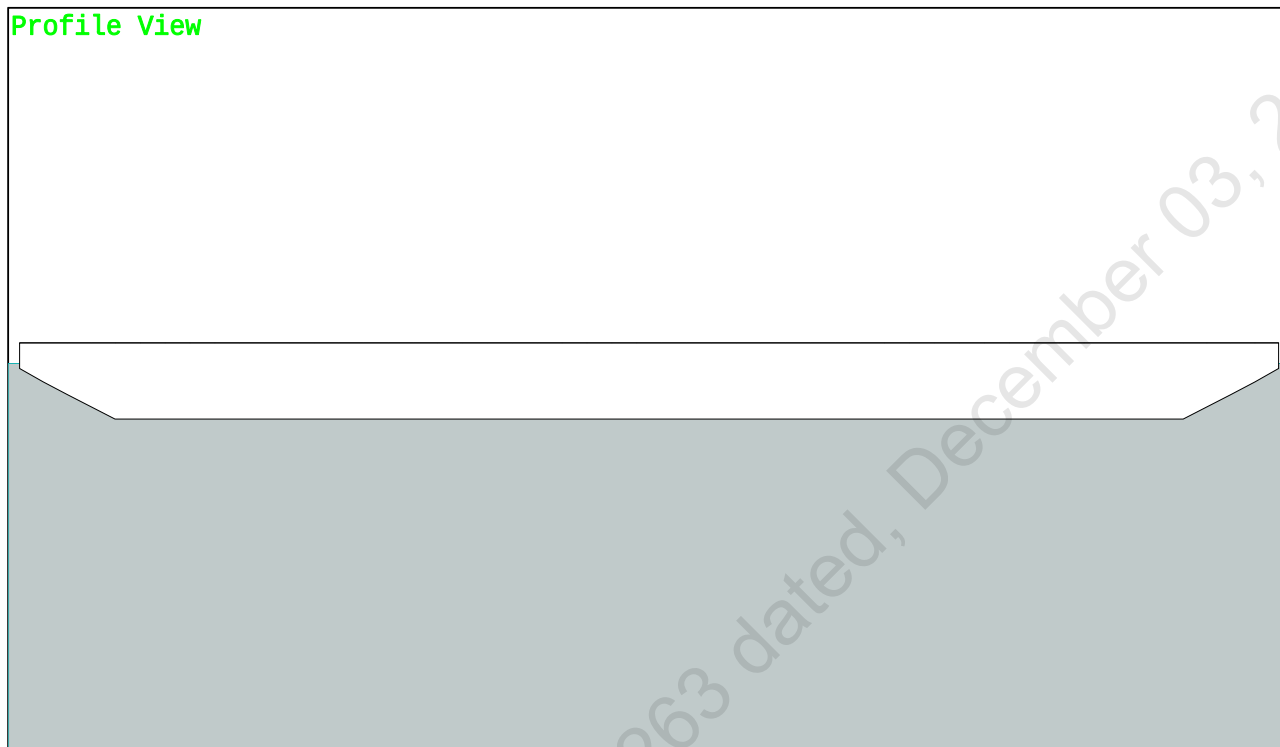
GHS 17.90

PONT00N

Condition 03 : Lightship + Deck CargoMaximum Deck Cargo Condition

Condition Graphic - Draft: 2.214 @ 50.000f, 2.210 @ 0.000 Heel: zero

Profile View



Plan View



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GHS 17.90

PONTOON

Condition 03 : Lightship + Deck Cargo
Maximum Deck Cargo Condition

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.214 @ 50.00f, 2.212 @ 25.00f, 2.210 @ 0.00

Trim: 0.000/50.000, Heel: zero

Part-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----	FSM-----
LIGHT SHIP	449.57	25.030f	0.000	3.000	
Deck Cargo	1,443.49	25.000f	0.000	4.000	
Total Weight----->	1,893.06	25.007f	0.000	3.763	
Load-----	SpGr-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----
Total Tanks----->		0.00			0.00
		Displ(MT)-----	LCB-----	TCB-----	VCB-----
HULL	1.025	1,893.06	25.007f	0.000	1.132

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025	900.4	25.000f	0.000	101.45	13.177
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
		9.23	37.41	98.82	10.546	

Distances in METERS.-----Moments in m.-MT.

SUMMARY OF LOADING

0.0 Cu.M. (0%) SALT WATER

1,443.49 MT Deck Cargo

DRAFT AT FP (M) = 2.214
DRAFT AT MS (M) = 2.212
DRAFT AT AP (M) = 2.210
DRAFT AT FWD MARK (M) = 2.214
DRAFT AT AFT MARK (M) = 2.210

HYDROSTATIC PROPERTIES

Trim: 0.000/50.000, No Heel, VCG = 3.763

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)-----	LCB-----	VCB-----	cm-----
2.212	1,893.06	25.007f	1.132	9.23
				25.000f
				37.41
				98.82
				10.546

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.214 @ 50.00f, 2.212 @ 25.00f, 2.210 @ 0.00

Trim: 0.000/50.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm-----	Pressure-----	Moment-----
HULL	40.5	0.405	1.478	0.05500	3.29
DECKCARGO.C	29.4	1.800	2.873	0.05500	4.65
Total wind heeling moment to starboard----->					7.94

Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT

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PONTOON
Condition 03 : Lightship + Deck Cargo
Maximum Deck Cargo Condition

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
LCG = 25.007f TCG = 0.000 VCG = 3.763

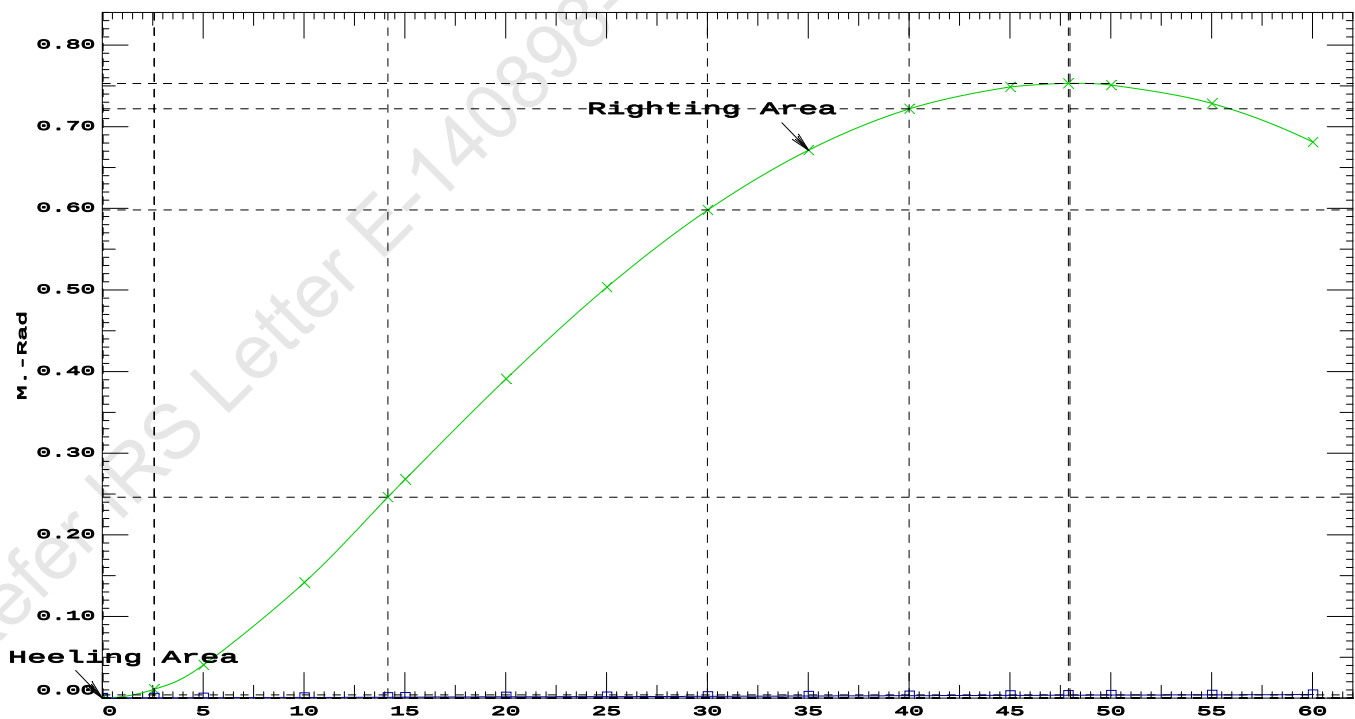
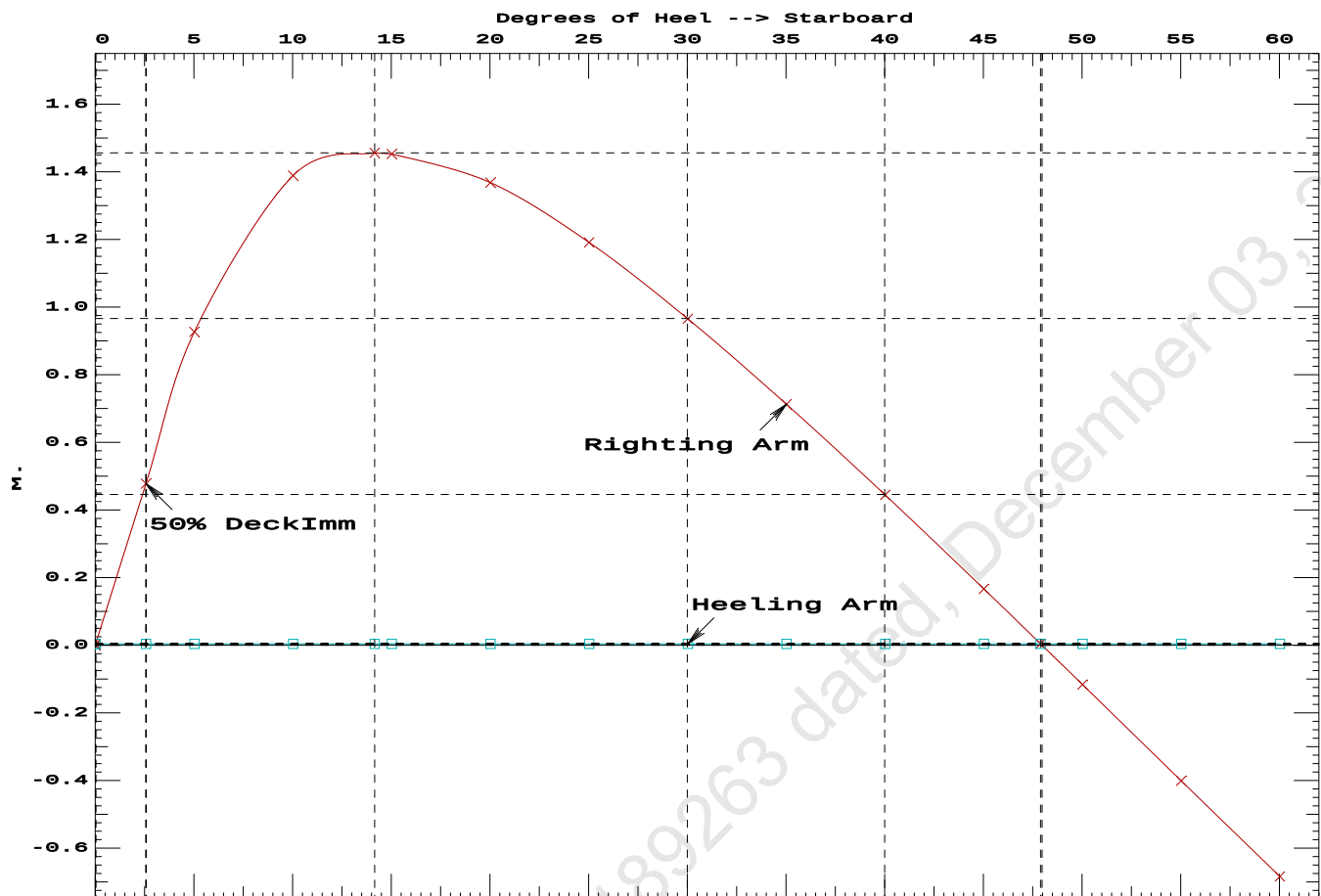
Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. 50% DeckImm Area	Res. 50% DeckImm Angle
2.198	0.00f	0.00	1,893.1	0.000	-0.004	0.0000	2.570
2.198	0.00f	0.02s	1,893.1	0.000	0.000	-0.0000	2.548
2.196	0.00f	2.57s	1,893.1	0.000	0.473	0.0105	0.000
2.187	0.00f	5.02s	1,893.1	0.000	0.923	0.0404	-2.452
2.280	0.01f	10.02s	1,893.1	0.000	1.385	0.1462	-7.452
2.462	0.01f	14.16s	1,892.9	0.000	1.452	0.2503	-11.587
2.506	0.01f	15.02s	1,893.1	0.000	1.449	0.2722	-12.452
2.775	0.01f	20.02s	1,893.1	0.000	1.364	0.3966	-17.452
3.036	0.01f	25.02s	1,893.1	0.000	1.187	0.5086	-22.452
3.273	0.02f	30.02s	1,893.1	0.000	0.961	0.6026	-27.452
3.486	0.02f	35.02s	1,893.1	0.000	0.708	0.6757	-32.452
3.672	0.02f	40.02s	1,893.1	0.000	0.440	0.7259	-37.452
3.829	0.02f	45.02s	1,893.1	0.000	0.162	0.7523	-42.452
3.907	0.02f	47.90s	1,893.1	0.000	0.000	0.7564	-45.330
3.958	0.03f	50.02s	1,893.1	0.000	-0.120	0.7541	-47.452
4.057	0.03f	55.02s	1,893.1	0.000	-0.405	0.7312	-52.452
4.125	0.03f	60.02s	1,893.1	0.000	-0.688	0.6835	-57.452

Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.

Note: No tank loads are present.

Note: The Residual Righting Arms shown above are in excess of the
wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 7.94 (constant)

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.2503 P
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg 47.88 P
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 2.55 P

01-12-21 09:39:35
GHS 17.90Cybermarine Technologies PTE. Ltd.
PONTOONCondition 03 : Lightship + Deck Cargo
Maximum Deck Cargo Condition

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

Profile View



Plan View



Tanks		
17 06_WBTK.C.100% SALT WATER Intact	19 07_WBTK.P.100% SALT WATER Intact	21 07_WBTK.S.100% SALT WATER Intact
	20 07_WBTK.C.100% SALT WATER Intact	

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GHS 17.90

PONTOON

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.199 @ 50.00f, 1.419 @ 25.00f, 1.639 @ 0.00

Trim: Aft 0.440/50.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			449.57	25.030f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	19.500a	0.000	34.602	
Total Fixed----->			889.57	16.944f	0.000	5.916	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			284.98	45.086f	0.000	1.697	352.57
Total Weight----->			1,174.55	23.772f	0.000	4.893	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,174.44	23.735f	0.000	0.724	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	862.4	24.615f	0.000	143.66	20.017*
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			8.84		32.76	139.49	15.848
Distances in METERS.-----				Moments in m.-MT			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

278.0 Cu.M. (11%) SALT WATER

400.00 MT Crane

40.00 MT Crane Load

DRAFT AT FP (M) = 1.199
DRAFT AT MS (M) = 1.419
DRAFT AT AP (M) = 1.639
DRAFT AT FWD MARK (M) = 1.235
DRAFT AT AFT MARK (M) = 1.604

HYDROSTATIC PROPERTIES

Trim: Aft 0.440/50.000, No Heel, VCG = 4.893

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)----	LCB-----	VCB-----	cm trim-----
1.423	1,174.44	23.735f	0.724	8.84
				24.615f
				32.76
				139.49
				15.848

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 352.6 m.-MT

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GHS 17.90

PONTOON

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.199 @ 50.00f, 1.419 @ 25.00f, 1.639 @ 0.00

Trim: Aft 0.440/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	79.3	0.812	1.511	0.05500	6.59
CRANE	57.0	3.961	4.660	0.05500	14.61
Total wind heeling moment to starboard-->					21.20
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.772f TCG = 0.000 VCG = 4.893

Free Surface Adjustment: 0.300

Adjusted CG: LCG = 23.775f TCG = 0.000 VCG = 5.193

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	in Heel	Res. 50% DeckImm Area	Angle
1.627	0.50a	0.00	1,174.6	0.000	-0.018	0.0000	4.419
1.627	0.50a	0.06s	1,174.6	0.000	0.000	-0.0000	4.358
1.616	0.50a	4.42s	1,174.6	0.000	1.214	0.0462	0.000
1.613	0.51a	5.06s	1,174.6	0.000	1.393	0.0608	-0.642
1.582	0.55a	10.06s	1,174.5	0.000	2.672	0.2390	-5.642
1.585	0.81a	15.06s	1,174.6	0.000	3.094	0.4975	-10.642
1.585	0.83a	15.33s	1,174.6	0.000	3.095	0.5122	-10.914
1.585	1.09a	20.06s	1,174.6	0.000	2.957	0.7653	-15.642
1.580	1.38a	25.06s	1,174.6	0.000	2.635	1.0106	-20.642
1.570	1.68a	30.06s	1,174.6	0.000	2.229	1.2234	-25.642
1.553	1.97a	35.06s	1,174.6	0.000	1.775	1.3984	-30.642
1.530	2.26a	40.06s	1,174.6	0.000	1.291	1.5324	-35.642
1.499	2.55a	45.06s	1,174.6	0.000	0.787	1.6232	-40.642
1.458	2.82a	50.06s	1,174.6	0.000	0.272	1.6695	-45.642
1.433	2.95a	52.68s	1,174.6	0.000	0.000	1.6758	-48.262
1.407	3.06a	55.06s	1,174.6	0.000	-0.248	1.6706	-50.642
1.342	3.28a	60.06s	1,174.6	0.000	-0.769	1.6263	-55.642
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.							

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

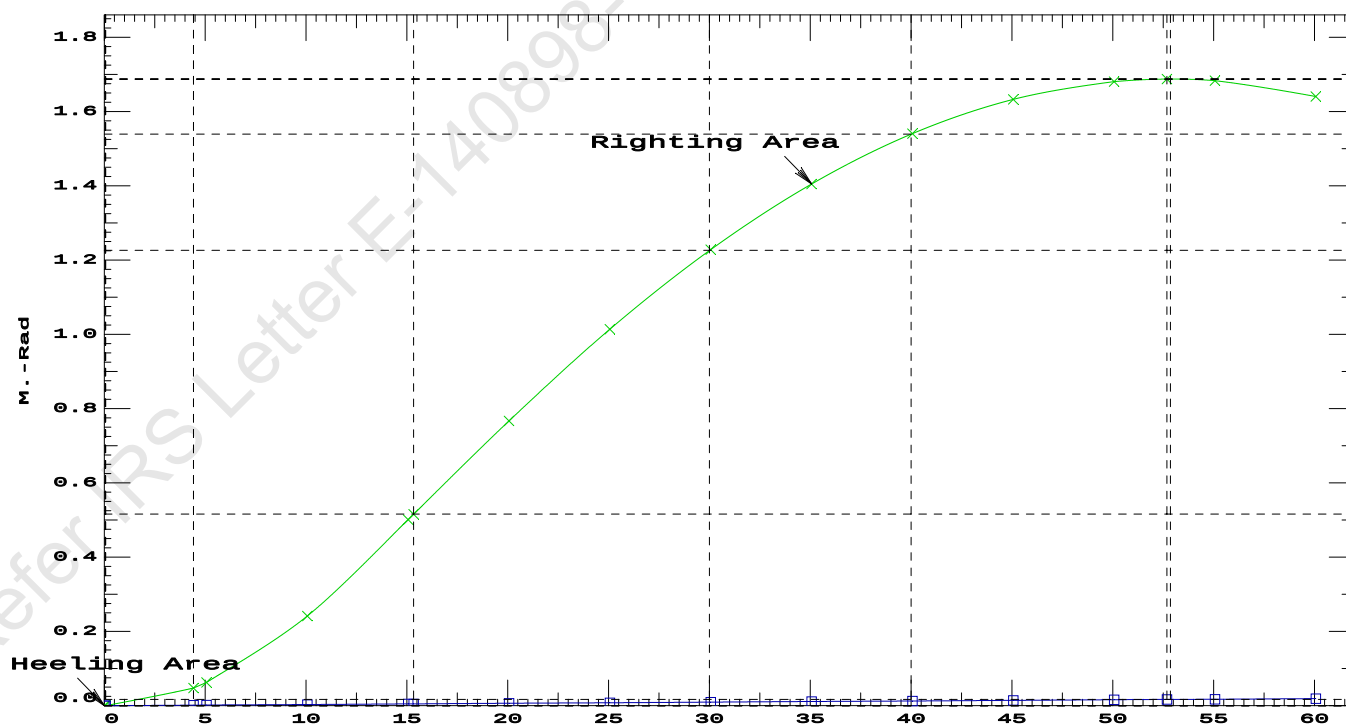
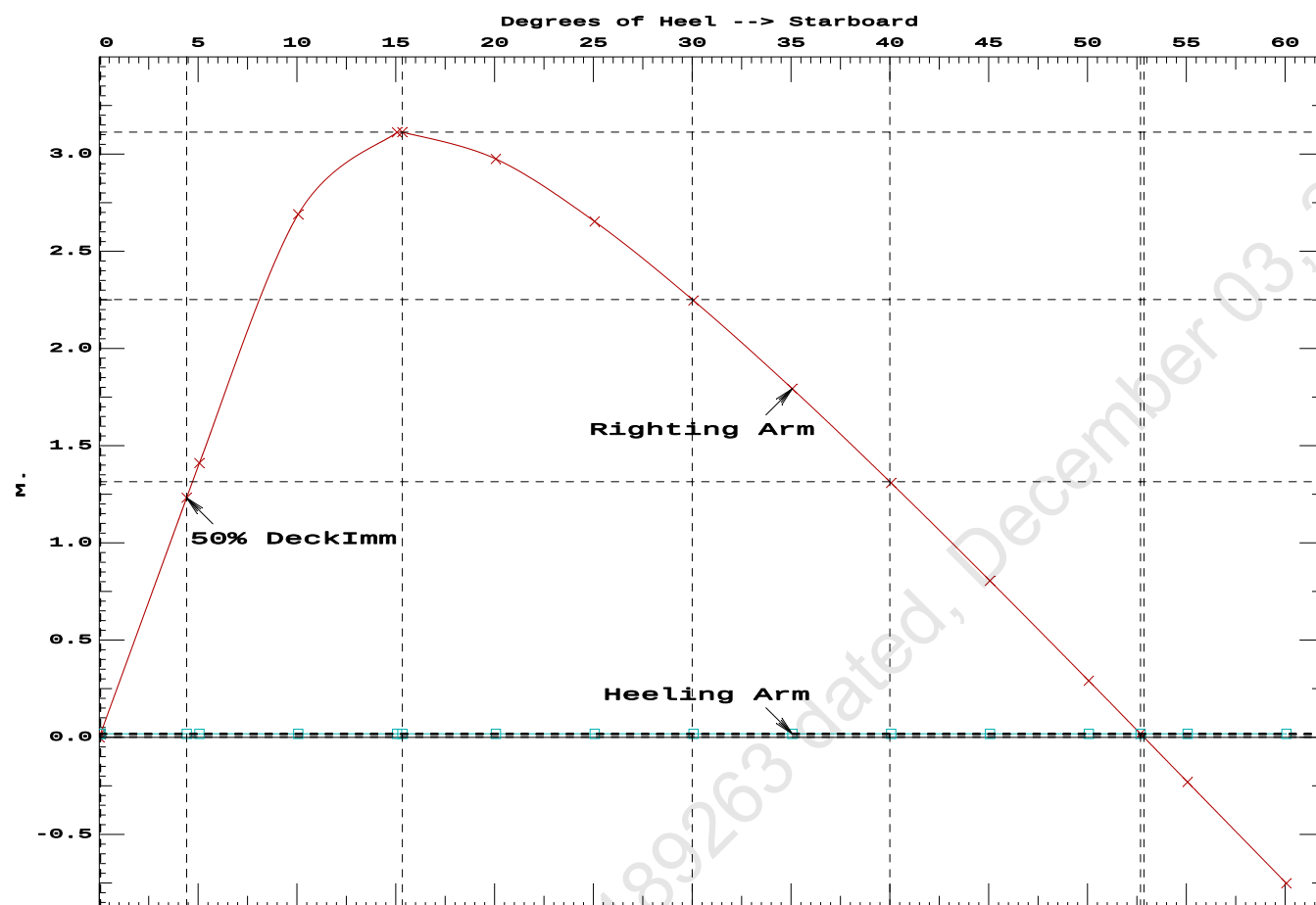
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.20 (constant)

LIM	IS CODE 2008 PART B CHAPTER-2 SEC.2	Min/Max	Attained
(1)	Residual Area from abs 0 deg to MaxRA	> 0.0800 m.-Rad	0.5122 P
(2)	Angle from Equilibrium to RAZero or Flood	> 20.00 deg	52.62 P
(3)	Angle from Equilibrium to 50% Dk Imm Angle	> 0.00 deg	4.36 P

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PONT00N

Condition 04 :Operating Condition of 400T CraneCrane Lifting 40.0T @ 30m Radius Aft

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PONTOON

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.772f TCG = 0.000 VCG = 4.893
Free Surface Adjustment: 0.300
Adjusted CG: LCG = 23.775f TCG = 0.000 VCG = 5.193

Origin Depth---	Degrees of Trim----	Displacement Heel----	Weight(MT)---	Residual Arms in Trim--	in Heel---	Res. Area
1.627	0.50a	0.01s	1,174.6	0.000	0.000	0.0000
1.614	0.51a	5.01s	1,174.6	0.000	1.393	0.0608
1.582	0.55a	10.01s	1,174.5	0.000	2.676	0.2391
1.585	0.81a	15.01s	1,174.6	0.000	3.107	0.4985
1.585	0.83a	15.34s	1,174.6	0.000	3.109	0.5165
1.585	1.09a	20.01s	1,174.6	0.000	2.973	0.7675
1.580	1.38a	25.01s	1,174.6	0.000	2.652	1.0142
1.570	1.68a	30.01s	1,174.6	0.000	2.246	1.2286
1.553	1.97a	35.01s	1,174.6	0.000	1.793	1.4052
1.530	2.26a	40.01s	1,174.6	0.000	1.309	1.5408
1.499	2.55a	45.01s	1,174.6	0.000	0.806	1.6332
1.459	2.81a	50.01s	1,174.6	0.000	0.291	1.6811
1.431	2.96a	52.81s	1,174.6	0.000	0.000	1.6883
1.407	3.06a	55.01s	1,174.6	0.000	-0.229	1.6839
1.343	3.28a	60.01s	1,174.6	0.000	-0.750	1.6411

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

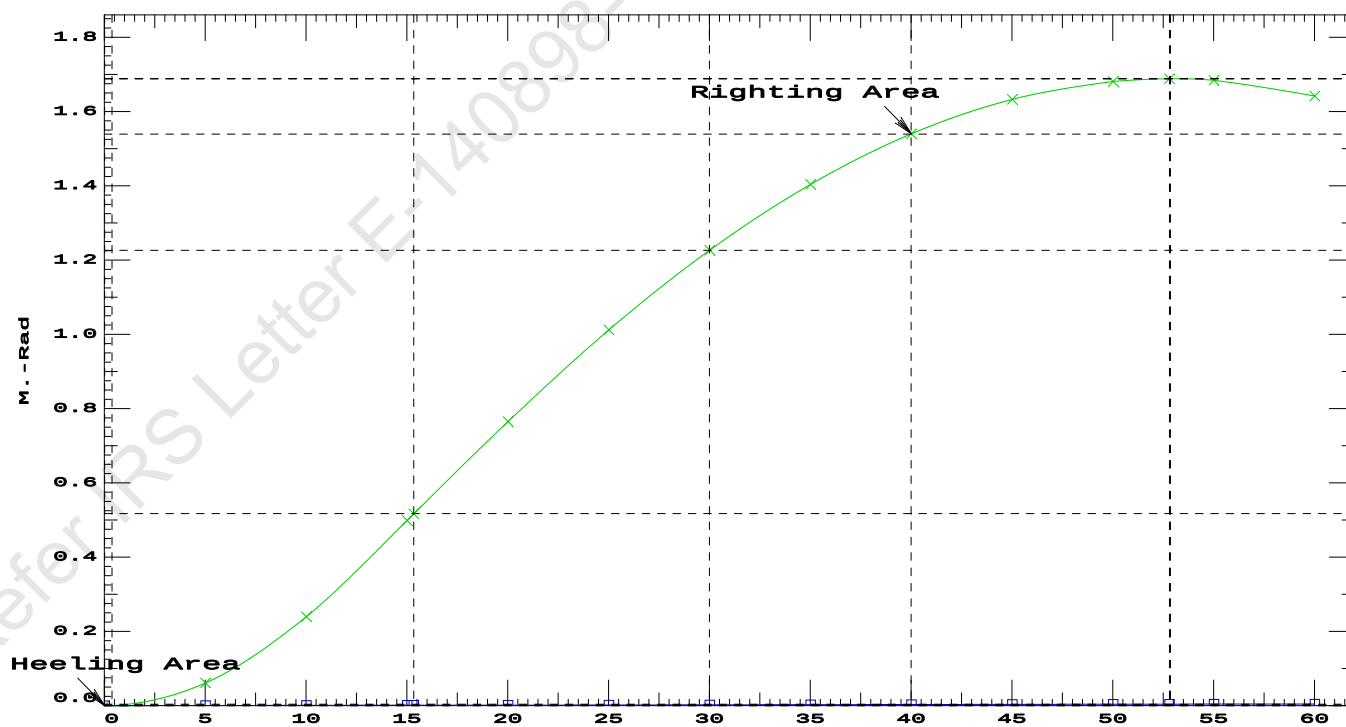
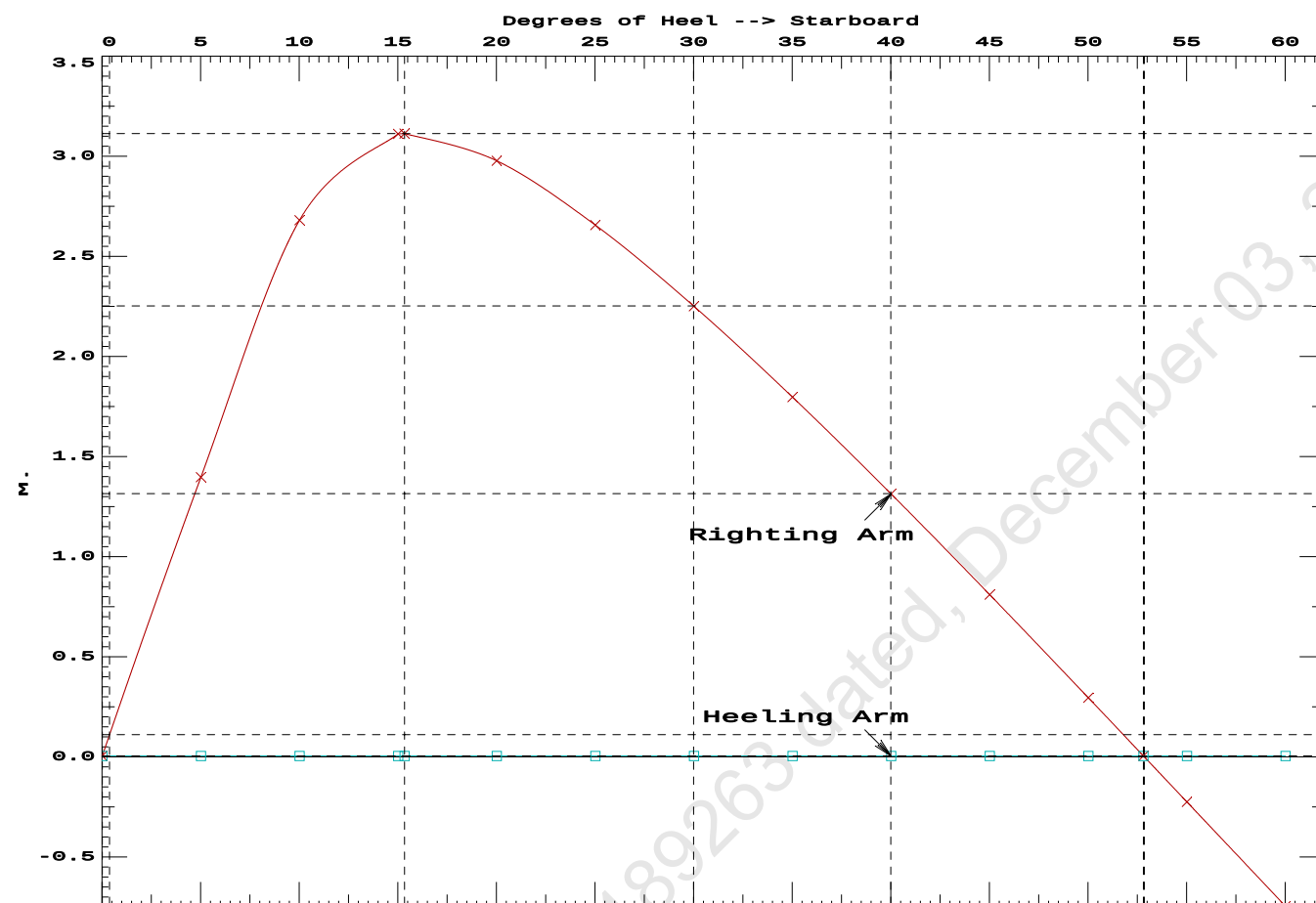
LIM-----STABILITY CRITERIA FOR CRANE OPERAT-----Min/Max-----Attained

(1) Absolute Angle at Equilibrium	<	15.00	deg	0.01 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	73025.3 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667		431.362 P

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PONT00N

Condition 04 :Operating Condition of 400T CraneCrane Lifting 40.0T @ 30m Radius Aft

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Trim: Aft 0.440/50.000, Heel: 0.00 deg.

Distances in METERS.-----Moments in m.-MT.

Angle Theta1 = 0.00

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After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.427 @ 50.00f, 1.376 @ 25.00f, 1.324 @ 0.00
Trim: Fwd 0.103/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----		
LIGHT SHIP	449.57	25.030f	0.000	3.000			
Crane	400.00	11.500f	0.000	6.325			
Total Fixed----->	849.57	18.660f	0.000	4.565			
Load-----	SpGr-----	Weight(MT)---- <td>LCG-----</td> <td>TCG-----</td> <td>VCG-----</td> <td>FSM-----</td>	LCG-----	TCG-----	VCG-----	FSM-----	
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			284.98	45.086f	0.000	1.697	352.57
Total Weight----->			1,134.55	25.298f	0.000	3.845	
			Displ(MT)---- <td>LCB-----</td> <td>TCB-----</td> <td>VCB-----</td> <td></td>	LCB-----	TCB-----	VCB-----	
HULL	1.025		1,134.56	25.304f	0.005s	0.695	

Righting Arms:			0.000	0.005s			
External Arms:			0.000	0.004s			
Residual Righting Arms:			0.000	0.001s			
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----	
Total Waterplane---->	1.025	849.0	25.028f	0.069s	141.96	20.425*	
		MT/cm					
		8.70					
Distances in METERS.-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.660f TCG = 0.000 VCG = 4.565

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.312	0.12f	0.00	1,134.6	0.000	0.001	0.0000
1.312	0.12f	0.07s	1,134.7	0.000	0.022	0.0000
1.297	0.12f	5.00s	1,134.6	0.000	1.562	0.0682
1.241	0.13f	10.00s	1,134.6	0.000	3.011	0.2686
1.081	0.19f	15.00s	1,134.6	0.000	3.594	0.5631
1.005	0.21f	17.25s	1,134.6	0.000	3.625	0.7049
0.908	0.25f	20.00s	1,134.6	0.000	3.580	0.8788
0.726	0.32f	25.00s	1,134.6	0.000	3.370	1.1845
0.537	0.38f	30.00s	1,134.6	0.000	3.066	1.4659

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 849.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 LARGE

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.660f TCG = 0.000 VCG = 4.565

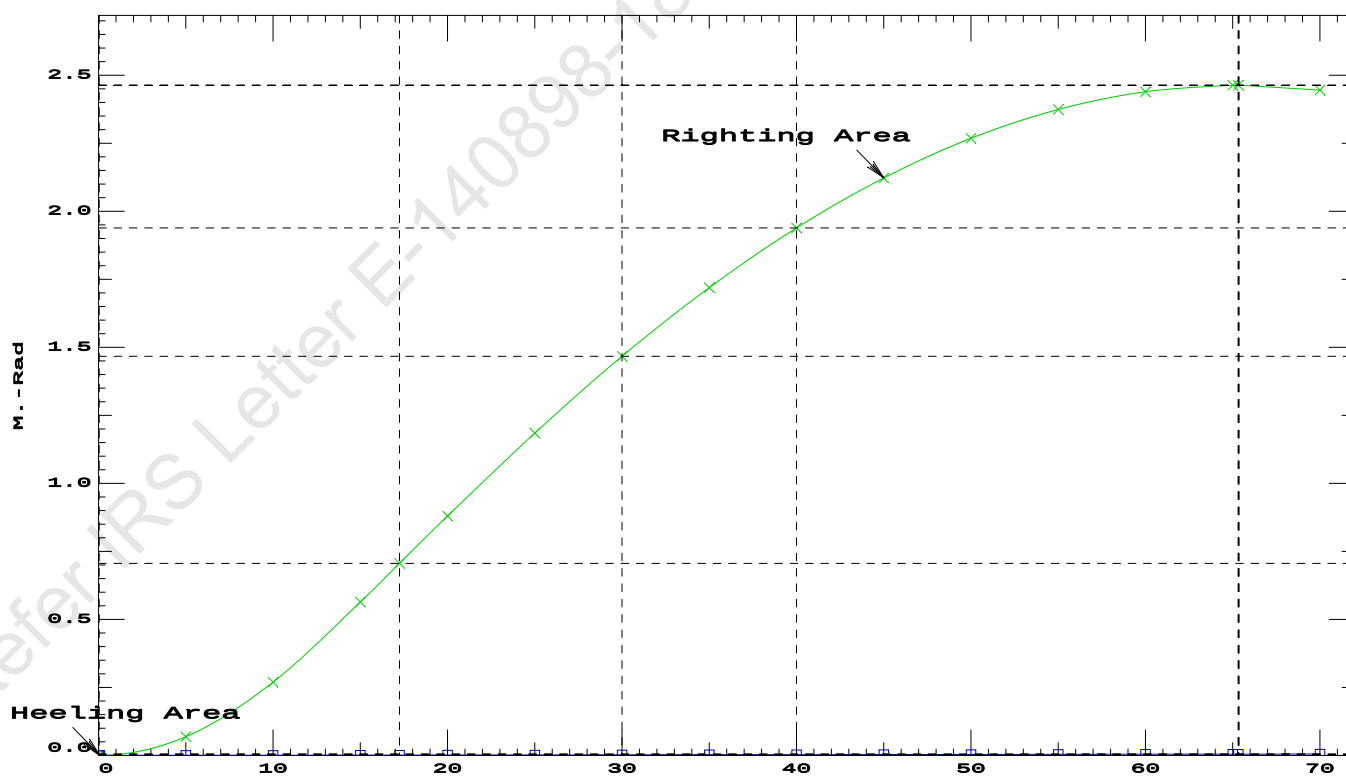
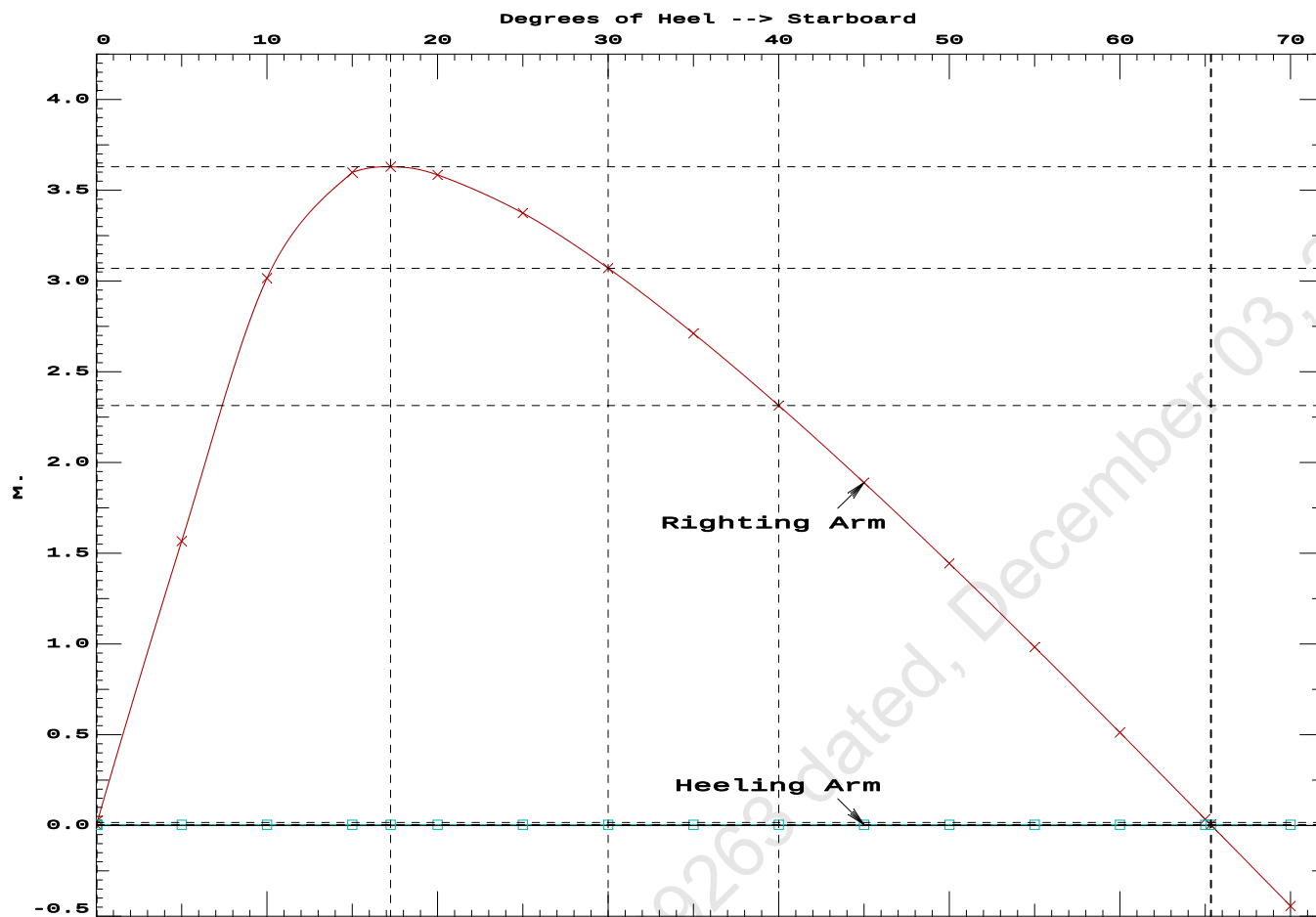
Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.312	0.12f	0.00	1,134.6	0.000	0.001	0.0000
1.312	0.12f	0.07s	1,134.7	0.000	0.022	0.0000
1.297	0.12f	5.00s	1,134.6	0.000	1.562	0.0682
1.241	0.13f	10.00s	1,134.6	0.000	3.011	0.2686
1.081	0.19f	15.00s	1,134.6	0.000	3.594	0.5631
1.005	0.21f	17.25s	1,134.6	0.000	3.625	0.7049
0.908	0.25f	20.00s	1,134.6	0.000	3.580	0.8788
0.726	0.32f	25.00s	1,134.6	0.000	3.370	1.1845
0.537	0.38f	30.00s	1,134.6	0.000	3.066	1.4659
0.344	0.45f	35.00s	1,134.6	0.000	2.707	1.7182
0.146	0.51f	40.00s	1,134.6	0.000	2.310	1.9374
-0.054	0.58f	45.00s	1,134.6	0.000	1.885	2.1206
-0.254	0.64f	50.00s	1,134.6	0.000	1.439	2.2658
-0.453	0.69f	55.00s	1,134.6	0.000	0.978	2.3714
-0.648	0.75f	60.00s	1,134.6	0.000	0.507	2.4363
-0.837	0.79f	65.00s	1,134.6	0.000	0.030	2.4598
-0.849	0.79f	65.32s	1,134.6	0.000	0.000	2.4599
-1.020	0.83f	70.00s	1,134.6	0.000	-0.448	2.4416

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 849.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 0.1 deg to RAZero > 20.00 deg 65.25 P

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PONT00N

01-12-21 10:00:51 Cybermarine Technologies PTE. Ltd.
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Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd

Condition Graphic - Draft: 1.390 @ 50.000f, 1.452 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

17 06_WBTK.C.100% SALT WATER Intact	19 07_WBTK.P.100% SALT WATER Intact	21 07_WBTK.S.100% SALT WATER Intact
	20 07_WBTK.C.100% SALT WATER Intact	

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PONTOON

Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.390 @ 50.00f, 1.421 @ 25.00f, 1.452 @ 0.00

Trim: Aft 0.062/50.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			449.57	25.030f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	11.500f	0.000	34.602	
Total Fixed----->			889.57	18.338f	0.000	5.916	
	Load----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			284.98	45.086f	0.000	1.697	352.57
Total Weight----->			1,174.55	24.828f	0.000	4.893	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,174.55	24.823f	0.000	0.718	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	863.3	24.940f	0.000	144.08	20.072*
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			8.85		32.87	139.91	15.898
Distances in METERS.-----				Moments in m.-MT			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

278.0 Cu.M. (11%) SALT WATER

400.00 MT Crane

40.00 MT Crane Load

DRAFT AT FP (M) = 1.390
DRAFT AT MS (M) = 1.421
DRAFT AT AP (M) = 1.452
DRAFT AT FWD MARK (M) = 1.395
DRAFT AT AFT MARK (M) = 1.447

HYDROSTATIC PROPERTIES

Trim: Aft 0.062/50.000, No Heel, VCG = 4.893

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----
1.421	1,174.55	24.823f	0.718	8.85
				24.940f
				32.87
				139.91
				15.898

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 352.6 m.-MT

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PONTOON

Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.390 @ 50.00f, 1.421 @ 25.00f, 1.452 @ 0.00

Trim: Aft 0.062/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	79.4	0.806	1.502	0.05500	6.56
CRANE	57.0	4.072	4.768	0.05500	14.95
Total wind heeling moment to starboard-->					21.51
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.828f TCG = 0.000 VCG = 4.893

Free Surface Adjustment: 0.300

Adjusted CG: LCG = 24.828f TCG = 0.000 VCG = 5.193

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. 50% DeckImm Area	Angle
1.440	0.07a	0.00	1,174.6	0.000	-0.018	0.0000	5.071
1.440	0.07a	0.04s	1,174.6	0.000	0.000	-0.0000	5.027
1.425	0.07a	5.04s	1,174.6	0.000	1.388	0.0606	0.027
1.425	0.07a	5.07s	1,174.6	0.000	1.396	0.0612	0.000
1.378	0.08a	10.04s	1,174.6	0.000	2.687	0.2391	-4.973
1.281	0.11a	15.04s	1,174.6	0.000	3.127	0.5002	-9.973
1.275	0.12a	15.35s	1,174.6	0.000	3.128	0.5167	-10.276
1.176	0.15a	20.04s	1,174.6	0.000	2.992	0.7710	-14.973
1.062	0.19a	25.04s	1,174.6	0.000	2.671	1.0194	-19.973
0.941	0.24a	30.04s	1,174.6	0.000	2.265	1.2354	-24.973
0.814	0.28a	35.04s	1,174.6	0.000	1.811	1.4136	-29.973
0.682	0.32a	40.04s	1,174.6	0.000	1.327	1.5507	-34.973
0.544	0.36a	45.04s	1,174.6	0.000	0.822	1.6447	-39.973
0.403	0.40a	50.04s	1,174.6	0.000	0.305	1.6939	-44.973
0.320	0.42a	52.96s	1,174.6	0.000	0.000	1.7017	-47.891
0.259	0.43a	55.04s	1,174.6	0.000	-0.218	1.6977	-49.973
0.113	0.46a	60.04s	1,174.6	0.000	-0.743	1.6558	-54.973
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.							

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

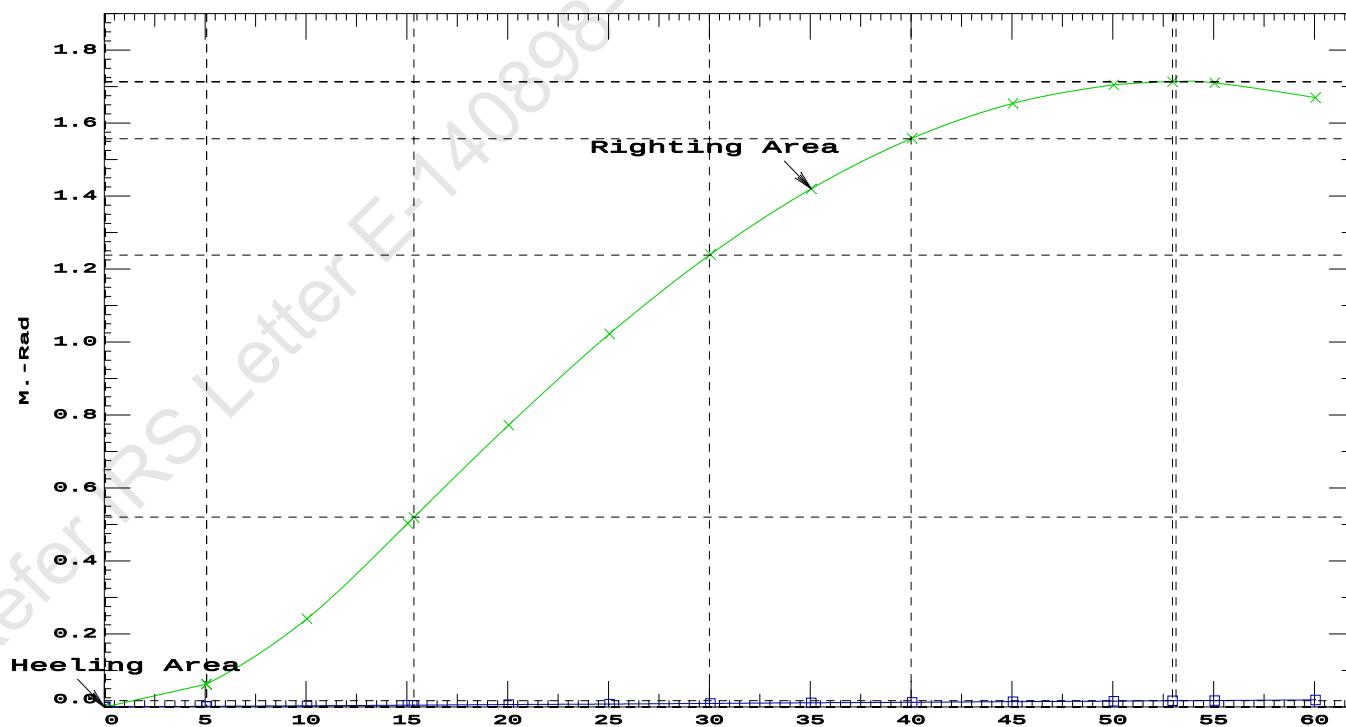
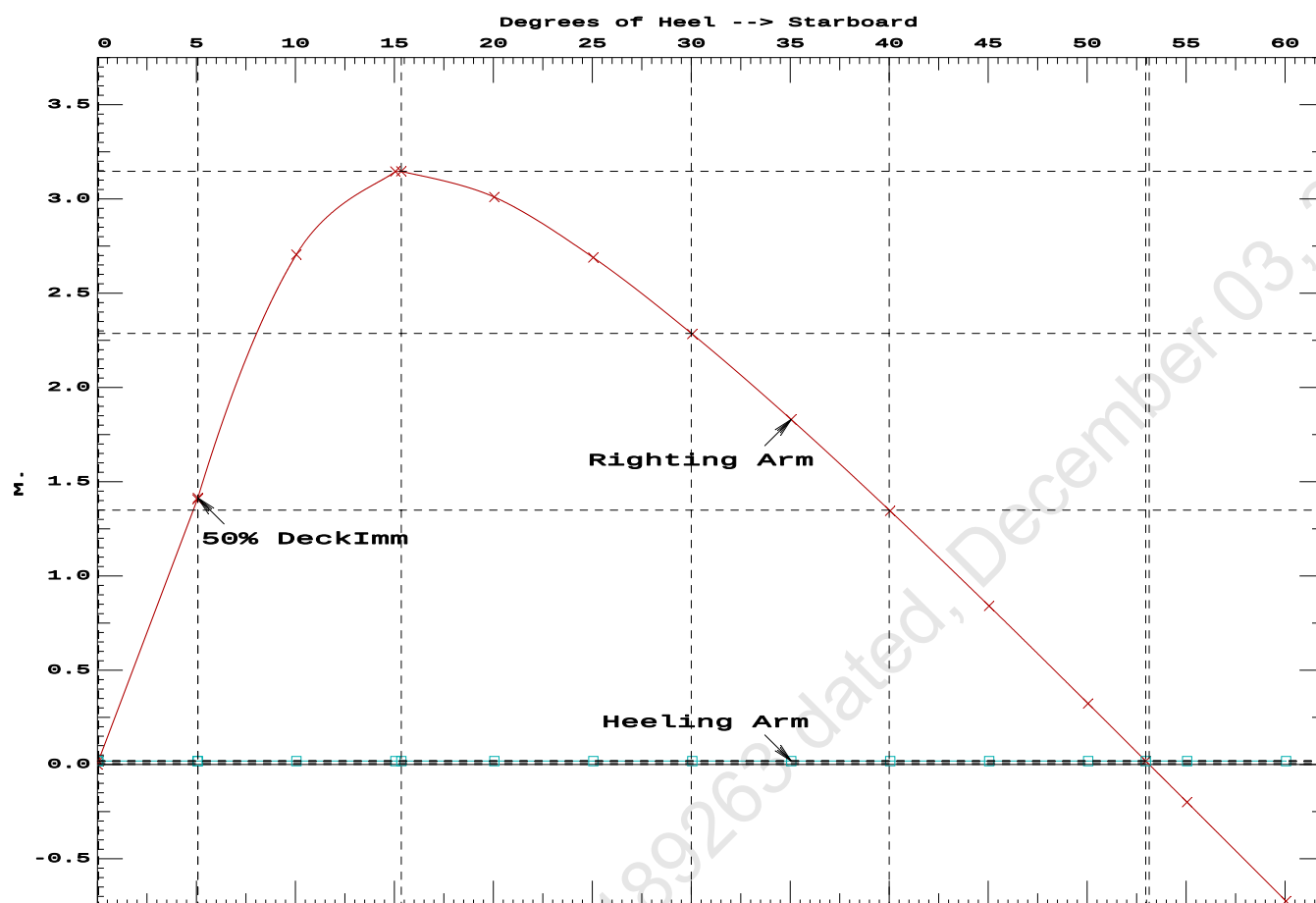
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.51 (constant)

LIM	IS CODE 2008 PART B CHAPTER-2 SEC.2	Min/Max	Attained
(1)	Residual Area from abs 0 deg to MaxRA	> 0.0800 m.-Rad	0.5167 P
(2)	Angle from Equilibrium to RAZero or Flood	> 20.00 deg	52.92 P
(3)	Angle from Equilibrium to 50% Dk Imm Angle	> 0.00 deg	5.03 P

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GHS 17.90

PONTOON

Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd



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GHS 17.90

PONTOON

Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.828f TCG = 0.000 VCG = 4.893
Free Surface Adjustment: 0.300
Adjusted CG: LCG = 24.828f TCG = 0.019p VCG = 5.192

Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---	Trim----	Heel----	Weight(MT)----	in Trim--in Heel----> Area
1.431	0.07a	3.59s	1,174.6	0.000 0.000 0.0000
1.398	0.07a	8.59s	1,174.6	0.000 1.352 0.0590
1.310	0.10a	13.59s	1,174.6	0.000 2.109 0.2194
1.271	0.12a	15.57s	1,174.6	0.000 2.143 0.2934
1.207	0.14a	18.59s	1,174.6	0.000 2.073 0.4055
1.096	0.18a	23.59s	1,174.6	0.000 1.790 0.5771
0.977	0.22a	28.59s	1,174.6	0.000 1.403 0.7172
0.852	0.26a	33.59s	1,174.6	0.000 0.960 0.8207
0.721	0.31a	38.59s	1,174.6	0.000 0.482 0.8838
0.590	0.35a	43.40s	1,174.6	0.000 0.000 0.9042
0.585	0.35a	43.59s	1,174.6	0.000 -0.019 0.9042
0.445	0.39a	48.59s	1,174.6	0.000 -0.534 0.8802
0.302	0.42a	53.59s	1,174.6	0.000 -1.057 0.8108
0.156	0.45a	58.59s	1,174.6	0.000 -1.584 0.6956
0.008	0.48a	63.59s	1,174.6	0.000 -2.108 0.5345

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1200.00

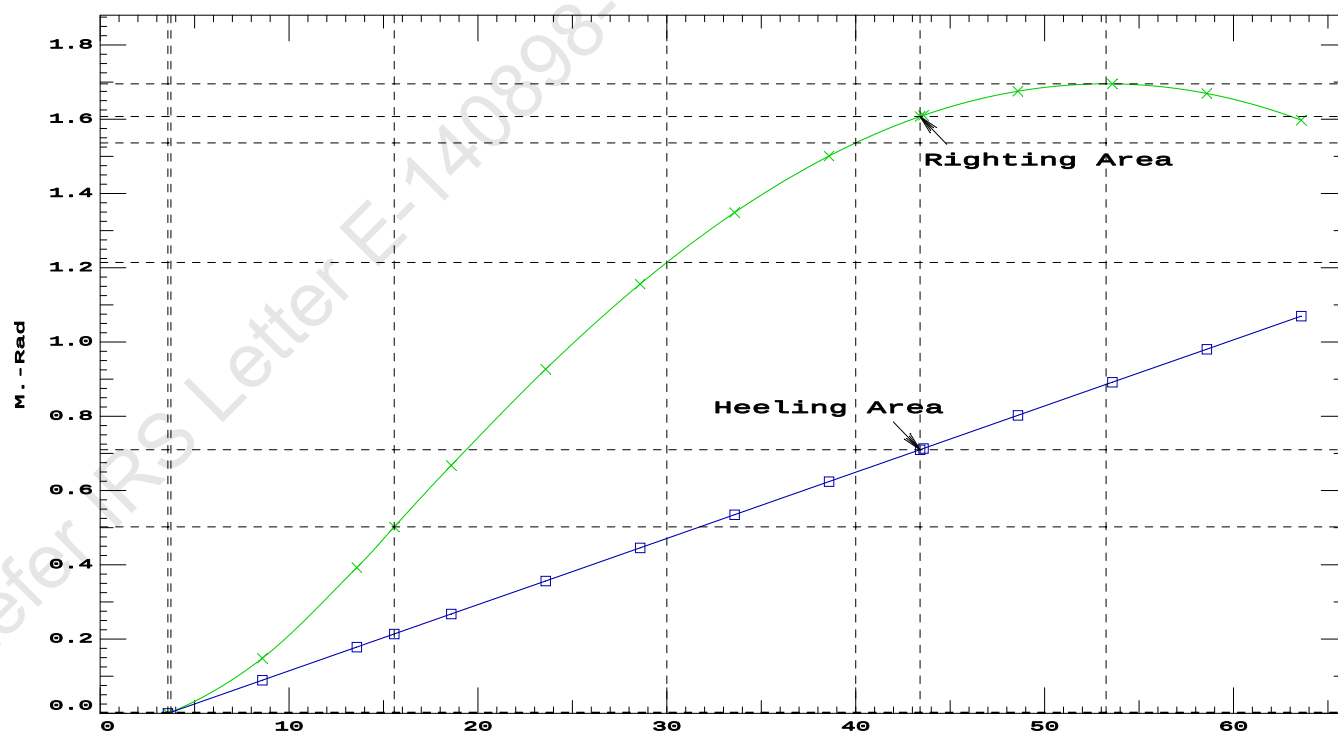
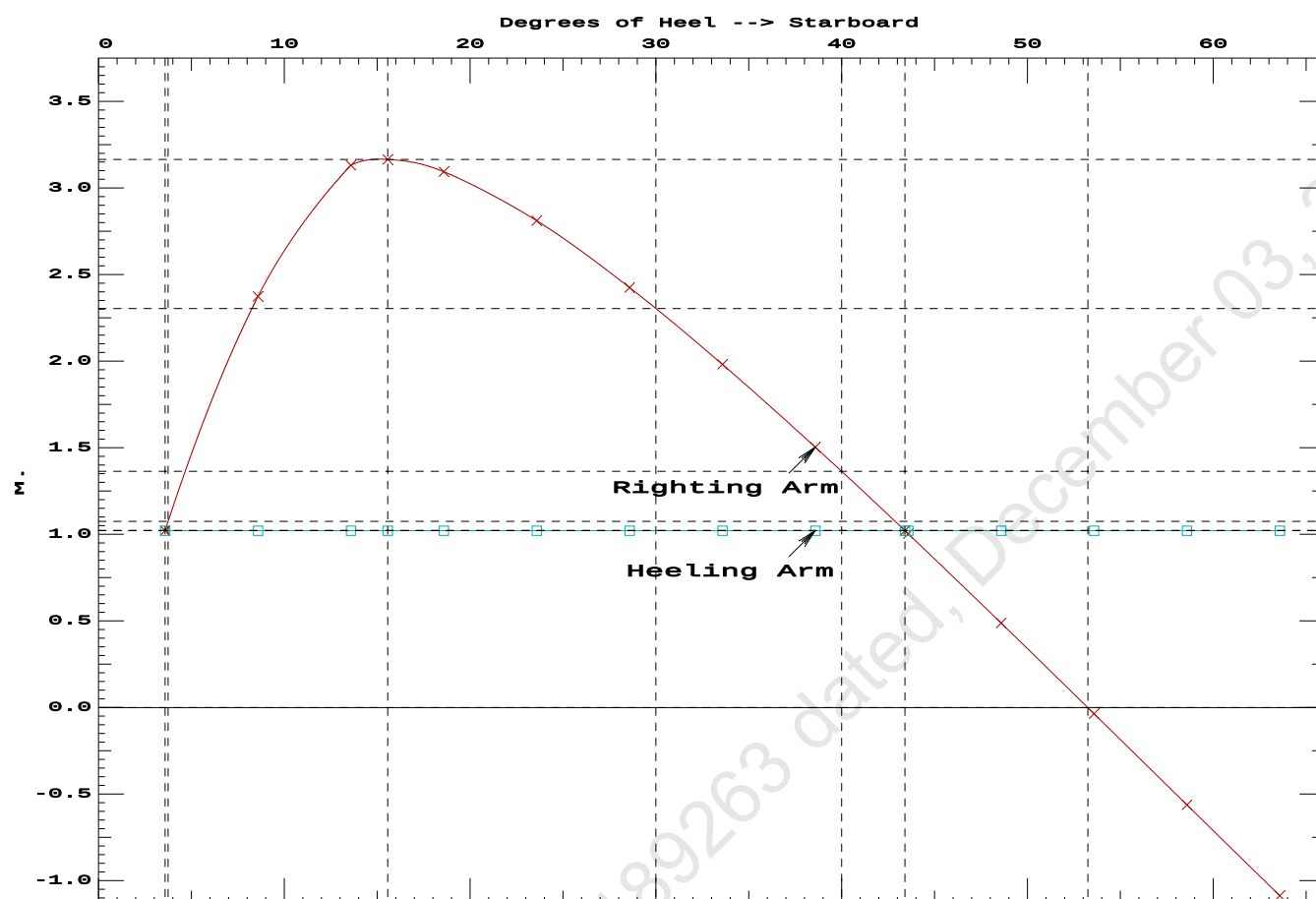
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	3.59 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	209.8 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	2.274 P

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GHS 17.90

PONT00N

Condition 05 : Operating Condition of 400T CraneCrane Lifting 40.0T @ 30m Radius Stbd

PONTOON

Crane Lifting 40.0T @ 30m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

Trim: Aft 0.061/50.000, Heel: Stbd 3.59 deg.

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025	862.7	24.934f	0.151s	143.58	20.082*
		MT/cm				
		8.84				

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 3.59

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After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.427 @ 50.00f, 1.376 @ 25.00f, 1.324 @ 0.00
Trim: Fwd 0.103/50.000, Heel: zero

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----			
LIGHT SHIP	449.57	25.030f	0.000	3.000			
Crane	400.00	11.500f	0.000	6.325			
Total Fixed----->	849.57	18.660f	0.000	4.565			
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----	
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			284.98	45.086f	0.000	1.697	352.57
Total Weight----->			1,134.55	25.298f	0.000	3.845	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL	1.025		1,134.53	25.304f	0.000	0.695	

Righting Arms:			0.000	0.000			
External Arms:			0.000	0.001			
Residual Righting Arms:			0.000	-0.001s			
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----	
Total Waterplane---->	1.025	860.0	25.098f	0.000	147.48	20.689*	
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----		
		8.81	32.75	144.33	17.539		
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 4073.64; t3: 30.0; t: 15.0
limmarg: 1503.78; t3: 15.0; t: 6.0
limmarg: 524.80; t3: 9.0; t: 2.4
limmarg: 238.48; t3: 6.6; t: 1.0
limmarg: 143.96; t3: 5.6; t: 0.4
limmarg: 110.39; t3: 5.2; t: 0.2
limmarg: 94.53; t3: 5.0; t: 0.1
limmarg: 86.86; t3: 4.9; t: 0.0
limmarg: 86.86; t3: 4.9; t: 0.0

Theta3 is 4.90 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.660f TCG = 0.000 VCG = 4.565

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.305	0.12f	3.59s	1,134.5	0.000	-1.125	0.0000
1.312	0.12f	0.00	1,134.6	0.000	0.001	-0.0352
1.298	0.12f	4.90p	1,134.6	0.000	1.536	0.0306
1.298	0.12f	5.00p	1,134.6	0.000	1.567	0.0333
1.241	0.13f	10.00p	1,134.6	0.000	3.016	0.2341
1.081	0.19f	15.00p	1,134.6	0.000	3.599	0.5291
1.005	0.21f	17.25p	1,134.6	0.000	3.630	0.6712
0.908	0.25f	20.00p	1,134.6	0.000	3.585	0.8453
0.726	0.32f	25.00p	1,134.6	0.000	3.375	1.1514
0.537	0.38f	30.00p	1,134.6	0.000	3.071	1.4333

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 849.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Res. Area Ratio from abs -3.6 deg to abs 4.9 > 1.000 1.869 P

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.660f TCG = 0.000 VCG = 4.565

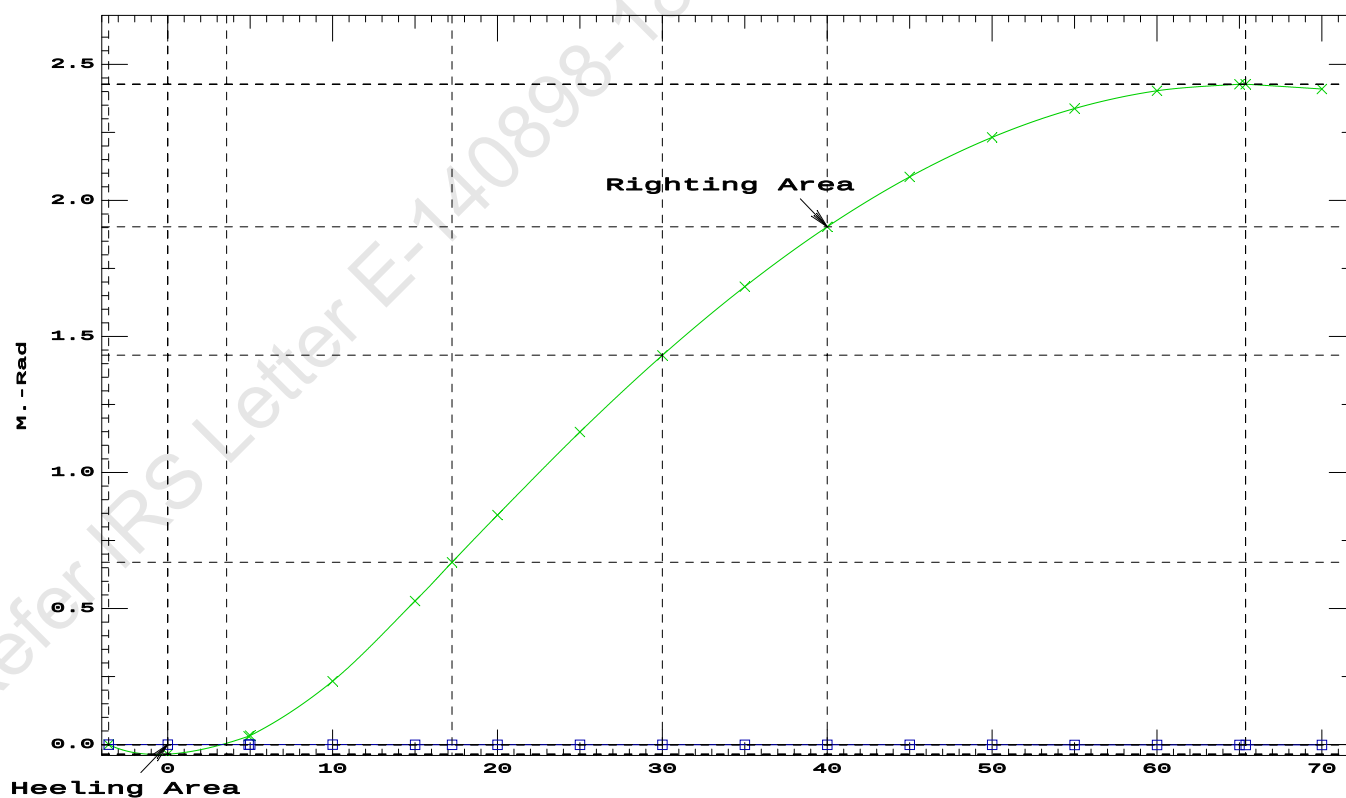
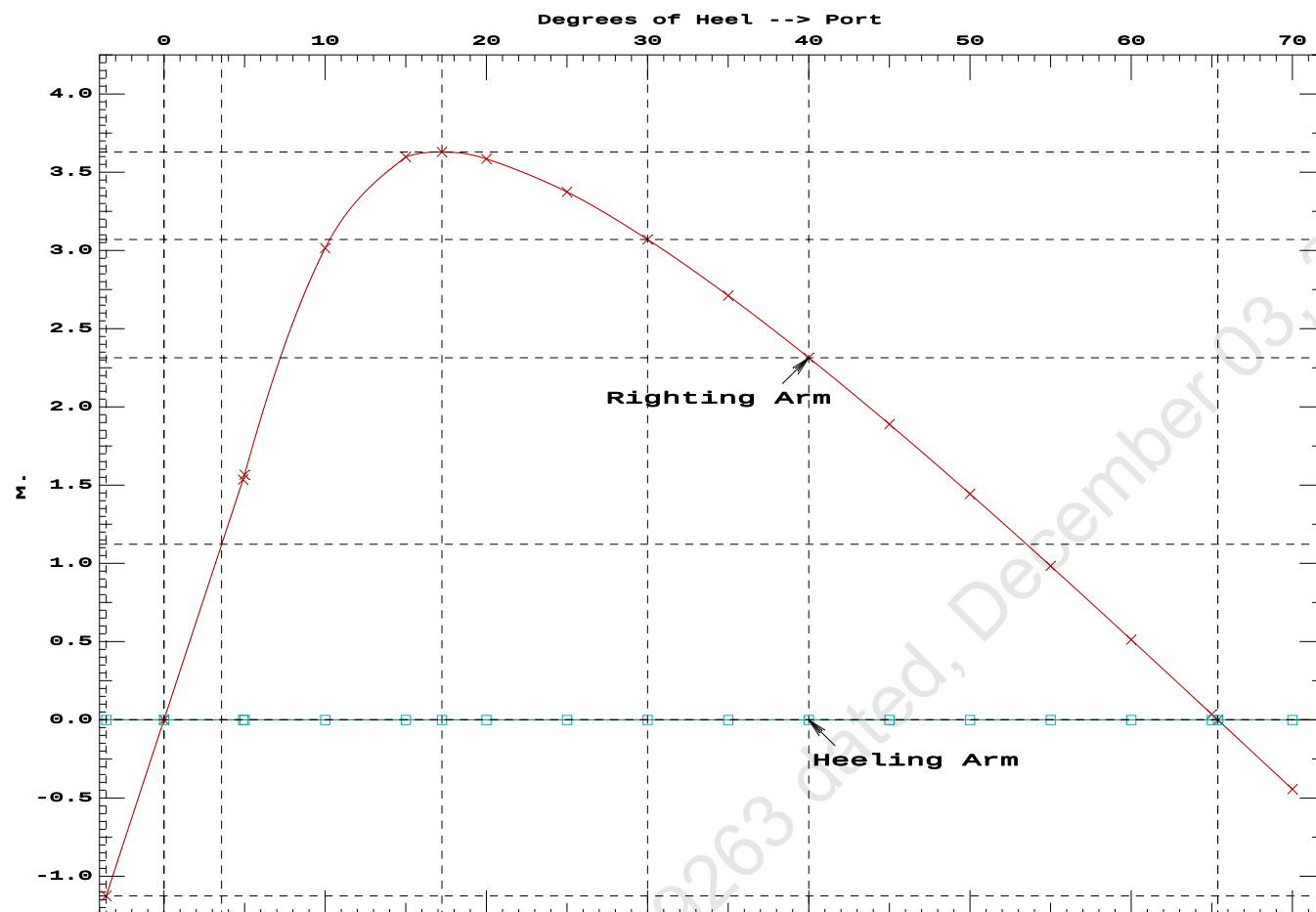
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.305	0.12f	3.59s	1,134.5	0.000	-1.125	0.0000
1.312	0.12f	0.00	1,134.6	0.000	0.001	-0.0352
1.298	0.12f	4.90p	1,134.6	0.000	1.536	0.0306
1.298	0.12f	5.00p	1,134.6	0.000	1.567	0.0333
1.241	0.13f	10.00p	1,134.6	0.000	3.016	0.2341
1.081	0.19f	15.00p	1,134.6	0.000	3.599	0.5291
1.005	0.21f	17.25p	1,134.6	0.000	3.630	0.6712
0.908	0.25f	20.00p	1,134.6	0.000	3.585	0.8453
0.726	0.32f	25.00p	1,134.6	0.000	3.375	1.1514
0.537	0.38f	30.00p	1,134.6	0.000	3.071	1.4333
0.344	0.45f	35.00p	1,134.6	0.000	2.712	1.6861
0.146	0.51f	40.00p	1,134.6	0.000	2.315	1.9057
-0.054	0.58f	45.00p	1,134.6	0.000	1.890	2.0894
-0.254	0.64f	50.00p	1,134.6	0.000	1.445	2.2350
-0.453	0.69f	55.00p	1,134.6	0.000	0.984	2.3411
-0.648	0.75f	60.00p	1,134.6	0.000	0.513	2.4065
-0.837	0.79f	65.00p	1,134.6	0.000	0.036	2.4305
-0.851	0.79f	65.37p	1,134.6	0.000	0.000	2.4306
-1.019	0.83f	70.00p	1,134.6	0.000	-0.443	2.4127

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 849.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Res. Area Ratio from abs -3.6 deg to RAZero > 1.000 70.086 P
(2) Angle from abs 4.9 deg to RAZero > 20.00 deg 60.47 P

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PONT00N

01-12-21 10:00:55 Cybermarine Technologies PTE. Ltd.
GHS 17.90 PONT00N

Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd

Condition Graphic - Draft: 1.578 @ 50.000f, 1.262 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

17 06_WBTK.C.100% SALT WATER Intact

19 07_WBTK.P.100% SALT WATER Intact

20 07_WBTK.C.100% SALT WATER Intact

21 07_WBTK.S.100% SALT WATER Intact

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PONTOON

Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.578 @ 50.00f, 1.420 @ 25.00f, 1.262 @ 0.00

Trim: Fwd 0.316/50.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			449.57	25.030f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	42.500f	0.000	34.602	
Total Fixed----->			889.57	19.732f	0.000	5.916	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			284.98	45.086f	0.000	1.697	352.57
Total Weight----->			1,174.55	25.883f	0.000	4.893	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,174.49	25.910f	0.000	0.721	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->		1.025	862.5	25.276f	0.000	143.70	20.022*
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			8.84		32.77	139.53	15.850
Distances in METERS.-----			Moments in m.-MT				

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

278.0 Cu.M. (11%) SALT WATER

400.00 MT Crane

40.00 MT Crane Load

DRAFT AT FP (M) = 1.578
DRAFT AT MS (M) = 1.420
DRAFT AT AP (M) = 1.262
DRAFT AT FWD MARK (M) = 1.553
DRAFT AT AFT MARK (M) = 1.287

HYDROSTATIC PROPERTIES

Trim: Fwd 0.316/50.000, No Heel, VCG = 4.893

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----
1.422	1,174.49	25.910f	0.721	8.84
				25.276f
				32.77
				139.53
				15.850

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 352.6 m.-MT

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PONTOON

Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.578 @ 50.00f, 1.420 @ 25.00f, 1.262 @ 0.00

Trim: Fwd 0.316/50.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm----	Pressure-----	Moment
HULL	79.3	0.809	1.506	0.05500	6.57
CRANE	57.0	4.186	4.882	0.05500	15.31
Total wind heeling moment to starboard----->					21.88
Distances in METERS.-----					Pressure in MT/Sqm.-----
					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.883f TCG = 0.000 VCG = 4.893

Free Surface Adjustment: 0.300

Adjusted CG: LCG = 25.882f TCG = 0.000 VCG = 5.193

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---	Trim----	Heel----	Weight(MT)---	Area-----
			in Trim--in Heel---	Angle
1.250	0.36f	0.00	1,174.5	0.0000
1.250	0.36f	0.06s	1,174.6	0.0000
1.237	0.36f	4.63s	1,174.6	0.0506
1.234	0.36f	5.06s	1,174.6	0.0608
1.172	0.39f	10.06s	1,174.6	0.2393
0.977	0.58f	15.06s	1,174.7	0.4992
0.965	0.59f	15.36s	1,174.6	0.5153
0.765	0.79f	20.06s	1,174.6	0.7684
0.542	1.00f	25.06s	1,174.6	1.0151
0.311	1.21f	30.06s	1,174.6	1.2294
0.073	1.42f	35.06s	1,174.6	1.4059
-0.169	1.63f	40.06s	1,174.6	1.5413
-0.413	1.83f	45.06s	1,174.6	1.6335
-0.655	2.03f	50.06s	1,174.6	1.6812
-0.786	2.13f	52.82s	1,174.6	1.6881
-0.892	2.21f	55.06s	1,174.6	1.6835
-1.120	2.36f	60.06s	1,174.6	1.6402
Distances in METERS.-----				Specific Gravity = 1.025.-----
				Area in m.-Rad.

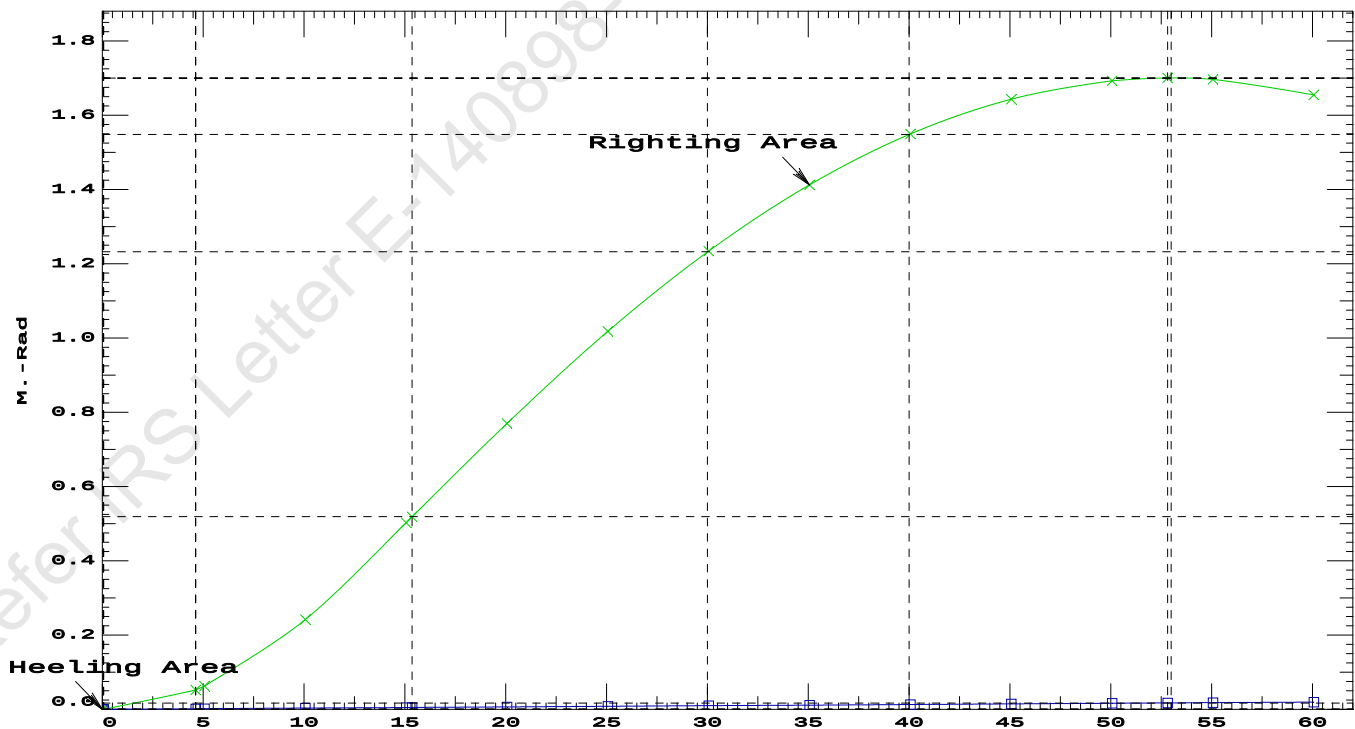
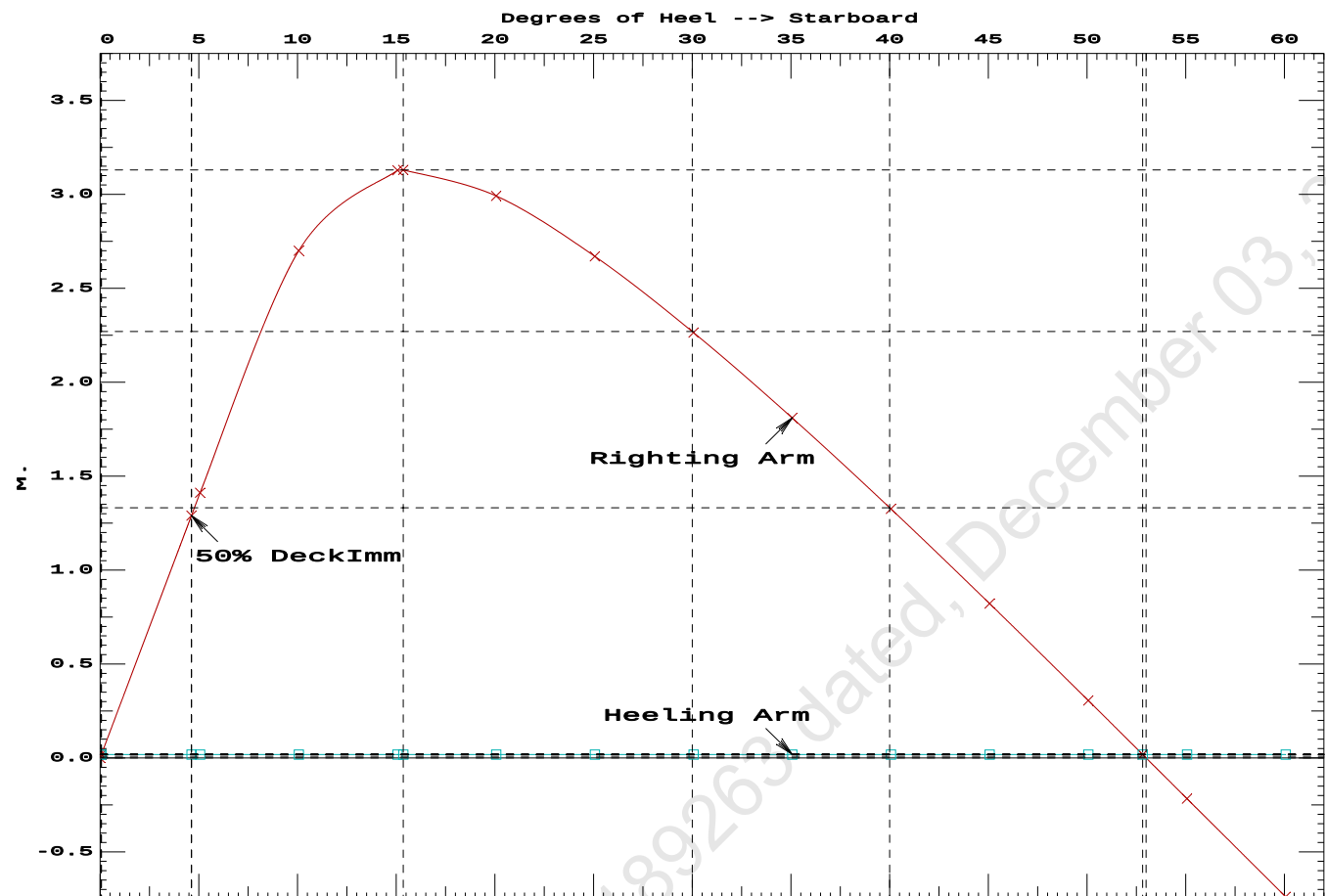
Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.88 (constant)

LIM-----	IS CODE 2008 PART B CHAPTER-2 SEC.2----	Min/Max-----	Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.5153 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	52.75 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	4.56 P

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Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd



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PONTOON

Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.883f TCG = 0.000 VCG = 4.893
Free Surface Adjustment: 0.300
Adjusted CG: LCG = 25.882f TCG = 0.000 VCG = 5.193

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.250	0.36f	0.00	1,174.6	0.000	0.001	0.0000
1.235	0.36f	5.00s	1,174.5	0.000	1.394	0.0608
1.173	0.39f	10.00s	1,174.6	0.000	2.687	0.2396
0.979	0.58f	15.00s	1,174.6	0.000	3.127	0.5006
0.965	0.59f	15.35s	1,174.6	0.000	3.129	0.5198
0.767	0.78f	20.00s	1,174.6	0.000	2.994	0.7714
0.545	0.99f	25.00s	1,174.6	0.000	2.674	1.0200
0.314	1.20f	30.00s	1,174.6	0.000	2.268	1.2362
0.076	1.42f	35.00s	1,174.6	0.000	1.815	1.4147
-0.166	1.63f	40.00s	1,174.6	0.000	1.331	1.5522
-0.410	1.83f	45.00s	1,174.6	0.000	0.827	1.6464
-0.652	2.03f	50.00s	1,174.6	0.000	0.311	1.6962
-0.794	2.13f	52.98s	1,174.6	0.000	0.000	1.7043
-0.889	2.20f	55.00s	1,174.6	0.000	-0.211	1.7006
-1.118	2.36f	60.00s	1,174.6	0.000	-0.734	1.6593

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

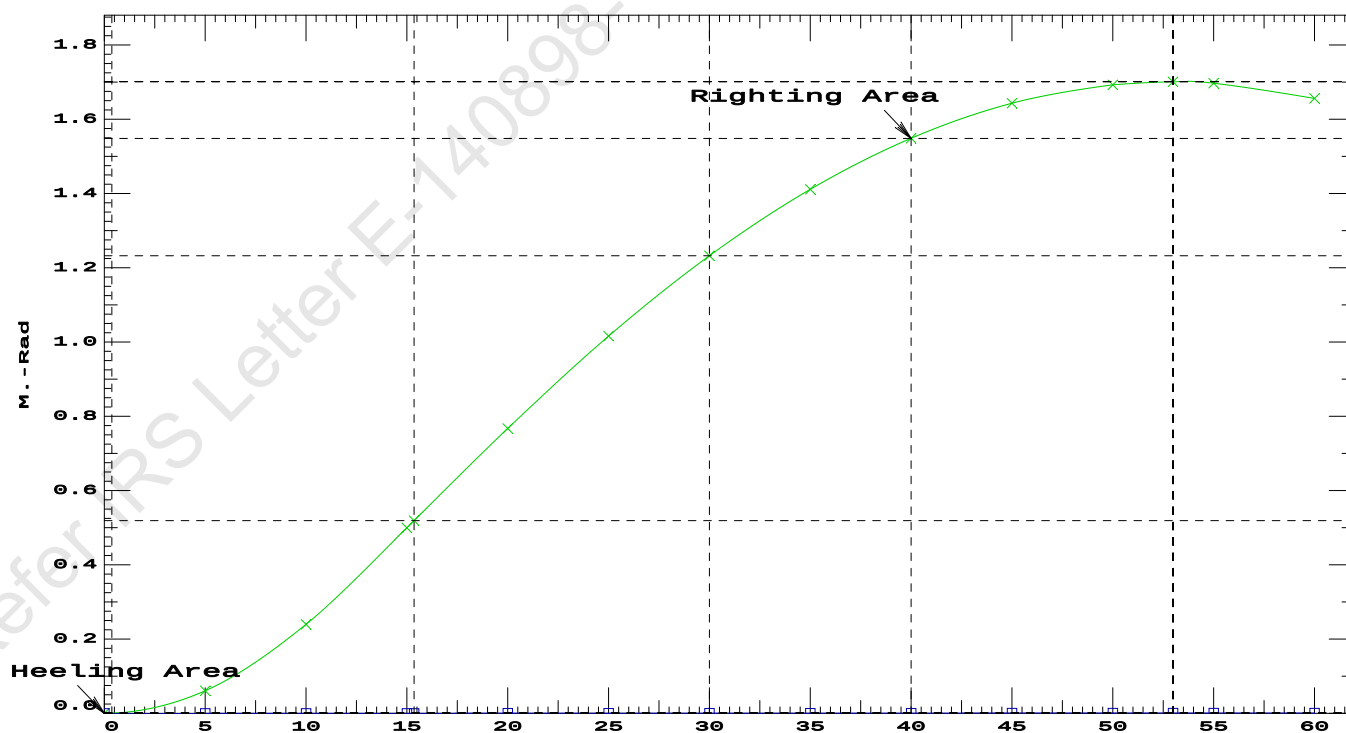
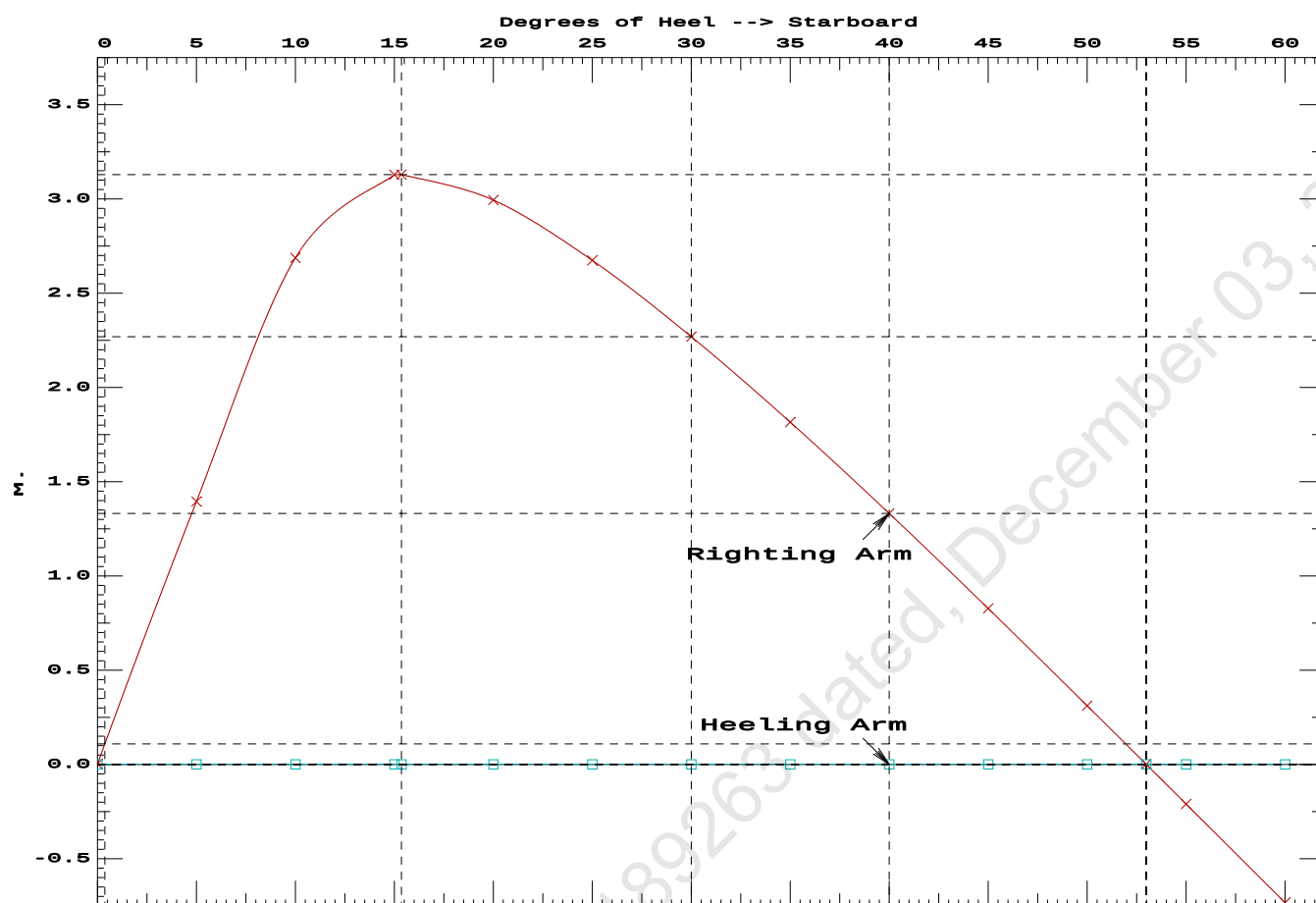
LIM-----STABILITY CRITERIA FOR CRANE OPERAT-----Min/Max-----Attained

(1) Absolute Angle at Equilibrium	<	15.00	deg	0.00	P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	367471	P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667		2165.72	P

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PONT00N

Condition 06 :Operating Condition of 400T CraneCrane Lifting 40.0T @ 30m Radius Fwd

PONTOON

Crane Lifting 40.0T @ 30m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

Trim: Fwd 0.316/50.000, Heel: Stbd 0.01 deg.

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----	1.025	859.2	25.351f	0.025s	142.08	19.916*
		MT/cm				
		8.81				

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.01

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GHS 17.90 PONTON

After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.427 @ 50.00f, 1.376 @ 25.00f, 1.324 @ 0.00
Trim: Fwd 0.103/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----			
LIGHT SHIP	449.57	25.030f	0.000	3.000			
Crane	400.00	11.500f	0.000	6.325			
Total Fixed----->	849.57	18.660f	0.000	4.565			
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----	
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			284.98	45.086f	0.000	1.697	352.57
Total Weight----->			1,134.55	25.298f	0.000	3.845	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL	1.025		1,134.56	25.304f	0.005s	0.695	

Righting Arms:			0.000	0.005s			
External Arms:			0.000	0.004s			
Residual Righting Arms:			0.000	0.001s			
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----	
Total Waterplane----->	1.025	849.0	25.028f	0.069s	141.96	20.425*	
		MT/cm					
		8.70					

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 347483712.00; t3: 30; t: 15.00
limmarg: 133522536.00; t3: 15; t: 13.50
limmarg: 1470186.25; t3: 2; t: 12.15
limmarg: ; t3: -10; t: 10.94
limmarg: 583269.63; t3: 1; t: 9.84
limmarg: ; t3: -9; t: 8.86
limmarg: ; t3: 0; t: 7.97
limmarg: 40973084.00; t3: 8; t: 7.17
limmarg: 456157.53; t3: 1; t: 6.46
limmarg: ; t3: -5; t: 5.81
limmarg: 434731.06; t3: 1; t: 5.23
limmarg: ; t3: -4; t: 4.71
limmarg: 335368.00; t3: 1; t: 4.24
limmarg: ; t3: -3; t: 3.81
limmarg: 434731.06; t3: 1; t: 3.43
limmarg: ; t3: -2; t: 3.09
limmarg: 781151.69; t3: 1; t: 2.78
limmarg: ; t3: -2; t: 2.50
limmarg: 168800.22; t3: 1; t: 2.25
limmarg: ; t3: -1; t: 2.03
limmarg: 698611.88; t3: 1; t: 1.82
limmarg: ; t3: -1; t: 1.64
limmarg: 273512.81; t3: 1; t: 1.48
limmarg: ; t3: 0; t: 1.33
limmarg: 1158020.63; t3: 1; t: 1.20
limmarg: ; t3: 0; t: 1.08
limmarg: 767070.50; t3: 1; t: 0.97
limmarg: 896.84; t3: 0; t: 0.87
limmarg: ; t3: -1; t: 0.79

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limmarg: ; t3: 0; t: 0.71
limmarg: 335368.00; t3: 1; t: 0.64
limmarg: 89397.04; t3: 0; t: 0.57
limmarg: ; t3: -1; t: 0.52
limmarg: ; t3: 0; t: 0.46
limmarg: 143531.22; t3: 0; t: 0.42
limmarg: ; t3: 0; t: 0.38
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limmarg: ; t3: 0; t: 0.30
limmarg: 63096.69; t3: 0; t: 0.27
limmarg: ; t3: 0; t: 0.25
limmarg: 44539.38; t3: 0; t: 0.22
limmarg: ; t3: 0; t: 0.20
limmarg: 29197.32; t3: 0; t: 0.18
limmarg: ; t3: 0; t: 0.16
limmarg: 19236.85; t3: 0; t: 0.15
limmarg: ; t3: 0; t: 0.13
limmarg: 13114.78; t3: 0; t: 0.12
limmarg: ; t3: 0; t: 0.11
limmarg: 9675.07; t3: 0; t: 0.10
limmarg: ; t3: 0; t: 0.09
limmarg: 6748.70; t3: 0; t: 0.08
limmarg: ; t3: 0; t: 0.07
limmarg: 4335.20; t3: 0; t: 0.06
limmarg: ; t3: 0; t: 0.06
limmarg: 3320.72; t3: 0; t: 0.05
limmarg: ; t3: 0; t: 0.05
limmarg: 2434.40; t3: 0; t: 0.04
limmarg: ; t3: 0; t: 0.04
limmarg: 1676.23; t3: 0; t: 0.03
limmarg: ; t3: 0; t: 0.03
limmarg: 896.84; t3: 0; t: 0.03
limmarg: ; t3: 0; t: 0.02
limmarg: 453.15; t3: 0; t: 0.02
limmarg: 0.00; t3: 0; t: 0.02

Theta3 is -0.02 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.660f TCG = 0.000 VCG = 4.565

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.312	0.12f	0.01s	1,134.6	0.000	-0.004	0.0000
1.312	0.12f	0.00	1,134.6	0.000	-0.001	-0.0000
1.297	0.12f	5.00p	1,134.6	0.000	1.570	0.0685
1.241	0.13f	10.00p	1,134.6	0.000	3.019	0.2696
1.081	0.19f	15.00p	1,134.6	0.000	3.603	0.5649
1.005	0.21f	17.25p	1,134.6	0.000	3.634	0.7073
0.908	0.25f	20.00p	1,134.6	0.000	3.589	0.8814
0.726	0.32f	25.00p	1,134.6	0.000	3.378	1.1878
0.538	0.38f	30.00p	1,134.6	0.000	3.075	1.4701

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 849.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0 > 1.000 1.000 P

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.660f TCG = 0.000 VCG = 4.565

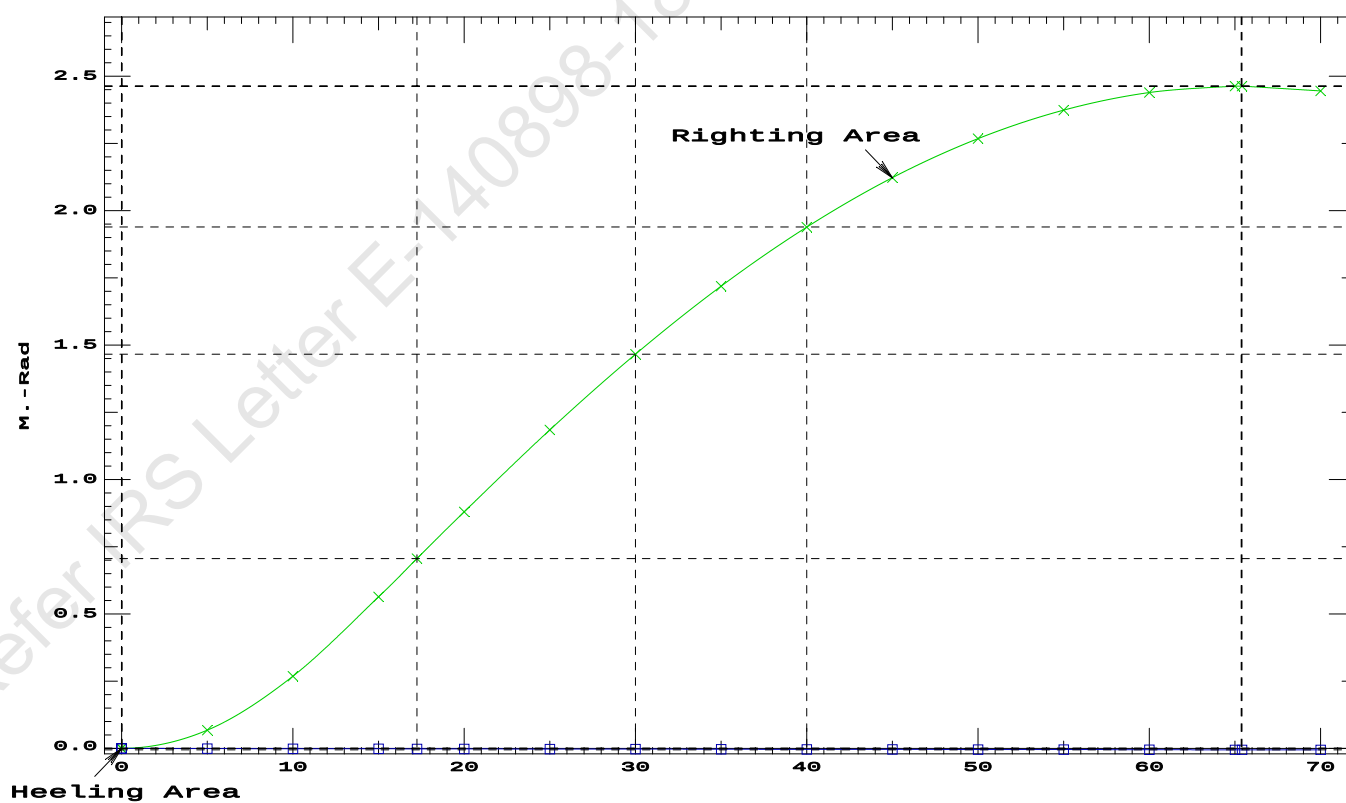
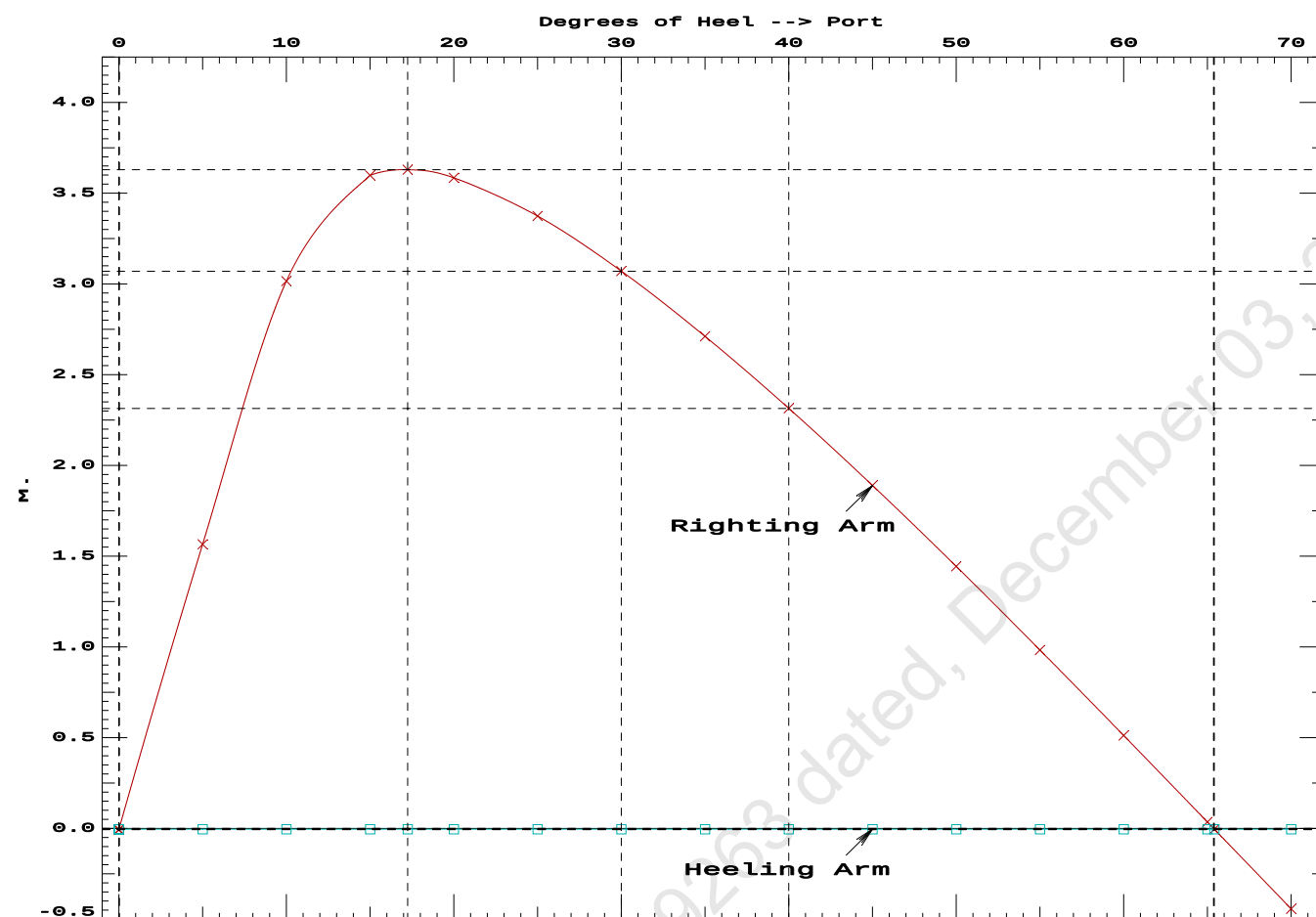
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.312	0.12f	0.01s	1,134.6	0.000	-0.004	0.0000
1.312	0.12f	0.00	1,134.6	0.000	-0.001	-0.0000
1.297	0.12f	5.00p	1,134.6	0.000	1.570	0.0685
1.241	0.13f	10.00p	1,134.6	0.000	3.019	0.2696
1.081	0.19f	15.00p	1,134.6	0.000	3.603	0.5649
1.005	0.21f	17.25p	1,134.6	0.000	3.634	0.7073
0.908	0.25f	20.00p	1,134.6	0.000	3.589	0.8814
0.726	0.32f	25.00p	1,134.6	0.000	3.378	1.1878
0.538	0.38f	30.00p	1,134.6	0.000	3.075	1.4701
0.344	0.45f	35.00p	1,134.6	0.000	2.716	1.7231
0.146	0.51f	40.00p	1,134.6	0.000	2.319	1.9431
-0.054	0.58f	45.00p	1,134.6	0.000	1.894	2.1271
-0.254	0.64f	50.00p	1,134.6	0.000	1.448	2.2731
-0.453	0.69f	55.00p	1,134.6	0.000	0.987	2.3794
-0.648	0.75f	60.00p	1,134.6	0.000	0.516	2.4451
-0.837	0.79f	65.00p	1,134.6	0.000	0.040	2.4694
-0.853	0.79f	65.41p	1,134.6	0.000	0.000	2.4696
-1.019	0.83f	70.00p	1,134.6	0.000	-0.439	2.4520

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 849.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 5837443 P
(2) Angle from abs 0 deg to RAZero > 20.00 deg 65.42 P

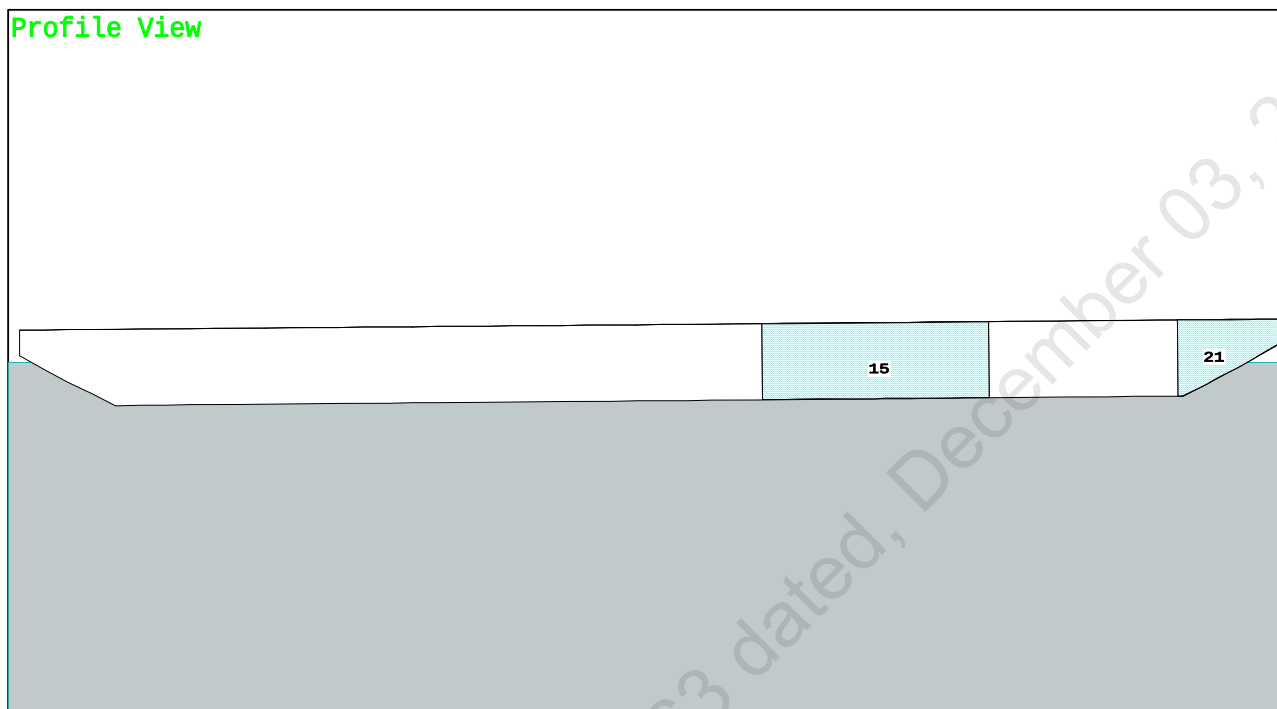
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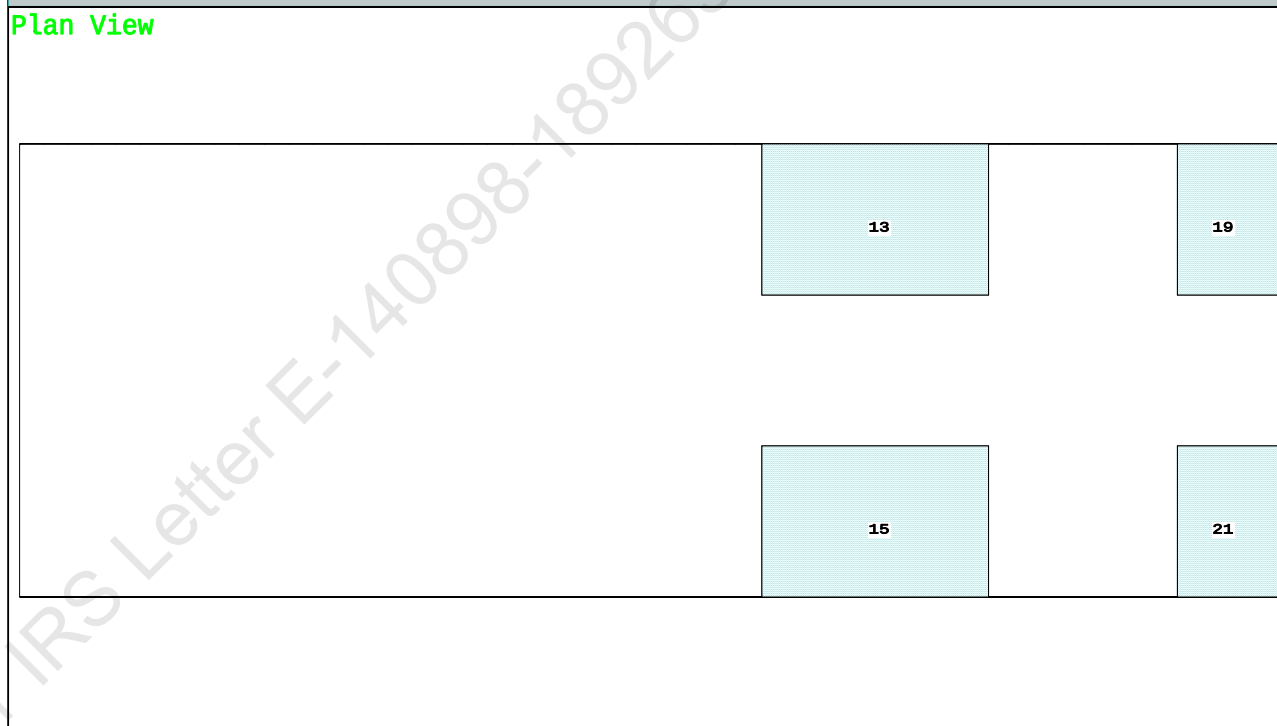
Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

Condition Graphic - Draft: 1.296 @ 50.000f, 1.740 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

13 05_WBTK.P.100% SALT WATER Intact

15 05_WBTK.S.100% SALT WATER Intact

19 07_WBTK.P.100% SALT WATER Intact

21 07_WBTK.S.100% SALT WATER Intact

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PONTOON

Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.296 @ 50.00f, 1.518 @ 25.00f, 1.740 @ 0.00

Trim: Aft 0.443/50.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			449.57	25.030f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Crane Load			40.00	23.500a	0.000	21.478	
Total Fixed----->			839.57	17.077f	0.000	5.266	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,261.97	23.815f	0.000	4.038	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,261.85	23.787f	0.000	0.775	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	869.0	24.615f	0.000	136.81	18.690*
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			8.91		33.70	133.55	15.426
Distances in METERS.-----				Moments in m.-MT			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

350.00 MT Crane

40.00 MT Crane Load

DRAFT AT FP (M) = 1.296
DRAFT AT MS (M) = 1.518
DRAFT AT AP (M) = 1.740
DRAFT AT FWD MARK (M) = 1.332
DRAFT AT AFT MARK (M) = 1.704

HYDROSTATIC PROPERTIES

Trim: Aft 0.443/50.000, No Heel, VCG = 4.038

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF-----	cm trim----	GML-----	GMT
1.521	1,261.85	23.787f	0.775	8.91	24.615f	33.70	133.55	15.426
Distances in METERS.-----				Specific Gravity = 1.025.-----				
				Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

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GHS 17.90

PONTOON

Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.296 @ 50.00f, 1.518 @ 25.00f, 1.740 @ 0.00

Trim: Aft 0.443/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	74.6	0.762	1.508	0.05500	6.19
CRANE	57.0	3.862	4.607	0.05500	14.44
Total wind heeling moment to starboard-->					20.63
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.815f TCG = 0.000 VCG = 4.038

Free Surface Adjustment: 0.373

Adjusted CG: LCG = 23.819f TCG = 0.000 VCG = 4.411

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. 50% DeckImm Area	Angle
1.728	0.51a	0.00	1,262.0	0.000	-0.016	0.0000 4.088
1.728	0.51a	0.05s	1,262.0	0.000	0.000	-0.0000 4.035
1.718	0.51a	4.09s	1,262.0	0.000	1.092	0.0385 0.000
1.715	0.51a	5.05s	1,262.0	0.000	1.352	0.0590 -0.965
1.692	0.56a	10.05s	1,261.9	0.000	2.617	0.2329 -5.965
1.749	0.83a	15.05s	1,262.0	0.000	3.062	0.4874 -10.965
1.760	0.89a	16.01s	1,262.0	0.000	3.069	0.5388 -11.925
1.802	1.12a	20.05s	1,262.0	0.000	2.978	0.7538 -15.965
1.847	1.42a	25.05s	1,262.0	0.000	2.719	1.0045 -20.965
1.885	1.72a	30.05s	1,262.0	0.000	2.380	1.2276 -25.965
1.914	2.02a	35.05s	1,262.0	0.000	1.995	1.4189 -30.965
1.933	2.31a	40.05s	1,262.0	0.000	1.578	1.5750 -35.965
1.943	2.60a	45.05s	1,262.0	0.000	1.141	1.6938 -40.965
1.940	2.88a	50.05s	1,262.0	0.000	0.690	1.7737 -45.965
1.923	3.13a	55.05s	1,262.0	0.000	0.230	1.8139 -50.965
1.909	3.25a	57.53s	1,262.0	0.000	0.000	1.8189 -53.446
1.890	3.36a	60.05s	1,262.0	0.000	-0.234	1.8138 -55.965
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.						

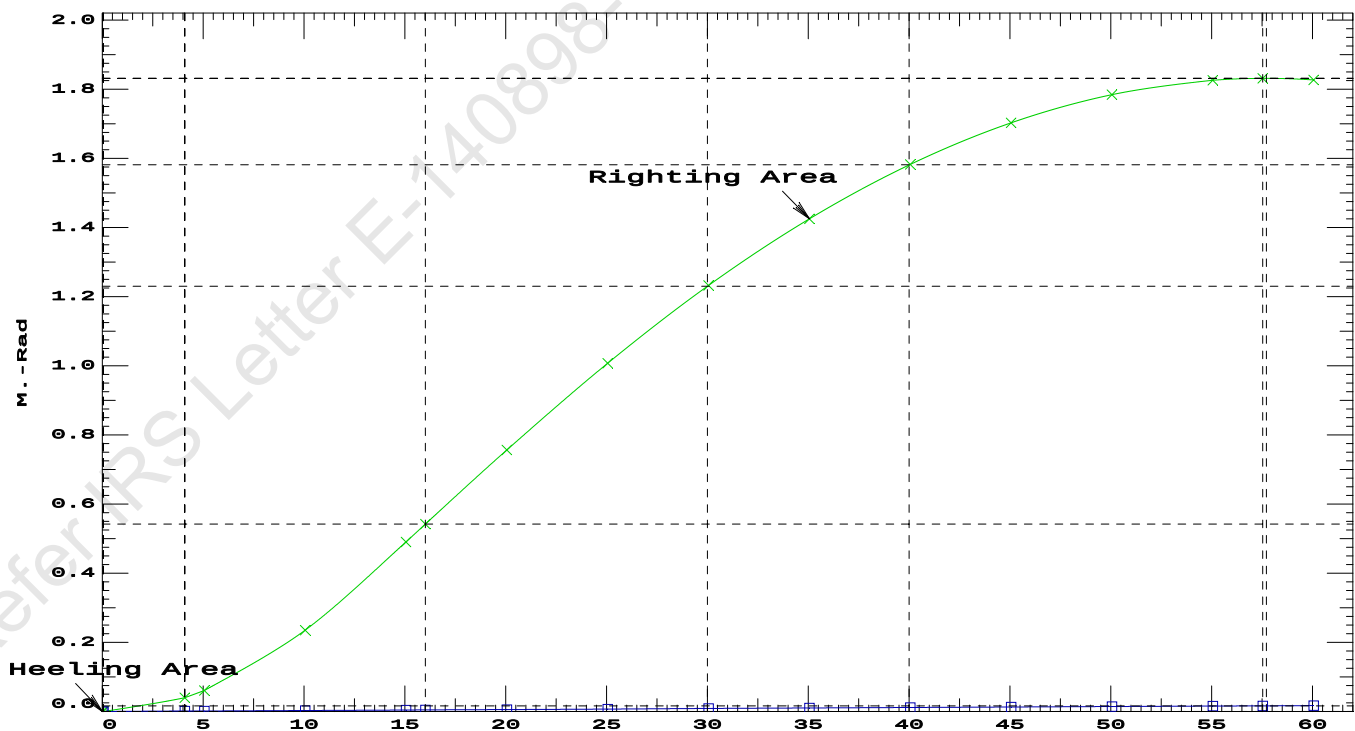
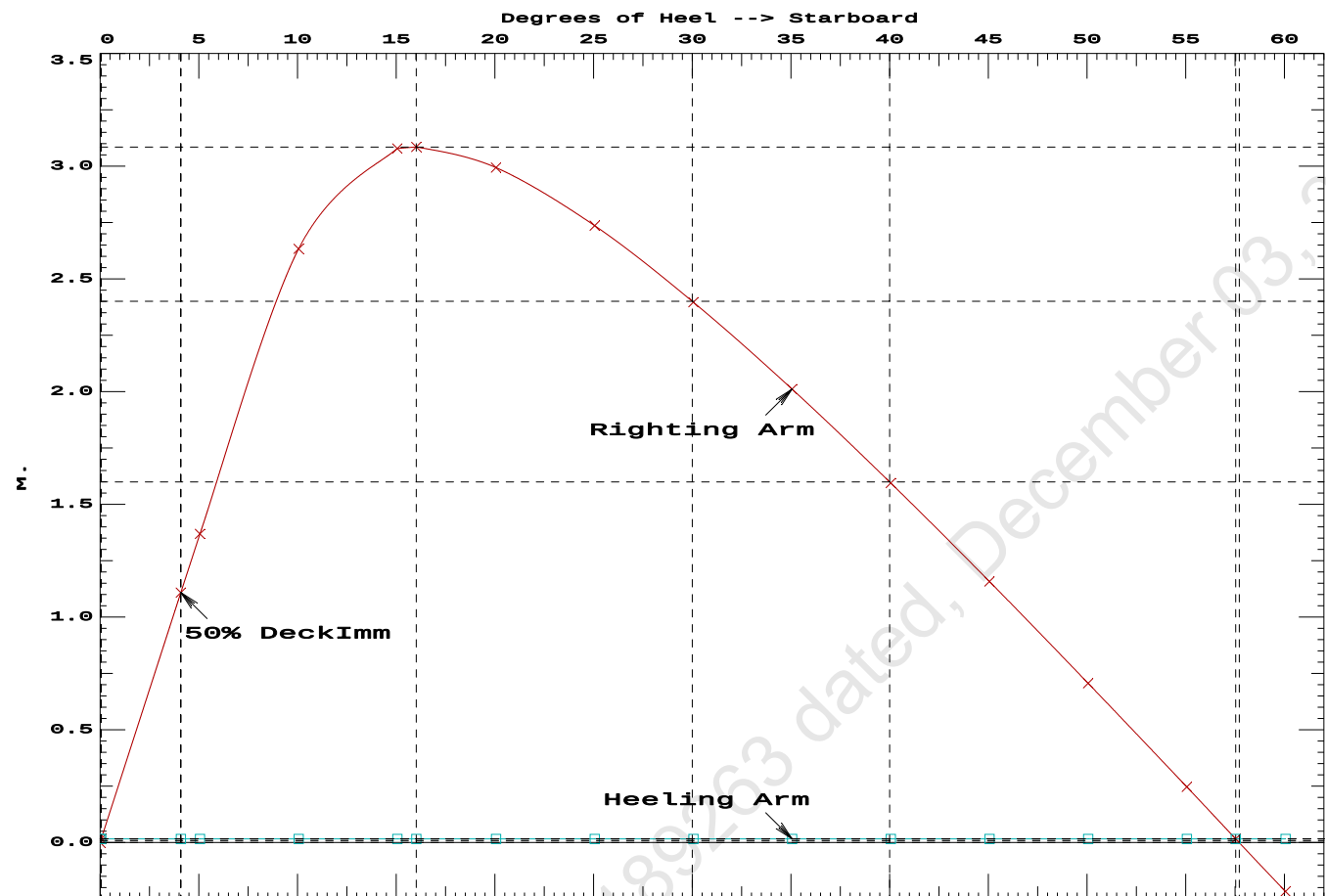
Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 20.63 (constant)

LIM	IS CODE 2008 PART B CHAPTER-2 SEC.2	Min/Max	Attained
(1)	Residual Area from abs 0 deg to MaxRA	> 0.0800 m.-Rad	0.5388 P
(2)	Angle from Equilibrium to RAZero or Flood	> 20.00 deg	57.48 P
(3)	Angle from Equilibrium to 50% Dk Imm Angle	> 0.00 deg	4.04 P

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Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft



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PONTOON

Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.815f TCG = 0.000 VCG = 4.038
Free Surface Adjustment: 0.373
Adjusted CG: LCG = 23.819f TCG = 0.000 VCG = 4.411

Origin Depth---	Degrees of Trim----	Displacement Heel----	Weight(MT)---	Residual Arms in Trim--	in Heel---	Res. Area
1.728	0.51a	0.00	1,262.0	0.000	-0.001	0.0000
1.715	0.51a	5.00s	1,262.0	0.000	1.353	0.0590
1.692	0.56a	10.00s	1,261.9	0.000	2.621	0.2330
1.749	0.83a	15.00s	1,262.0	0.000	3.077	0.4885
1.760	0.89a	16.02s	1,262.0	0.000	3.084	0.5434
1.801	1.12a	20.00s	1,262.0	0.000	2.995	0.7563
1.847	1.42a	25.00s	1,262.0	0.000	2.738	1.0086
1.884	1.71a	30.00s	1,262.0	0.000	2.400	1.2333
1.913	2.01a	35.00s	1,262.0	0.000	2.014	1.4262
1.933	2.31a	40.00s	1,262.0	0.000	1.598	1.5841
1.943	2.60a	45.00s	1,262.0	0.000	1.161	1.7046
1.940	2.88a	50.00s	1,262.0	0.000	0.710	1.7863
1.923	3.13a	55.00s	1,262.0	0.000	0.250	1.8283
1.908	3.26a	57.70s	1,262.0	0.000	0.000	1.8342
1.891	3.36a	60.00s	1,262.0	0.000	-0.213	1.8299

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

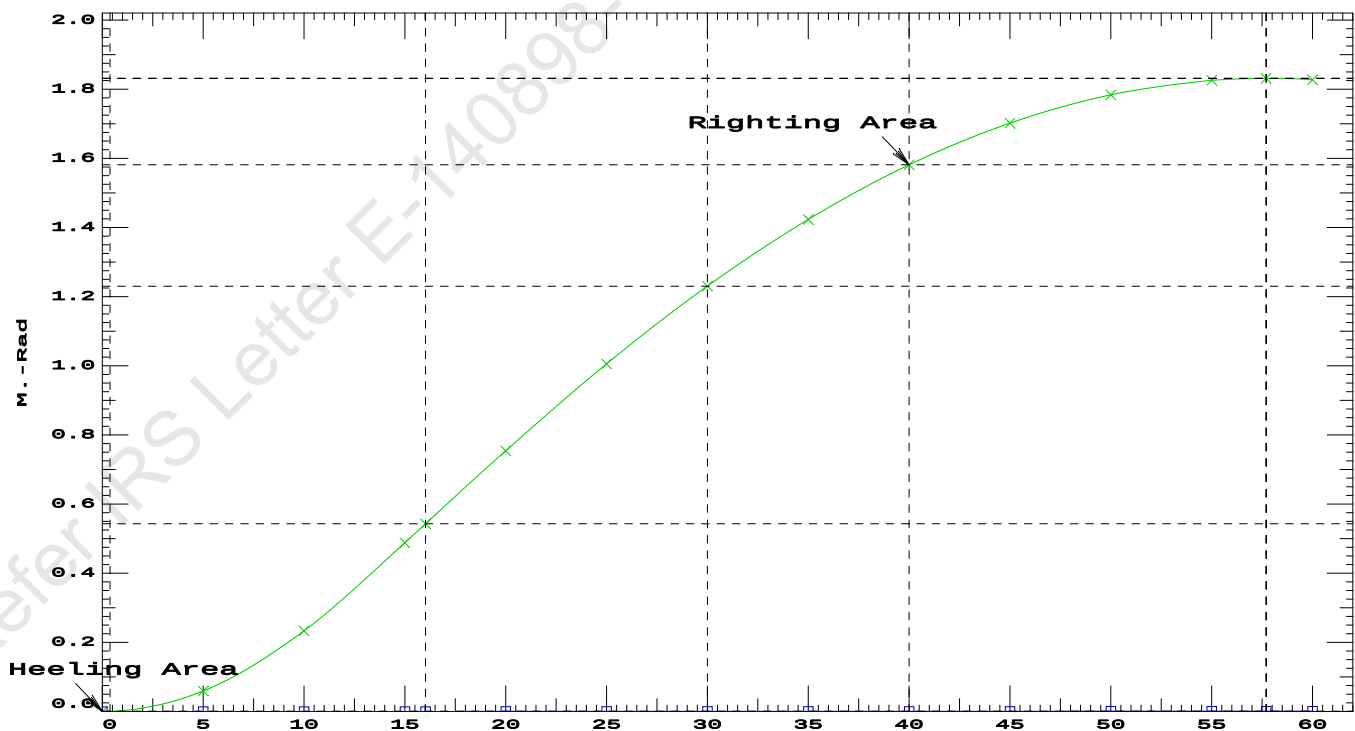
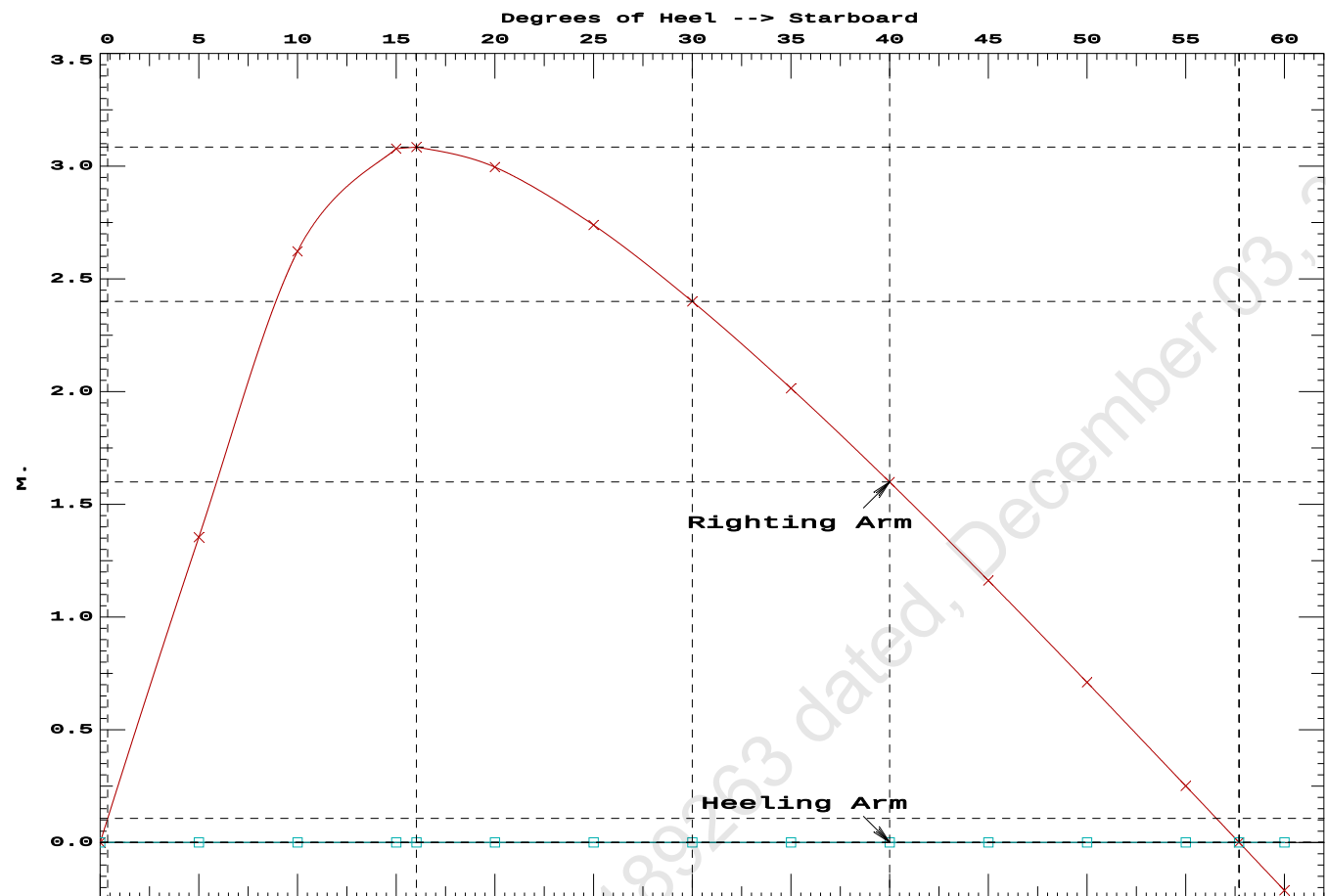
LIM-----STABILITY CRITERIA FOR CRANE OPERAT-----Min/Max-----Attained

(1) Absolute Angle at Equilibrium	<	15.00	deg	0.00	P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	389187	P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667		2299.49	P

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GHS 17.90

PONTOON

Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft



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GHS 17.90 PONTON

Trim: Aft 0.443/50.000, Heel: zero

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.541 @ 50.00f, 1.475 @ 25.00f, 1.408 @ 0.00
Trim: Fwd 0.133/50.000, Heel: zero

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----		
LIGHT SHIP	449.57	25.030f	0.000	3.000			
Crane	350.00	11.500f	0.000	6.325			
Total Fixed----->	799.57	19.107f	0.000	4.455			
Load-----	SpGr-----	Weight(MT)---- <td>LCG-----</td> <td>TCG-----</td> <td>VCG-----</td> <td>FSM-----</td>	LCG-----	TCG-----	VCG-----	FSM-----	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,221.97	25.364f	0.000	3.467	
			Displ(MT)---- <td>LCB-----</td> <td>TCB-----</td> <td>VCB-----</td> <td></td>	LCB-----	TCB-----	VCB-----	
HULL	1.025		1,221.96	25.372f	0.000	0.746	

Righting Arms:			0.000	0.000			
External Arms:			0.000	0.001			
Residual Righting Arms:			0.000	-0.001s			
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----	
Total Waterplane----->	1.025	866.4	25.106f	0.000	139.99	19.249*	
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----		
		8.88	33.55	137.27	16.529		
Distances in METERS.-----							
-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 0.9; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.2; t: 0.12
limmarg: INFINITY; t3: 0.1; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.0; t: 0.01

Theta3 is 0.03 degrees, as demonstrated below.

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GHS 17.90 PONTON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 19.107f TCG = 0.000 VCG = 4.455

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.396	0.15f	0.00	1,222.0	0.000	-0.001	0.0000
1.396	0.15f	0.03s	1,222.0	0.000	0.010	0.0000
1.381	0.15f	5.00s	1,222.0	0.000	1.484	0.0648
1.331	0.16f	10.00s	1,222.0	0.000	2.892	0.2563
1.209	0.25f	15.00s	1,222.0	0.000	3.461	0.5396
1.143	0.29f	17.56s	1,221.9	0.000	3.497	0.6954
1.076	0.33f	20.00s	1,222.0	0.000	3.464	0.8439
0.933	0.42f	25.00s	1,222.0	0.000	3.283	1.1406
0.782	0.50f	30.00s	1,222.0	0.000	3.015	1.4160

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 799.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0 > 1.000 LARGE

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GHS 17.90 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 19.107f TCG = 0.000 VCG = 4.455

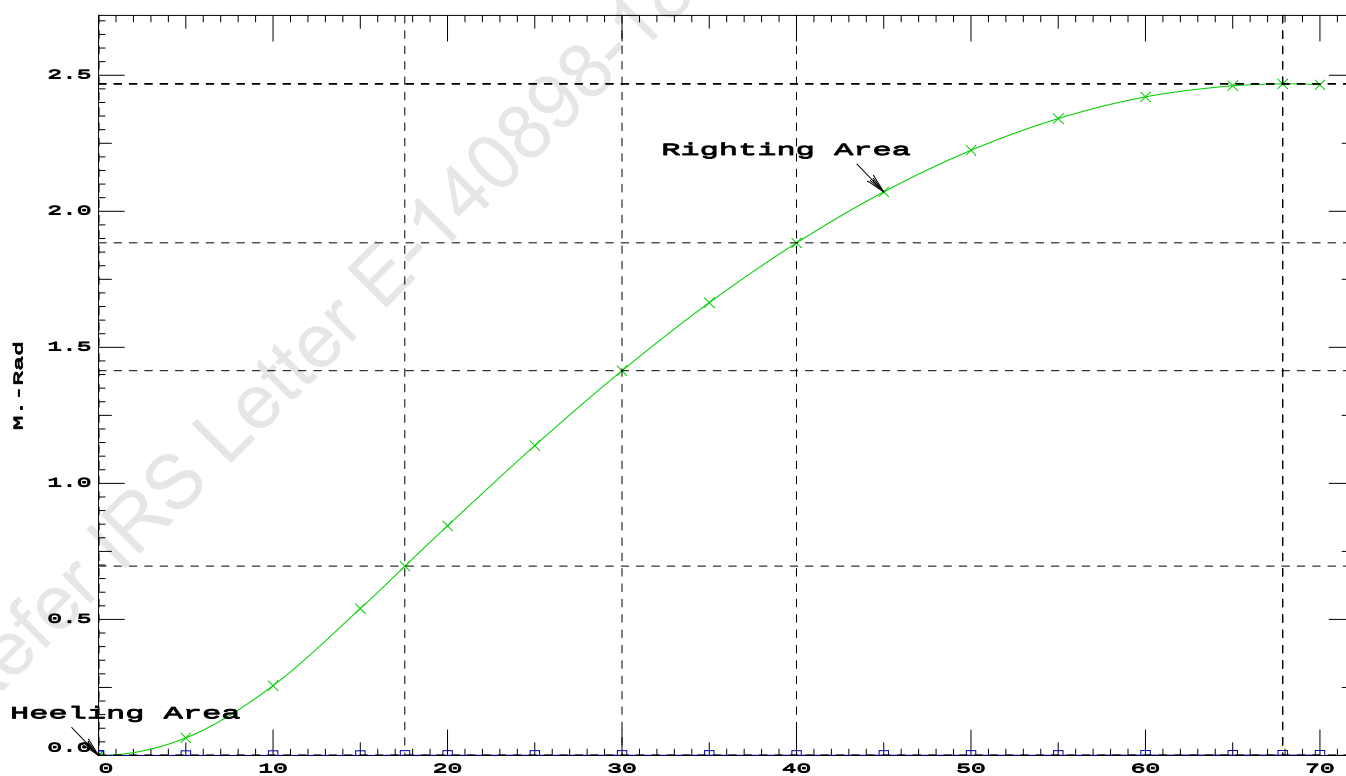
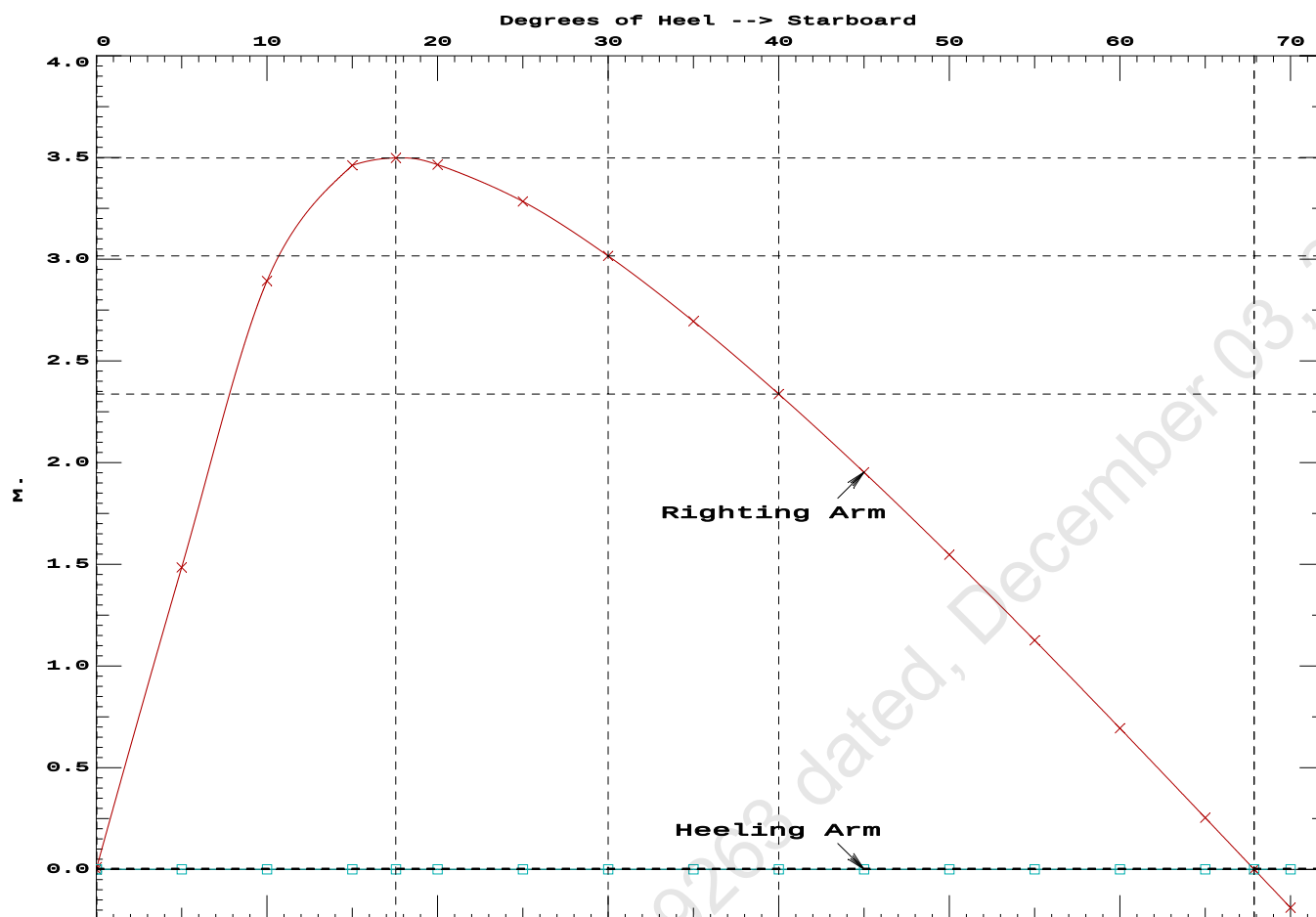
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.396	0.15f	0.00	1,222.0	0.000	-0.001	0.0000
1.396	0.15f	0.03s	1,222.0	0.000	0.010	0.0000
1.381	0.15f	5.00s	1,222.0	0.000	1.484	0.0648
1.331	0.16f	10.00s	1,222.0	0.000	2.892	0.2563
1.209	0.25f	15.00s	1,222.0	0.000	3.461	0.5396
1.143	0.29f	17.56s	1,221.9	0.000	3.497	0.6954
1.076	0.33f	20.00s	1,222.0	0.000	3.464	0.8439
0.933	0.42f	25.00s	1,222.0	0.000	3.283	1.1406
0.782	0.50f	30.00s	1,222.0	0.000	3.015	1.4160
0.623	0.59f	35.00s	1,222.0	0.000	2.694	1.6655
0.457	0.67f	40.00s	1,222.0	0.000	2.337	1.8853
0.287	0.76f	45.00s	1,222.0	0.000	1.952	2.0726
0.113	0.84f	50.00s	1,222.0	0.000	1.546	2.2254
-0.062	0.91f	55.00s	1,222.0	0.000	1.125	2.3420
-0.236	0.98f	60.00s	1,222.0	0.000	0.693	2.4214
-0.409	1.04f	65.00s	1,222.0	0.000	0.253	2.4628
-0.505	1.07f	67.86s	1,222.0	0.000	0.000	2.4691
-0.576	1.09f	70.00s	1,222.0	0.000	-0.189	2.4656

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 799.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 0 deg to RAZero > 20.00 deg 67.83 P

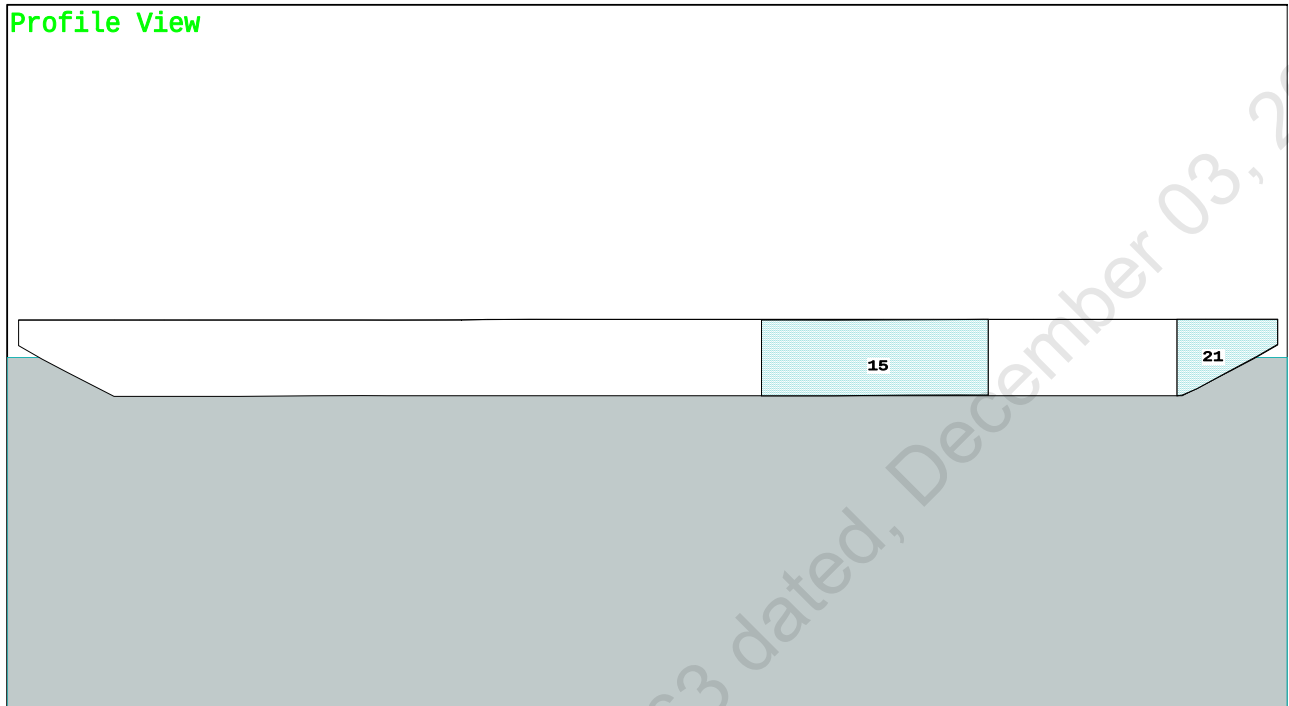
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GHS 17.90Cybermarine Technologies PTE. Ltd.
PONT00N

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GHS 17.90 PONT00N

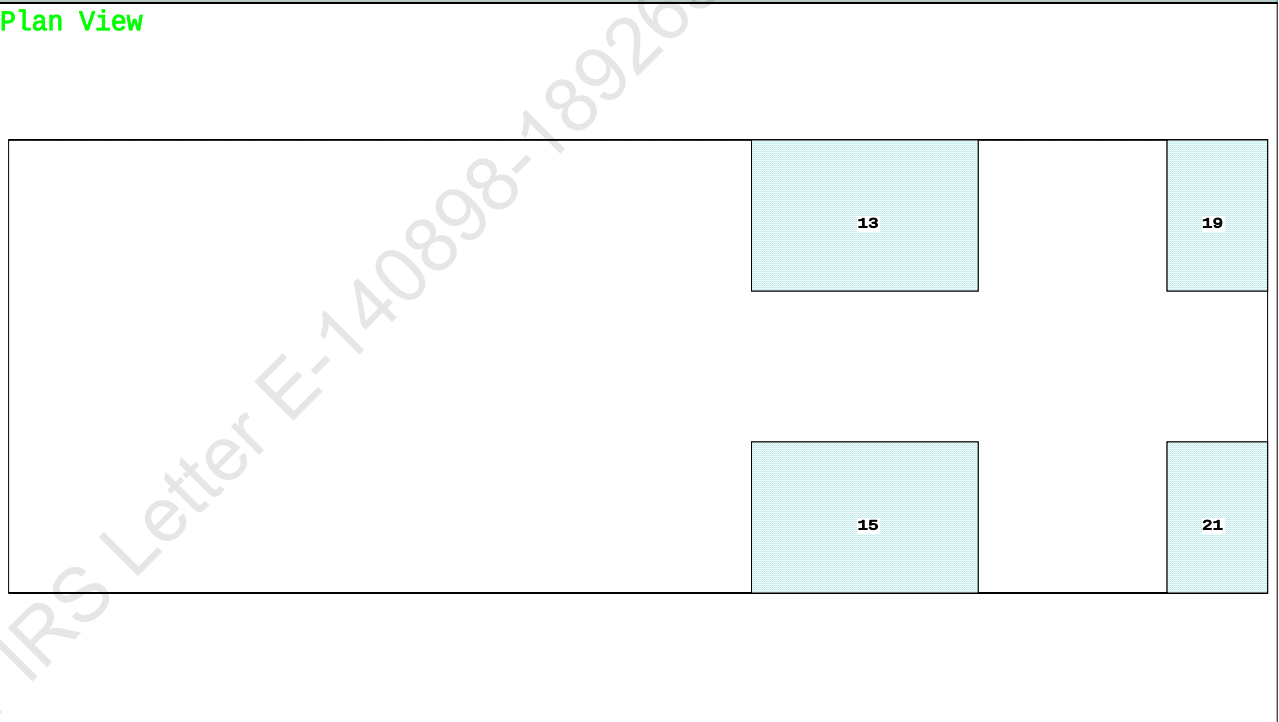
Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

Condition Graphic - Draft: 1.506 @ 50.000f, 1.534 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

13 05_WBTK.P.100% SALT WATER Intact	15 05_WBTK.S.100% SALT WATER Intact	21 07_WBTK.S.100% SALT WATER Intact
	19 07_WBTK.P.100% SALT WATER Intact	

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GHS 17.90

PONTOON

Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.506 @ 50.00f, 1.520 @ 25.00f, 1.534 @ 0.00

Trim: Aft 0.028/50.000, Heel: zero

Part-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----	FSM-----
LIGHT SHIP	449.57	25.030f	0.000	3.000	
Crane	350.00	11.500f	0.000	6.325	
Crane Load	40.00	11.500f	0.000	21.478	
Total Fixed----->	839.57	18.745f	0.000	5.266	
Load-----	SpGr-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s
Total Tanks----->		422.40	37.208f	0.000	1.596
Total Weight----->		1,261.97	24.925f	0.000	4.038
		Displ(MT)-----	LCB-----	TCB-----	VCB-----
HULL	1.025	1,261.97	24.923f	0.000	0.769

Righting Arms:		0.000	0.000		
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----
Total Waterplane----->	1.025	869.0	24.976f	0.000	136.79
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----
		8.91	33.70	133.52	15.415
Distances in METERS.-----				Moments in m.-MT.	

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

350.00 MT Crane

40.00 MT Crane Load

DRAFT AT FP (M) = 1.506
DRAFT AT MS (M) = 1.520
DRAFT AT AP (M) = 1.534
DRAFT AT FWD MARK (M) = 1.508
DRAFT AT AFT MARK (M) = 1.532

HYDROSTATIC PROPERTIES

Trim: Aft 0.028/50.000, No Heel, VCG = 4.038

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)-----	LCB-----	VCB-----	cm-----
1.520	1,261.97	24.923f	0.769	8.91
				24.976f
				33.70
				133.52
				15.415

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

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GHS 17.90

PONTOON

Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.506 @ 50.00f, 1.520 @ 25.00f, 1.534 @ 0.00

Trim: Aft 0.028/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	74.7	0.755	1.497	0.05500	6.15
CRANE	57.0	3.984	4.726	0.05500	14.82
Total wind heeling moment to starboard-->					20.96
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.925f TCG = 0.000 VCG = 4.038

Free Surface Adjustment: 0.373

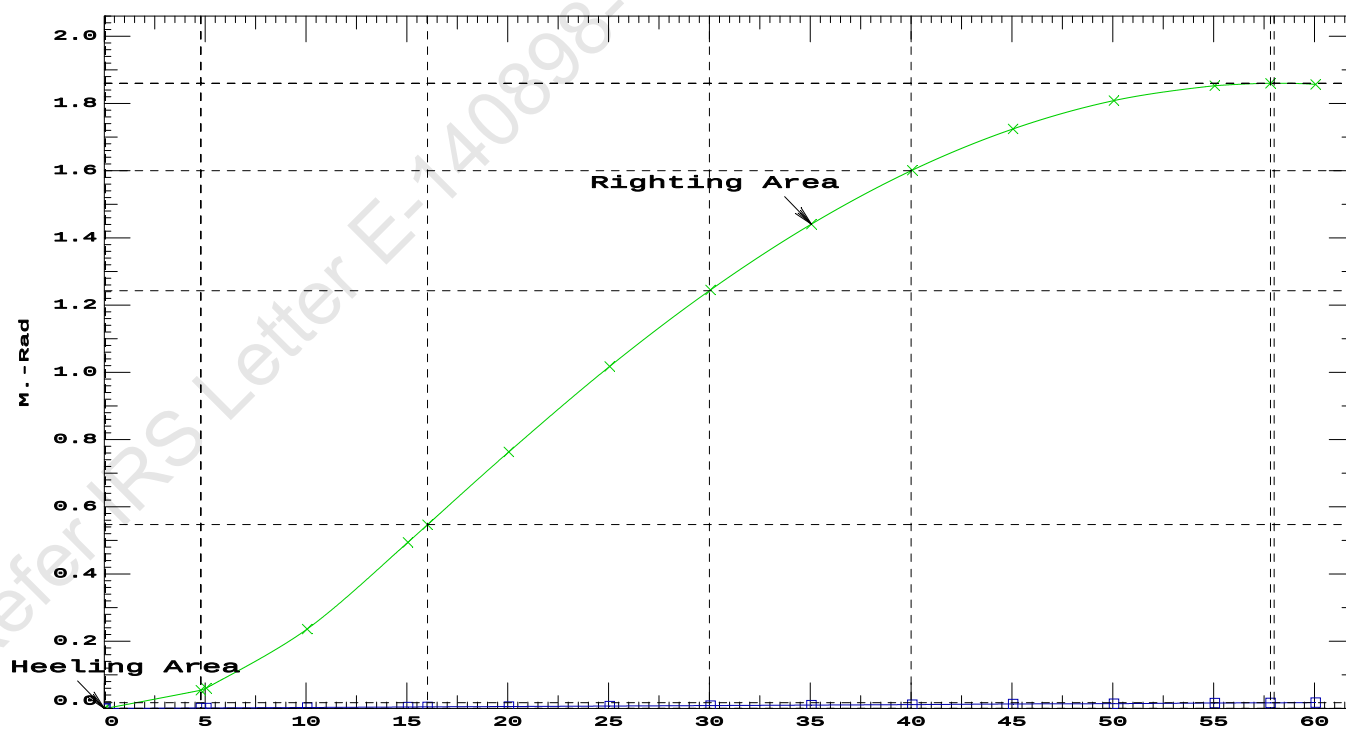
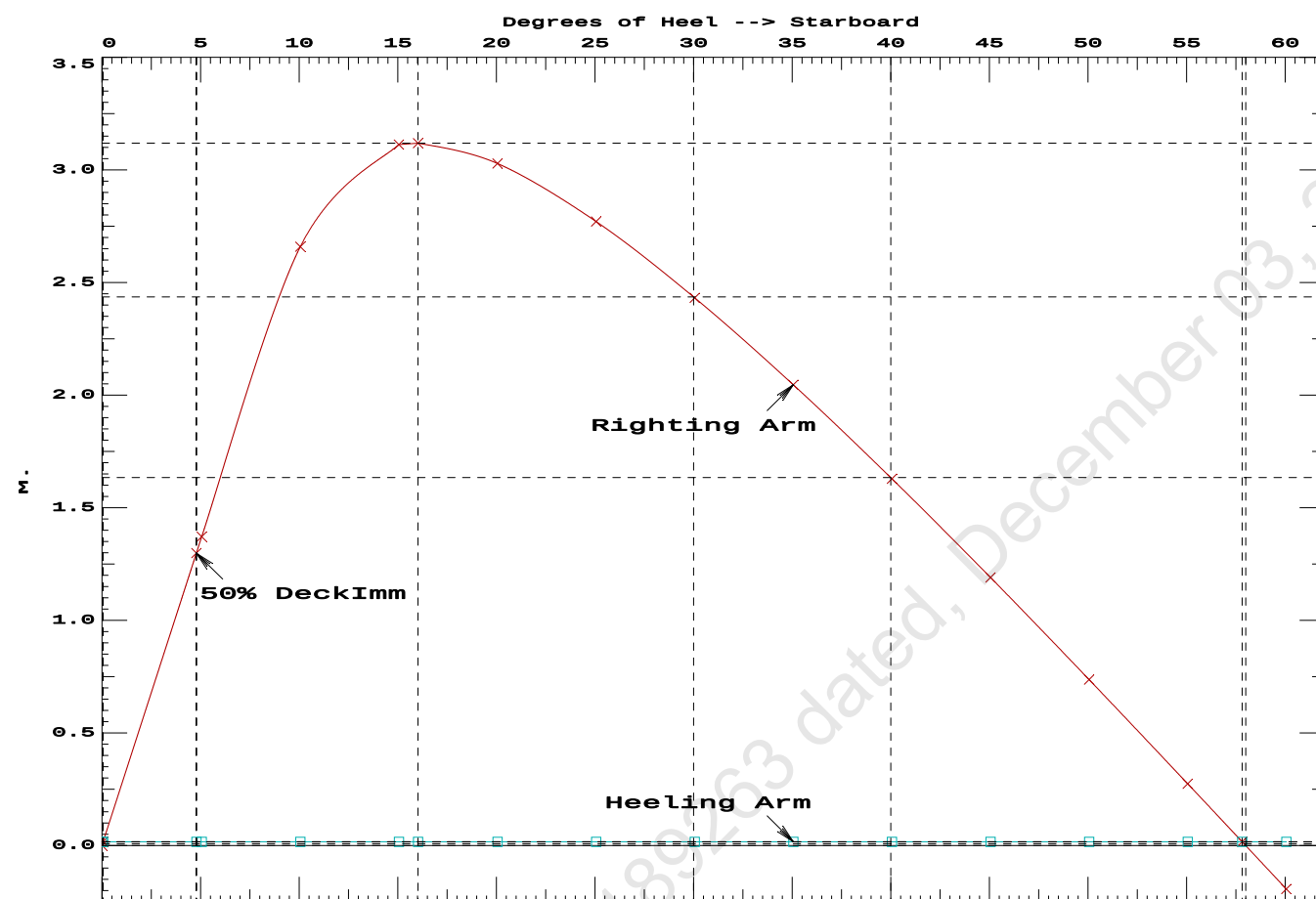
Adjusted CG: LCG = 24.925f TCG = 0.000 VCG = 4.411

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	in Heel	Res. 50% DeckImm Area	Angle
1.522	0.03a	0.00	1,262.0	0.000	-0.016	0.0000	4.788
1.522	0.03a	0.06s	1,262.0	0.000	0.000	-0.0000	4.730
1.508	0.03a	4.79s	1,262.2	0.000	1.282	0.0529	0.000
1.507	0.03a	5.06s	1,261.8	0.000	1.355	0.0591	-0.270
1.464	0.03a	10.06s	1,262.0	0.000	2.643	0.2340	-5.270
1.409	0.05a	15.06s	1,262.0	0.000	3.096	0.4913	-10.270
1.397	0.06a	16.02s	1,262.0	0.000	3.102	0.5436	-11.236
1.343	0.07a	20.06s	1,262.0	0.000	3.012	0.7607	-15.270
1.268	0.09a	25.06s	1,262.0	0.000	2.754	1.0145	-20.270
1.183	0.11a	30.06s	1,262.0	0.000	2.415	1.2406	-25.270
1.090	0.13a	35.06s	1,262.0	0.000	2.029	1.4349	-30.270
0.988	0.15a	40.06s	1,262.0	0.000	1.612	1.5940	-35.270
0.880	0.17a	45.06s	1,262.0	0.000	1.174	1.7157	-40.270
0.765	0.18a	50.06s	1,262.0	0.000	0.720	1.7984	-45.270
0.644	0.20a	55.06s	1,262.0	0.000	0.258	1.8412	-50.270
0.575	0.21a	57.82s	1,262.0	0.000	0.000	1.8474	-53.030
0.518	0.21a	60.06s	1,262.0	0.000	-0.210	1.8433	-55.270
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.							

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 20.96 (constant)

LIM	IS CODE 2008 PART B CHAPTER-2 SEC.2	Min/Max	Attained
(1)	Residual Area from abs 0 deg to MaxRA	> 0.0800 m.-Rad	0.5436 P
(2)	Angle from Equilibrium to RAZero or Flood	> 20.00 deg	57.76 P
(3)	Angle from Equilibrium to 50% Dk Imm Angle	> 0.00 deg	4.73 P

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GHS 17.90Cybermarine Technologies PTE. Ltd.
PONT00NCondition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

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GHS 17.90

PONTOON

Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.925f TCG = 0.000 VCG = 4.038

Free Surface Adjustment: 0.373

Adjusted CG: LCG = 24.925f TCG = 0.026p VCG = 4.410

Origin Depth---	Degrees of Trim----	Displacement Heel----	Weight(MT)---	Residual Arms in Trim--	Res. in Heel---	Area -->
1.512	0.03a	3.99s	1,262.0	0.000	0.000	0.0000
1.476	0.03a	8.99s	1,262.0	0.000	1.317	0.0575
1.421	0.05a	13.99s	1,262.0	0.000	2.001	0.2113
1.395	0.06a	16.16s	1,262.0	0.000	2.034	0.2882
1.358	0.07a	18.99s	1,262.0	0.000	1.981	0.3882
1.285	0.09a	23.99s	1,262.0	0.000	1.749	0.5536
1.202	0.11a	28.99s	1,262.0	0.000	1.422	0.6927
1.110	0.12a	33.99s	1,262.0	0.000	1.044	0.8007
1.011	0.14a	38.99s	1,262.0	0.000	0.631	0.8740
0.904	0.16a	43.99s	1,262.0	0.000	0.195	0.9102
0.855	0.17a	46.16s	1,262.0	0.000	0.000	0.9139
0.790	0.18a	48.99s	1,262.0	0.000	-0.257	0.9076
0.671	0.20a	53.99s	1,262.0	0.000	-0.720	0.8650
0.546	0.21a	58.99s	1,262.0	0.000	-1.188	0.7818
0.417	0.22a	63.99s	1,262.0	0.000	-1.658	0.6576

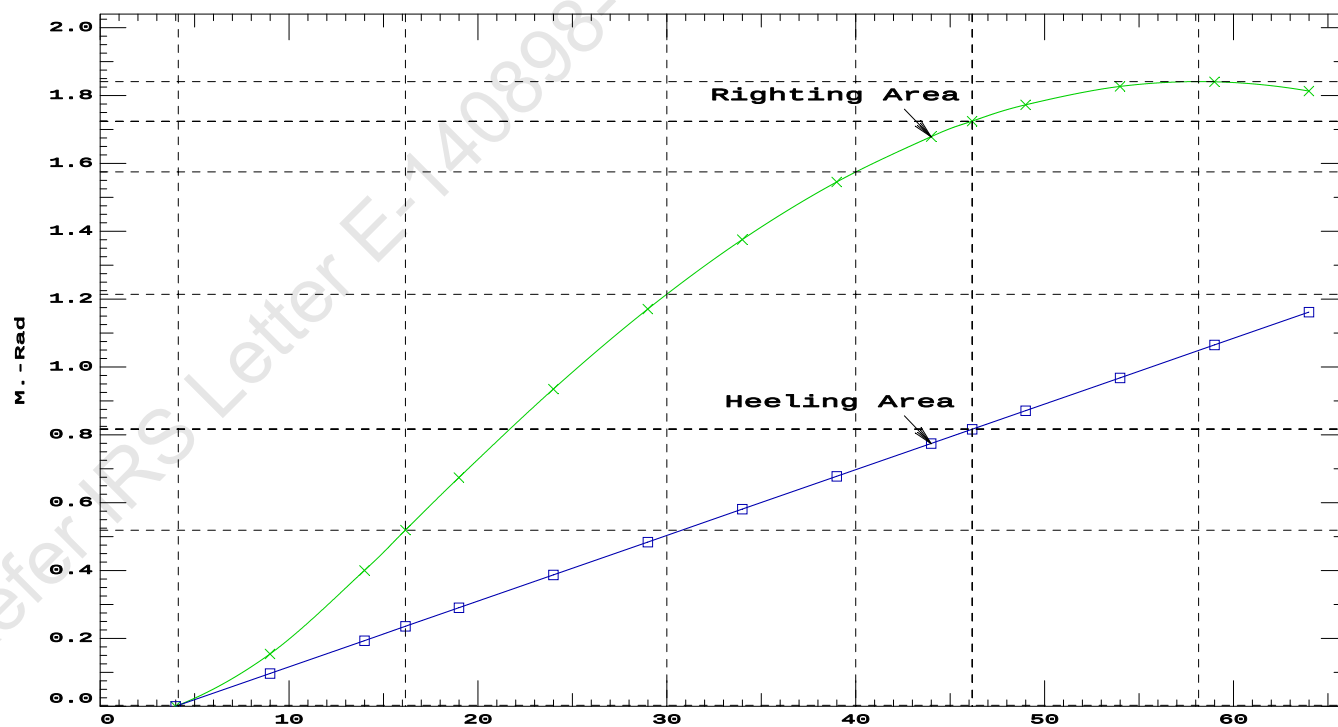
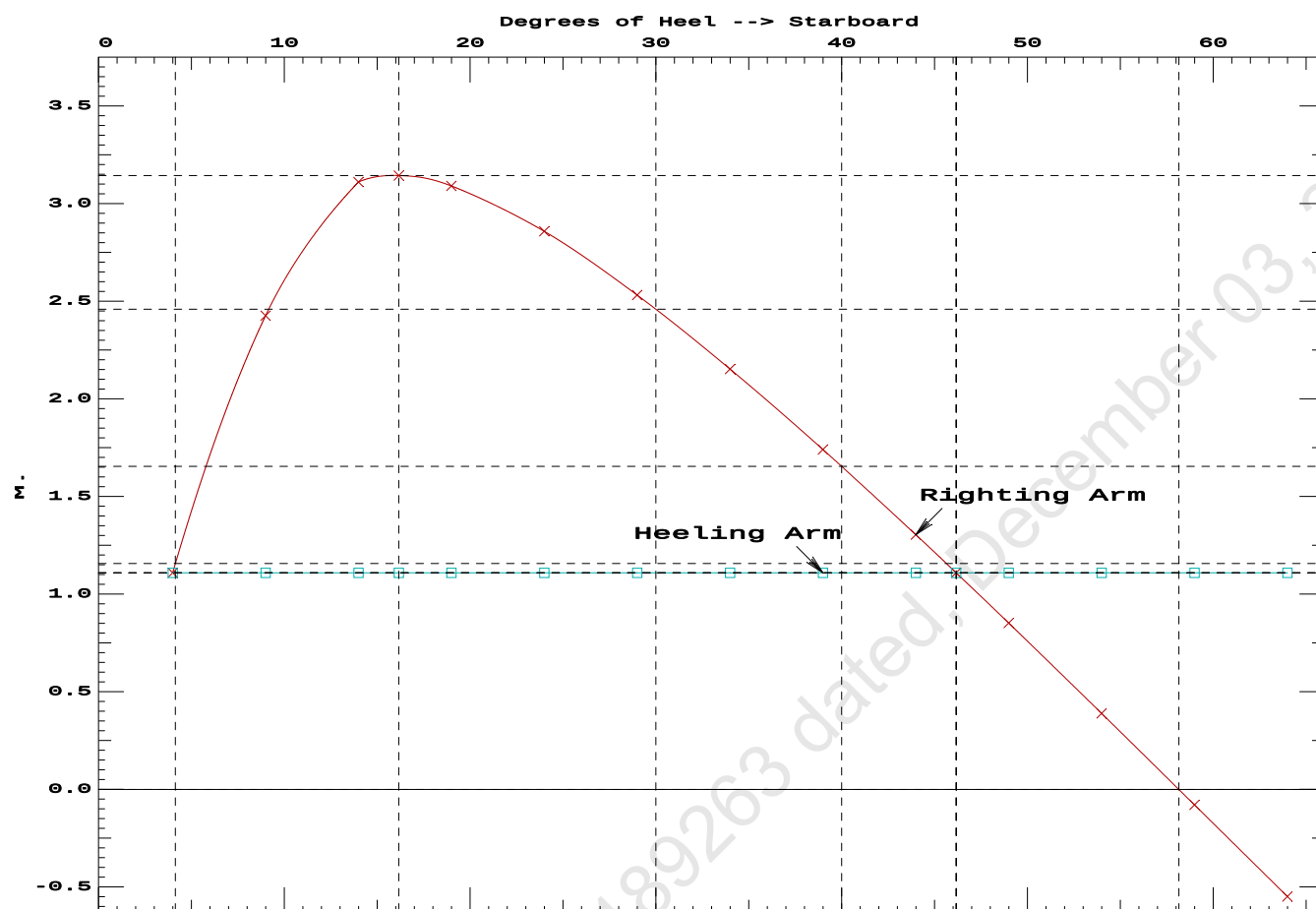
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1400.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	3.99 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	183.4 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	2.119 P

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GHS 17.90Cybermarine Technologies PTE. Ltd.
PONT00NCondition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

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GHS 17.90 PONTON

Condition 08 : Operating Condition of 350T CraneCrane Lifting 40.0T @ 35m Radius Stbd

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.499 @ 50.00f, 1.513 @ 25.00f, 1.527 @ 0.00

Trim: Aft 0.028/50.000, Heel: Stbd 3.99 deg.

Part-----	Weight(MT)	LCG	TCG	VCG	
LIGHT SHIP	449.57	25.030f	0.000	3.000	
Crane	350.00	11.500f	0.000	6.325	
Crane Load	40.00	11.500f	0.000	21.478	
Total Fixed----->	839.57	18.745f	0.000	5.266	
Load-----	SpGr	Weight(MT)	LCG	TCG	VCG
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s
Total Tanks----->		422.40	37.208f	0.000	1.596
Total Weight----->		1,261.97	24.925f	0.000	4.038
		Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,261.97	24.923f	1.337s	0.816

Righting Arms:		0.000	1.109s		
External Arms:		0.000	1.109s		
Residual Righting Arms:		0.000	0.000s		
Part-----	SpGr	WPA	LCF	TCF	BML
Total Waterplane----->	1.025	869.4	24.974f	0.154s	136.66
		MT/cm			
		8.91			

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 3.99

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After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.541 @ 50.00f, 1.475 @ 25.00f, 1.408 @ 0.00
Trim: Fwd 0.133/50.000, Heel: zero

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
LIGHT SHIP	449.57	25.030f	0.000	3.000
Crane	350.00	11.500f	0.000	6.325
Total Fixed----->	799.57	19.107f	0.000	4.455

Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880
Total Tanks----->			422.40	37.208f	0.000	1.596
Total Weight----->			1,221.97	25.364f	0.000	3.467

Displ(MT)----	LCB-----	TCB-----	VCB-----
HULL	1.025	1,221.96	25.372f
		0.000	0.746

Righting Arms:	0.000	0.000
External Arms:	0.000	0.001
Residual Righting Arms:	0.000	-0.001s

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->	1.025	866.4	25.106f	0.000	139.99	19.249*
		MT/cm-----	m.-MT/cm-----		GML-----	GMT-----
		8.88	33.55		137.27	16.529

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 3332.70; t3: 30.0; t: 15.0
limmarg: 1208.33; t3: 15.0; t: 7.5
limmarg: 252.79; t3: 7.5; t: 3.8
limmarg: -13.74; t3: 3.7; t: 1.9
limmarg: 97.10; t3: 5.6; t: 0.9
limmarg: 38.94; t3: 4.7; t: 0.5
limmarg: 11.06; t3: 4.2; t: 0.2
limmarg: 0.78; t3: 4.0; t: 0.1
limmarg: -4.19; t3: 3.9; t: 0.1
limmarg: 0.78; t3: 4.0; t: 0.0
limmarg: 0.78; t3: 4.0; t: 0.0

Theta3 is 4.00 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 19.107f TCG = 0.000 VCG = 4.455

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.386	0.15f	3.99s	1,222.0	0.000	-1.186	0.0000
1.396	0.15f	0.00	1,222.0	0.000	0.001	-0.0413
1.386	0.15f	4.00p	1,222.0	0.000	1.191	0.0003
1.381	0.15f	5.00p	1,222.0	0.000	1.485	0.0237
1.331	0.16f	10.00p	1,222.0	0.000	2.894	0.2152
1.209	0.25f	15.00p	1,222.0	0.000	3.463	0.4986
1.143	0.29f	17.56p	1,221.9	0.000	3.499	0.6545
1.076	0.33f	20.00p	1,222.0	0.000	3.466	0.8031
0.933	0.42f	25.00p	1,222.0	0.000	3.285	1.0999
0.782	0.50f	30.00p	1,222.0	0.000	3.017	1.3755

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 799.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs -4 deg to abs 4 > 1.000 1.008 P

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GHS 17.90 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 19.107f TCG = 0.000 VCG = 4.455

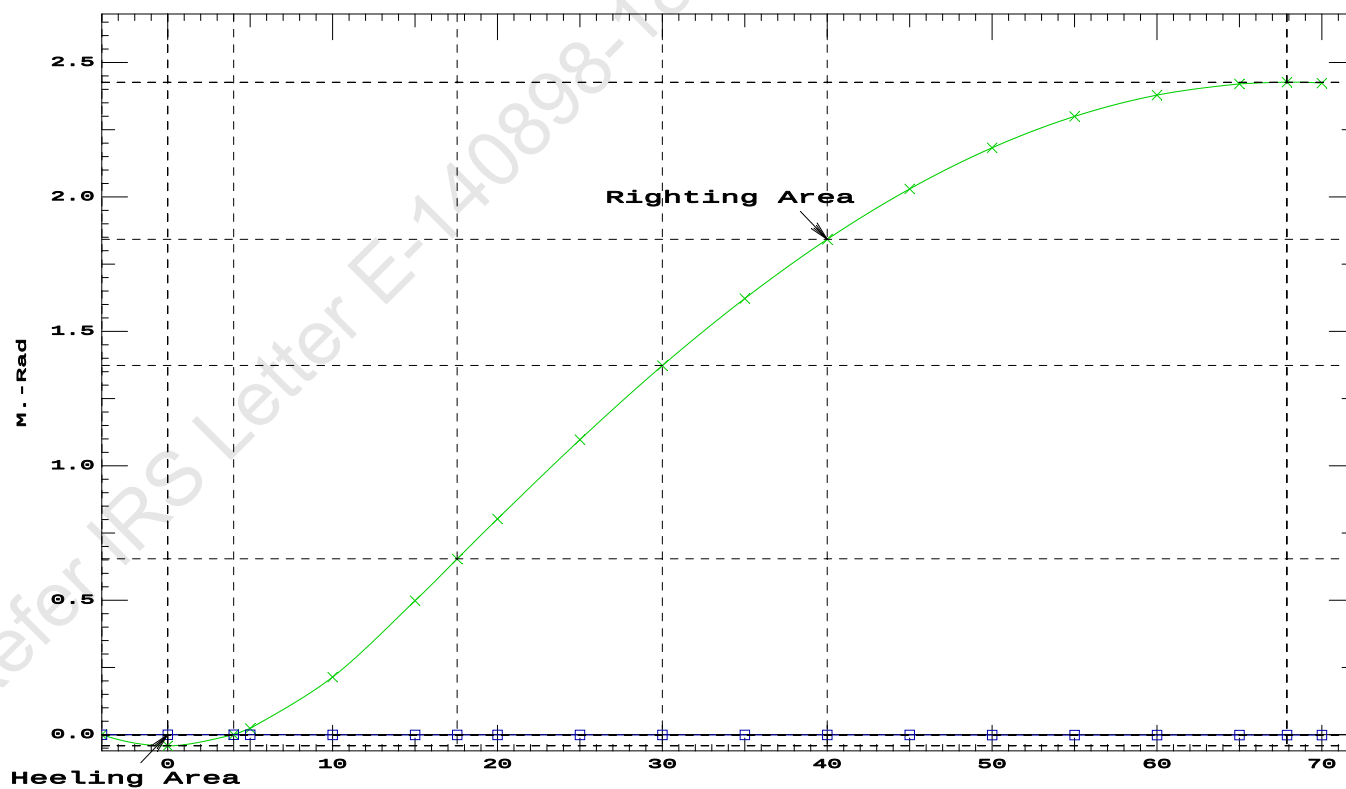
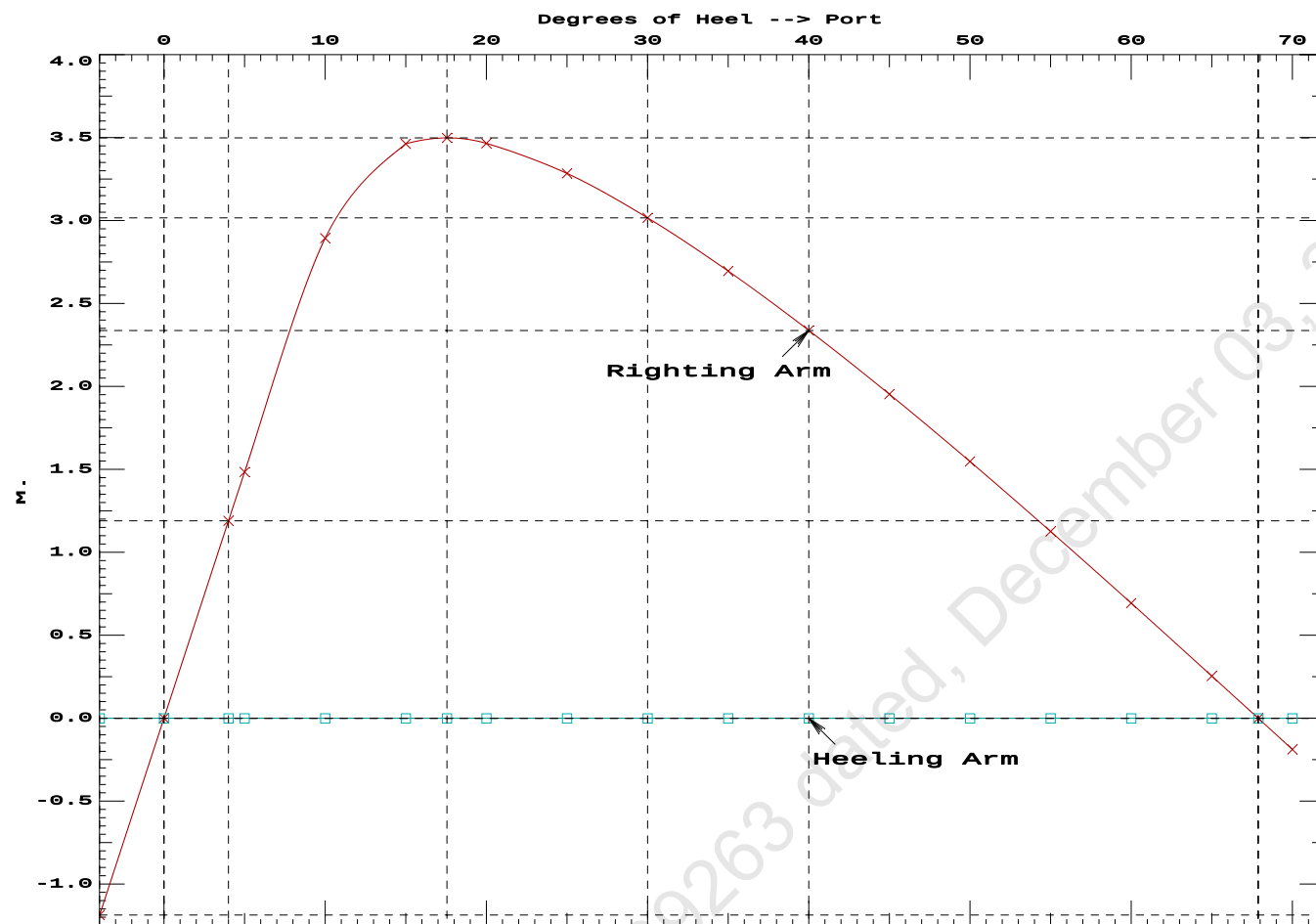
Origin Depth	Degrees of Trim	Displacement Heel	Residual Arms Weight(MT)	Res. in Trim	Res. in Heel	Area
1.386	0.15f	3.99s	1,222.0	0.000	-1.186	0.0000
1.396	0.15f	0.00	1,222.0	0.000	0.001	-0.0413
1.386	0.15f	4.00p	1,222.0	0.000	1.191	0.0003
1.381	0.15f	5.00p	1,222.0	0.000	1.485	0.0237
1.331	0.16f	10.00p	1,222.0	0.000	2.894	0.2152
1.209	0.25f	15.00p	1,222.0	0.000	3.463	0.4986
1.143	0.29f	17.56p	1,221.9	0.000	3.499	0.6545
1.076	0.33f	20.00p	1,222.0	0.000	3.466	0.8031
0.933	0.42f	25.00p	1,222.0	0.000	3.285	1.0999
0.782	0.50f	30.00p	1,222.0	0.000	3.017	1.3755
0.623	0.59f	35.00p	1,222.0	0.000	2.696	1.6251
0.457	0.67f	40.00p	1,222.0	0.000	2.338	1.8450
0.287	0.76f	45.00p	1,222.0	0.000	1.953	2.0325
0.113	0.84f	50.00p	1,222.0	0.000	1.548	2.1854
-0.062	0.91f	55.00p	1,222.0	0.000	1.127	2.3022
-0.236	0.98f	60.00p	1,222.0	0.000	0.694	2.3817
-0.409	1.04f	65.00p	1,222.0	0.000	0.255	2.4232
-0.506	1.07f	67.88p	1,222.0	0.000	0.000	2.4296
-0.576	1.09f	70.00p	1,222.0	0.000	-0.188	2.4261

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 799.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs -4 deg to RAZero > 1.000 59.868 P
(2) Angle from abs 4 deg to RAZero > 20.00 deg 63.88 P

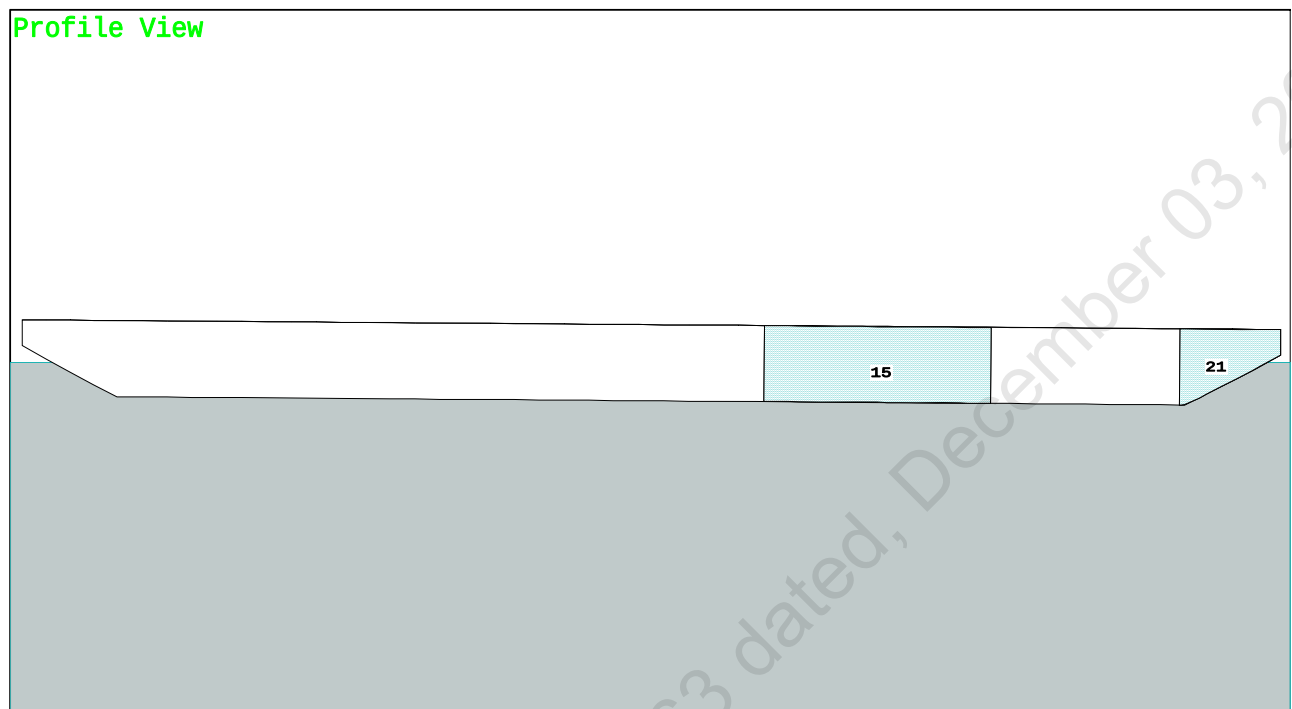
01-12-21 09:53:42
GHS 17.90Cybermarine Technologies PTE. Ltd.
PONT00N

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GHS 17.90 PONT00N

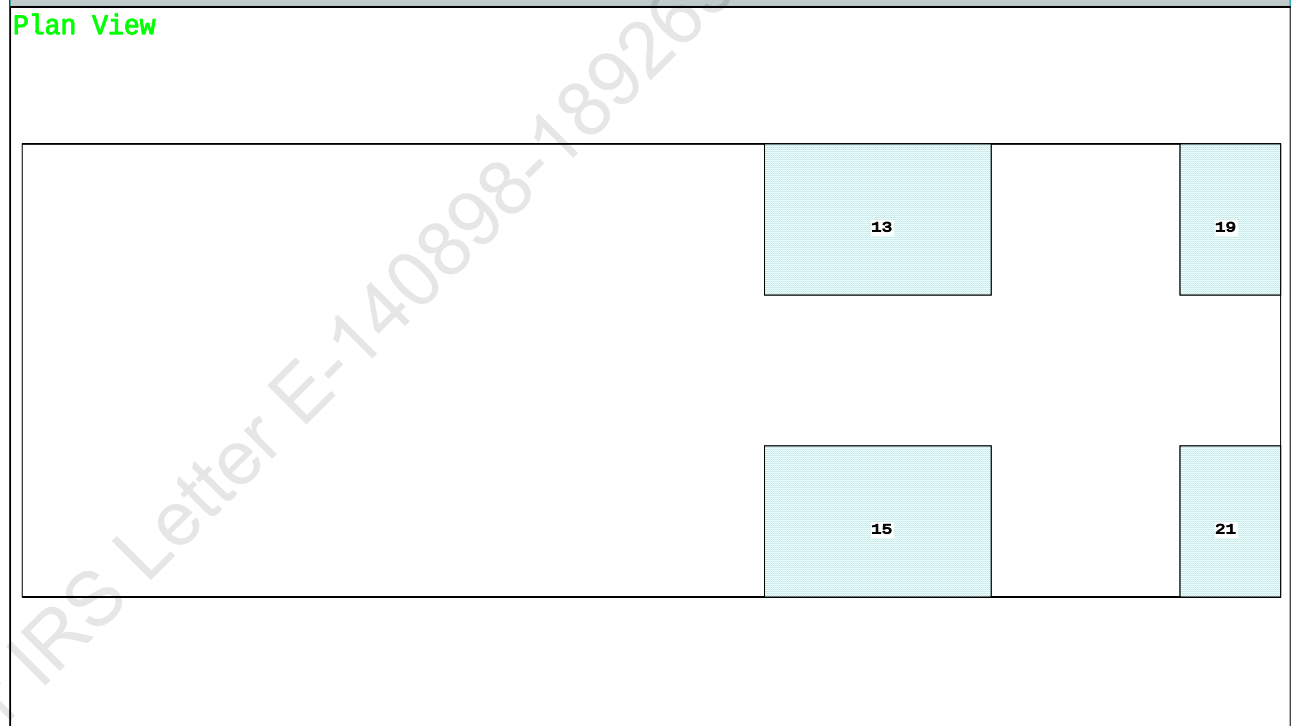
Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

Condition Graphic - Draft: 1.712 @ 50.000f, 1.325 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

13 05_WBTK.P.100% SALT WATER Intact

15 05_WBTK.S.100% SALT WATER Intact

19 07_WBTK.P.100% SALT WATER Intact

21 07_WBTK.S.100% SALT WATER Intact

01-12-21 09:55:20 Cybermarine Technologies PTE. Ltd.
GHS 17.90

PONTOON

Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.712 @ 50.00f, 1.518 @ 25.00f, 1.325 @ 0.00

Trim: Fwd 0.387/50.000, Heel: zero

Part-----			Weight(MT)-----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			449.57	25.030f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Crane Load			40.00	46.500f	0.000	21.478	
Total Fixed----->			839.57	20.413f	0.000	5.266	
	Load-----	SpGr-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,261.97	26.034f	0.000	4.038	
			Displ(MT)-----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,261.88	26.060f	0.000	0.773	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	869.0	25.334f	0.000	136.83	18.691*
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			8.91		33.71	133.56	15.427
Distances in METERS.-----				Moments in m.-MT-----			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

350.00 MT Crane

40.00 MT Crane Load

DRAFT AT FP (M) = 1.712
DRAFT AT MS (M) = 1.518
DRAFT AT AP (M) = 1.325
DRAFT AT FWD MARK (M) = 1.681
DRAFT AT AFT MARK (M) = 1.356

HYDROSTATIC PROPERTIES

Trim: Fwd 0.387/50.000, No Heel, VCG = 4.038

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)-----	LCB-----	VCB-----	cm-----	LCF-----	cm trim-----	GML-----	GMT
1.521	1,261.88	26.060f	0.773	8.91	25.334f	33.71	133.56	15.427
Distances in METERS.-----				Specific Gravity = 1.025.-----				
				Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

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PONTOON

Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.712 @ 50.00f, 1.518 @ 25.00f, 1.325 @ 0.00

Trim: Fwd 0.387/50.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm----	Pressure-----	Moment
HULL	74.6	0.761	1.505	0.05500	6.18
CRANE	57.0	4.109	4.854	0.05500	15.22
Total wind heeling moment to starboard----->					21.39
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 26.034f TCG = 0.000 VCG = 4.038

Free Surface Adjustment: 0.373

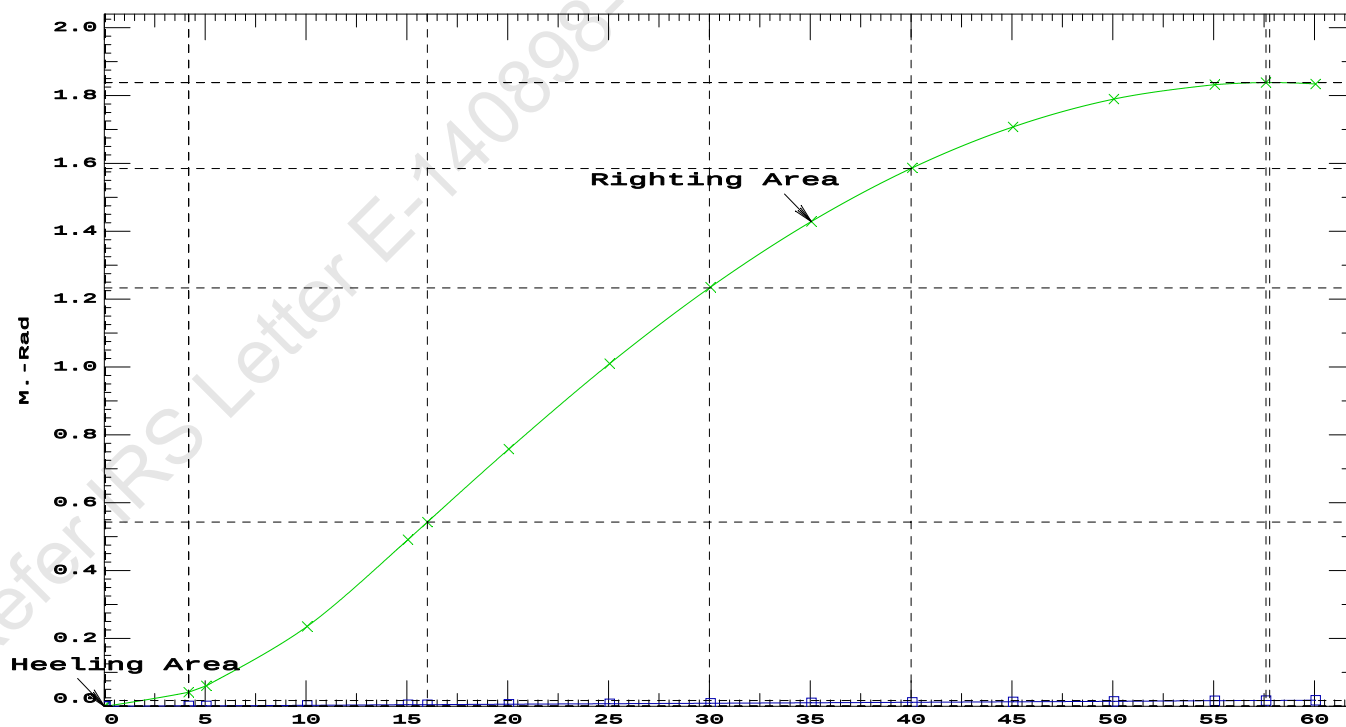
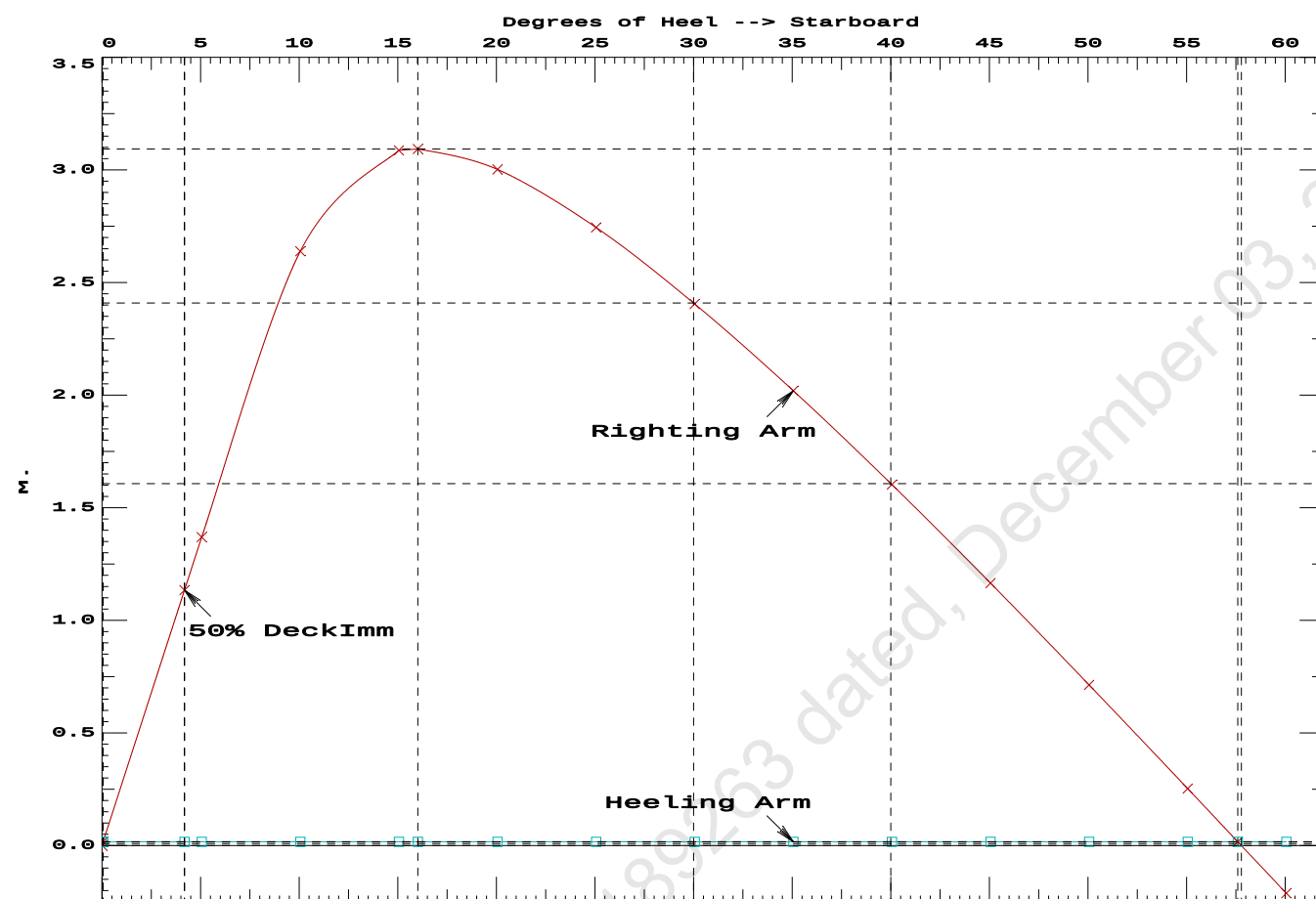
Adjusted CG: LCG = 26.031f TCG = 0.000 VCG = 4.411

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---	Trim----	Heel----	Weight(MT)---	Area-----
			in Trim--in Heel---	Angle
1.313	0.44f	0.00	1,261.9	0.0000 -0.017 0.0000 4.180
1.313	0.44f	0.05s	1,262.0	0.0000 0.000 -0.0000 4.125
1.302	0.45f	4.18s	1,262.0	0.0000 1.117 0.0402 0.000
1.296	0.45f	5.05s	1,262.0	0.0000 1.352 0.0590 -0.875
1.237	0.49f	10.05s	1,262.0	0.0000 2.622 0.2331 -5.875
1.068	0.73f	15.05s	1,262.0	0.0000 3.070 0.4882 -10.875
1.034	0.78f	16.02s	1,262.0	0.0000 3.076 0.5397 -11.836
0.884	0.98f	20.05s	1,262.0	0.0000 2.985 0.7553 -15.875
0.687	1.24f	25.05s	1,262.0	0.0000 2.727 1.0067 -20.875
0.480	1.50f	30.05s	1,262.0	0.0000 2.388 1.2304 -25.875
0.264	1.76f	35.05s	1,262.0	0.0000 2.002 1.4223 -30.875
0.042	2.02f	40.05s	1,262.0	0.0000 1.586 1.5791 -35.875
-0.185	2.27f	45.05s	1,262.0	0.0000 1.148 1.6986 -40.875
-0.413	2.52f	50.05s	1,262.0	0.0000 0.696 1.7792 -45.875
-0.637	2.74f	55.05s	1,262.0	0.0000 0.236 1.8199 -50.875
-0.749	2.84f	57.60s	1,262.0	0.0000 0.000 1.8251 -53.416
-0.856	2.94f	60.05s	1,262.0	0.0000 -0.229 1.8202 -55.875
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.39 (constant)

LIM-----	IS CODE 2008 PART B CHAPTER-2 SEC.2----	Min/Max-----	Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.5397 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	57.54 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	4.13 P

01-12-21 09:55:20
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PONT00NCondition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

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GHS 17.90

PONTOON

Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 26.034f TCG = 0.000 VCG = 4.038
Free Surface Adjustment: 0.373
Adjusted CG: LCG = 26.031f TCG = 0.000 VCG = 4.411

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.313	0.44f	0.00	1,262.0	0.000	0.001	0.0000
1.297	0.45f	5.00s	1,262.0	0.000	1.353	0.0591
1.238	0.48f	10.00s	1,262.0	0.000	2.626	0.2332
1.070	0.73f	15.00s	1,262.0	0.000	3.084	0.4893
1.033	0.78f	16.02s	1,262.0	0.000	3.091	0.5444
0.886	0.98f	20.00s	1,262.0	0.000	3.002	0.7577
0.690	1.24f	25.00s	1,262.0	0.000	2.745	1.0107
0.483	1.50f	30.00s	1,262.0	0.000	2.407	1.2361
0.267	1.76f	35.00s	1,262.0	0.000	2.022	1.4297
0.044	2.02f	40.00s	1,262.0	0.000	1.605	1.5881
-0.183	2.27f	45.00s	1,262.0	0.000	1.168	1.7093
-0.410	2.51f	50.00s	1,262.0	0.000	0.716	1.7916
-0.635	2.74f	55.00s	1,262.0	0.000	0.256	1.8341
-0.757	2.85f	57.76s	1,262.0	0.000	0.000	1.8403
-0.854	2.93f	60.00s	1,262.0	0.000	-0.208	1.8362

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

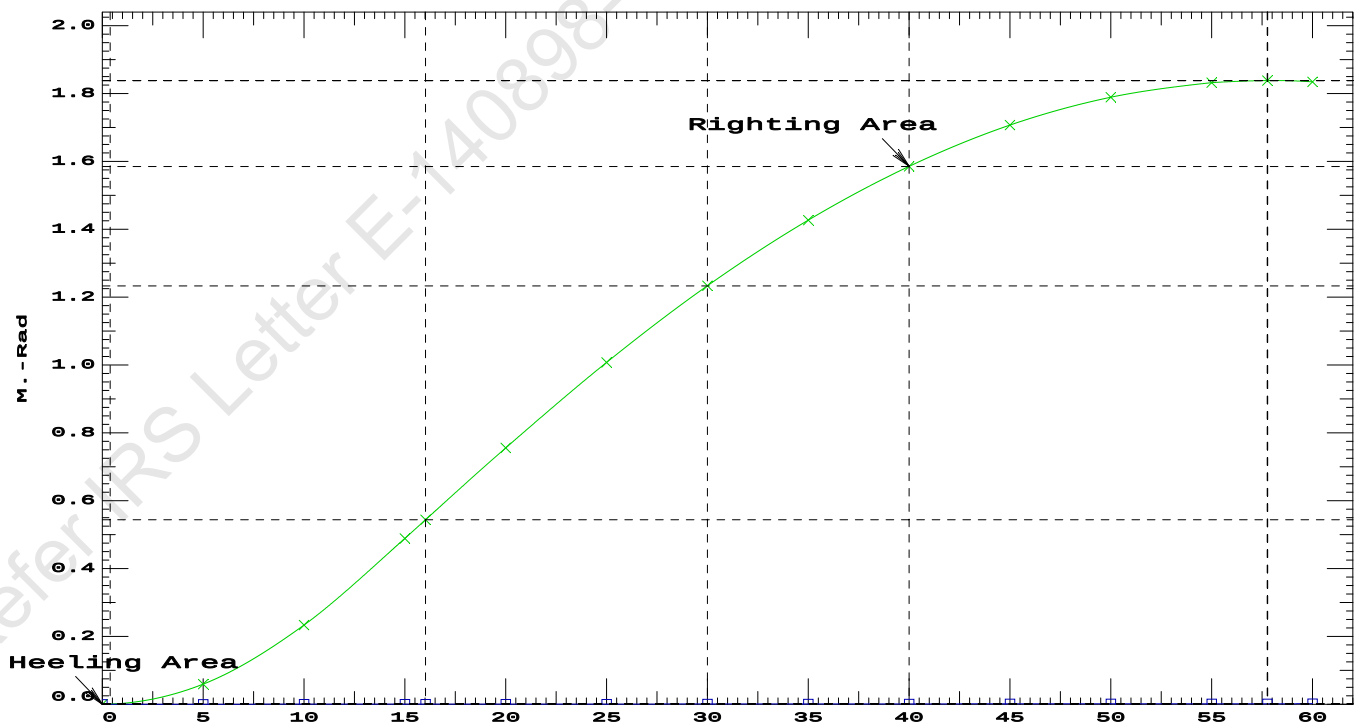
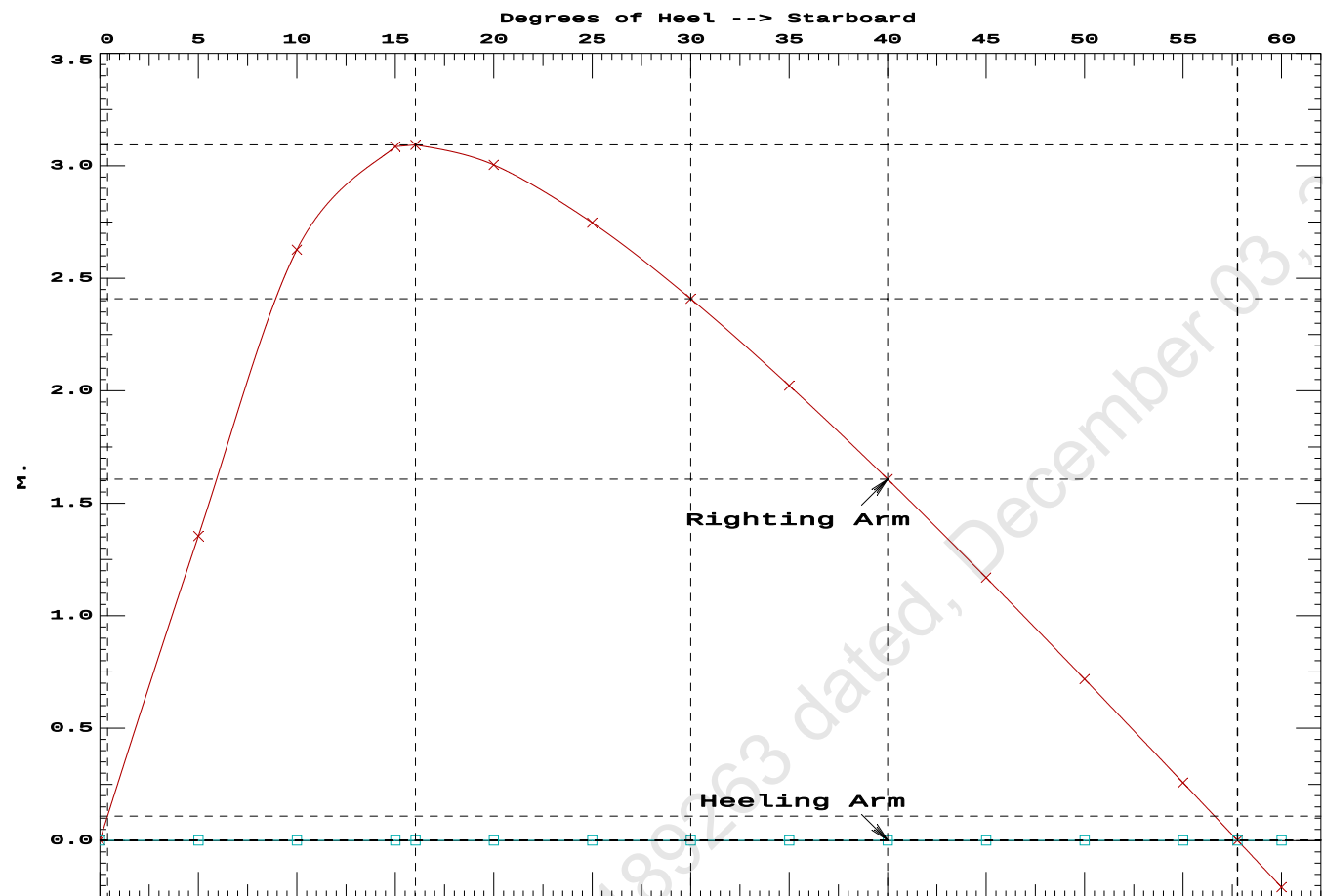
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 2.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	195056 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	1152.90 P

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Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd



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GHS 17.90 PONTON

Trim: Fwd 0.387/50.000, Heel: zero

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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GHS 17.90 PONT00N

After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.541 @ 50.00f, 1.475 @ 25.00f, 1.408 @ 0.00
Trim: Fwd 0.133/50.000, Heel: zero

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
LIGHT SHIP	449.57	25.030f	0.000	3.000
Crane	350.00	11.500f	0.000	6.325
Total Fixed----->	799.57	19.107f	0.000	4.455

Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880
Total Tanks----->			422.40	37.208f	0.000	1.596
Total Weight----->			1,221.97	25.364f	0.000	3.467

Displ(MT)----	LCB-----	TCB-----	VCB-----
HULL	1.025	1,221.96	25.371f
		0.000	0.746

Righting Arms:	0.000	0.000
External Arms:	0.000	0.001
Residual Righting Arms:	0.000	-0.001s

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->	1.025	866.4	25.106f	0.000	139.99	19.249*
		MT/cm-----	m.-MT/cm-----		GML-----	GMT-----
		8.88	33.55		137.27	16.529

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30; t: 15.00
limmarg: INFINITY; t3: 15; t: 7.50
limmarg: INFINITY; t3: 8; t: 3.75
limmarg: INFINITY; t3: 4; t: 1.88
limmarg: INFINITY; t3: 2; t: 0.94
limmarg: INFINITY; t3: 1; t: 0.47
limmarg: INFINITY; t3: 1; t: 0.23
limmarg: INFINITY; t3: 1; t: 0.12
limmarg: INFINITY; t3: 1; t: 0.06
limmarg: INFINITY; t3: 1; t: 0.03
limmarg: INFINITY; t3: 1; t: 0.01

Theta3 is 0.97 degrees, as demonstrated below.

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GHS 17.90 PONTON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 19.107f TCG = 0.000 VCG = 4.455

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.396	0.15f	0.00	1,222.0	0.000	-0.001	0.0000
1.394	0.15f	0.97s	1,222.0	0.000	0.290	0.0024
1.381	0.15f	5.00s	1,222.0	0.000	1.484	0.0648
1.331	0.16f	10.00s	1,222.0	0.000	2.892	0.2563
1.209	0.25f	15.00s	1,222.0	0.000	3.461	0.5396
1.143	0.29f	17.56s	1,221.9	0.000	3.497	0.6954
1.076	0.33f	20.00s	1,222.0	0.000	3.464	0.8439
0.933	0.42f	25.00s	1,222.0	0.000	3.283	1.1406
0.782	0.50f	30.00s	1,222.0	0.000	3.015	1.4161

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 799.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 1 > 1.000 LARGE

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GHS 17.90 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 19.107f TCG = 0.000 VCG = 4.455

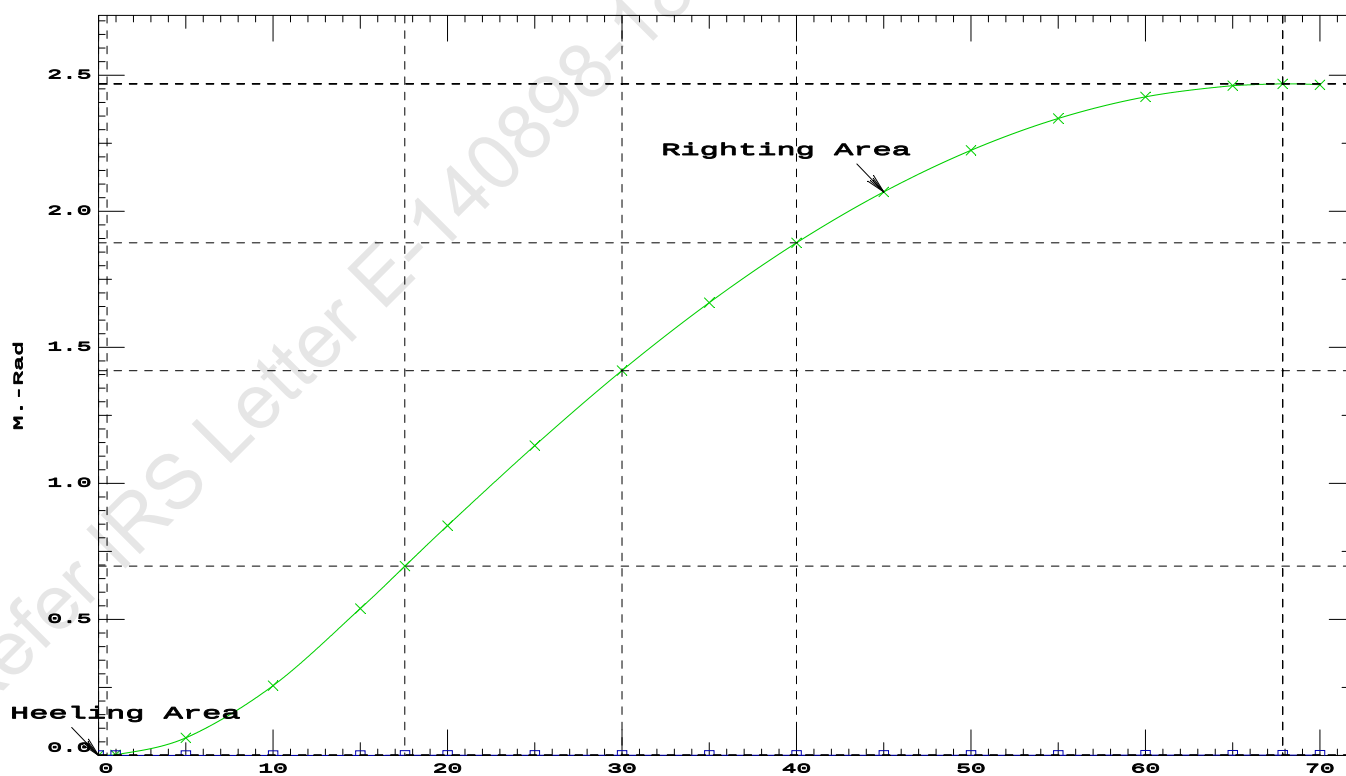
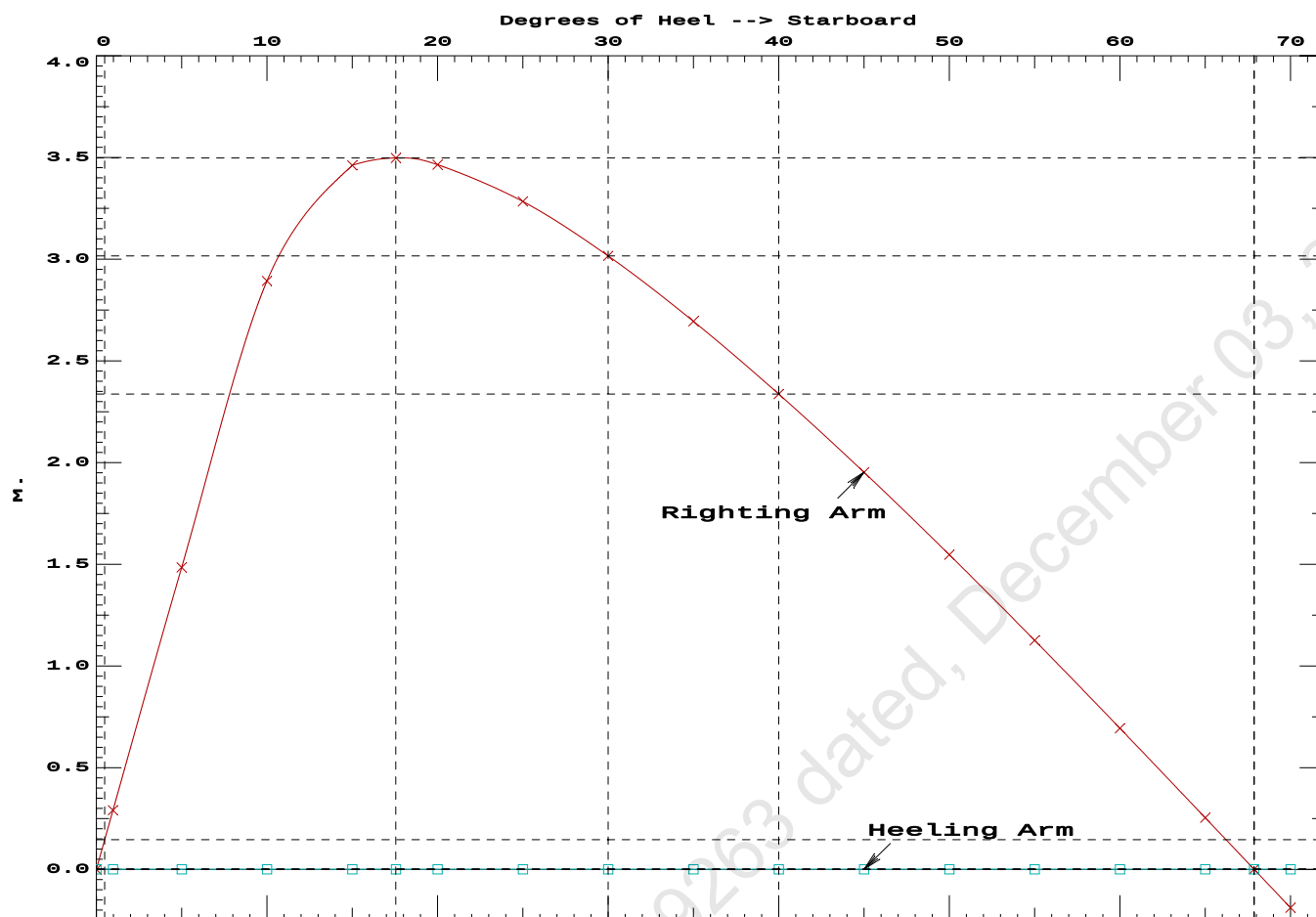
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.396	0.15f	0.00	1,222.0	0.000	-0.001	0.0000
1.394	0.15f	0.97s	1,222.0	0.000	0.290	0.0024
1.381	0.15f	5.00s	1,222.0	0.000	1.484	0.0648
1.331	0.16f	10.00s	1,222.0	0.000	2.892	0.2563
1.209	0.25f	15.00s	1,222.0	0.000	3.461	0.5396
1.143	0.29f	17.56s	1,221.9	0.000	3.497	0.6954
1.076	0.33f	20.00s	1,222.0	0.000	3.464	0.8439
0.933	0.42f	25.00s	1,222.0	0.000	3.283	1.1406
0.782	0.50f	30.00s	1,222.0	0.000	3.015	1.4161
0.623	0.59f	35.00s	1,222.0	0.000	2.694	1.6656
0.457	0.67f	40.00s	1,222.0	0.000	2.337	1.8853
0.287	0.76f	45.00s	1,222.0	0.000	1.952	2.0726
0.113	0.84f	50.00s	1,222.0	0.000	1.546	2.2254
-0.062	0.91f	55.00s	1,222.0	0.000	1.125	2.3421
-0.236	0.98f	60.00s	1,222.0	0.000	0.693	2.4215
-0.409	1.04f	65.00s	1,222.0	0.000	0.253	2.4628
-0.505	1.07f	67.86s	1,222.0	0.000	0.000	2.4691
-0.576	1.09f	70.00s	1,222.0	0.000	-0.189	2.4656

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 799.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 1 deg to RAZero > 20.00 deg 66.89 P

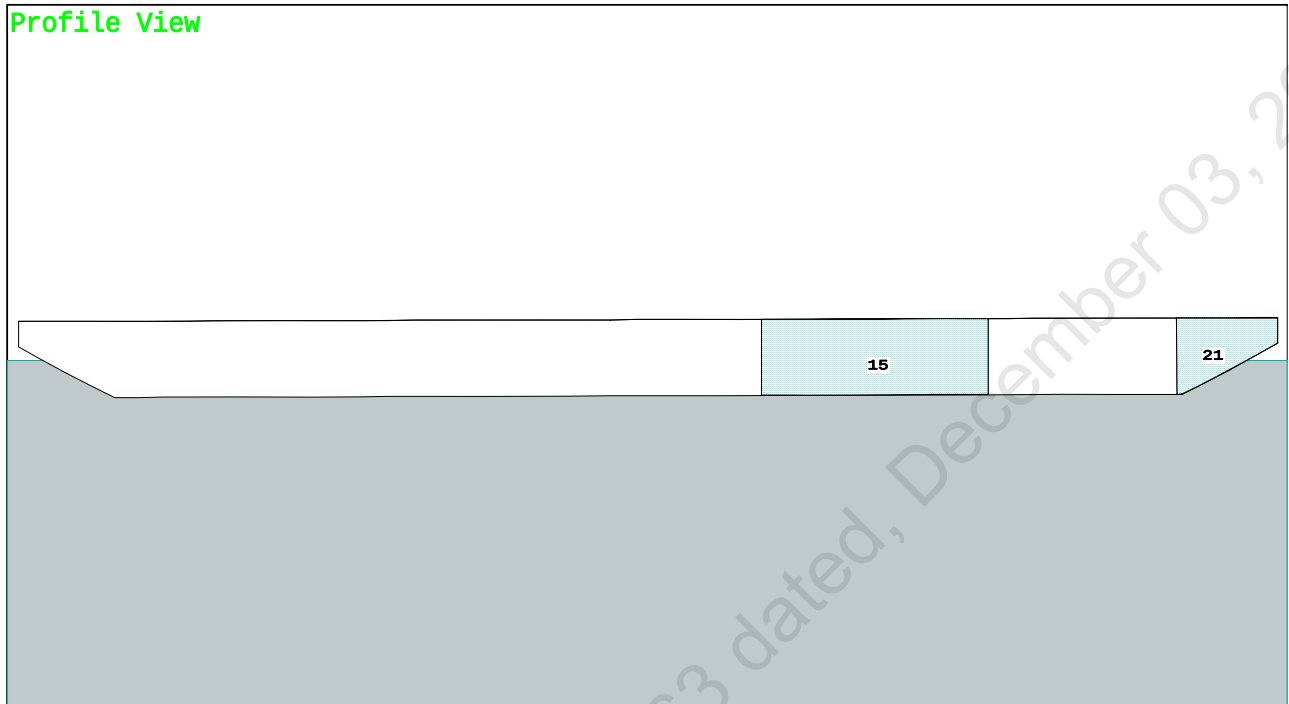
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GHS 17.90 PONT00N

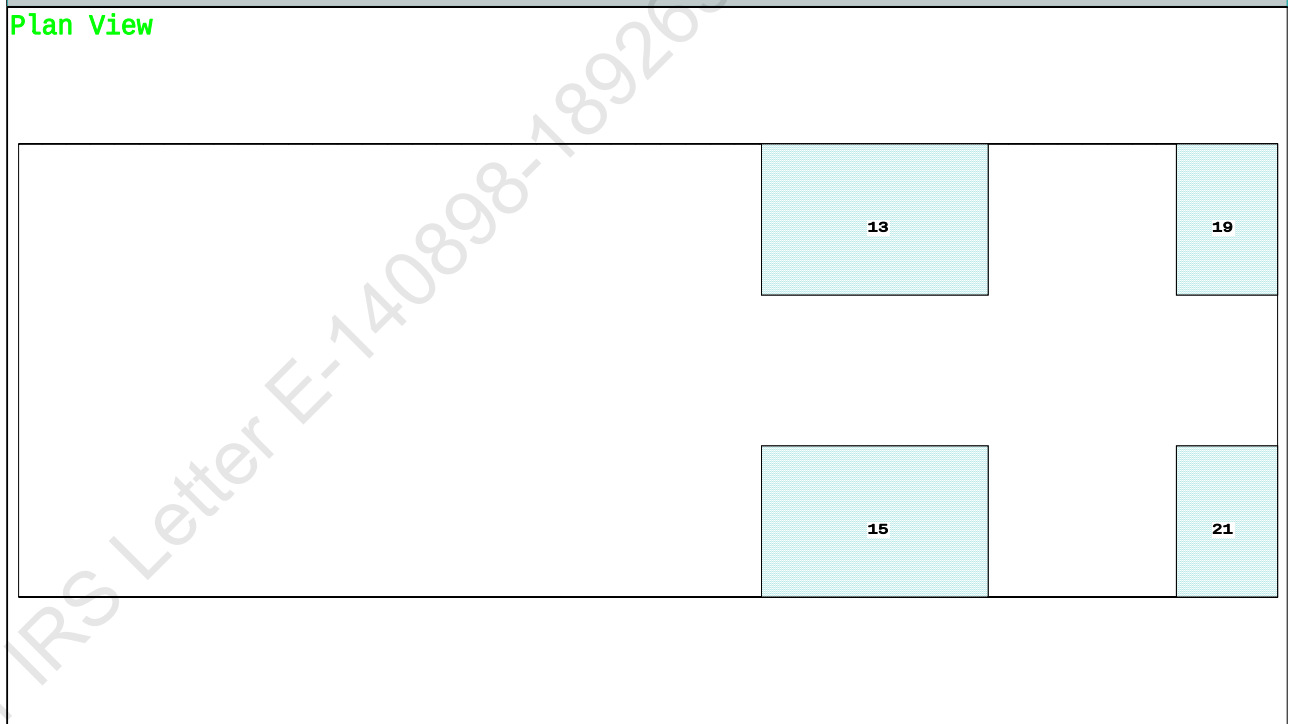
Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

Condition Graphic - Draft: 1.326 @ 50.000f, 1.480 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

13 05_WBTK.P.100% SALT WATER Intact	15 05_WBTK.S.100% SALT WATER Intact	21 07_WBTK.S.100% SALT WATER Intact
	19 07_WBTK.P.100% SALT WATER Intact	

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PONTOON

Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.326 @ 50.00f, 1.403 @ 25.00f, 1.480 @ 0.00

Trim: Aft 0.155/50.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			449.57	25.030f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	10.500a	0.000	76.977	
Total Fixed----->			736.17	17.310f	0.000	7.425	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,158.57	24.565f	0.000	5.300	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,158.54	24.551f	0.000	0.709	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	861.5	24.866f	0.000	145.18	20.179*
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			8.83		32.57	140.59	15.589
Distances in METERS.-----				Moments in m.-MT			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

250.00 MT Crane

36.60 MT Crane Load

DRAFT AT FP (M) = 1.326
DRAFT AT MS (M) = 1.403
DRAFT AT AP (M) = 1.480
DRAFT AT FWD MARK (M) = 1.338
DRAFT AT AFT MARK (M) = 1.468

HYDROSTATIC PROPERTIES

Trim: Aft 0.155/50.000, No Heel, VCG = 5.300

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----
1.403	1,158.54	24.551f	0.709	8.83
				24.866f
				32.57
				140.59
				15.589

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

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GHS 17.90

PONTOON

Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.326 @ 50.00f, 1.403 @ 25.00f, 1.480 @ 0.00

Trim: Aft 0.155/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	80.3	0.816	1.503	0.05500	6.63
CRANE	57.0	4.063	4.750	0.05500	14.89
Total wind heeling moment to starboard-->					21.52
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.565f TCG = 0.000 VCG = 5.300

Free Surface Adjustment: 0.406

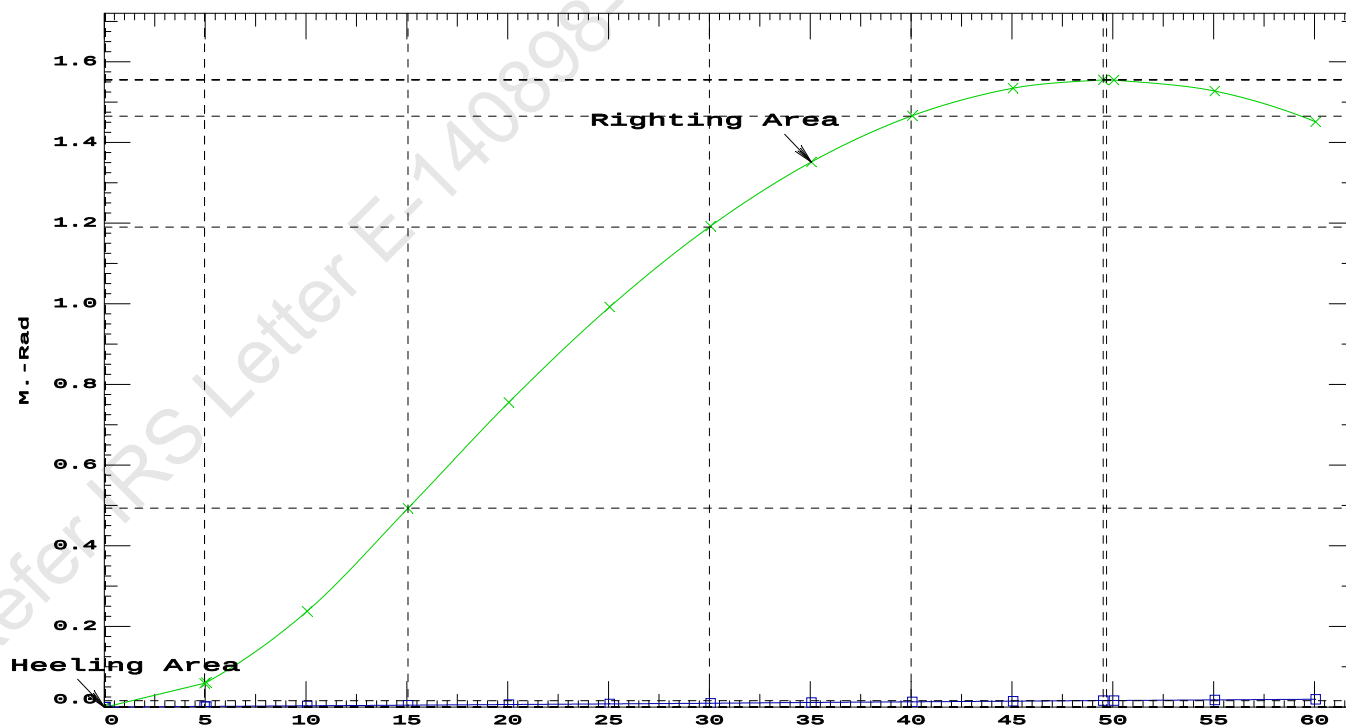
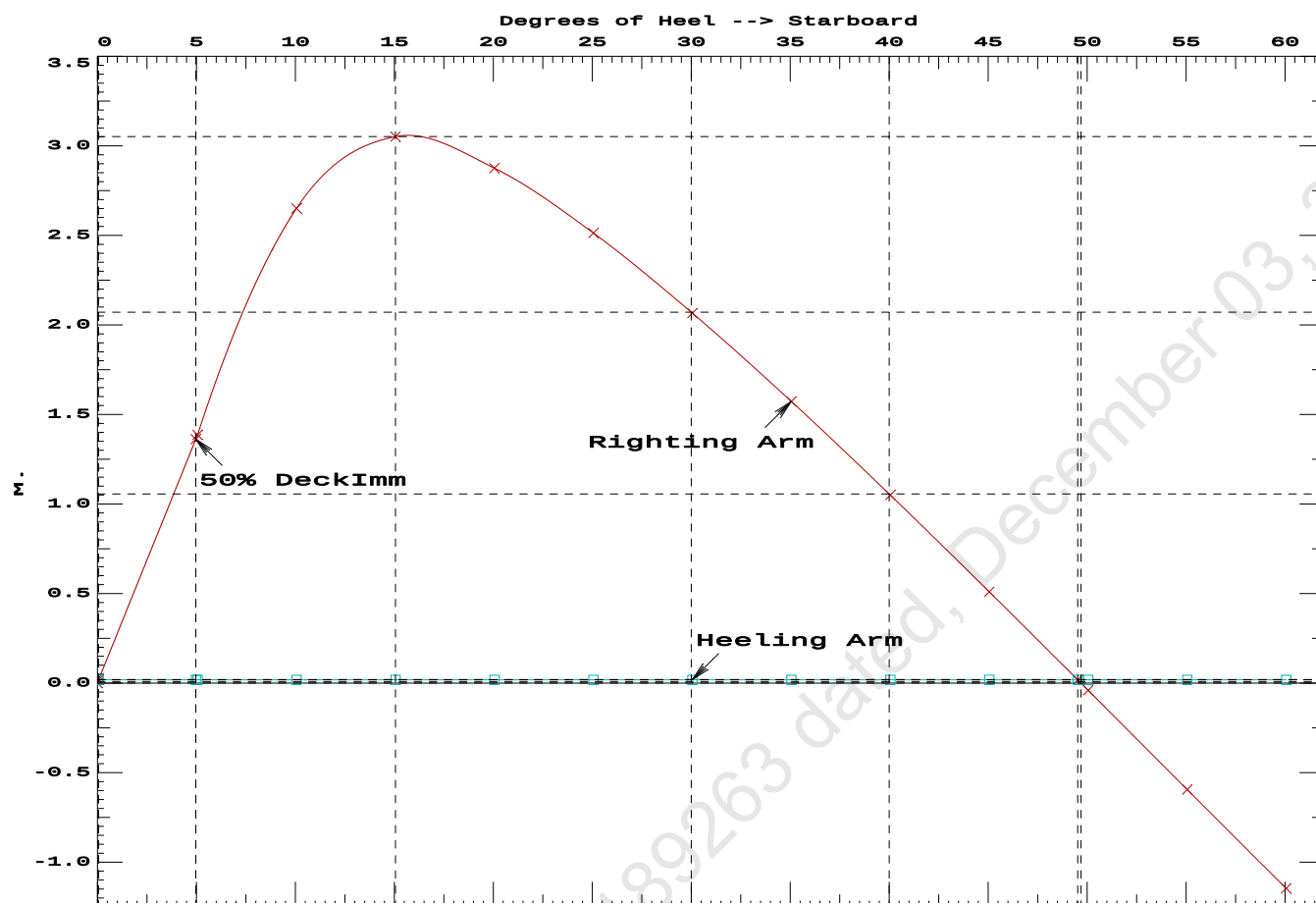
Adjusted CG: LCG = 24.566f TCG = 0.000 VCG = 5.706

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Res. 50% DeckImm Area	Angle
1.468	0.18a	0.00	1,158.6	0.000 -0.018	0.0000	4.966
1.468	0.18a	0.06s	1,158.6	0.000 0.000	-0.0000	4.911
1.454	0.18a	4.97s	1,158.6	0.000 1.343	0.0576	0.000
1.454	0.18a	5.06s	1,158.6	0.000 1.368	0.0597	-0.089
1.409	0.19a	10.06s	1,158.6	0.000 2.633	0.2350	-5.089
1.327	0.28a	15.06s	1,158.6	0.000 3.033	0.4895	-10.089
1.239	0.38a	20.06s	1,158.6	0.000 2.857	0.7507	-15.089
1.144	0.49a	25.06s	1,158.6	0.000 2.494	0.9855	-20.089
1.042	0.59a	30.06s	1,158.6	0.000 2.048	1.1843	-25.089
0.935	0.70a	35.06s	1,158.6	0.000 1.554	1.3418	-30.089
0.822	0.80a	40.06s	1,158.6	0.000 1.031	1.4548	-35.089
0.704	0.90a	45.06s	1,158.6	0.000 0.491	1.5214	-40.089
0.595	0.99a	49.52s	1,158.6	0.000 0.000	1.5406	-44.555
0.582	1.00a	50.06s	1,158.6	0.000 -0.059	1.5403	-45.089
0.455	1.08a	55.06s	1,158.6	0.000 -0.613	1.5110	-50.089
0.323	1.16a	60.06s	1,158.6	0.000 -1.165	1.4334	-55.089
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.						

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.52 (constant)

LIM	IS CODE 2008 PART B CHAPTER-2 SEC.2	Min/Max	Attained
(1)	Residual Area from abs 0 deg to MaxRA	> 0.0800 m.-Rad	0.4895 P
(2)	Angle from Equilibrium to RAZero or Flood	> 20.00 deg	49.47 P
(3)	Angle from Equilibrium to 50% Dk Imm Angle	> 0.00 deg	4.91 P

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GHS 17.90Cybermarine Technologies PTE. Ltd.
PONT00NCondition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

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GHS 17.90

PONTOON

Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.565f TCG = 0.000 VCG = 5.300
Free Surface Adjustment: 0.406
Adjusted CG: LCG = 24.566f TCG = 0.000 VCG = 5.706

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.468	0.18a	0.01s	1,158.6	0.000	0.000	0.0000
1.454	0.18a	5.01s	1,158.6	0.000	1.369	0.0597
1.410	0.19a	10.01s	1,158.6	0.000	2.637	0.2353
1.328	0.28a	15.01s	1,158.6	0.000	3.047	0.4907
1.240	0.38a	20.01s	1,158.6	0.000	2.873	0.7533
1.145	0.49a	25.01s	1,158.6	0.000	2.512	0.9896
1.043	0.59a	30.01s	1,158.6	0.000	2.066	1.1900
0.936	0.70a	35.01s	1,158.6	0.000	1.573	1.3491
0.823	0.80a	40.01s	1,158.6	0.000	1.050	1.4638
0.705	0.90a	45.01s	1,158.6	0.000	0.510	1.5320
0.592	0.99a	49.65s	1,158.6	0.000	0.000	1.5527
0.583	1.00a	50.01s	1,158.6	0.000	-0.040	1.5526
0.456	1.08a	55.01s	1,158.6	0.000	-0.594	1.5250
0.325	1.16a	60.01s	1,158.6	0.000	-1.146	1.4490

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

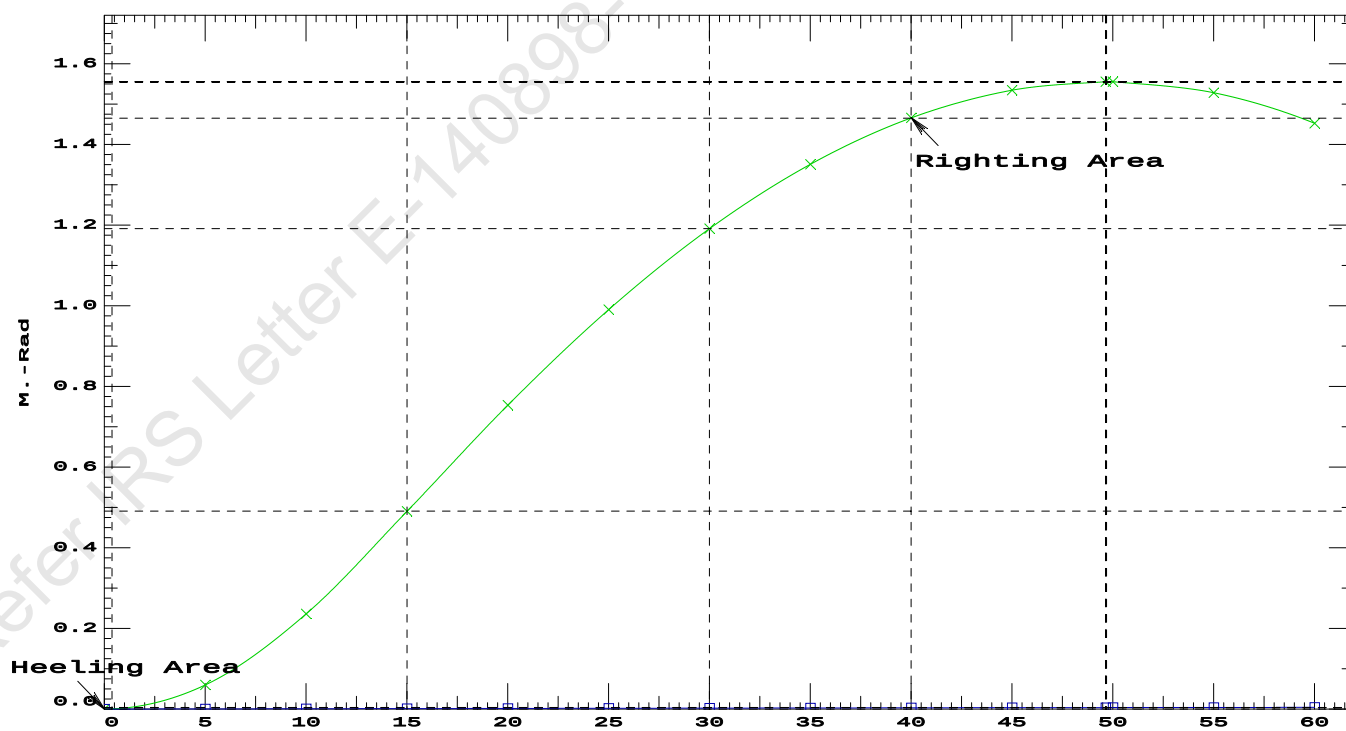
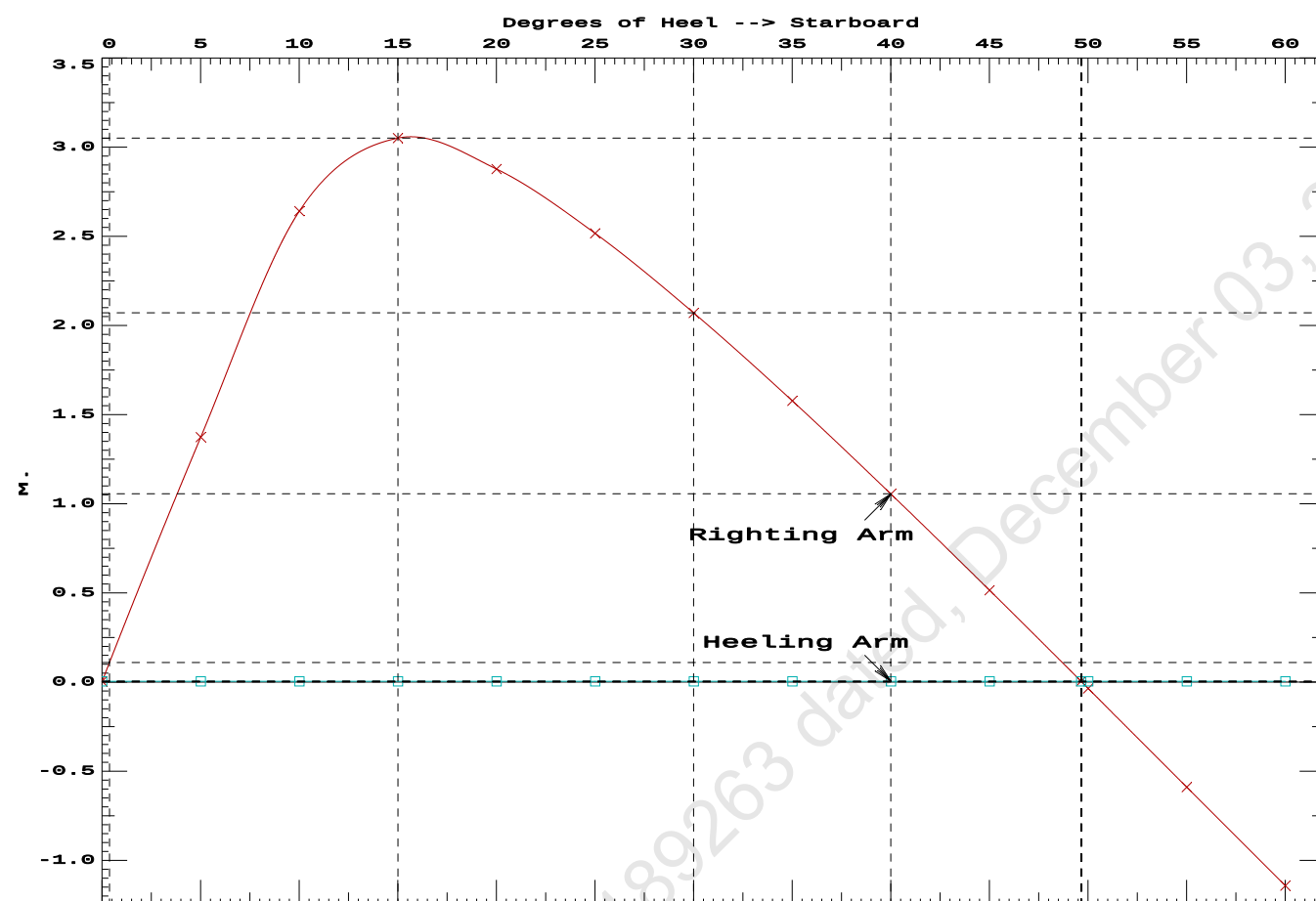
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM	STABILITY CRITERIA FOR CRANE OPERAT	Min/Max	Attained
(1)	Absolute Angle at Equilibrium	< 15.00 deg	0.01 P
(2)	Rise in Abs. RA from Equilibrium to MaxRA	> 66.7 %	70601.3 P
(3)	Abs Ratio from Equilibrium to RAZero or Flood	> 1.667	416.291 P

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GHS 17.90 PONT00N

Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft



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GHS 17.90 PONTON

Condition 10 : Operating Condition of 250T CraneCrane Lifting 36.6T @ 18m Radius Aft

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.326 @ 50.00f, 1.403 @ 25.00f, 1.480 @ 0.00

Trim: Aft 0.155/50.000, Heel: Stbd 0.01 deg.

Part-----	Weight(MT)	LCG	TCG	VCG	
LIGHT SHIP	449.57	25.030f	0.000	3.000	
Crane	250.00	7.500f	0.000	5.200	
Crane Load	36.60	10.500a	0.000	76.977	
Total Fixed----->	736.17	17.310f	0.000	7.425	
Load-----	SpGr	Weight(MT)	LCG	TCG	VCG
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s
Total Tanks----->		422.40	37.208f	0.000	1.596
Total Weight----->		1,158.57	24.565f	0.000	5.300
		Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,158.57	24.551f	0.005s	0.709

Righting Arms:		0.000	0.004s		
External Arms:		0.000	0.004s		
Residual Righting Arms:		0.000	0.000		
Part-----	SpGr	WPA	LCF	TCF	BML
Total Waterplane----->	1.025	857.3	24.747f	0.029s	143.07
		MT/cm			
		8.79			

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.01

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GHS 17.90 PONTON

After hook load drop							
WEIGHT and DISPLACEMENT and WATERPLANE STATUS							
USK draft: 1.482 @ 50.00f, 1.361 @ 25.00f, 1.240 @ 0.00							
Trim: Fwd 0.242/50.000, Heel: 0.00 deg.							
Part-----	Weight(MT)----		LCG-----	TCG-----	VCG-----		
LIGHT SHIP		449.57	25.030f	0.000	3.000		
Crane		250.00	7.500f	0.000	5.200		
Total Fixed----->		699.57	18.765f	0.000	3.786		
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,121.97	25.709f	0.000	2.962	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,121.97	25.720f	0.001s	0.689	

	Righting Arms:			0.000	0.001s		
	External Arms:			0.000	0.001		
	Residual Righting Arms:			0.000	0.000		
Part-----	SpGr-----		WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		856.0	25.286f	0.019s	147.09	20.675*
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			8.77		32.50	144.82	18.402
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

```
limmarg: 704283328.00; t3: 30.0; t: 15.00
limmarg: 263907072.00; t3: 15.0; t: 7.50
limmarg: 71671560.00; t3: 7.5; t: 3.75
limmarg: 18000424.00; t3: 3.8; t: 1.88
limmarg: 4734250.00; t3: 1.9; t: 0.94
limmarg: 1189493.00; t3: 1.0; t: 0.47
limmarg: 366010.47; t3: 0.5; t: 0.23
limmarg: 97000.89; t3: 0.3; t: 0.12
limmarg: 43957.37; t3: 0.2; t: 0.06
limmarg: 26990.98; t3: 0.1; t: 0.03
limmarg: 7067.33; t3: 0.1; t: 0.01
```

Theta3 is 0.07 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.765f TCG = 0.000 VCG = 3.786

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.228	0.28f	0.01s	1,122.0	0.000	-0.003	0.0000
1.228	0.28f	0.00	1,122.0	0.000	0.000	-0.0000
1.228	0.28f	0.07p	1,122.0	0.000	0.026	0.0000
1.213	0.28f	5.00p	1,121.9	0.000	1.662	0.0726
1.149	0.30f	10.00p	1,122.0	0.000	3.193	0.2854
0.951	0.43f	15.00p	1,122.0	0.000	3.852	0.5992
0.795	0.54f	18.61p	1,122.2	0.000	3.924	0.8459
0.733	0.58f	20.00p	1,122.0	0.000	3.914	0.9410
0.508	0.73f	25.00p	1,122.0	0.000	3.775	1.2790
0.276	0.88f	30.00p	1,122.0	0.000	3.539	1.5988

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 699.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 71.673 P

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.765f TCG = 0.000 VCG = 3.786

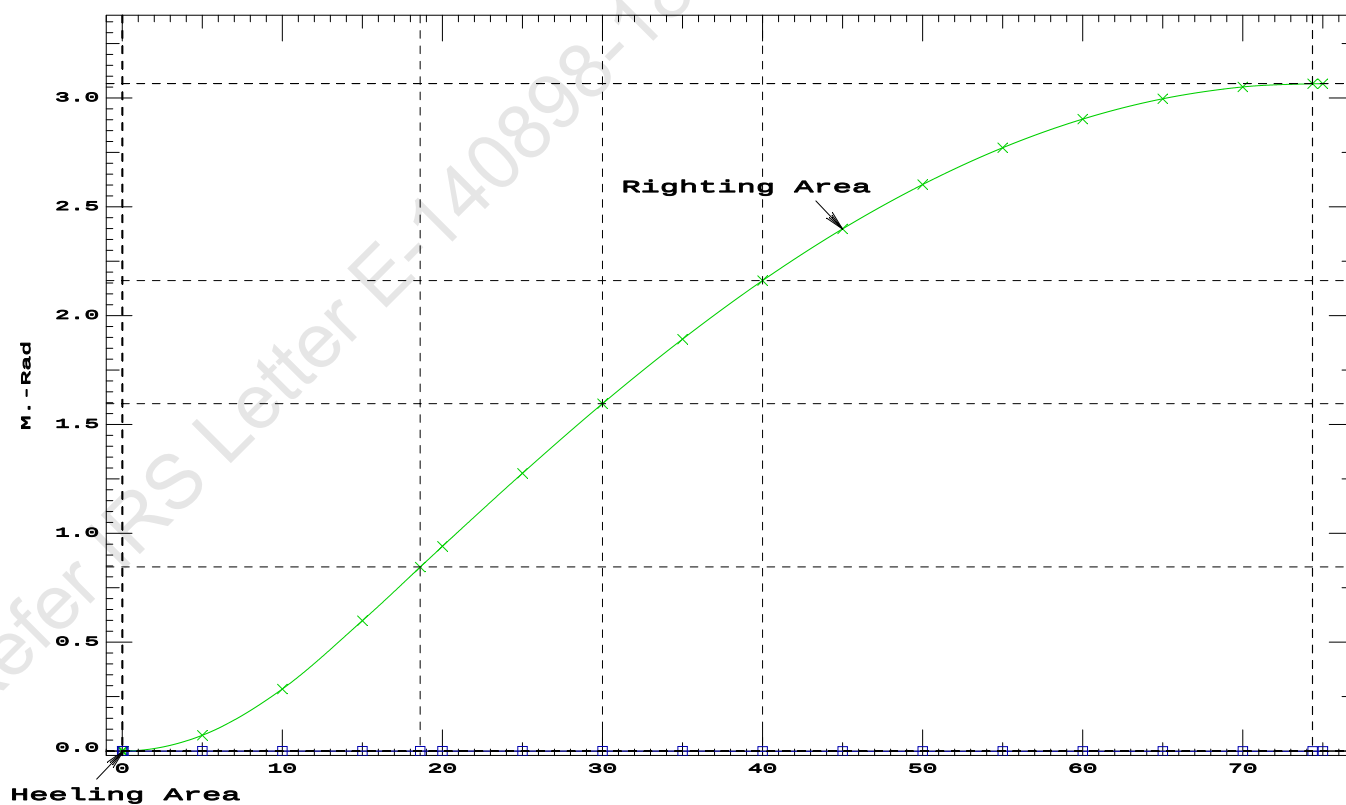
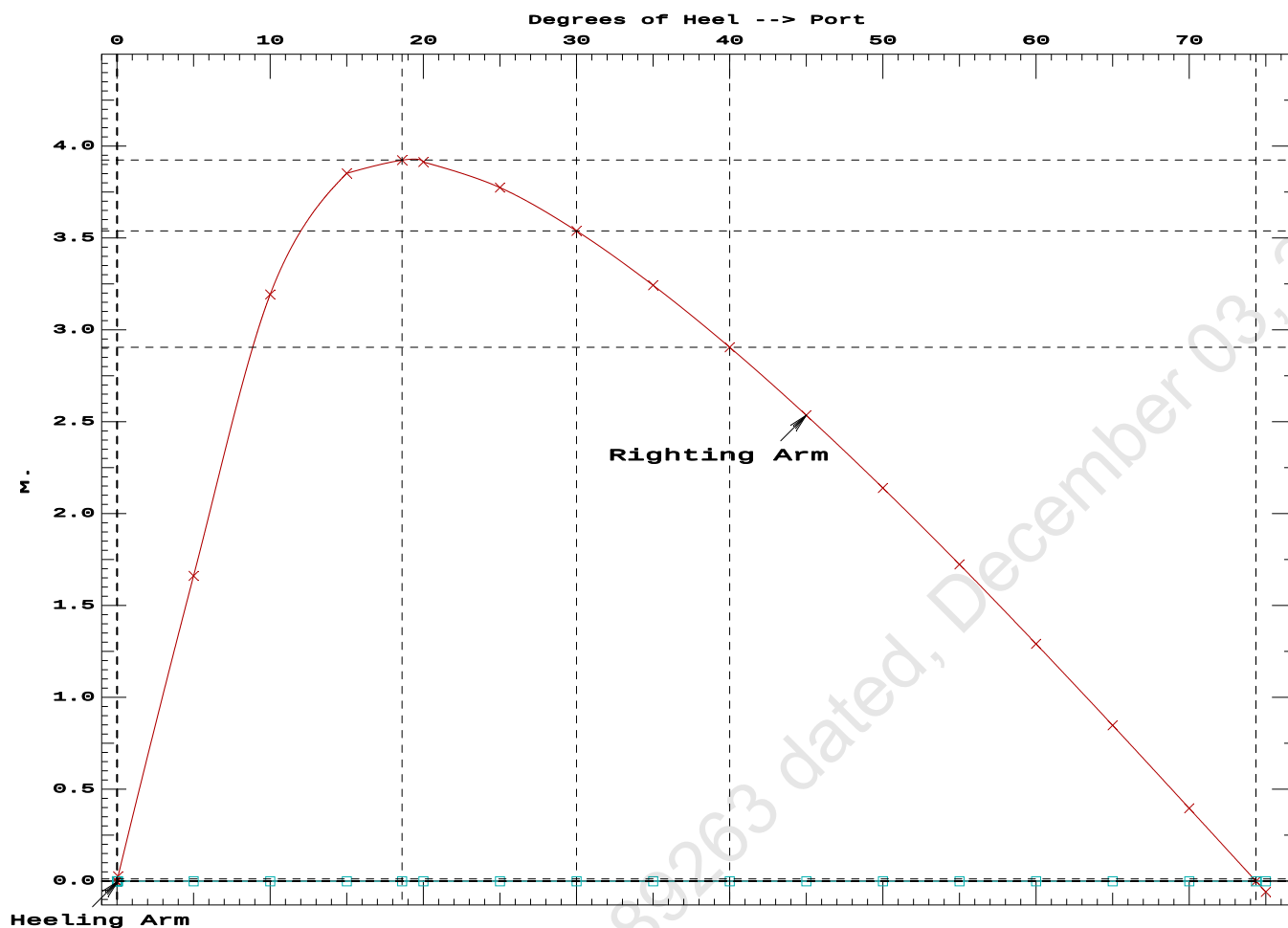
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.228	0.28f	0.01s	1,122.0	0.000	-0.003	0.0000
1.228	0.28f	0.00	1,122.0	0.000	0.000	-0.0000
1.228	0.28f	0.07p	1,122.0	0.000	0.026	0.0000
1.213	0.28f	5.00p	1,121.9	0.000	1.662	0.0726
1.149	0.30f	10.00p	1,122.0	0.000	3.193	0.2854
0.951	0.43f	15.00p	1,122.0	0.000	3.852	0.5992
0.795	0.54f	18.61p	1,122.2	0.000	3.924	0.8459
0.733	0.58f	20.00p	1,122.0	0.000	3.914	0.9410
0.508	0.73f	25.00p	1,122.0	0.000	3.775	1.2790
0.276	0.88f	30.00p	1,122.0	0.000	3.539	1.5988
0.039	1.03f	35.00p	1,122.0	0.000	3.243	1.8952
-0.201	1.18f	40.00p	1,122.0	0.000	2.906	2.1638
-0.442	1.33f	45.00p	1,122.0	0.000	2.536	2.4014
-0.682	1.47f	50.00p	1,122.0	0.000	2.140	2.6056
-0.917	1.60f	55.00p	1,122.0	0.000	1.724	2.7744
-1.147	1.72f	60.00p	1,122.0	0.000	1.292	2.9061
-1.366	1.83f	65.00p	1,122.0	0.000	0.848	2.9995
-1.574	1.91f	70.00p	1,122.0	0.000	0.397	3.0539
-1.744	1.98f	74.36p	1,122.0	0.000	0.000	3.0691
-1.768	1.98f	75.00p	1,122.0	0.000	-0.058	3.0688

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 699.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 0.1 deg to RAZero > 20.00 deg 74.29 P

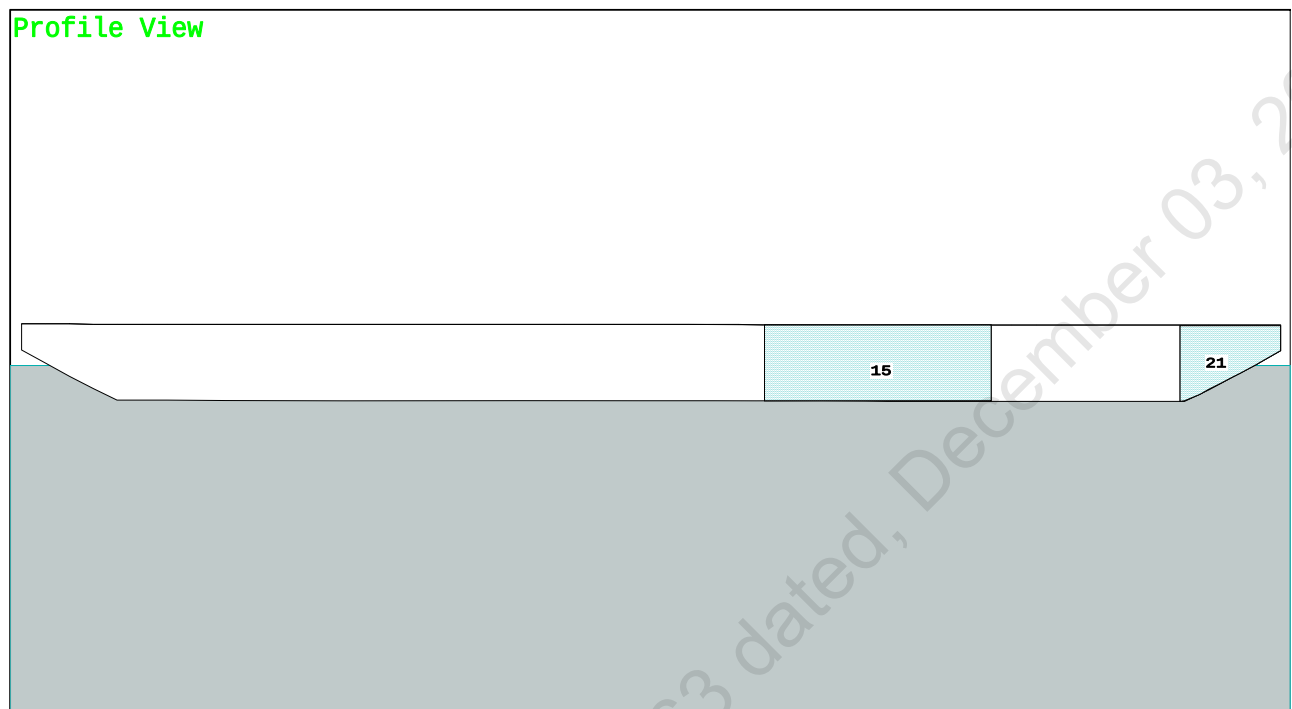
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PONTOON

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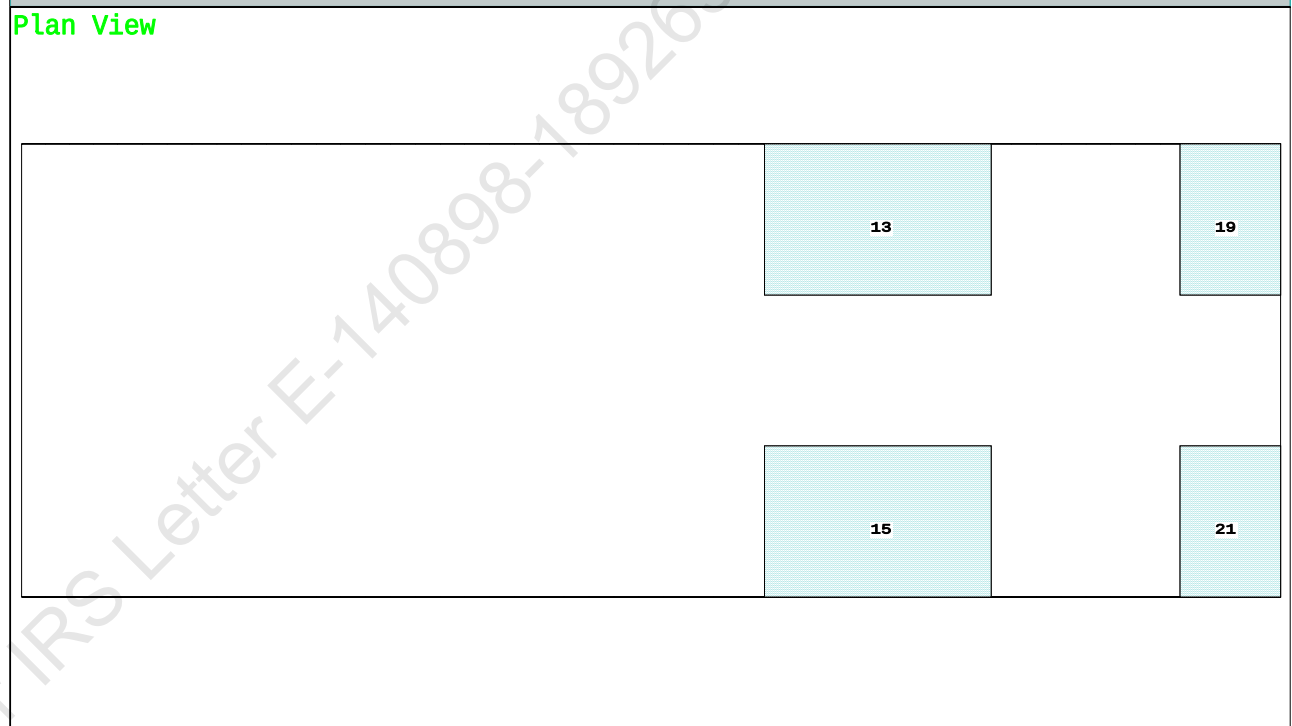
Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

Condition Graphic - Draft: 1.427 @ 50.000f, 1.379 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

13 05_WBTK.P.100% SALT WATER Intact

15 05_WBTK.S.100% SALT WATER Intact
19 07_WBTK.P.100% SALT WATER Intact

21 07_WBTK.S.100% SALT WATER Intact

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PONTOON

Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.427 @ 50.00f, 1.403 @ 25.00f, 1.379 @ 0.00

Trim: Fwd 0.048/50.000, Heel: zero

Part-----			Weight(MT)-----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			449.57	25.030f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	7.500f	0.000	76.977	
Total Fixed----->			736.17	18.205f	0.000	7.425	
	Load-----	SpGr-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,158.57	25.134f	0.000	5.300	
			Displ(MT)-----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,158.53	25.139f	0.000	0.709	

	Righting Arms:			0.001f	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	861.9	25.046f	0.000	145.40	20.211*
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			8.83		32.63	140.81	15.620
Distances in METERS.-----				Moments in m.-MT			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

250.00 MT Crane

36.60 MT Crane Load

DRAFT AT FP (M) = 1.427
DRAFT AT MS (M) = 1.403
DRAFT AT AP (M) = 1.379
DRAFT AT FWD MARK (M) = 1.423
DRAFT AT AFT MARK (M) = 1.383

HYDROSTATIC PROPERTIES

Trim: Fwd 0.048/50.000, No Heel, VCG = 5.300

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)-----	LCB-----	VCB-----	cm-----	LCF-----	cm trim-----	GML-----	GMT
1.403	1,158.53	25.139f	0.709	8.83	25.046f	32.63	140.81	15.620
Distances in METERS.-----				Specific Gravity = 1.025.-----				
				Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

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GHS 17.90

PONTOON

Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.427 @ 50.00f, 1.403 @ 25.00f, 1.379 @ 0.00

Trim: Fwd 0.048/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	80.3	0.815	1.502	0.05500	6.63
CRANE	57.0	4.123	4.810	0.05500	15.08
Total wind heeling moment to starboard-->					21.71
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.134f TCG = 0.000 VCG = 5.300

Free Surface Adjustment: 0.406

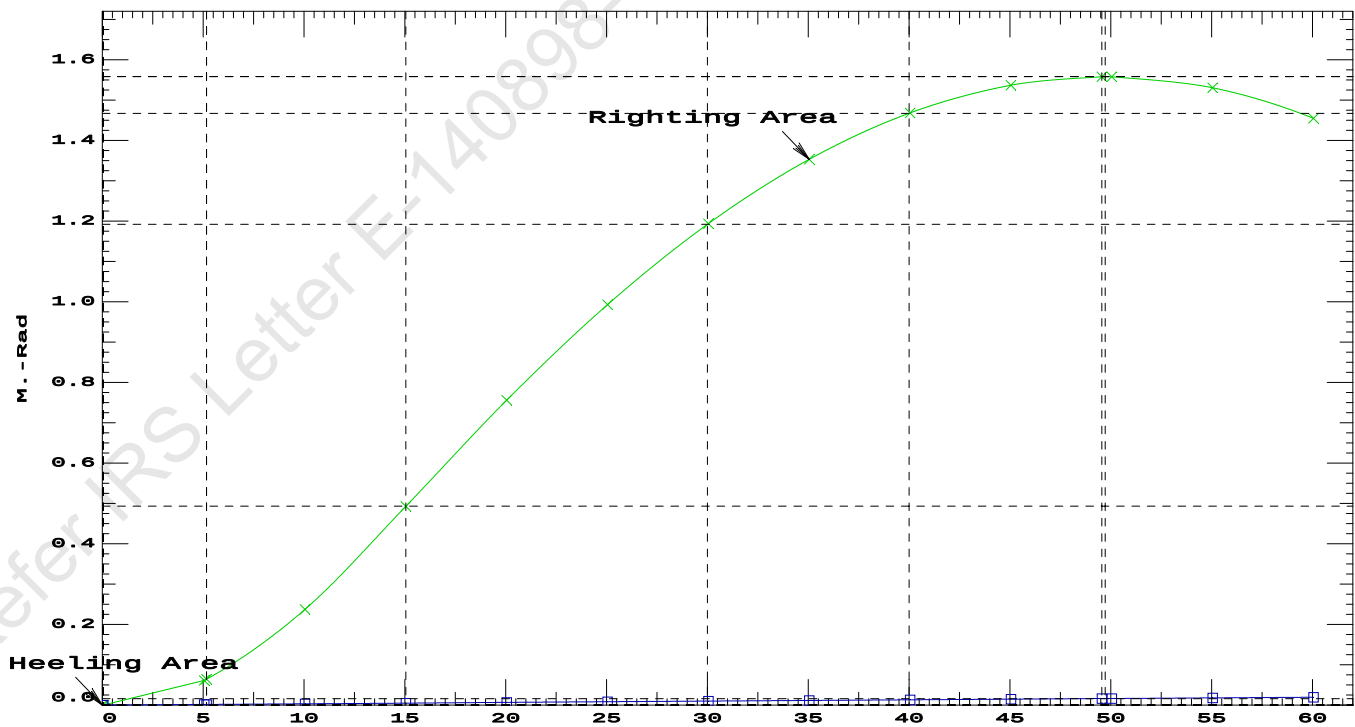
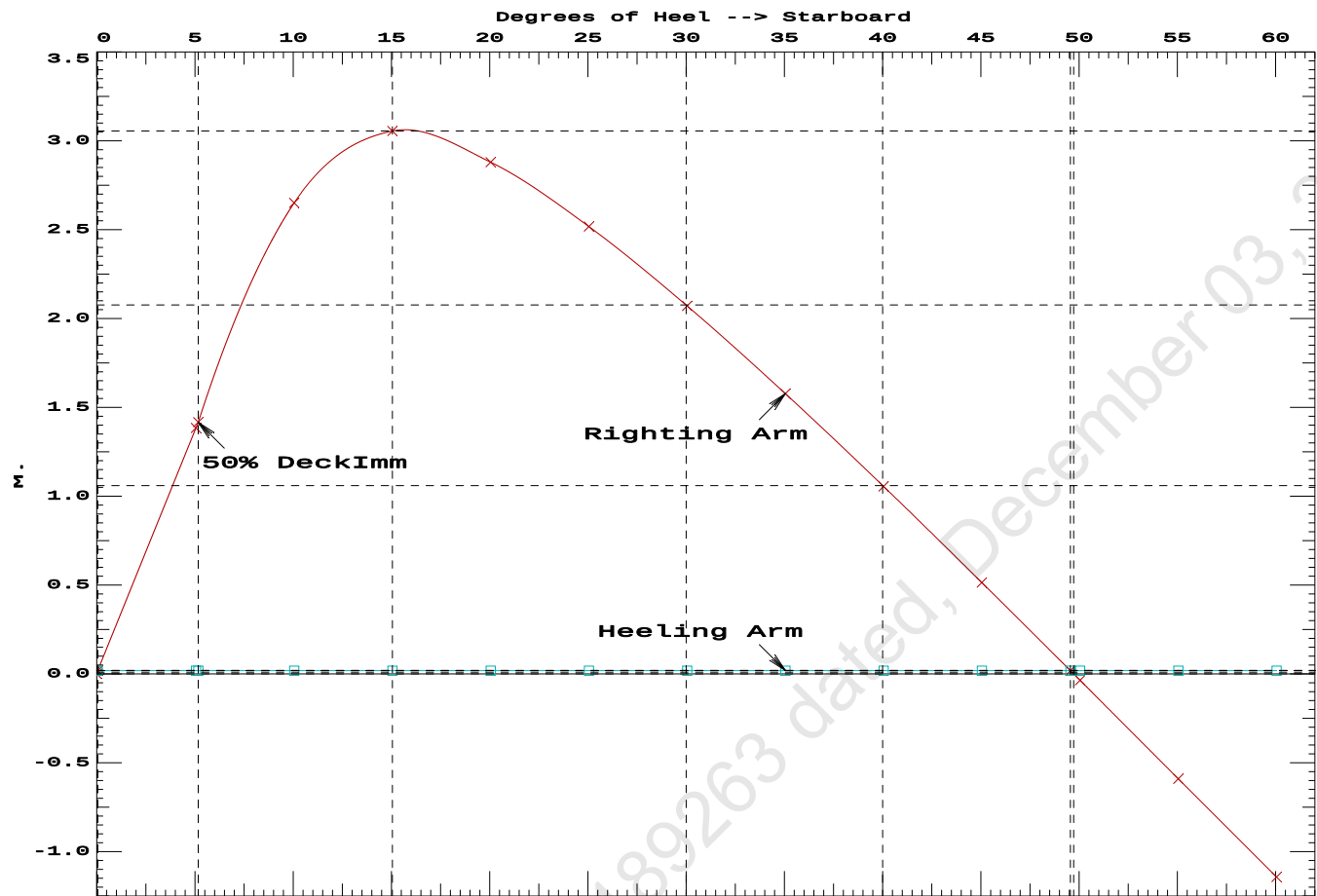
Adjusted CG: LCG = 25.133f TCG = 0.000 VCG = 5.706

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	in Heel	Res. 50% DeckImm Area	Angle
1.367	0.05f	0.00	1,158.5	0.000	-0.018	0.0000	5.166
1.367	0.05f	0.05s	1,158.6	0.000	0.000	-0.0000	5.118
1.352	0.05f	5.05s	1,158.6	0.000	1.366	0.0596	0.118
1.352	0.05f	5.17s	1,158.6	0.000	1.398	0.0624	0.000
1.300	0.06f	10.05s	1,158.6	0.000	2.632	0.2348	-4.882
1.166	0.09f	15.05s	1,158.6	0.000	3.037	0.4896	-9.882
1.020	0.12f	20.05s	1,158.6	0.000	2.861	0.7511	-14.882
0.866	0.15f	25.05s	1,158.6	0.000	2.499	0.9863	-19.882
0.705	0.18f	30.05s	1,158.6	0.000	2.052	1.1855	-24.882
0.538	0.21f	35.05s	1,158.6	0.000	1.559	1.3434	-29.882
0.366	0.25f	40.05s	1,158.6	0.000	1.036	1.4569	-34.882
0.191	0.28f	45.05s	1,158.6	0.000	0.496	1.5238	-39.882
0.031	0.30f	49.55s	1,158.6	0.000	0.000	1.5434	-44.386
0.014	0.31f	50.05s	1,158.6	0.000	-0.055	1.5431	-44.882
-0.163	0.33f	55.05s	1,158.6	0.000	-0.609	1.5142	-49.882
-0.338	0.36f	60.05s	1,158.6	0.000	-1.162	1.4369	-54.882
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.							

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.71 (constant)

LIM	IS CODE 2008 PART B CHAPTER-2 SEC.2	Min/Max	Attained
(1)	Residual Area from abs 0 deg to MaxRA	> 0.0800 m.-Rad	0.4896 P
(2)	Angle from Equilibrium to RAZero or Flood	> 20.00 deg	49.50 P
(3)	Angle from Equilibrium to 50% Dk Imm Angle	> 0.00 deg	5.12 P

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GHS 17.90Cybermarine Technologies PTE. Ltd.
PONT00NCondition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

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GHS 17.90

PONTOON

Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.134f TCG = 0.000 VCG = 5.300
Free Surface Adjustment: 0.406
Adjusted CG: LCG = 25.133f TCG = 0.014p VCG = 5.705

Origin Depth---	Degrees of Trim----	Displacement Heel----	Weight(MT)---	Residual Arms in Trim--	in Heel---	Res. Area
1.364	0.06f	2.02s	1,158.6	0.000	0.000	0.0000
1.339	0.05f	7.02s	1,158.7	0.000	1.359	0.0593
1.249	0.07f	12.02s	1,158.6	0.000	2.384	0.2250
1.159	0.09f	15.27s	1,158.7	0.000	2.499	0.3657
1.109	0.10f	17.02s	1,158.6	0.000	2.466	0.4414
0.960	0.13f	22.02s	1,158.6	0.000	2.195	0.6488
0.803	0.16f	27.02s	1,158.6	0.000	1.793	0.8238
0.640	0.19f	32.02s	1,158.6	0.000	1.324	0.9602
0.470	0.23f	37.02s	1,158.6	0.000	0.817	1.0539
0.297	0.26f	42.02s	1,158.6	0.000	0.285	1.1022
0.205	0.27f	44.64s	1,158.6	0.000	0.000	1.1088
0.121	0.29f	47.02s	1,158.6	0.000	-0.261	1.1034
-0.056	0.32f	52.02s	1,158.6	0.000	-0.814	1.0565
-0.232	0.34f	57.02s	1,158.6	0.000	-1.370	0.9612
-0.407	0.36f	62.02s	1,158.6	0.000	-1.921	0.8176

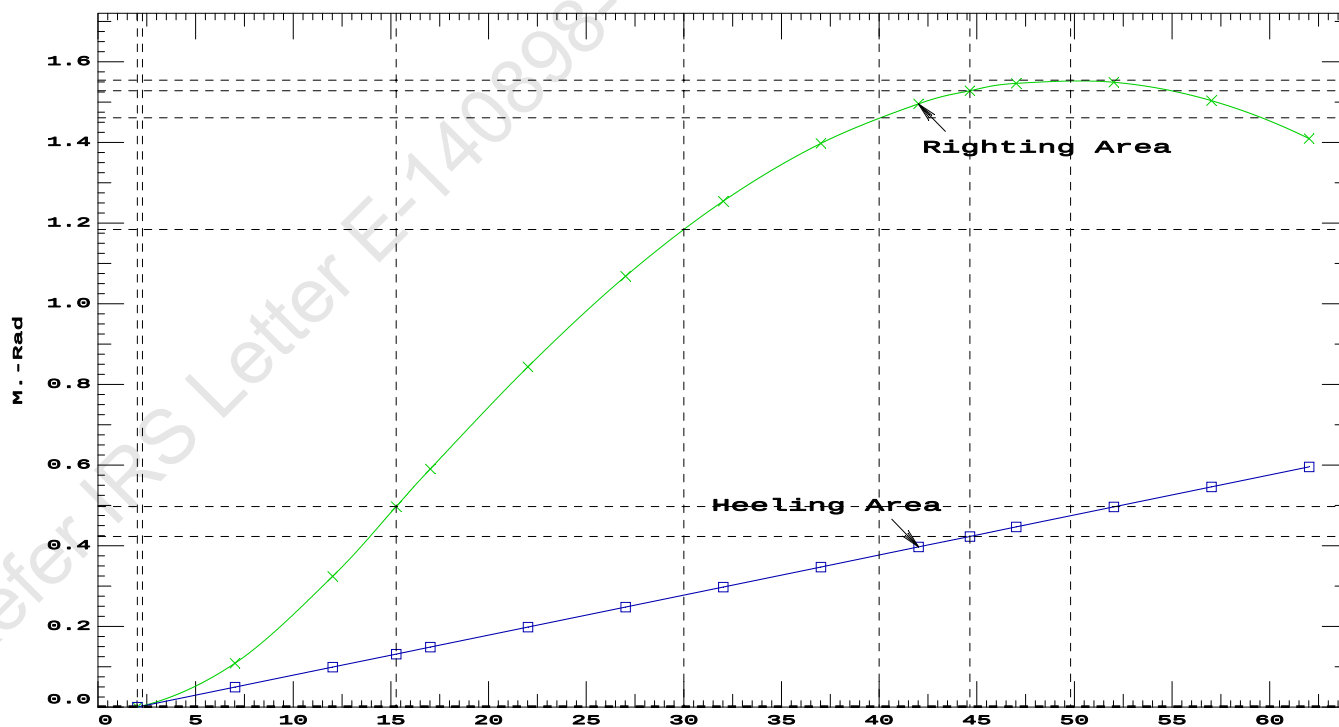
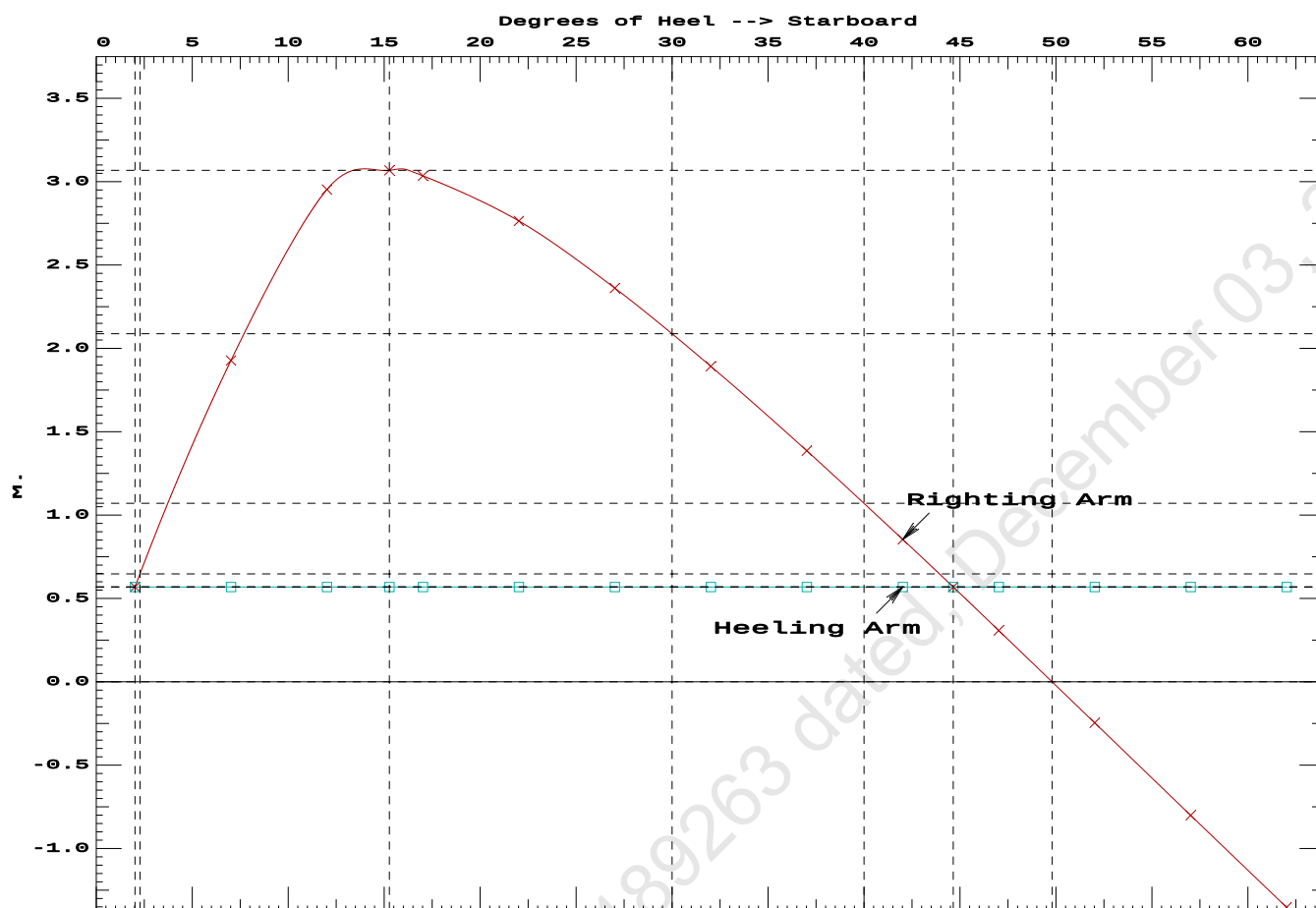
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 658.80

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	2.02 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	439.5 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	3.621 P

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PONT00NCondition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

PONTOON

Crane Lifting 36.6T @ 18m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

Trim: Fwd 0.048/50.000, Heel: Stbd 2.02 deg.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 2.02

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After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.482 @ 50.00f, 1.361 @ 25.00f, 1.240 @ 0.00
Trim: Fwd 0.242/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----			
LIGHT SHIP	449.57	25.030f	0.000	3.000			
Crane	250.00	7.500f	0.000	5.200			
Total Fixed----->	699.57	18.765f	0.000	3.786			
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,121.97	25.709f	0.000	2.962	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL	1.025		1,121.97	25.720f	0.001s	0.689	

Righting Arms:			0.000	0.001s			
External Arms:			0.000	0.001			
Residual Righting Arms:			0.000	0.000			
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----	
Total Waterplane---->	1.025	856.0	25.286f	0.019s	147.09	20.675*	
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----		
		8.77	32.50	144.82	18.402		
Distances in METERS.-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 13398.21; t3: 30.00; t: 15.00
limmarg: 4958.21; t3: 15.00; t: 7.35
limmarg: 1328.83; t3: 7.65; t: 3.60
limmarg: 302.27; t3: 4.05; t: 1.76
limmarg: 28.91; t3: 2.29; t: 0.86
limmarg: -49.59; t3: 1.43; t: 0.42
limmarg: -15.78; t3: 1.85; t: 0.21
limmarg: 4.37; t3: 2.06; t: 0.10
limmarg: -5.49; t3: 1.96; t: 0.05
limmarg: -0.62; t3: 2.01; t: 0.02
limmarg: 1.36; t3: 2.03; t: 0.01

Theta3 is 2.03 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.765f TCG = 0.000 VCG = 3.786

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.224	0.28f	2.02s	1,122.0	0.000	-0.672	0.0000
1.228	0.28f	0.00	1,122.0	0.000	0.000	-0.0118
1.224	0.28f	2.03p	1,122.0	0.000	0.678	0.0002
1.213	0.28f	5.00p	1,121.9	0.000	1.662	0.0608
1.149	0.30f	10.00p	1,122.0	0.000	3.193	0.2734
0.951	0.43f	15.00p	1,122.0	0.000	3.852	0.5872
0.795	0.54f	18.61p	1,122.2	0.000	3.924	0.8339
0.733	0.58f	20.00p	1,122.0	0.000	3.914	0.9291
0.508	0.73f	25.00p	1,122.0	0.000	3.775	1.2670
0.276	0.88f	30.00p	1,122.0	0.000	3.539	1.5868

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 699.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs -2 deg to abs 2 > 1.000 1.014 P

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GHS 17.90 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.765f TCG = 0.000 VCG = 3.786

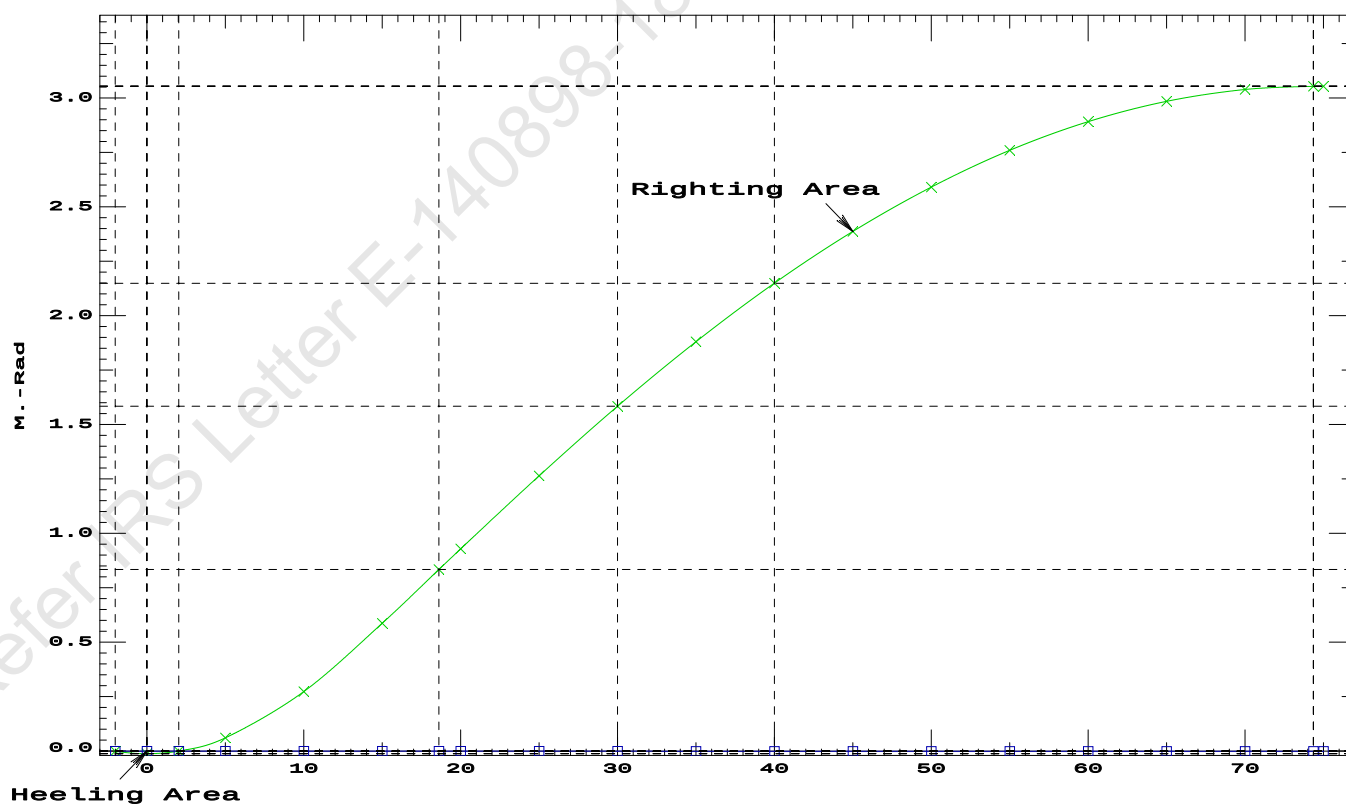
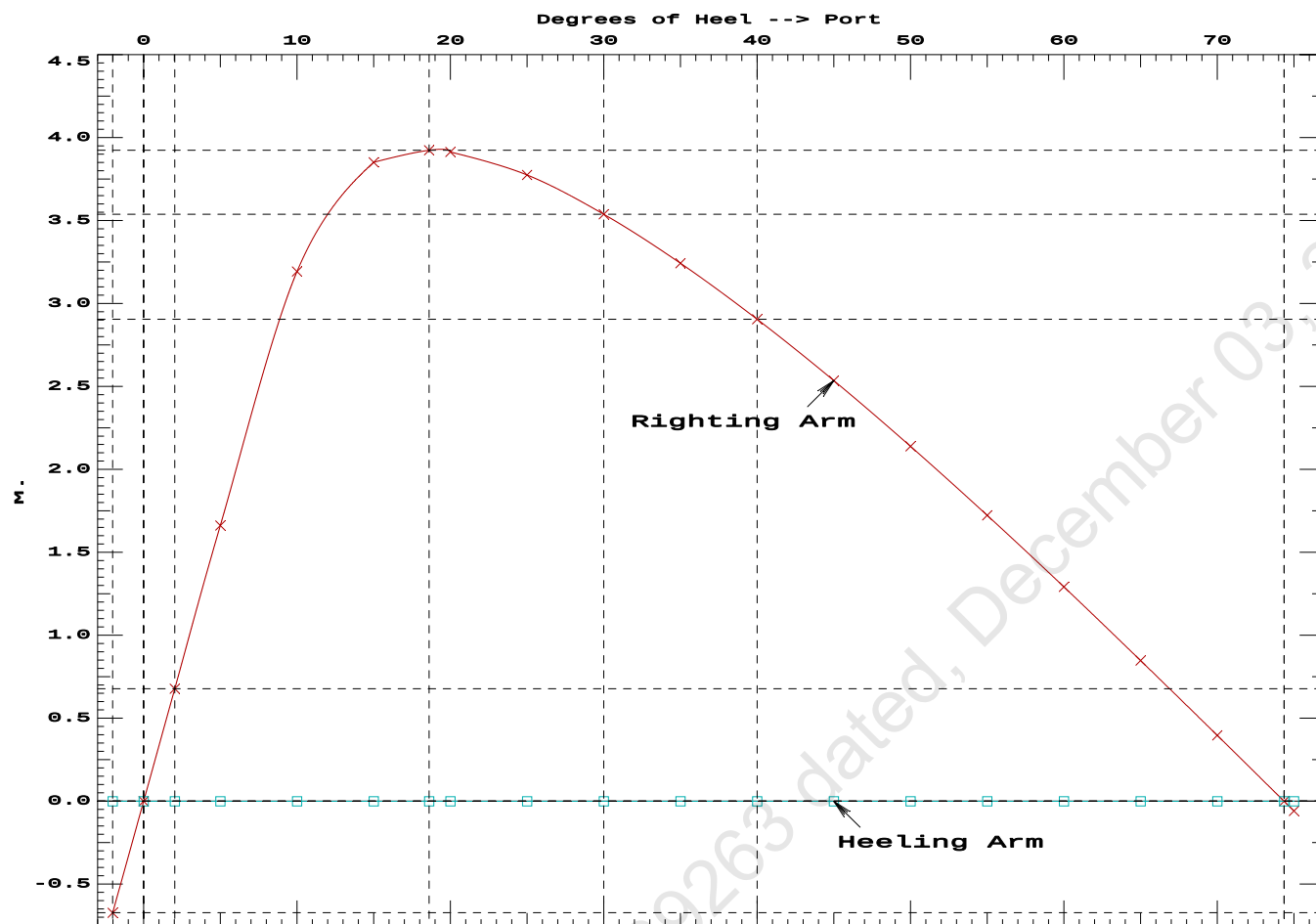
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.224	0.28f	2.02s	1,122.0	0.000	-0.672	0.0000
1.228	0.28f	0.00	1,122.0	0.000	0.000	-0.0118
1.224	0.28f	2.03p	1,122.0	0.000	0.678	0.0002
1.213	0.28f	5.00p	1,121.9	0.000	1.662	0.0608
1.149	0.30f	10.00p	1,122.0	0.000	3.193	0.2734
0.951	0.43f	15.00p	1,122.0	0.000	3.852	0.5872
0.795	0.54f	18.61p	1,122.2	0.000	3.924	0.8339
0.733	0.58f	20.00p	1,122.0	0.000	3.914	0.9291
0.508	0.73f	25.00p	1,122.0	0.000	3.775	1.2670
0.276	0.88f	30.00p	1,122.0	0.000	3.539	1.5868
0.039	1.03f	35.00p	1,122.0	0.000	3.243	1.8832
-0.201	1.18f	40.00p	1,122.0	0.000	2.906	2.1518
-0.442	1.33f	45.00p	1,122.0	0.000	2.536	2.3895
-0.682	1.47f	50.00p	1,122.0	0.000	2.140	2.5937
-0.917	1.60f	55.00p	1,122.0	0.000	1.724	2.7624
-1.147	1.72f	60.00p	1,122.0	0.000	1.292	2.8941
-1.366	1.83f	65.00p	1,122.0	0.000	0.848	2.9876
-1.574	1.91f	70.00p	1,122.0	0.000	0.397	3.0420
-1.744	1.98f	74.36p	1,122.0	0.000	0.000	3.0571
-1.768	1.98f	75.00p	1,122.0	0.000	-0.058	3.0568

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 699.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs -2 deg to RAZero > 1.000 259.119 P
(2) Angle from abs 2 deg to RAZero > 20.00 deg 72.33 P

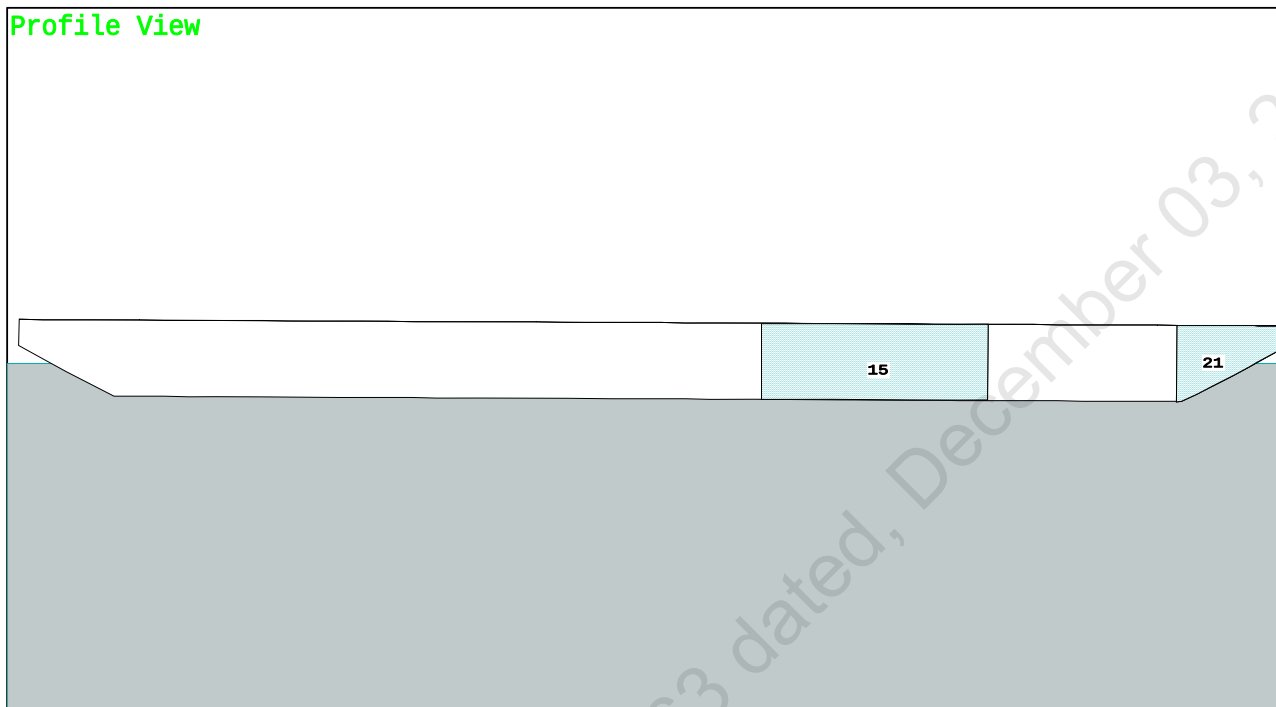
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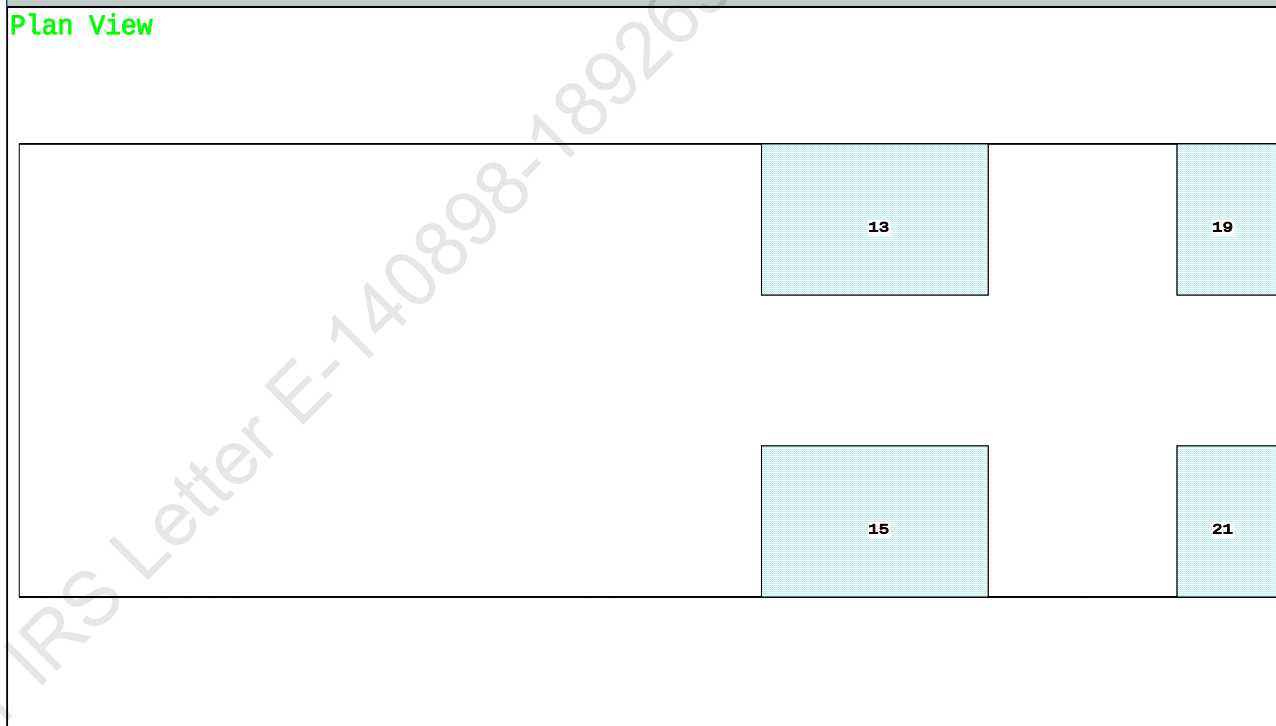
Condition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd

Condition Graphic - Draft: 1.527 @ 50.000f, 1.278 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

13 05_WBTK.P.100% SALT WATER Intact	15 05_WBTK.S.100% SALT WATER Intact	21 07_WBTK.S.100% SALT WATER Intact
	19 07_WBTK.P.100% SALT WATER Intact	

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PONTOON

Condition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.527 @ 50.00f, 1.403 @ 25.00f, 1.278 @ 0.00

Trim: Fwd 0.250/50.000, Heel: zero

Part-----			Weight(MT)-----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			449.57	25.030f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	25.500f	0.000	76.977	
Total Fixed----->			736.17	19.100f	0.000	7.425	
	Load-----	SpGr-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,158.57	25.702f	0.000	5.300	
			Displ(MT)-----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,158.52	25.725f	0.000	0.710	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	861.3	25.218f	0.000	145.11	20.171*
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			8.83		32.56	140.52	15.581
Distances in METERS.-----				Moments in m.-MT-----			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

250.00 MT Crane

36.60 MT Crane Load

DRAFT AT FP (M) = 1.527
DRAFT AT MS (M) = 1.403
DRAFT AT AP (M) = 1.278
DRAFT AT FWD MARK (M) = 1.507
DRAFT AT AFT MARK (M) = 1.298

HYDROSTATIC PROPERTIES

Trim: Fwd 0.250/50.000, No Heel, VCG = 5.300

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)-----	LCB-----	VCB-----	cm-----
1.404	1,158.52	25.725f	0.710	8.83
				25.218f
				32.56
				140.52
				15.581

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

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PONTOON

Condition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.527 @ 50.00f, 1.403 @ 25.00f, 1.278 @ 0.00

Trim: Fwd 0.250/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	80.2	0.817	1.505	0.05500	6.64
CRANE	57.0	4.184	4.871	0.05500	15.27
Total wind heeling moment to starboard-->					21.91
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.702f TCG = 0.000 VCG = 5.300

Free Surface Adjustment: 0.406

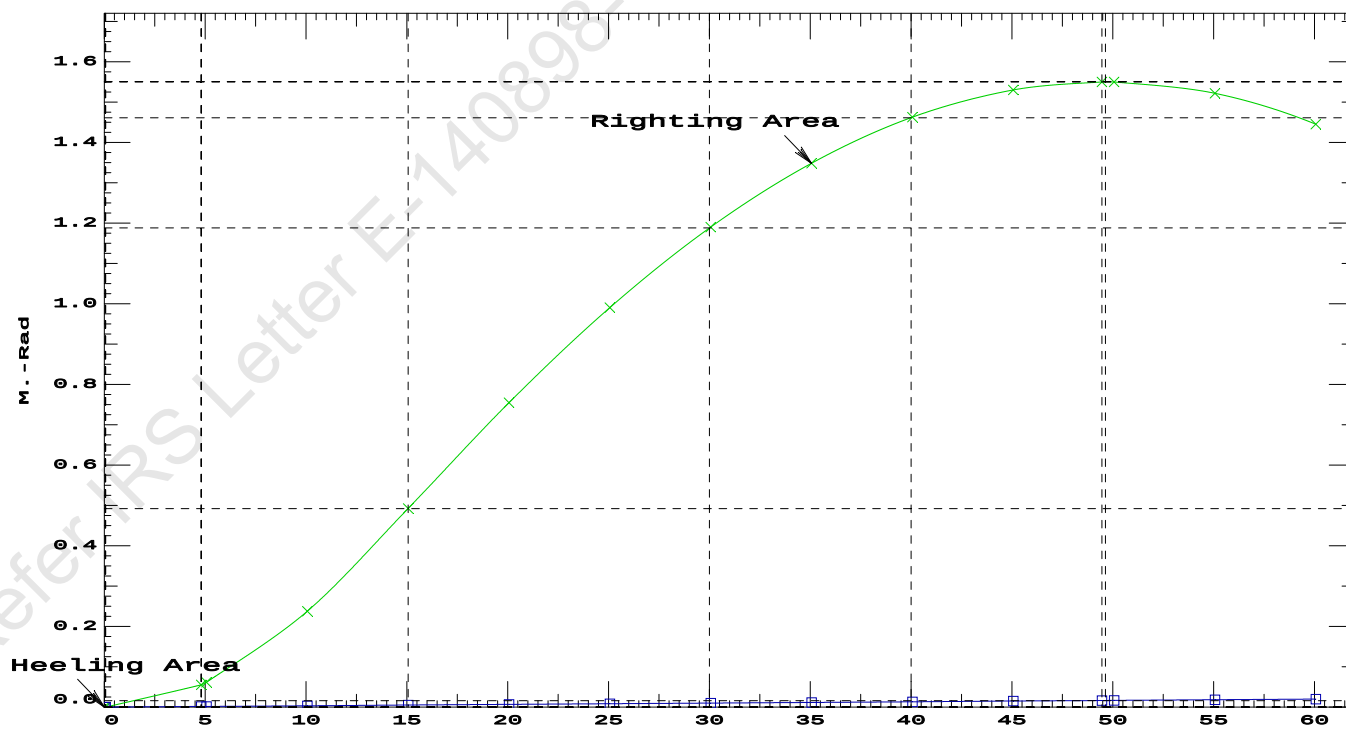
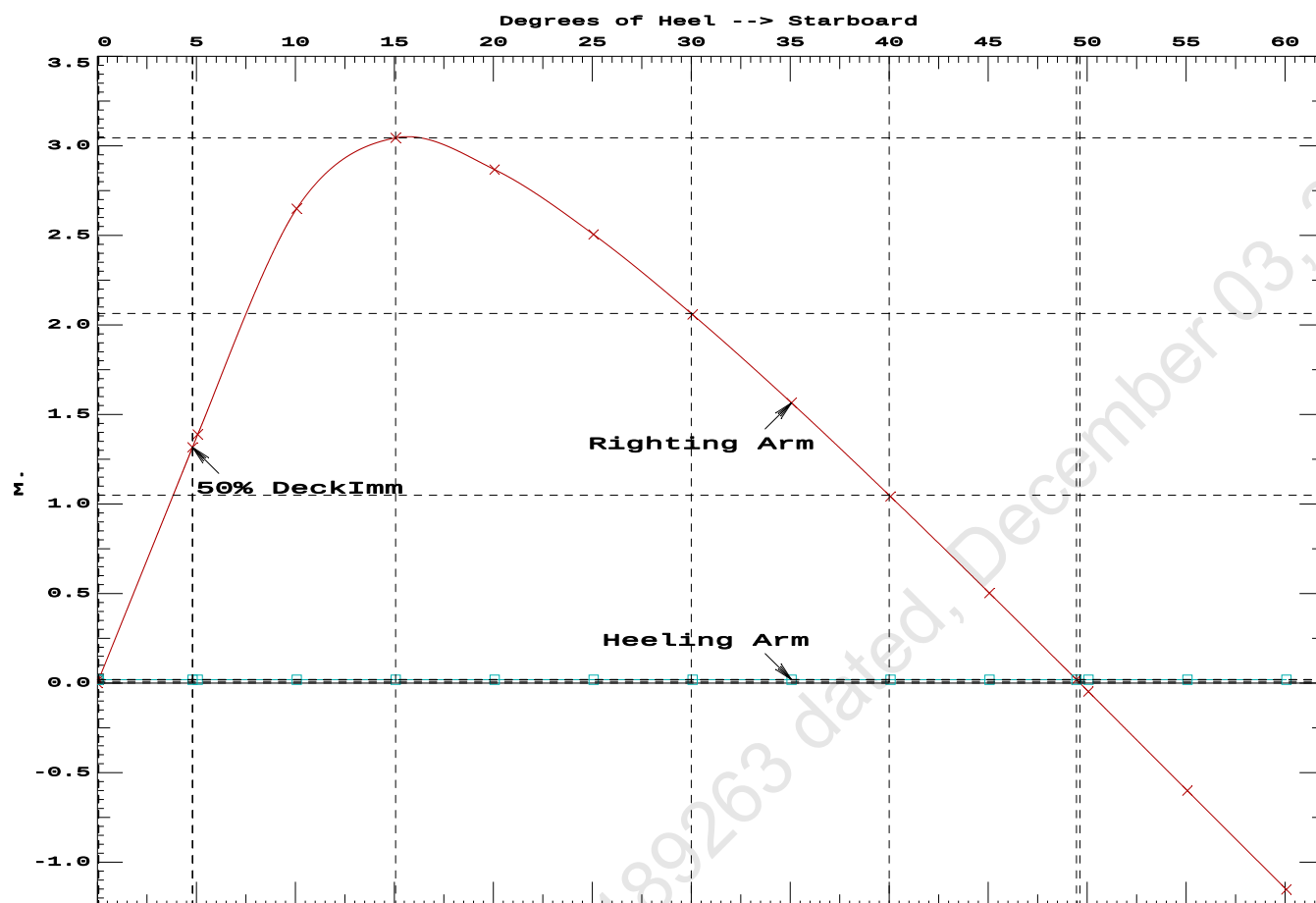
Adjusted CG: LCG = 25.700f TCG = 0.000 VCG = 5.706

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. 50% DeckImm Area	Angle
1.266	0.29f	0.00	1,158.5	0.000	-0.018	4.800
1.265	0.29f	0.06s	1,158.6	0.000	0.000	4.738
1.252	0.29f	4.80s	1,158.6	0.000	1.298	0.000
1.250	0.29f	5.06s	1,158.6	0.000	1.369	-0.262
1.189	0.31f	10.06s	1,158.6	0.000	2.630	-5.262
1.003	0.46f	15.06s	1,158.6	0.000	3.026	-10.262
0.800	0.62f	20.06s	1,158.6	0.000	2.849	-15.262
0.588	0.79f	25.06s	1,158.6	0.000	2.486	-20.262
0.367	0.95f	30.06s	1,158.6	0.000	2.039	-25.262
0.140	1.12f	35.06s	1,158.6	0.000	1.546	-30.262
-0.091	1.29f	40.06s	1,158.6	0.000	1.023	-35.262
-0.324	1.45f	45.06s	1,158.6	0.000	0.483	-40.262
-0.527	1.59f	49.46s	1,158.6	0.000	0.000	-44.660
-0.555	1.61f	50.06s	1,158.6	0.000	-0.067	-45.262
-0.781	1.75f	55.06s	1,158.6	0.000	-0.620	-50.262
-1.000	1.87f	60.06s	1,158.6	0.000	-1.171	-55.262
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.						

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.91 (constant)

LIM	IS CODE 2008 PART B CHAPTER-2 SEC.2	Min/Max	Attained
(1)	Residual Area from abs 0 deg to MaxRA	> 0.0800 m.-Rad	0.4891 P
(2)	Angle from Equilibrium to RAZero or Flood	> 20.00 deg	49.40 P
(3)	Angle from Equilibrium to 50% Dk Imm Angle	> 0.00 deg	4.74 P

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GHS 17.90Cybermarine Technologies PTE. Ltd.
PONT00NCondition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd

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PONTOON

Condition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.702f TCG = 0.000 VCG = 5.300
Free Surface Adjustment: 0.406
Adjusted CG: LCG = 25.700f TCG = 0.000 VCG = 5.706

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.266	0.29f	0.01s	1,158.6	0.000	0.000	0.0000
1.251	0.29f	5.01s	1,158.5	0.000	1.370	0.0598
1.190	0.31f	10.01s	1,158.6	0.000	2.634	0.2352
1.005	0.46f	15.01s	1,158.6	0.000	3.040	0.4902
0.802	0.62f	20.01s	1,158.6	0.000	2.866	0.7521
0.590	0.78f	25.01s	1,158.6	0.000	2.505	0.9878
0.369	0.95f	30.01s	1,158.6	0.000	2.058	1.1875
0.142	1.12f	35.01s	1,158.6	0.000	1.565	1.3460
-0.089	1.29f	40.01s	1,158.6	0.000	1.043	1.4600
-0.322	1.45f	45.01s	1,158.6	0.000	0.503	1.5276
-0.533	1.59f	49.59s	1,158.6	0.000	0.000	1.5478
-0.553	1.61f	50.01s	1,158.6	0.000	-0.047	1.5476
-0.779	1.75f	55.01s	1,158.6	0.000	-0.600	1.5194
-0.998	1.87f	60.01s	1,158.6	0.000	-1.151	1.4430

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

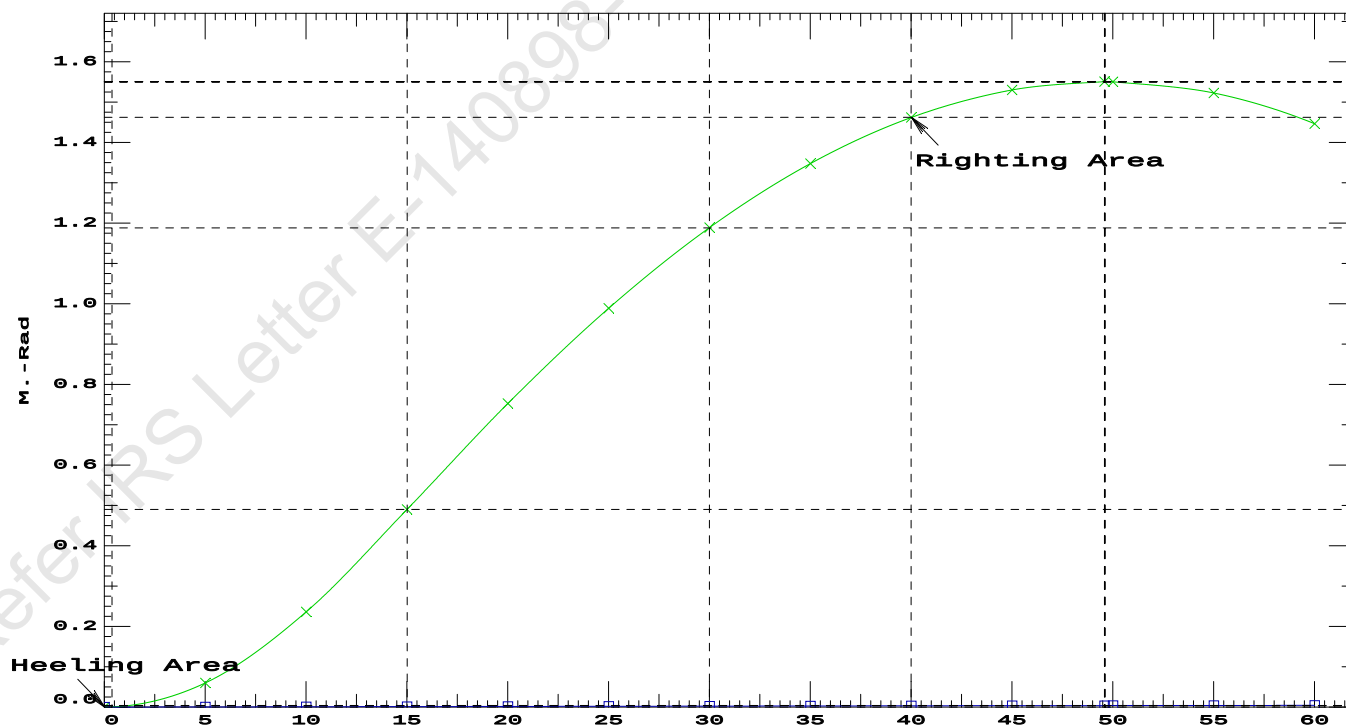
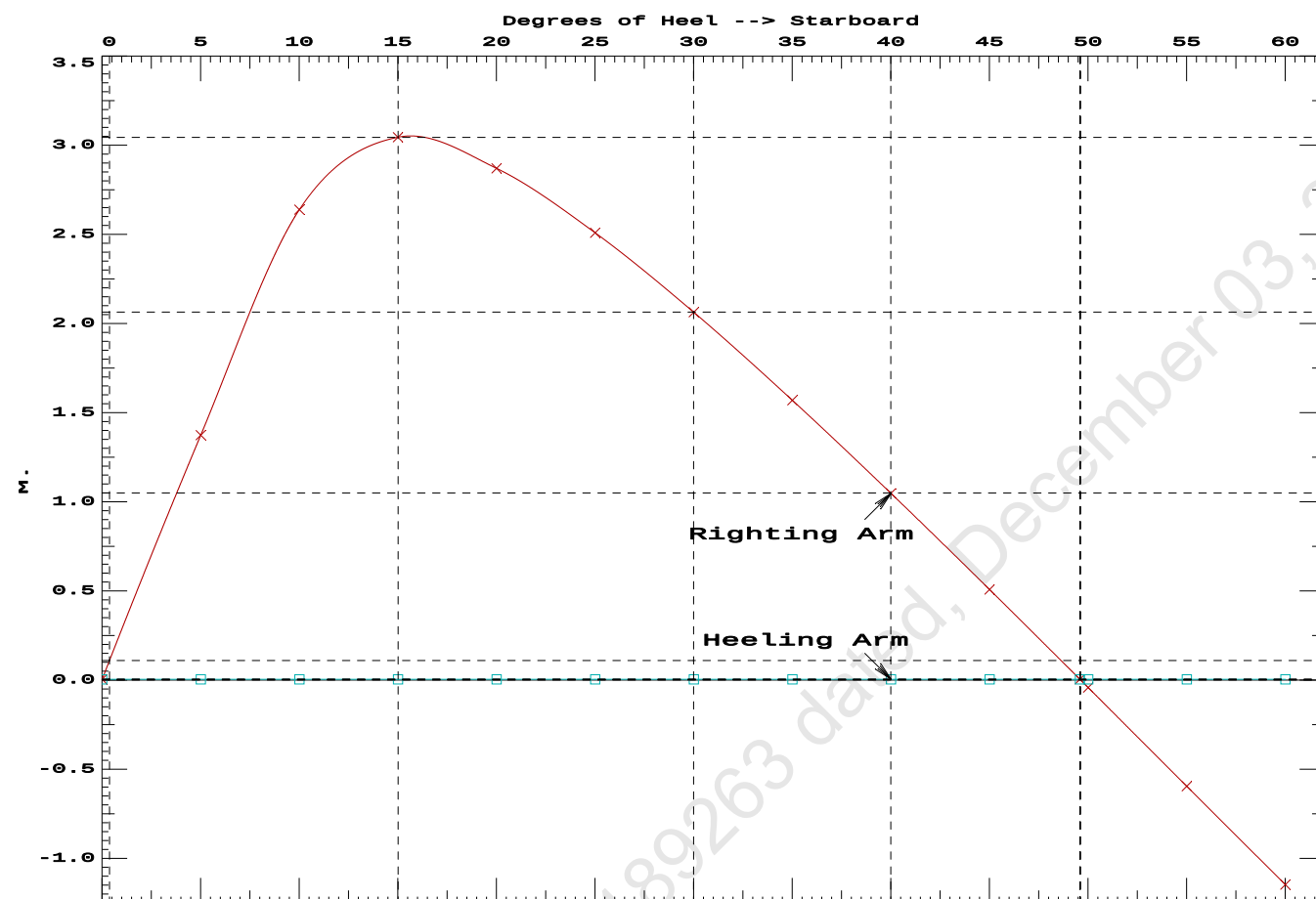
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM	STABILITY CRITERIA FOR CRANE OPERAT	Min/Max	Attained
(1)	Absolute Angle at Equilibrium	< 15.00 deg	0.01 P
(2)	Rise in Abs. RA from Equilibrium to MaxRA	> 66.7 %	70443.6 P
(3)	Abs Ratio from Equilibrium to RAZero or Flood	> 1.667	415.464 P

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PONTOON

Condition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd



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Condition 12 : Operating Condition of 250T CraneCrane Lifting 36.6T @ 18m Radius Fwd

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.527 @ 50.00f, 1.403 @ 25.00f, 1.278 @ 0.00

Trim: Fwd 0.250/50.000, Heel: Stbd 0.01 deg.

Part-----	Weight(MT)	LCG	TCG	VCG	
LIGHT SHIP	449.57	25.030f	0.000	3.000	
Crane	250.00	7.500f	0.000	5.200	
Crane Load	36.60	25.500f	0.000	76.977	
Total Fixed----->	736.17	19.100f	0.000	7.425	
Load-----	SpGr	Weight(MT)	LCG	TCG	VCG
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s
Total Tanks----->		422.40	37.208f	0.000	1.596
Total Weight----->		1,158.57	25.702f	0.000	5.300
		Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,158.57	25.725f	0.005s	0.710

Righting Arms:		0.000	0.004s		
External Arms:		0.000	0.004s		
Residual Righting Arms:		0.000	0.000		
Part-----	SpGr	WPA	LCF	TCF	BML
Total Waterplane----->	1.025	858.0	25.307f	0.024s	143.45
		MT/cm			
		8.79			

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.01

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After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.482 @ 50.00f, 1.361 @ 25.00f, 1.240 @ 0.00
Trim: Fwd 0.243/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
LIGHT SHIP	449.57	25.030f	0.000	3.000
Crane	250.00	7.500f	0.000	5.200
Total Fixed----->	699.57	18.765f	0.000	3.786

Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880
Total Tanks----->			422.40	37.208f	0.000	1.596
Total Weight----->			1,121.97	25.709f	0.000	2.962

Displ(MT)----	LCB-----	TCB-----	VCB-----
HULL	1.025	1,121.97	25.720f
		0.001s	0.689

Righting Arms:	0.000	0.001s
External Arms:	0.000	0.001
Residual Righting Arms:	0.000	0.000s

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->	1.025	856.0	25.286f	0.019s	147.09	20.675*
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
		8.77	32.50	144.82	18.402	

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 385304704.00; t3: 30; t: 15.00
limmarg: 144378208.00; t3: 15; t: 7.50
limmarg: 39210836.00; t3: 8; t: 3.75
limmarg: 12638143.00; t3: 4; t: 1.88
limmarg: 3155977.25; t3: 2; t: 0.94
limmarg: 792598.88; t3: 1; t: 0.47
limmarg: 200160.44; t3: 1; t: 0.23
limmarg: 419754.09; t3: 1; t: 0.12
limmarg: 547303.00; t3: 1; t: 0.06
limmarg: 624020.63; t3: 1; t: 0.03
limmarg: 664271.56; t3: 1; t: 0.01

Theta3 is 0.97 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.765f TCG = 0.000 VCG = 3.786

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.228	0.28f	0.01s	1,122.0	0.000	-0.004	0.0000
1.228	0.28f	0.00	1,122.0	0.000	-0.000	-0.0000
1.225	0.28f	0.97p	1,122.0	0.000	0.326	0.0028
1.213	0.28f	5.00p	1,122.0	0.000	1.662	0.0726
1.149	0.30f	10.00p	1,122.0	0.000	3.193	0.2854
0.951	0.43f	15.00p	1,122.0	0.000	3.852	0.5992
0.795	0.54f	18.61p	1,122.2	0.000	3.924	0.8459
0.733	0.58f	20.00p	1,122.0	0.000	3.914	0.9410
0.508	0.73f	25.00p	1,122.0	0.000	3.775	1.2789
0.276	0.88f	30.00p	1,122.0	0.000	3.539	1.5988

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 699.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 1 > 1.000 6643.72 P

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.765f TCG = 0.000 VCG = 3.786

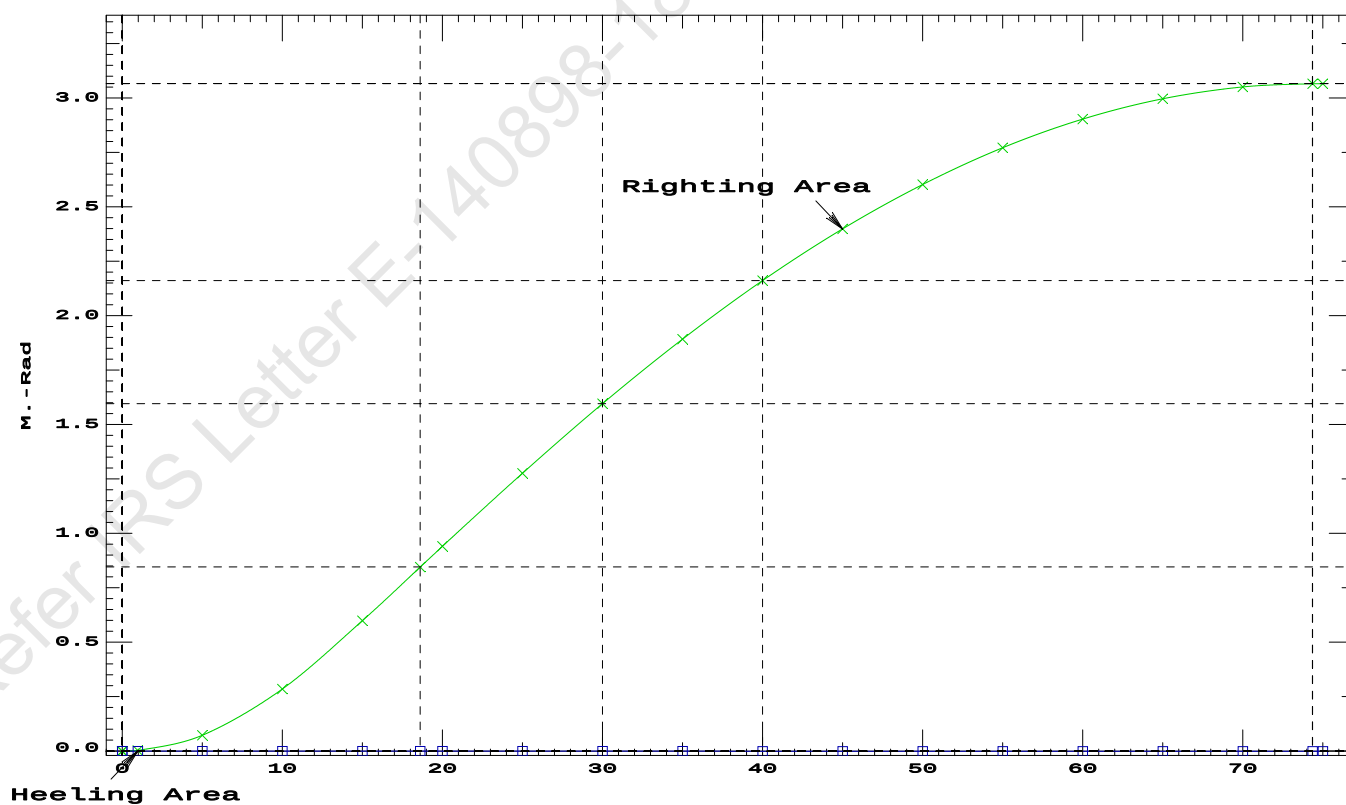
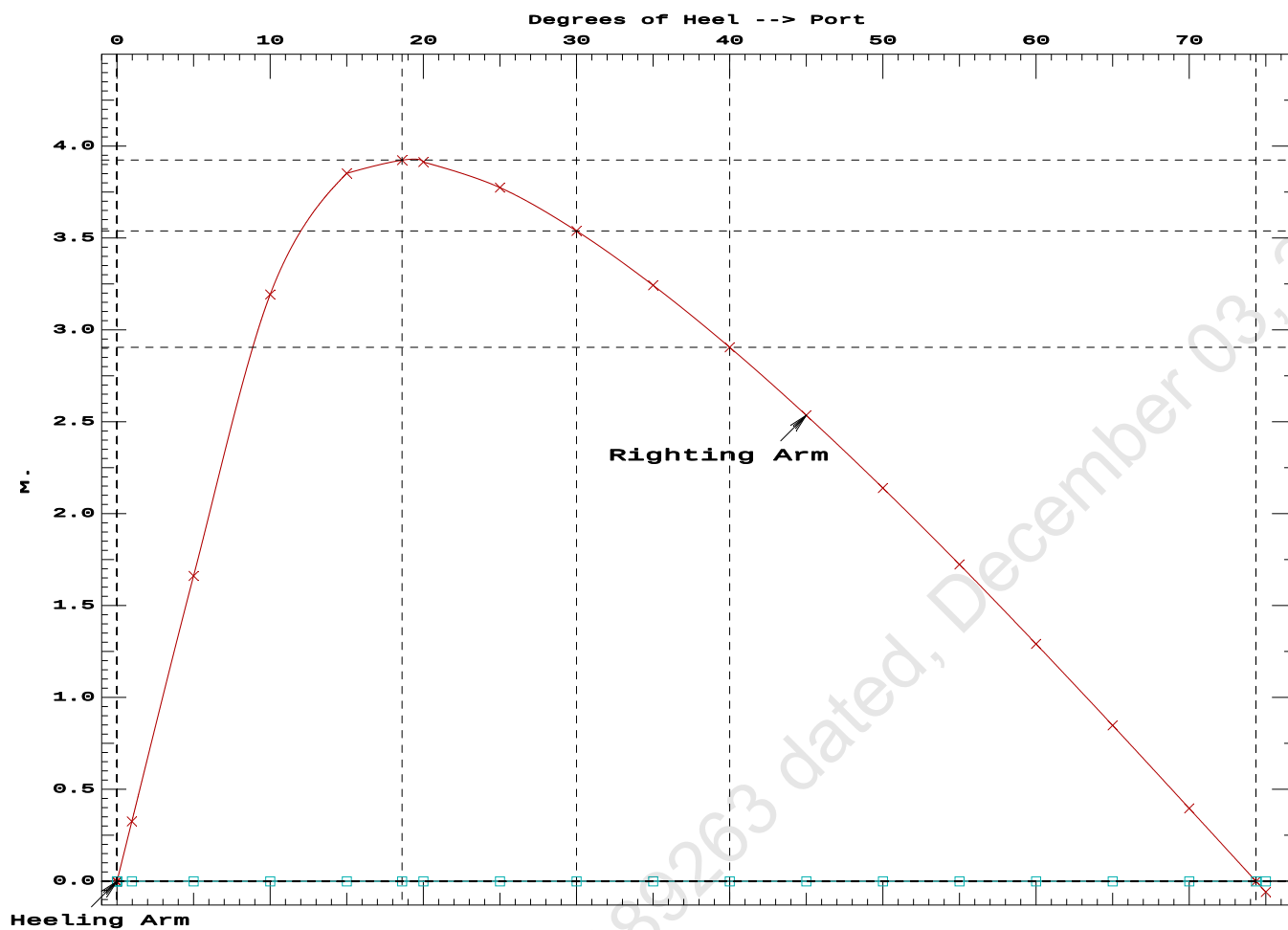
Origin Depth	Degrees of Trim	Displacement Heel	Residual Arms Weight(MT)	Res. in Trim	Res. in Heel	Area
1.228	0.28f	0.01s	1,122.0	0.000	-0.004	0.0000
1.228	0.28f	0.00	1,122.0	0.000	-0.000	-0.0000
1.225	0.28f	0.97p	1,122.0	0.000	0.326	0.0028
1.213	0.28f	5.00p	1,122.0	0.000	1.662	0.0726
1.149	0.30f	10.00p	1,122.0	0.000	3.193	0.2854
0.951	0.43f	15.00p	1,122.0	0.000	3.852	0.5992
0.795	0.54f	18.61p	1,122.2	0.000	3.924	0.8459
0.733	0.58f	20.00p	1,122.0	0.000	3.914	0.9410
0.508	0.73f	25.00p	1,122.0	0.000	3.775	1.2789
0.276	0.88f	30.00p	1,122.0	0.000	3.539	1.5988
0.039	1.03f	35.00p	1,122.0	0.000	3.243	1.8951
-0.201	1.18f	40.00p	1,122.0	0.000	2.906	2.1638
-0.442	1.33f	45.00p	1,122.0	0.000	2.536	2.4014
-0.682	1.47f	50.00p	1,122.0	0.000	2.140	2.6056
-0.917	1.60f	55.00p	1,122.0	0.000	1.724	2.7744
-1.147	1.72f	60.00p	1,122.0	0.000	1.292	2.9061
-1.366	1.83f	65.00p	1,122.0	0.000	0.848	2.9995
-1.574	1.91f	70.00p	1,122.0	0.000	0.397	3.0539
-1.744	1.98f	74.36p	1,122.0	0.000	0.000	3.0691
-1.768	1.98f	75.00p	1,122.0	0.000	-0.058	3.0688

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 699.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 7397060 P
(2) Angle from abs 1 deg to RAZero > 20.00 deg 73.39 P

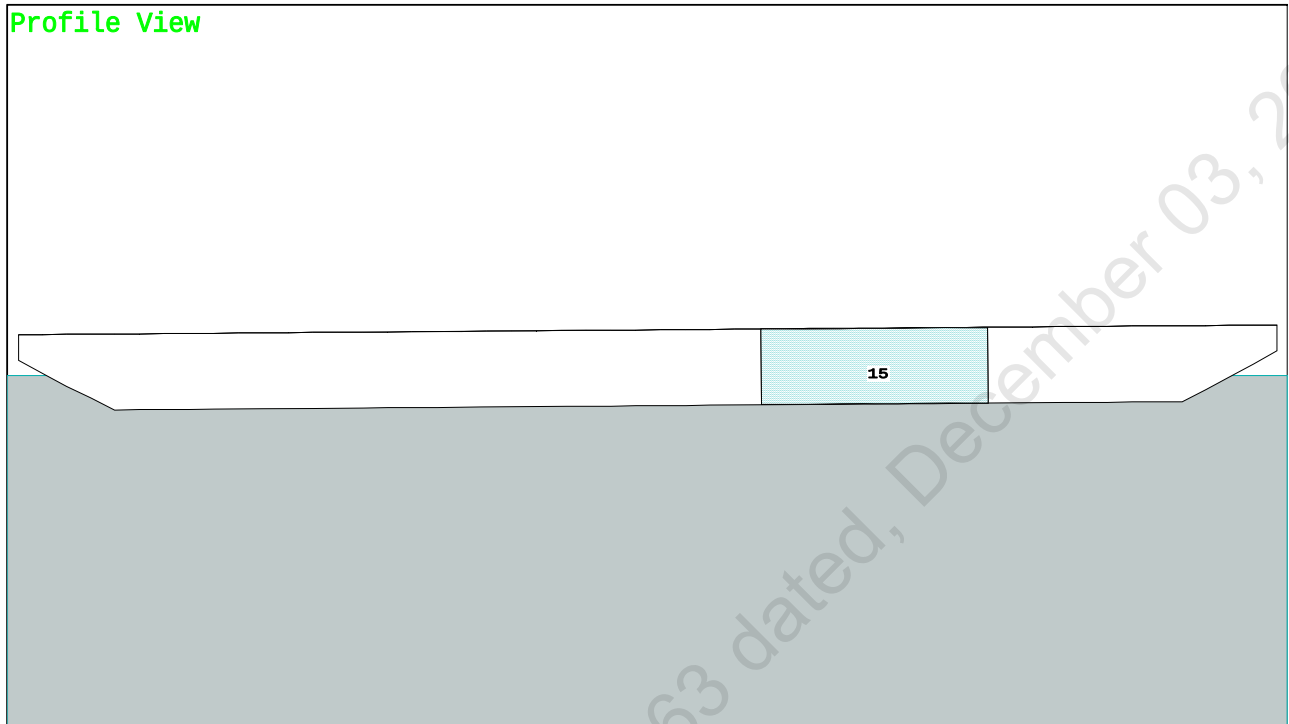
01-12-21 09:45:37
GHS 17.90Cybermarine Technologies PTE. Ltd.
PONTOON

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GHS 17.90 PONT00N

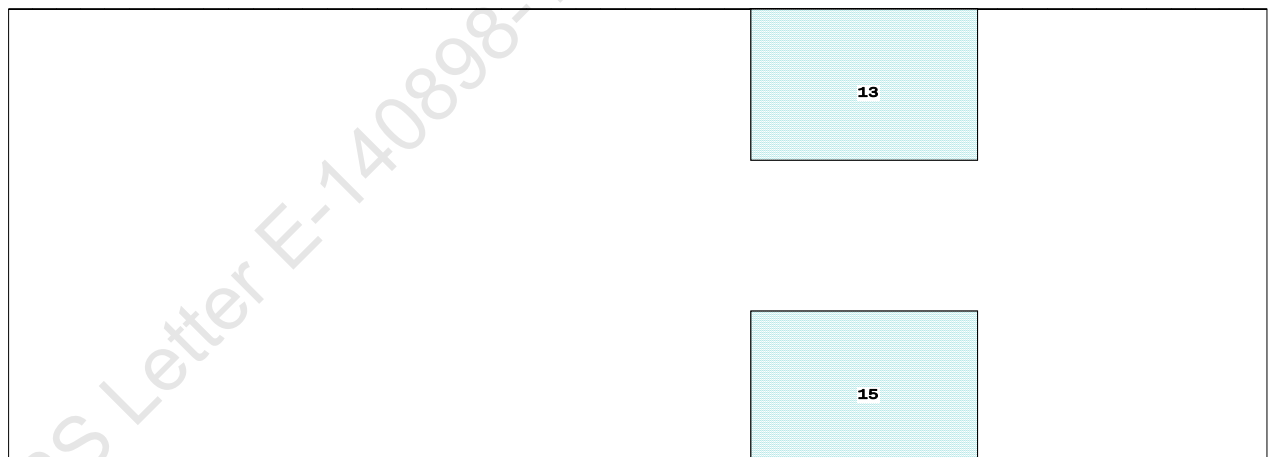
Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft

Condition Graphic - Draft: 0.997 @ 50.000f, 1.386 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

13 05_WBTK.P.100% SALT WATER Intact 15 05_WBTK.S.100% SALT WATER Intact

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PONTOON

Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 0.997 @ 50.00f, 1.191 @ 25.00f, 1.386 @ 0.00

Trim: Aft 0.389/50.000, Heel: zero

Part-----	Weight(MT)	LCG	TCG	VCG			
LIGHT SHIP	449.57	25.030f	0.000	3.000			
Crane	150.00	7.500f	0.000	5.200			
Crane Load	51.30	4.500a	0.000	33.218			
Total Fixed----->	650.87	18.663f	0.000	5.889			
Load-----	SpGr-----	Weight(MT)	LCG	TCG	VCG	FSM	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks----->			323.20	34.000f	0.000	1.509	325.46
Total Weight----->			974.07	23.752f	0.000	4.436	
		Displ(MT)	LCB	TCB	VCB		
HULL	1.025	974.07	23.722f	0.000	0.604		

Righting Arms:			0.000	0.000			
Part-----	SpGr-----	WPA	LCF	TCF	BML	BMT	
Total Waterplane---->	1.025	847.5	24.653f	0.000	164.40	23.754*	
		MT/cm	m	MT/cm	GML	GMT	
		8.69		31.28	160.57	19.922	
Distances in METERS.-----			Moments in m.-MT.				

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

315.3 Cu.M. (13%) SALT WATER

150.00 MT Crane

51.30 MT Crane Load

DRAFT AT FP (M) = 0.997
DRAFT AT MS (M) = 1.191
DRAFT AT AP (M) = 1.386
DRAFT AT FWD MARK (M) = 1.028
DRAFT AT AFT MARK (M) = 1.355

HYDROSTATIC PROPERTIES

Trim: Aft 0.389/50.000, No Heel, VCG = 4.436

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/	
Draft---	Weight(MT)	LCB	VCB	cm	LCF
1.194	974.07	23.722f	0.604	8.69	24.653f
					31.28
					160.57
					19.922
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.					
Trim is per 50.00m.					

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 325.5 m.-MT

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PONTOON

Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 0.997 @ 50.00f, 1.191 @ 25.00f, 1.386 @ 0.00

Trim: Aft 0.389/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	90.2	0.928	1.517	0.05500	7.53
CRANE	57.0	4.205	4.793	0.05500	15.03
Total wind heeling moment to starboard-->					22.56
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.752f TCG = 0.000 VCG = 4.436

Free Surface Adjustment: 0.334

Adjusted CG: LCG = 23.754f TCG = 0.000 VCG = 4.770

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. 50% DeckImm Area	Angle
1.374	0.45a	0.00	974.06	0.000	-0.023	0.0000	5.316
1.373	0.45a	0.06s	974.07	0.000	0.000	-0.0000	5.259
1.359	0.45a	5.06s	974.06	0.000	1.742	0.0760	0.259
1.357	0.45a	5.32s	974.01	0.000	1.830	0.0841	0.000
1.310	0.51a	10.06s	973.96	0.000	3.192	0.2938	-4.741
1.172	0.68a	15.06s	974.07	0.000	3.714	0.6012	-9.741
1.129	0.75a	16.68s	974.07	0.000	3.735	0.7065	-11.361
1.038	0.91a	20.06s	974.07	0.000	3.656	0.9257	-14.741
0.900	1.14a	25.06s	974.07	0.000	3.375	1.2350	-19.741
0.759	1.38a	30.06s	974.07	0.000	2.990	1.5135	-24.741
0.618	1.62a	35.06s	974.07	0.000	2.546	1.7555	-29.741
0.476	1.85a	40.06s	974.07	0.000	2.061	1.9568	-34.741
0.333	2.08a	45.06s	974.07	0.000	1.550	2.1146	-39.741
0.190	2.30a	50.06s	974.07	0.000	1.019	2.2268	-44.741
0.046	2.50a	55.06s	974.07	0.000	0.476	2.2921	-49.741
-0.080	2.66a	59.39s	974.07	0.000	0.000	2.3102	-54.074
-0.099	2.68a	60.06s	974.07	0.000	-0.074	2.3097	-54.741
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.							

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

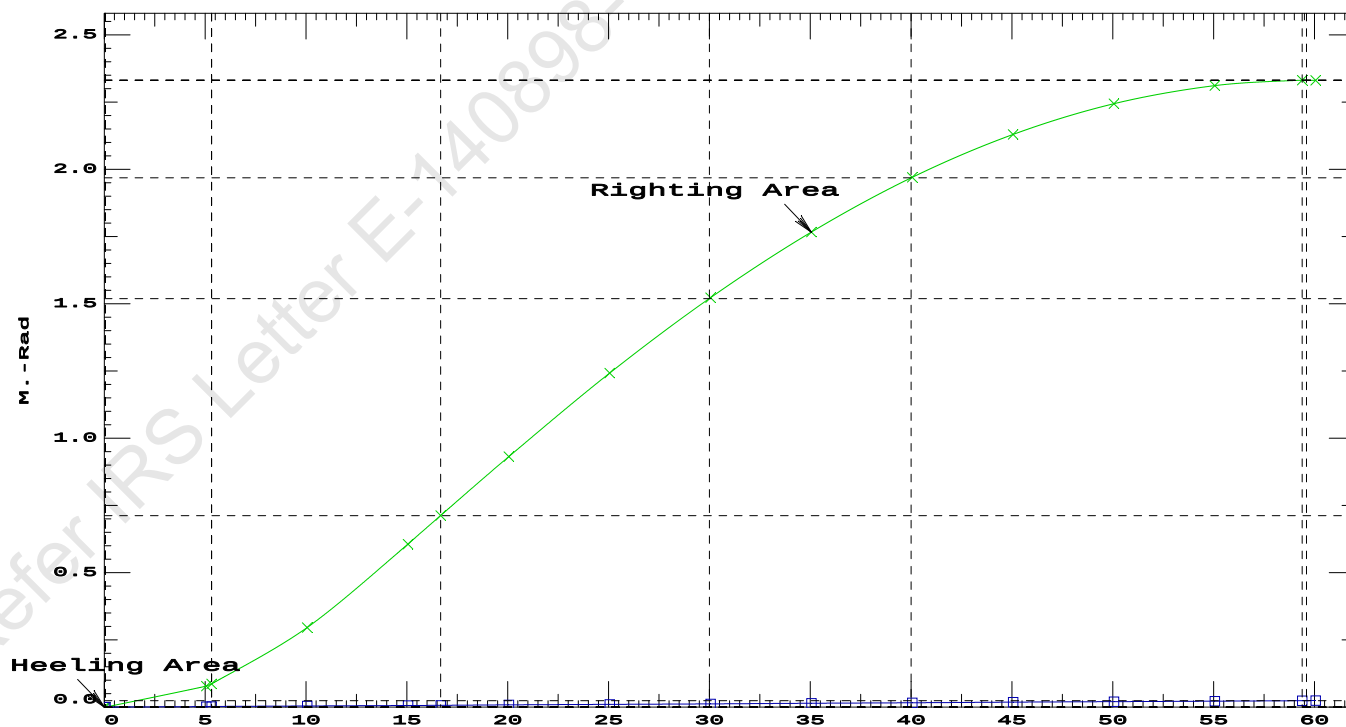
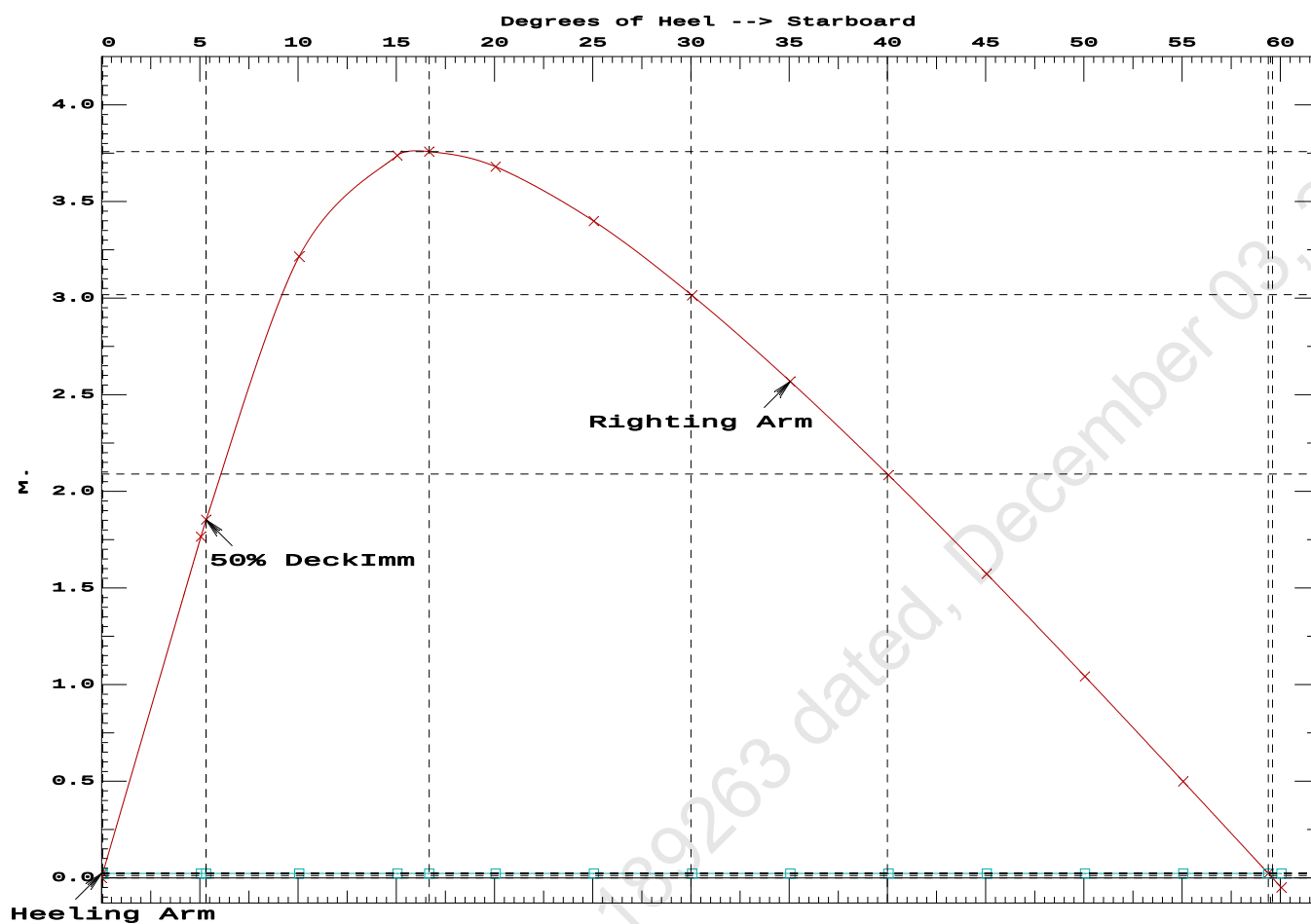
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 22.56 (constant)

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2-----Min/Max-----Attained

(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.7065 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	59.33 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	5.26 P

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Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft



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GHS 17.90

PONTOON

Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.752f TCG = 0.000 VCG = 4.436
Free Surface Adjustment: 0.334
Adjusted CG: LCG = 23.754f TCG = 0.000 VCG = 4.770

Origin Depth---	Trim----	Degrees of Heel----	Displacement Weight(MT)---	Residual Arms in Trim--in Heel----	Res. Area
1.374	0.45a	0.01s	974.07	0.000 0.000	0.0000
1.359	0.45a	5.01s	974.07	0.000 1.743	0.0760
1.311	0.51a	10.01s	973.96	0.000 3.200	0.2938
1.174	0.67a	15.01s	974.07	0.000 3.731	0.6023
1.129	0.75a	16.68s	974.07	0.000 3.753	0.7118
1.039	0.91a	20.01s	974.07	0.000 3.676	0.9286
0.901	1.14a	25.01s	974.07	0.000 3.396	1.2397
0.761	1.38a	30.01s	974.07	0.000 3.012	1.5201
0.619	1.62a	35.01s	974.07	0.000 2.568	1.7640
0.477	1.85a	40.01s	974.07	0.000 2.084	1.9673
0.334	2.08a	45.01s	974.07	0.000 1.573	2.1271
0.191	2.30a	50.01s	974.07	0.000 1.042	2.2413
0.047	2.50a	55.01s	974.07	0.000 0.499	2.3087
-0.085	2.66a	59.55s	974.06	0.000 0.000	2.3285
-0.098	2.68a	60.01s	974.07	0.000 -0.050	2.3283

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

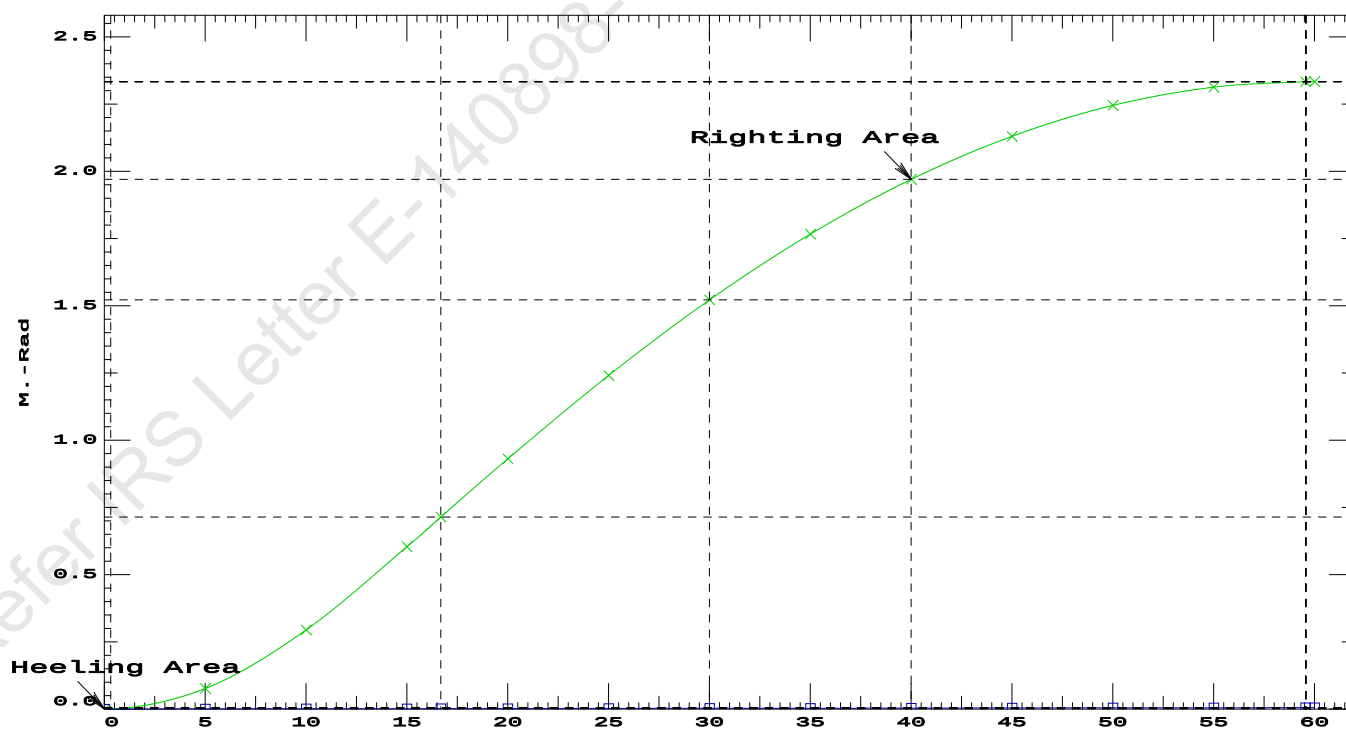
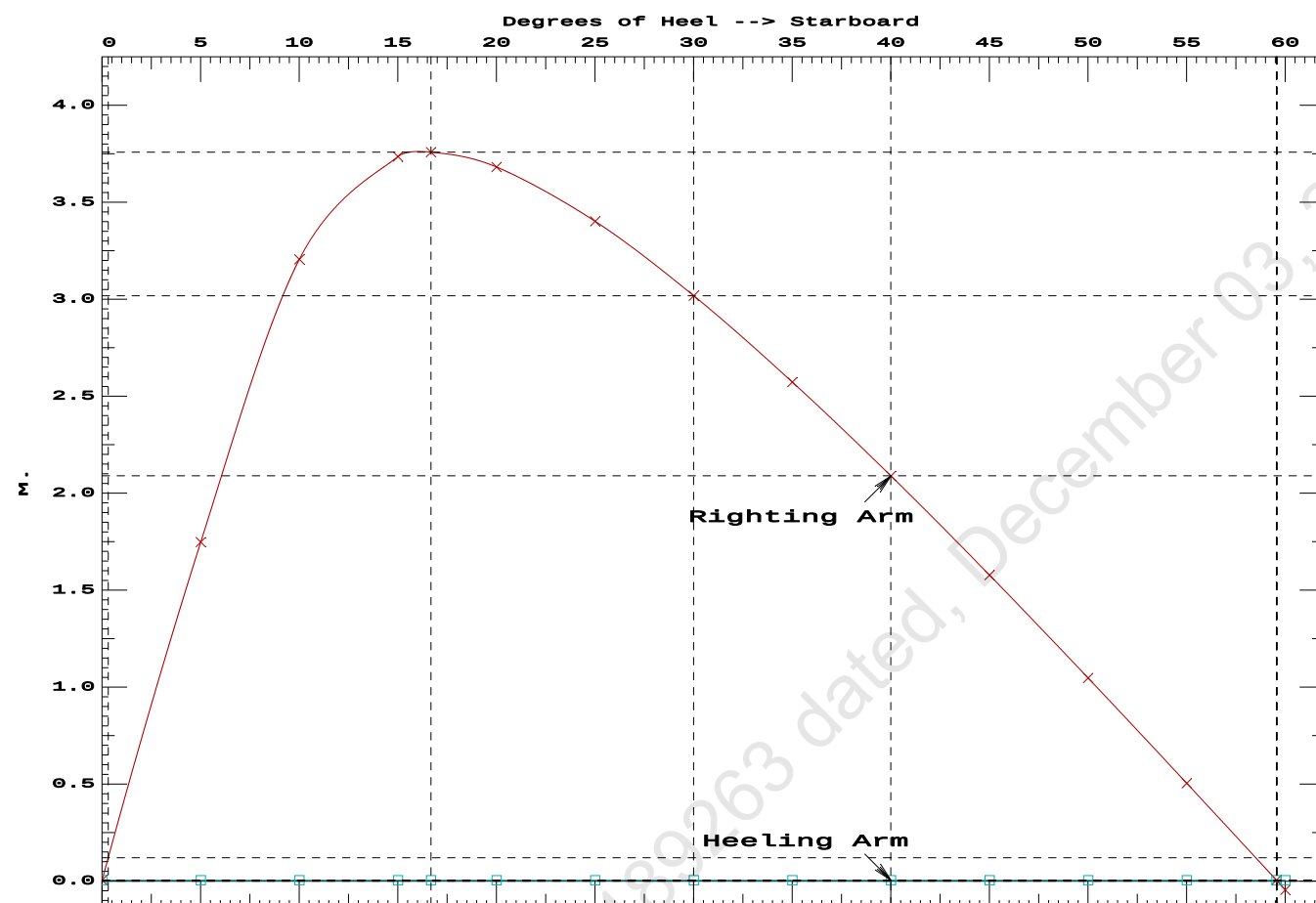
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.01 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	73106.6 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	437.483 P

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Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft



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Condition 13 : Operating Condition of 150T Crane

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 0.997 @ 50.00f, 1.191 @ 25.00f, 1.386 @ 0.00

Trim: Aft 0.389/50.000, Heel: zero

Part-----			Weight(MT)	---LCG---	---TCG---	---VCG---	
LIGHT SHIP			449.57	25.030f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	4.500a	0.000	33.218	
Total Fixed----->			650.87	18.663f	0.000	5.889	
	Load-----	SpGr-----	Weight(MT)	---LCG---	---TCG---	---VCG---	---FSM---
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks----->			323.20	34.000f	0.000	1.509	325.46
Total Weight----->			974.07	23.752f	0.000	4.436	
			Displ(MT)	---LCB---	---TCB---	---VCB---	
HULL		1.025	974.06	23.722f	0.000	0.604	

	Righting Arms:			0.000	0.000		
	External Arms:			0.000	0.001s		
	Residual Righting Arms:			0.000	-0.001s		
Part-----		SpGr-----	WPA-----	---LCF---	---TCF---	---BML---	---BMT---
Total Waterplane---->	1.025		847.5	24.653f	0.000	164.40	23.754*
			MT/cm-----	---m---	MT/cm-----	---GML---	---GMT---
			8.69		31.28	160.57	19.922
Distances in METERS.-----				-----Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.181 @ 50.00f, 1.133 @ 25.00f, 1.085 @ 0.00
Trim: Fwd 0.096/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----			
LIGHT SHIP	449.57	25.030f	0.000	3.000			
Crane	150.00	7.500f	0.000	5.200			
Total Fixed----->	599.57	20.644f	0.000	3.550			
Load-----	SpGr-----	Weight(MT)---- <td>LCG-----</td> <td>TCG-----</td> <td>VCG-----</td> <td>FSM-----</td>	LCG-----	TCG-----	VCG-----	FSM-----	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks----->			323.20	34.000f	0.000	1.509	325.46
Total Weight----->			922.77	25.322f	0.000	2.836	
		Displ(MT)---- <td>LCB-----</td> <td>TCB-----</td> <td>VCB-----</td> <td></td> <td></td>	LCB-----	TCB-----	VCB-----		
HULL	1.025	922.77	25.326f	0.001s	0.569		

Righting Arms:			0.000	0.001s			
External Arms:			0.000	0.001s			
Residual Righting Arms:			0.000	0.000			
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----	
Total Waterplane---->	1.025	841.1	25.057f	0.017s	169.66	24.823*	
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----		
		8.62	30.89	167.39	22.556		
Distances in METERS.-----							
-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 20.644f TCG = 0.000 VCG = 3.550

Origin Depth	Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.073	0.11f	0.00	922.76	0.000	0.000	0.0000
1.073	0.11f	0.07s	922.82	0.000	0.028	0.0000
1.059	0.11f	5.00s	922.77	0.000	2.016	0.0880
0.969	0.13f	10.00s	922.76	0.000	3.673	0.3388
0.720	0.16f	15.00s	922.77	0.000	4.382	0.6972
0.470	0.21f	19.34s	922.77	0.000	4.507	1.0363
0.431	0.22f	20.00s	922.77	0.000	4.505	1.0886
0.138	0.27f	25.00s	922.77	0.000	4.385	1.4792
-0.158	0.33f	30.00s	922.77	0.000	4.149	1.8524

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 599.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 LARGE

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 20.644f TCG = 0.000 VCG = 3.550

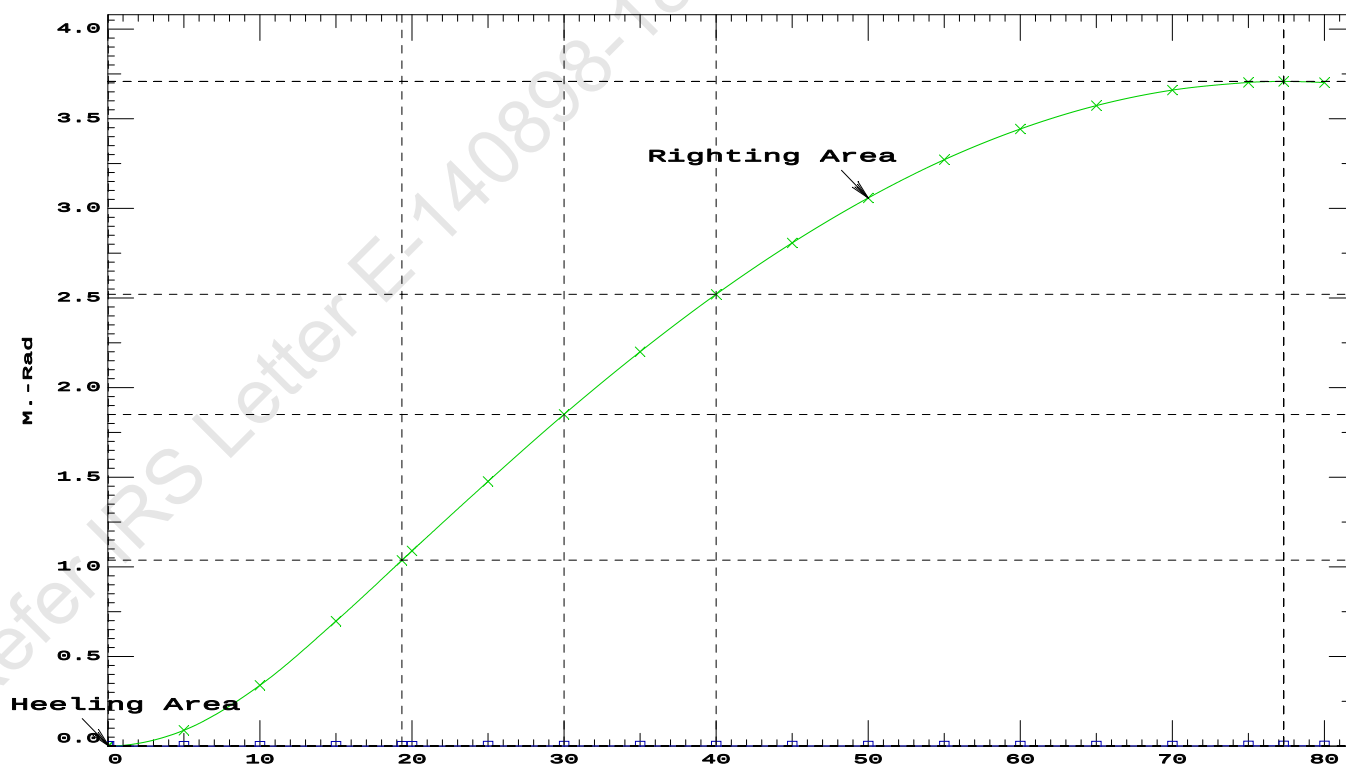
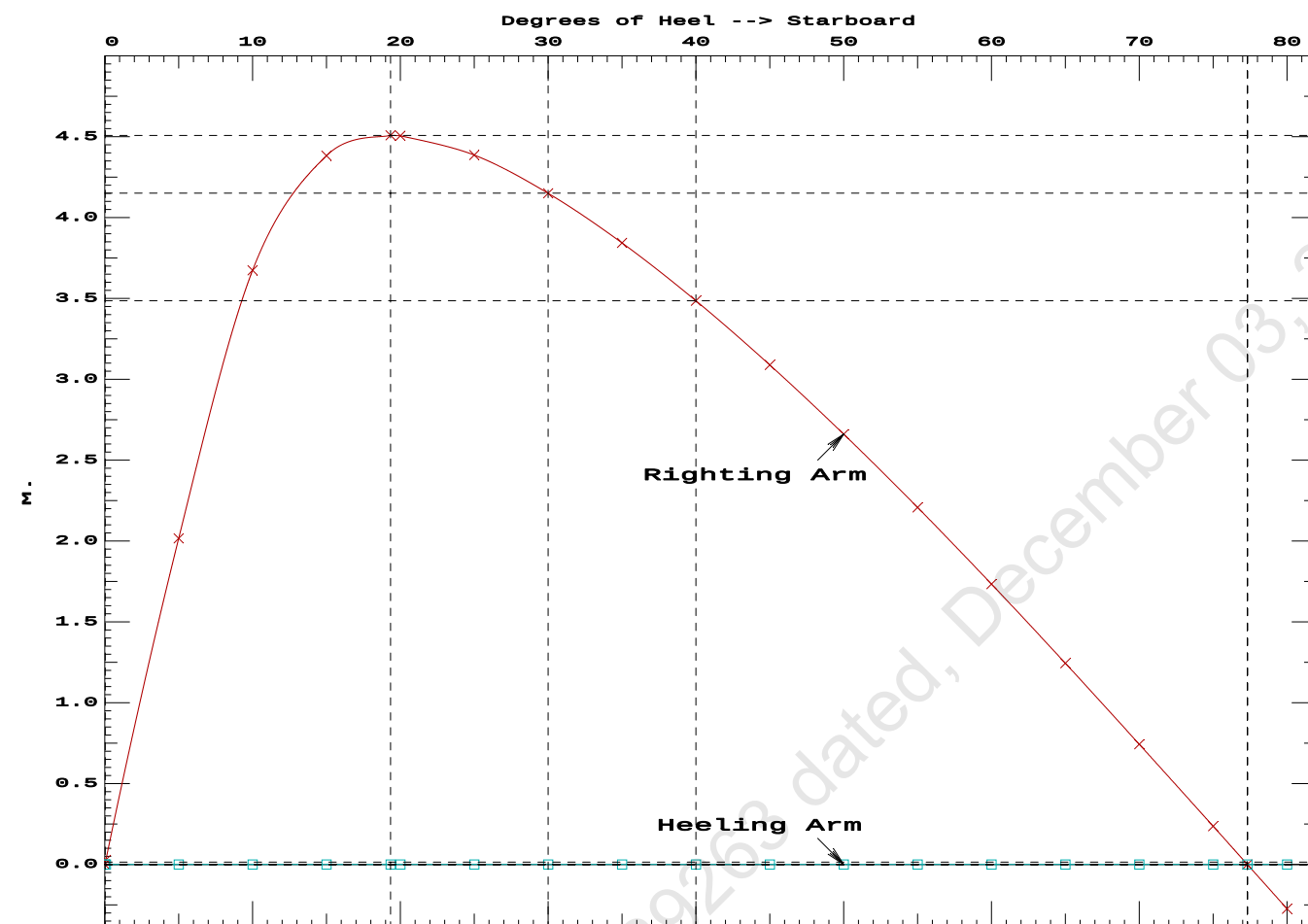
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.073	0.11f	0.00	922.76	0.000	0.000	0.0000
1.073	0.11f	0.07s	922.82	0.000	0.028	0.0000
1.059	0.11f	5.00s	922.77	0.000	2.016	0.0880
0.969	0.13f	10.00s	922.76	0.000	3.673	0.3388
0.720	0.16f	15.00s	922.77	0.000	4.382	0.6972
0.470	0.21f	19.34s	922.77	0.000	4.507	1.0363
0.431	0.22f	20.00s	922.77	0.000	4.505	1.0886
0.138	0.27f	25.00s	922.77	0.000	4.385	1.4792
-0.158	0.33f	30.00s	922.77	0.000	4.149	1.8524
-0.453	0.38f	35.00s	922.77	0.000	3.842	2.2016
-0.745	0.44f	40.00s	922.77	0.000	3.485	2.5217
-1.032	0.49f	45.00s	922.77	0.000	3.088	2.8088
-1.313	0.54f	50.00s	922.77	0.000	2.660	3.0598
-1.584	0.59f	55.00s	922.77	0.000	2.207	3.2724
-1.843	0.63f	60.00s	922.77	0.000	1.733	3.4444
-2.088	0.67f	65.00s	922.77	0.000	1.244	3.5744
-2.316	0.70f	70.00s	922.77	0.000	0.743	3.6612
-2.526	0.73f	75.00s	922.77	0.000	0.236	3.7040
-2.617	0.74f	77.32s	922.77	0.000	0.000	3.7088
-2.716	0.74f	80.00s	922.77	0.000	-0.274	3.7023

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 599.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 0.1 deg to RAZero > 20.00 deg 77.25 P

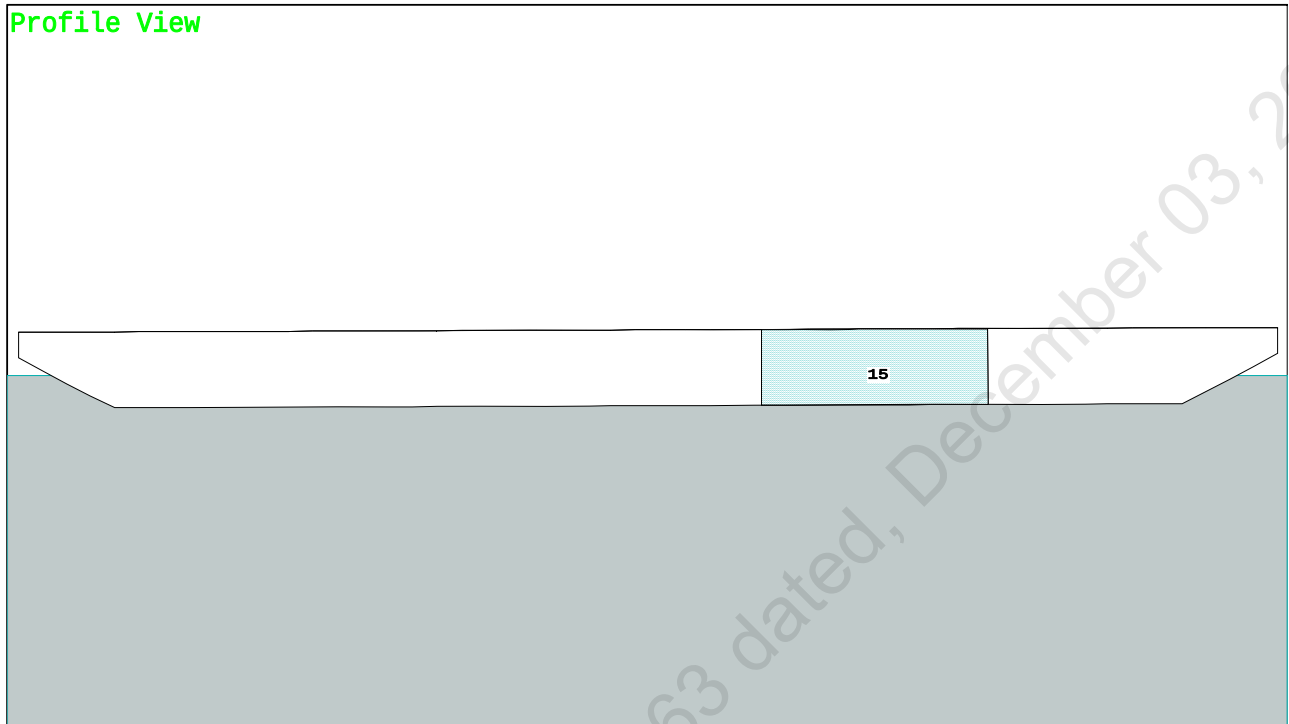
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GHS 17.90 PONT00N

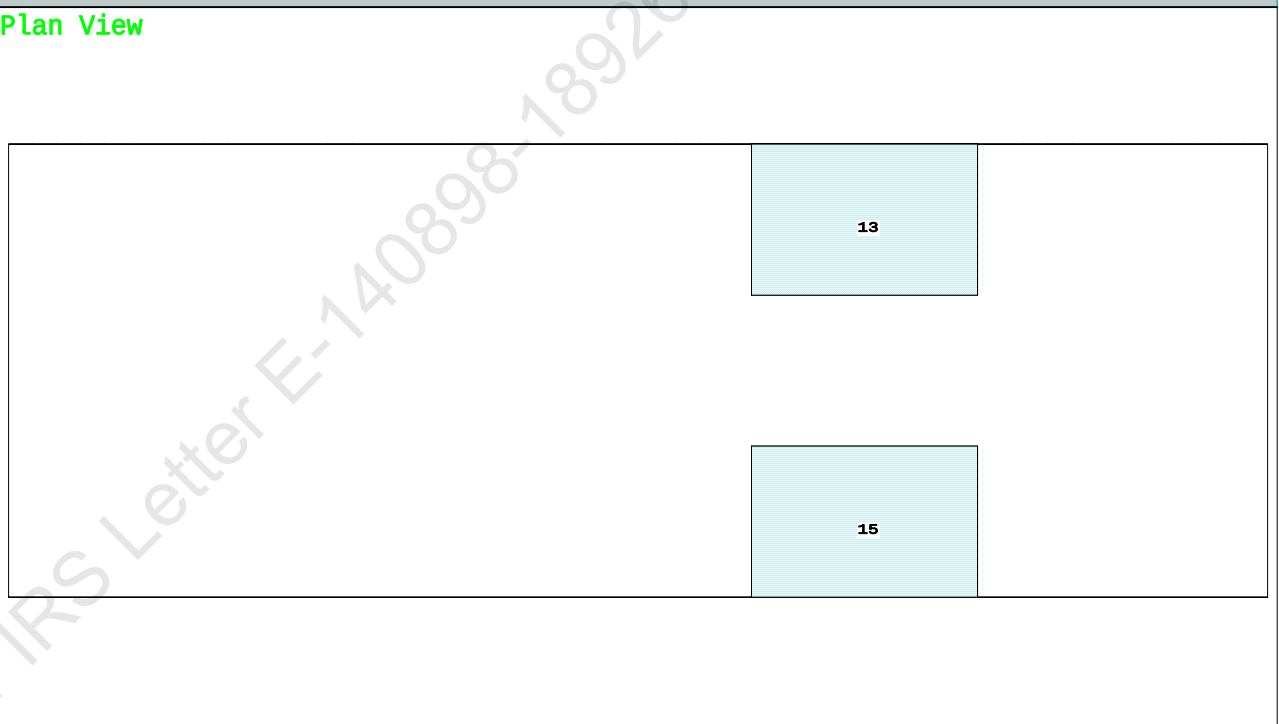
Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

Condition Graphic - Draft: 1.096 @ 50.000f, 1.288 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

13 05_WBTK.P.100% SALT WATER Intact 15 05_WBTK.S.100% SALT WATER Intact

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PONTOON

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.096 @ 50.00f, 1.192 @ 25.00f, 1.288 @ 0.00

Trim: Aft 0.192/50.000, Heel: zero

Part-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----	FSM-----
LIGHT SHIP	449.57	25.030f	0.000	3.000	
Crane	150.00	7.500f	0.000	5.200	
Crane Load	51.30	7.500f	0.000	33.218	
Total Fixed----->	650.87	19.608f	0.000	5.889	
Load-----	SpGr-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s
Total Tanks----->		323.20	34.000f	0.000	1.509
Total Weight----->		974.07	24.384f	0.000	4.436
		Displ(MT)-----	LCB-----	TCB-----	VCB-----
HULL	1.025	974.03	24.369f	0.000	0.601

Righting Arms:		0.000	0.000		
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----
Total Waterplane---->	1.025	847.2	24.829f	0.000	164.24
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----
		8.68	31.25	160.40	19.899
Distances in METERS.-----				Moments in m.-MT.	

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

315.3 Cu.M. (13%) SALT WATER

150.00 MT Crane

51.30 MT Crane Load

DRAFT AT FP (M) = 1.096
DRAFT AT MS (M) = 1.192
DRAFT AT AP (M) = 1.288
DRAFT AT FWD MARK (M) = 1.111
DRAFT AT AFT MARK (M) = 1.273

HYDROSTATIC PROPERTIES

Trim: Aft 0.192/50.000, No Heel, VCG = 4.436

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)-----	LCB-----	VCB-----	cm-----
1.193	974.03	24.369f	0.601	8.68
				24.829f
				31.25
				160.40
				19.899

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 325.5 m.-MT

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PONTOON

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.096 @ 50.00f, 1.192 @ 25.00f, 1.288 @ 0.00

Trim: Aft 0.192/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	90.2	0.926	1.511	0.05500	7.49
CRANE	57.0	4.263	4.848	0.05500	15.20
Total wind heeling moment to starboard-->					22.69
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.384f TCG = 0.000 VCG = 4.436

Free Surface Adjustment: 0.334

Adjusted CG: LCG = 24.385f TCG = 0.000 VCG = 4.770

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Res. 50% DeckImm Area	Angle
1.276	0.22a	0.00	974.06	0.000 -0.023	0.0000	5.746
1.276	0.22a	0.06s	974.07	0.000 0.000	-0.0000	5.684
1.261	0.22a	5.06s	974.07	0.000 1.745	0.0761	0.684
1.257	0.22a	5.75s	974.04	0.000 1.977	0.0984	0.000
1.200	0.25a	10.06s	974.04	0.000 3.203	0.2949	-4.316
1.023	0.33a	15.06s	974.07	0.000 3.736	0.6029	-9.316
0.963	0.37a	16.68s	974.07	0.000 3.757	0.7088	-10.935
0.837	0.45a	20.06s	974.07	0.000 3.678	0.9293	-14.316
0.647	0.57a	25.06s	974.07	0.000 3.397	1.2406	-19.316
0.454	0.68a	30.06s	974.07	0.000 3.012	1.5210	-24.316
0.260	0.80a	35.06s	974.07	0.000 2.567	1.7649	-29.316
0.066	0.92a	40.06s	974.07	0.000 2.083	1.9681	-34.316
-0.127	1.03a	45.06s	974.07	0.000 1.570	2.1277	-39.316
-0.318	1.14a	50.06s	974.07	0.000 1.038	2.2416	-44.316
-0.506	1.24a	55.06s	974.07	0.000 0.493	2.3086	-49.316
-0.671	1.32a	59.53s	974.06	0.000 0.000	2.3279	-53.788
-0.691	1.32a	60.06s	974.07	0.000 -0.058	2.3276	-54.316
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.						

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

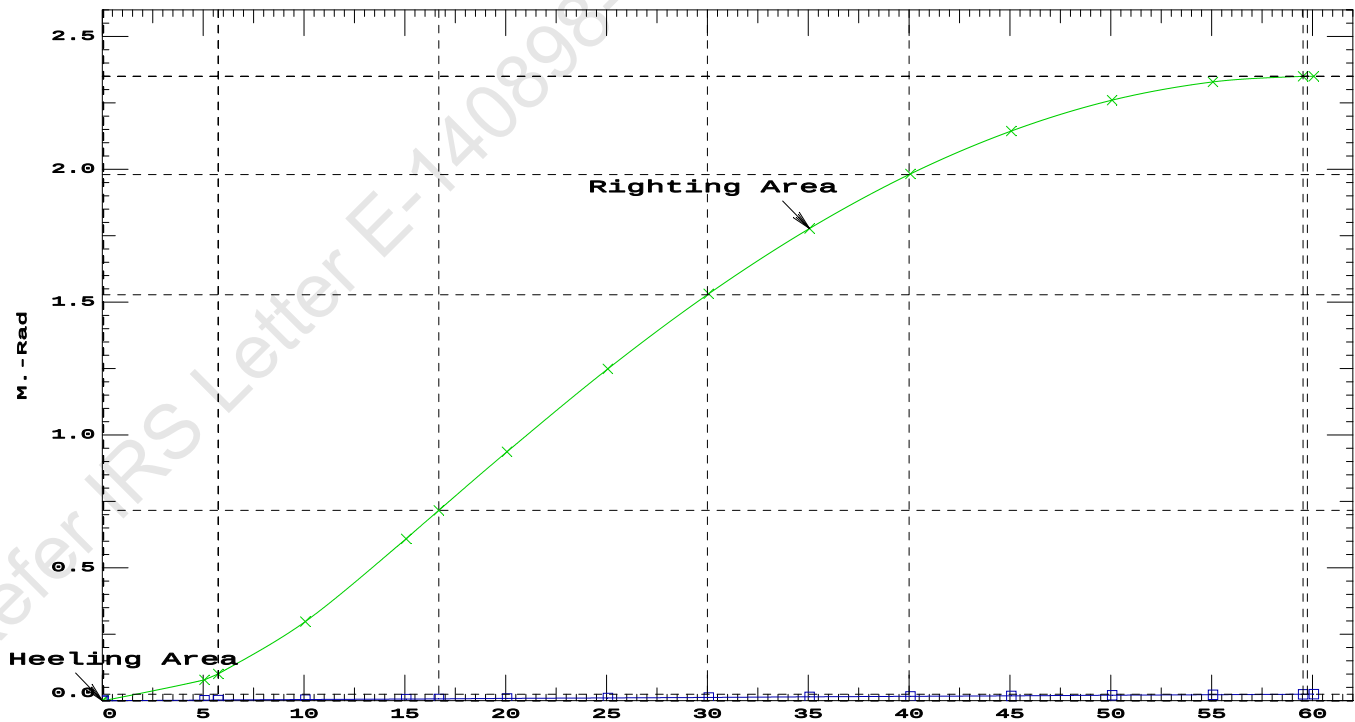
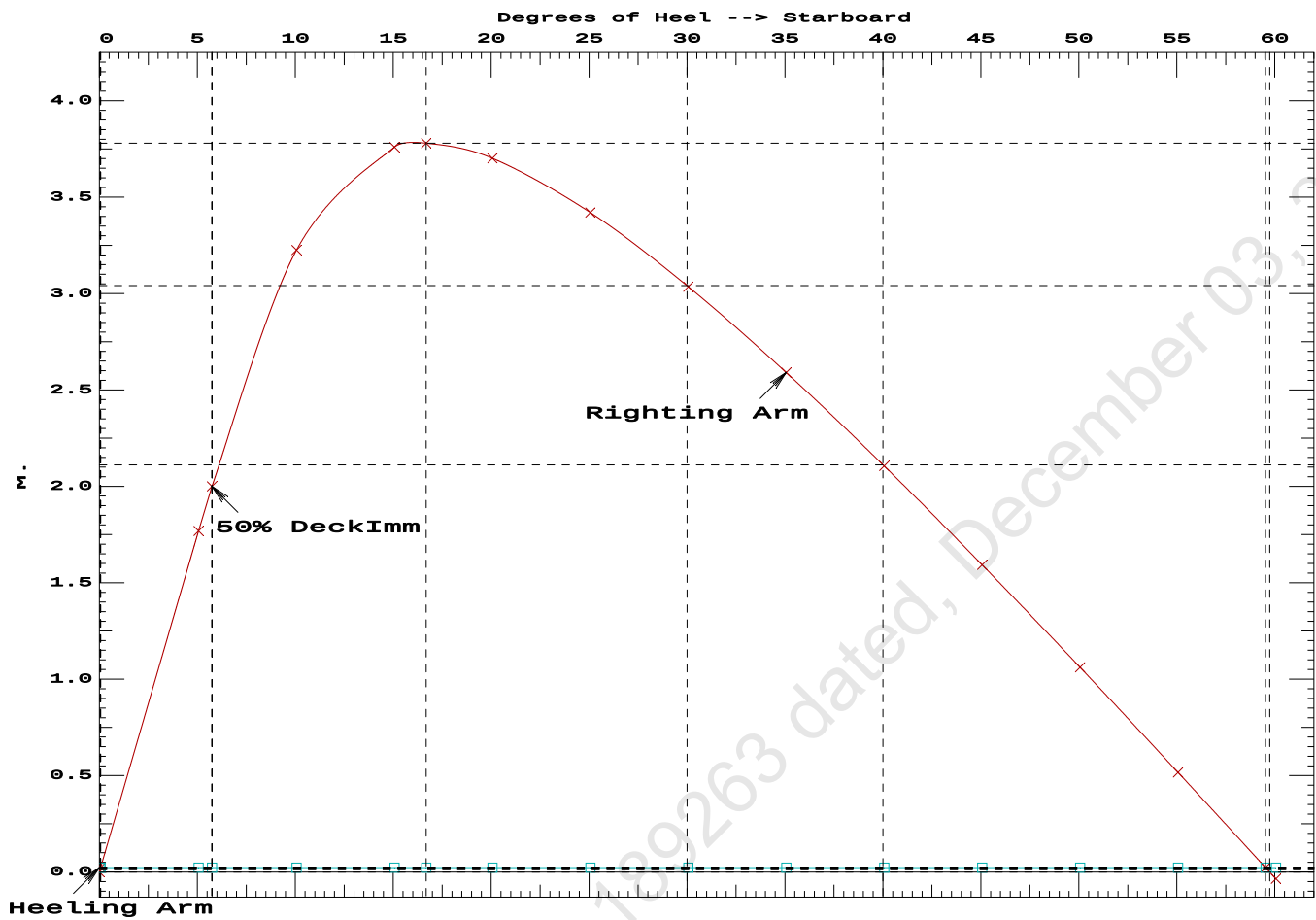
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 22.69 (constant)

LIM	IS CODE 2008 PART B CHAPTER-2 SEC.2	Min/Max	Attained
(1)	Residual Area from abs 0 deg to MaxRA	> 0.0800 m.-Rad	0.7088 P
(2)	Angle from Equilibrium to RAZero or Flood	> 20.00 deg	59.47 P
(3)	Angle from Equilibrium to 50% Dk Imm Angle	> 0.00 deg	5.68 P

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PONTOON

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd



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PONTOON

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.384f TCG = 0.000 VCG = 4.436
Free Surface Adjustment: 0.334
Adjusted CG: LCG = 24.385f TCG = 0.010p VCG = 4.770

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.272	0.22a	1.77s	974.06	0.000	0.000	0.0000
1.249	0.22a	6.77s	974.07	0.000	1.715	0.0748
1.144	0.27a	11.77s	974.02	0.000	2.879	0.2793
0.968	0.37a	16.55s	974.07	0.000	3.158	0.5368
0.960	0.37a	16.77s	974.07	0.000	3.157	0.5489
0.773	0.49a	21.77s	974.07	0.000	2.999	0.8207
0.581	0.61a	26.77s	974.07	0.000	2.675	1.0695
0.388	0.72a	31.77s	974.07	0.000	2.266	1.2857
0.194	0.84a	36.77s	974.07	0.000	1.805	1.4637
0.000	0.96a	41.77s	974.07	0.000	1.309	1.5999
-0.192	1.07a	46.77s	974.07	0.000	0.789	1.6916
-0.382	1.17a	51.77s	974.07	0.000	0.251	1.7371
-0.468	1.22a	54.07s	974.07	0.000	0.000	1.7422
-0.569	1.27a	56.77s	974.07	0.000	-0.297	1.7352
-0.753	1.35a	61.77s	974.07	0.000	-0.851	1.6851

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 615.60

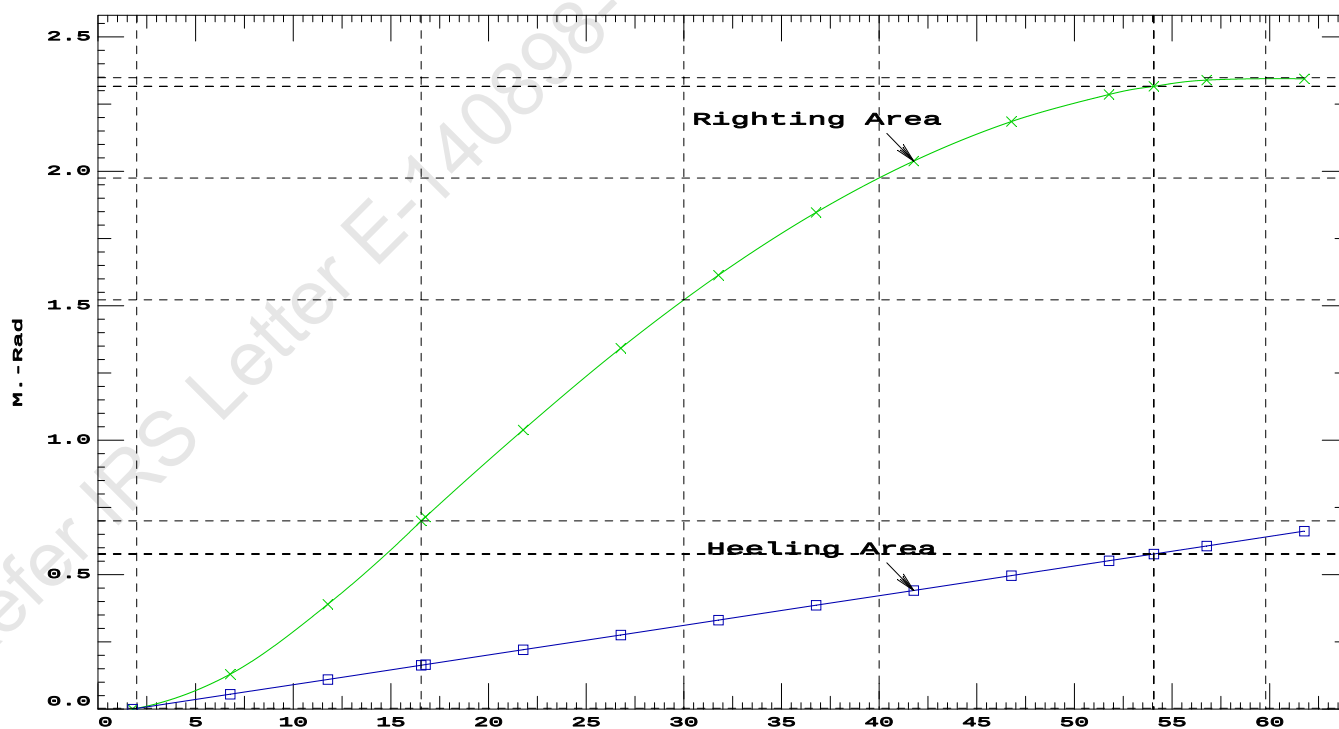
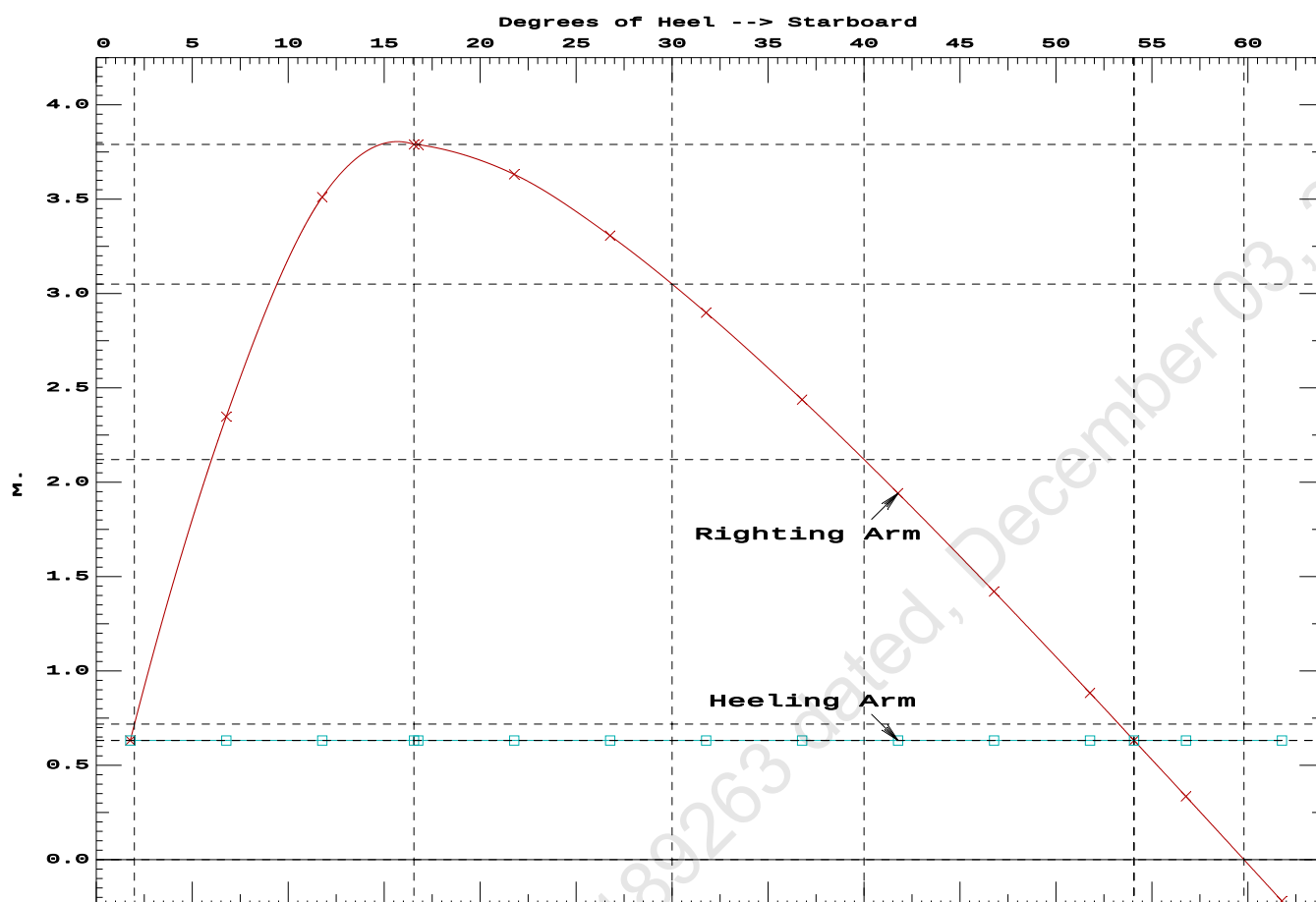
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT-----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	1.77 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	499.6 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	4.020 P

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PONTOON

**Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd**



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GHS 17.90 PONTON

Condition 14 : Operating Condition of 150T Crane**Crane Lifting 51.3T @ 12m Radius Stbd**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.095 @ 50.00f, 1.190 @ 25.00f, 1.285 @ 0.00

Trim: Aft 0.190/50.000, Heel: Stbd 1.77 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			449.57	25.030f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	7.500f	0.000	33.218	
Total Fixed----->			650.87	19.608f	0.000	5.889	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks----->			323.20	34.000f	0.000	1.509	325.46
Total Weight----->			974.07	24.384f	0.000	4.436	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	974.06	24.369f	0.750s	0.612	

	Righting Arms:			0.000	0.632s		
	External Arms:			0.000	0.632s		
	Residual Righting Arms:			0.000	0.000s		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		846.5	24.852f	0.101s	163.82	23.721*
			MT/cm				
			8.68				

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 1.77

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GHS 17.90 PONTON

After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.181 @ 50.00f, 1.133 @ 25.00f, 1.085 @ 0.00
Trim: Fwd 0.096/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----			
LIGHT SHIP	449.57	25.030f	0.000	3.000			
Crane	150.00	7.500f	0.000	5.200			
Total Fixed----->	599.57	20.644f	0.000	3.550			
Load-----	SpGr-----	Weight(MT)---- <td>LCG-----</td> <td>TCG-----</td> <td>VCG-----</td> <td>FSM-----</td>	LCG-----	TCG-----	VCG-----	FSM-----	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks----->			323.20	34.000f	0.000	1.509	325.46
Total Weight----->			922.77	25.322f	0.000	2.836	
			Displ(MT)---- <td>LCB-----</td> <td>TCB-----</td> <td>VCB-----</td> <td></td>	LCB-----	TCB-----	VCB-----	
HULL	1.025		922.77	25.326f	0.001s	0.569	

Righting Arms:			0.000	0.001s			
External Arms:			0.000	0.001s			
Residual Righting Arms:			0.000	0.000			
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----	
Total Waterplane---->	1.025	841.1	25.057f	0.017s	169.66	24.822*	
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----		
		8.62	30.89	167.39	22.556		
Distances in METERS.-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 16632.50; t3: 30.00; t: 15.00
limmarg: 6198.26; t3: 15.00; t: 7.50
limmarg: 1679.85; t3: 7.50; t: 3.75
limmarg: 349.31; t3: 3.75; t: 1.88
limmarg: 12.46; t3: 1.87; t: 0.94
limmarg: -72.09; t3: 0.93; t: 0.47
limmarg: -36.86; t3: 1.40; t: 0.23
limmarg: -14.48; t3: 1.63; t: 0.12
limmarg: -1.47; t3: 1.75; t: 0.06
limmarg: 5.38; t3: 1.81; t: 0.03
limmarg: 1.93; t3: 1.78; t: 0.01

Theta3 is 1.78 degrees, as demonstrated below.

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RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 20.644f TCG = 0.000 VCG = 3.550

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.071	0.11f	1.77s	922.75	0.000	-0.719	0.0000
1.073	0.11f	0.00	922.76	0.000	0.000	-0.0111
1.071	0.11f	1.78p	922.77	0.000	0.726	0.0002
1.059	0.11f	5.00p	922.76	0.000	2.017	0.0774
0.969	0.13f	10.00p	922.76	0.000	3.675	0.3279
0.720	0.16f	15.00p	922.77	0.000	4.384	0.6865
0.470	0.21f	19.34p	922.77	0.000	4.510	1.0259
0.431	0.22f	20.00p	922.77	0.000	4.507	1.0781
0.138	0.27f	25.00p	922.77	0.000	4.388	1.4688
-0.157	0.33f	30.00p	922.77	0.000	4.152	1.8423

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 599.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Res. Area Ratio from abs -1.8 deg to abs 1.8 > 1.000 1.019 P

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GHS 17.90 PONTON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 20.644f TCG = 0.000 VCG = 3.550

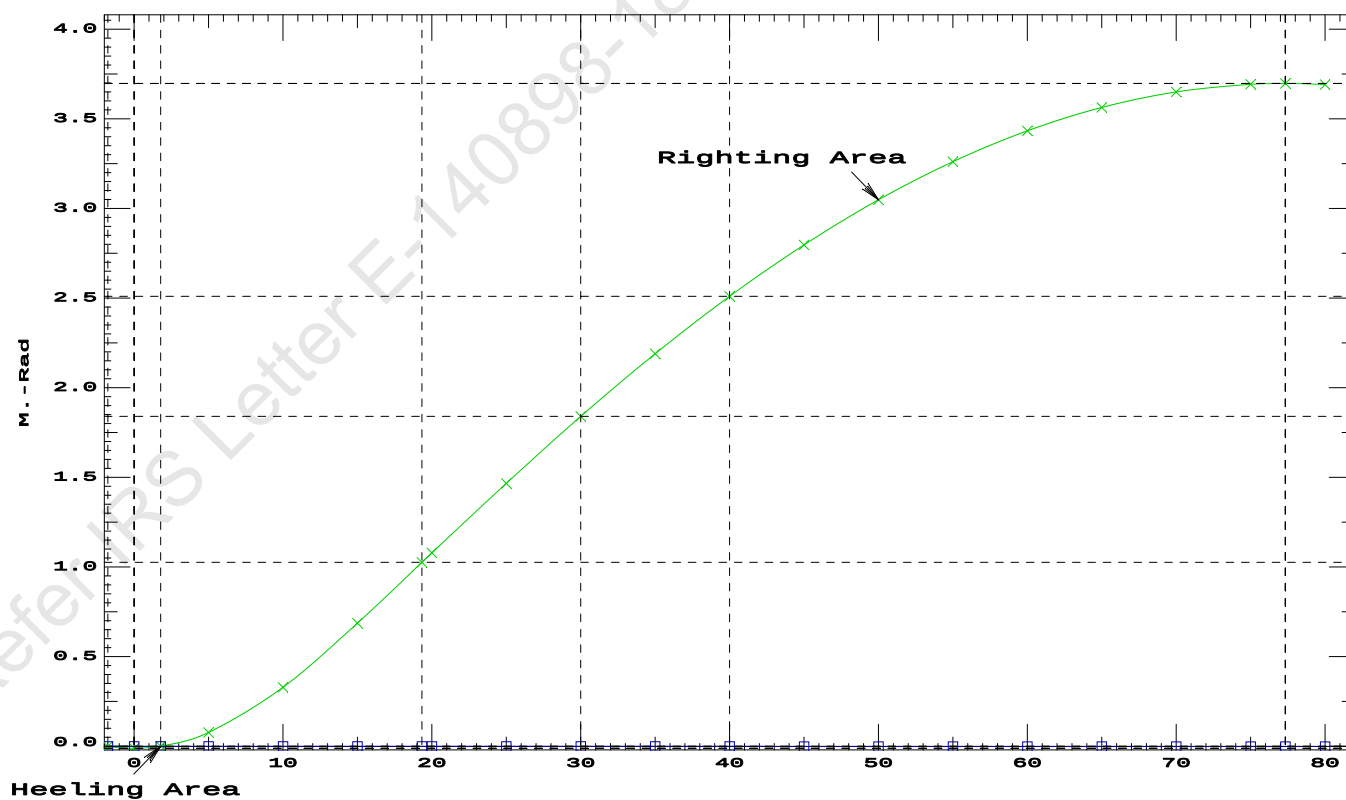
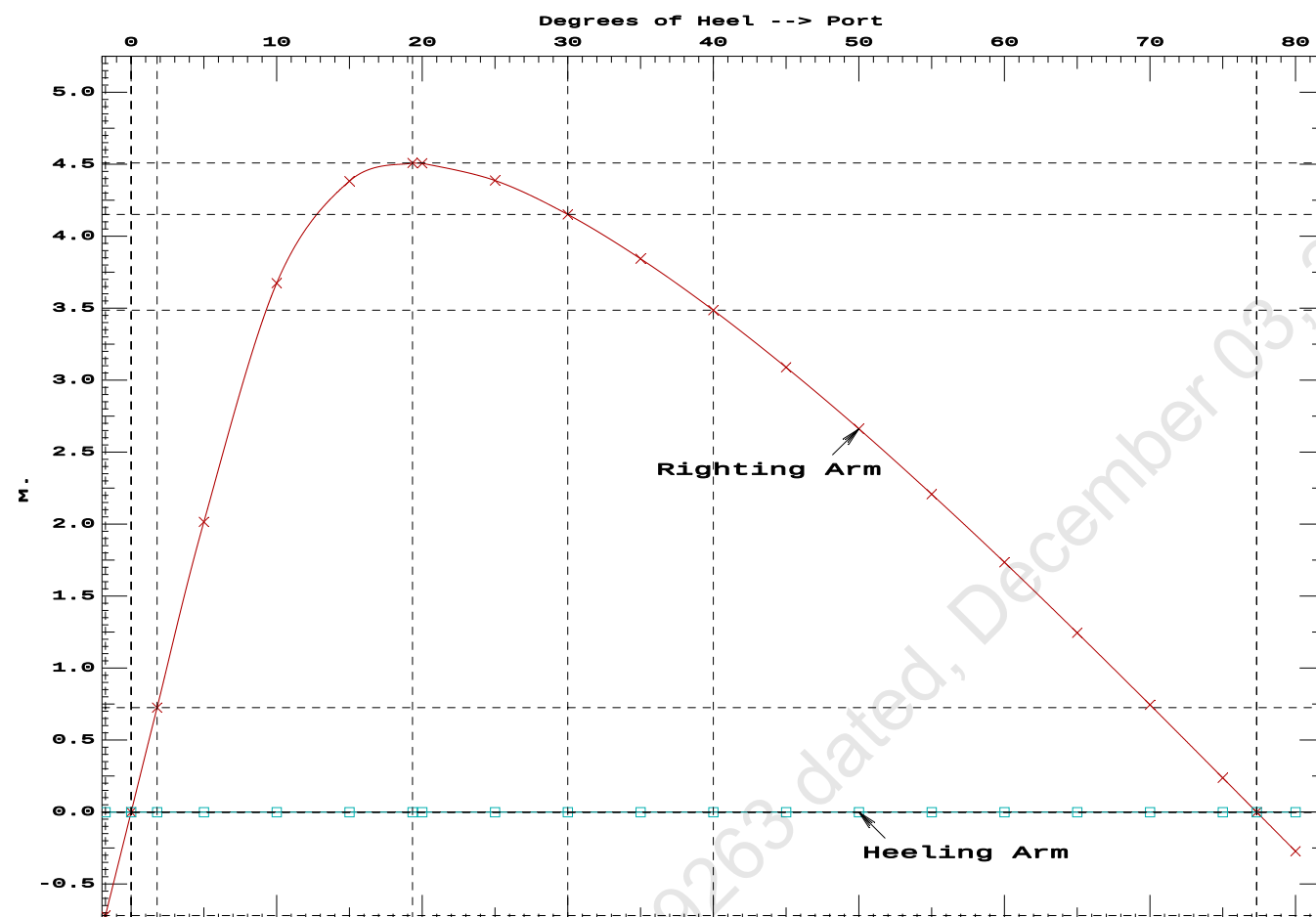
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.071	0.11f	1.77s	922.75	0.000	-0.719	0.0000
1.073	0.11f	0.00	922.76	0.000	0.000	-0.0111
1.071	0.11f	1.78p	922.77	0.000	0.726	0.0002
1.059	0.11f	5.00p	922.76	0.000	2.017	0.0774
0.969	0.13f	10.00p	922.76	0.000	3.675	0.3279
0.720	0.16f	15.00p	922.77	0.000	4.384	0.6865
0.470	0.21f	19.34p	922.77	0.000	4.510	1.0259
0.431	0.22f	20.00p	922.77	0.000	4.507	1.0781
0.138	0.27f	25.00p	922.77	0.000	4.388	1.4688
-0.157	0.33f	30.00p	922.77	0.000	4.152	1.8423
-0.452	0.38f	35.00p	922.77	0.000	3.845	2.1917
-0.745	0.44f	40.00p	922.77	0.000	3.487	2.5120
-1.032	0.49f	45.00p	922.77	0.000	3.091	2.7993
-1.313	0.54f	50.00p	922.77	0.000	2.663	3.0505
-1.584	0.59f	55.00p	922.77	0.000	2.209	3.2633
-1.843	0.63f	60.00p	922.77	0.000	1.735	3.4356
-2.087	0.67f	65.00p	922.77	0.000	1.246	3.5658
-2.316	0.70f	70.00p	922.77	0.000	0.746	3.6528
-2.526	0.73f	75.00p	922.77	0.000	0.238	3.6958
-2.618	0.74f	77.34p	922.77	0.000	0.000	3.7006
-2.716	0.74f	80.00p	922.77	0.000	-0.272	3.6943

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 599.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Res. Area Ratio from abs -1.8 deg to RAzero > 1.000 335.079 P
(2) Angle from abs 1.8 deg to RAzero > 20.00 deg 75.56 P

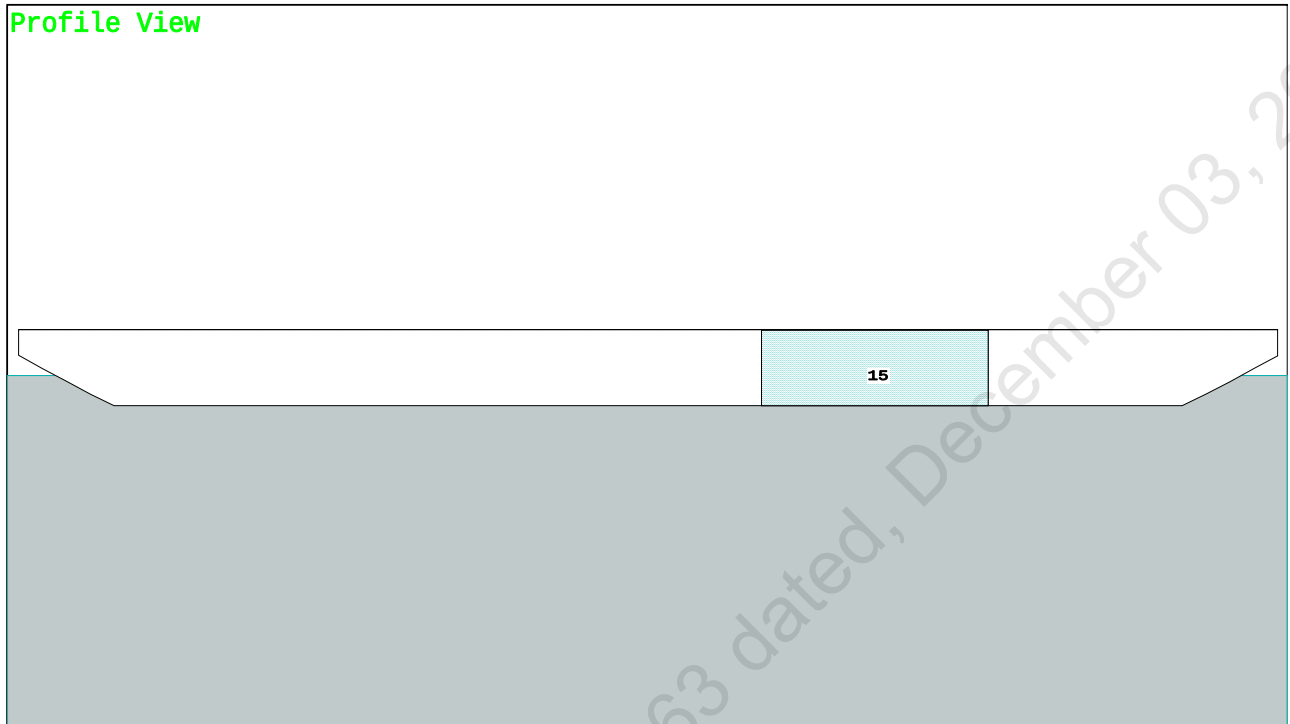
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GHS 17.90Cybermarine Technologies PTE. Ltd.
PONT00N

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GHS 17.90 PONT00N

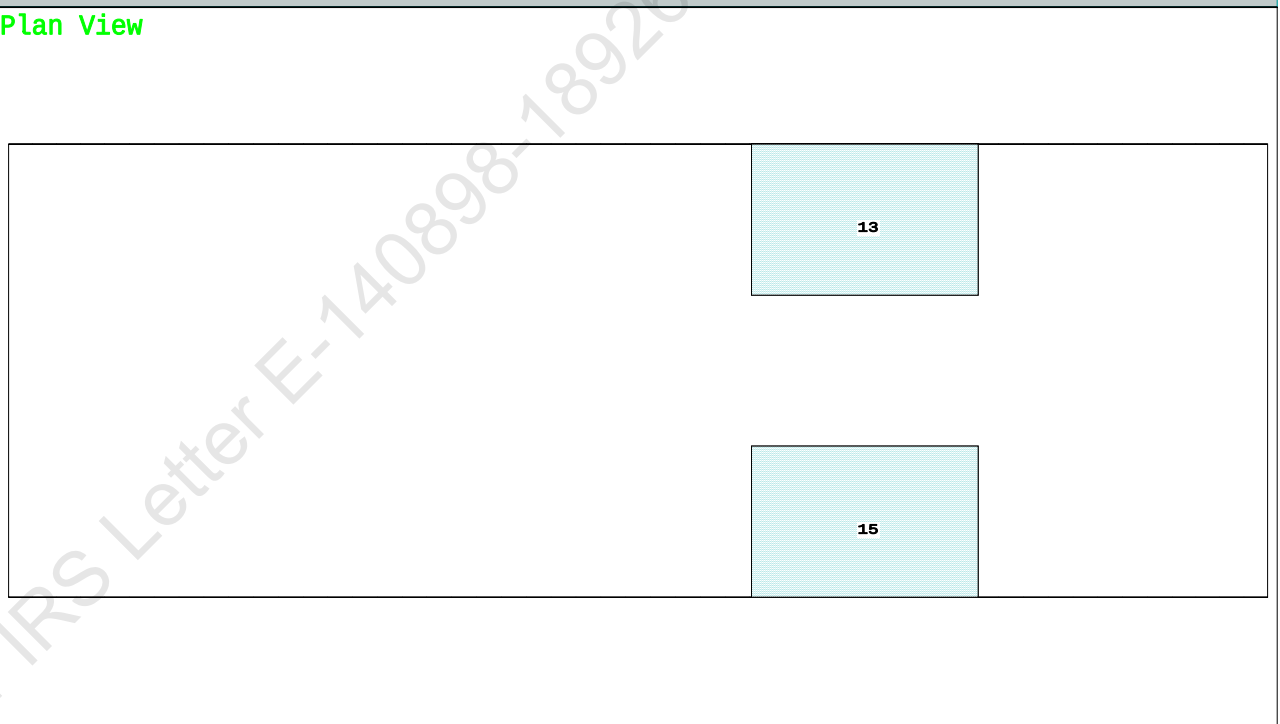
Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

Condition Graphic - Draft: 1.195 @ 50.000f, 1.190 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

13 05_WBTK.P.100% SALT WATER Intact 15 05_WBTK.S.100% SALT WATER Intact

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PONTOON

Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.195 @ 50.00f, 1.192 @ 25.00f, 1.190 @ 0.00

Trim: Fwd 0.005/50.000, Heel: zero

Part-----	Weight(MT)	LCG	TCG	VCG		
LIGHT SHIP	449.57	25.030f	0.000	3.000		
Crane	150.00	7.500f	0.000	5.200		
Crane Load	51.30	19.500f	0.000	33.218		
Total Fixed----->	650.87	20.554f	0.000	5.889		
Load-----	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509
Total Tanks----->			323.20	34.000f	0.000	1.509
Total Weight----->			974.07	25.016f	0.000	4.436
		Displ(MT)	LCB	TCB	VCB	
HULL	1.025	974.07	25.017f	0.000	0.599	

Righting Arms:			0.001f	0.000		
Part-----	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	847.1	25.005f	0.000	164.18	23.727*
		MT/cm	m	MT/cm	GML	GMT
		8.68		31.24	160.34	19.891
Distances in METERS.-----				Moments in m.-MT.		

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

315.3 Cu.M. (13%) SALT WATER

150.00 MT Crane

51.30 MT Crane Load

DRAFT AT FP (M) = 1.195
DRAFT AT MS (M) = 1.192
DRAFT AT AP (M) = 1.190
DRAFT AT FWD MARK (M) = 1.195
DRAFT AT AFT MARK (M) = 1.190

HYDROSTATIC PROPERTIES

Trim: Fwd 0.005/50.000, No Heel, VCG = 4.436

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)	LCB	VCB	cm	LCF	cm trim	GML	GMT
1.192	974.07	25.017f	0.599	8.68	25.005f	31.24	160.34	19.891
Distances in METERS.-----				Specific Gravity = 1.025.-----Moment in m.-MT.				
				Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 325.5 m.-MT

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PONTOON

Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.195 @ 50.00f, 1.192 @ 25.00f, 1.190 @ 0.00

Trim: Fwd 0.005/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	90.2	0.925	1.509	0.05500	7.48
CRANE	57.0	4.321	4.905	0.05500	15.38
Total wind heeling moment to starboard-->					22.86
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.016f TCG = 0.000 VCG = 4.436

Free Surface Adjustment: 0.334

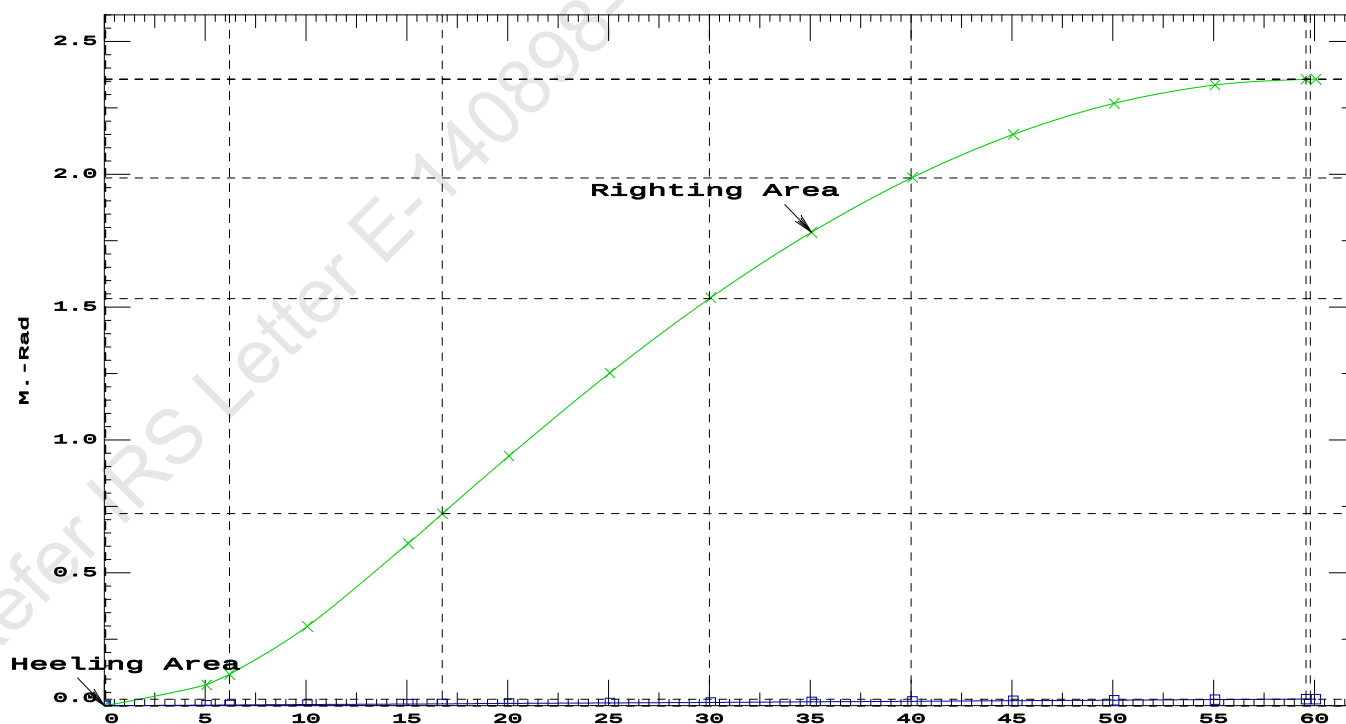
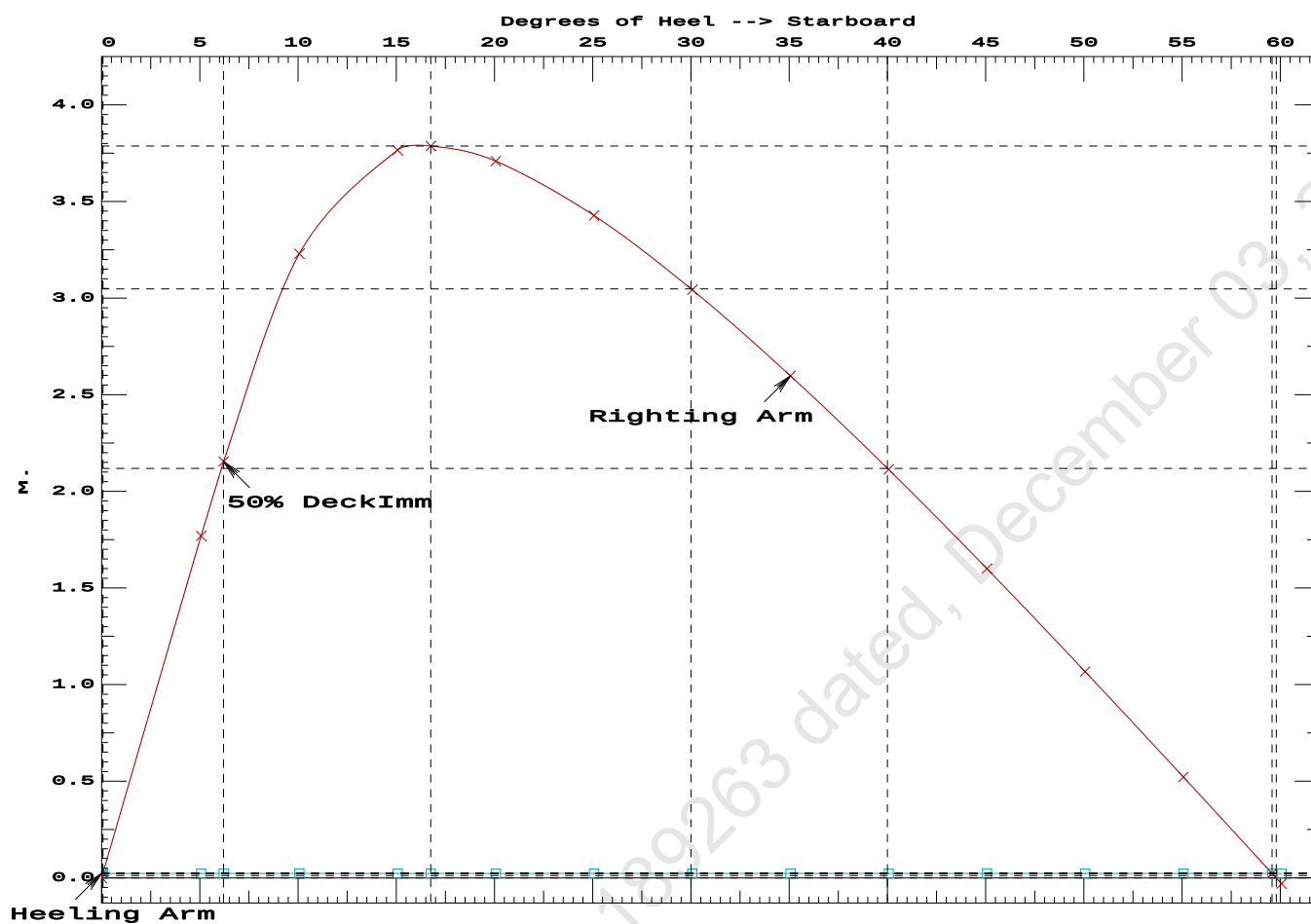
Adjusted CG: LCG = 25.015f TCG = 0.000 VCG = 4.770

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. 50% DeckImm Area	Angle
1.178	0.01f	0.00	973.99	0.000	-0.023	0.0000
1.178	0.01f	0.06s	974.06	0.000	0.000	-0.0000
1.163	0.01f	5.06s	974.07	0.000	1.746	0.0762
1.155	0.01f	6.21s	974.04	0.000	2.131	0.1148
1.089	0.01f	10.06s	974.13	0.000	3.206	0.2956
0.874	0.01f	15.06s	974.23	0.000	3.743	0.6040
0.794	0.01f	16.75s	974.07	0.000	3.763	0.7149
0.636	0.01f	20.06s	974.07	0.000	3.685	0.9312
0.394	0.01f	25.06s	974.07	0.000	3.404	1.2430
0.148	0.02f	30.06s	974.07	0.000	3.019	1.5241
-0.098	0.02f	35.06s	974.07	0.000	2.574	1.7686
-0.344	0.02f	40.06s	974.07	0.000	2.089	1.9724
-0.587	0.03f	45.06s	974.07	0.000	1.577	2.1325
-0.826	0.03f	50.06s	974.07	0.000	1.044	2.2470
-1.059	0.03f	55.06s	974.07	0.000	0.499	2.3145
-1.262	0.03f	59.58s	974.07	0.000	0.000	2.3342
-1.283	0.03f	60.06s	974.06	0.000	-0.054	2.3339
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.						

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 22.86 (constant)

LIM	IS CODE 2008 PART B CHAPTER-2 SEC.2	Min/Max	Attained
(1)	Residual Area from abs 0 deg to MaxRA	> 0.0800 m.-Rad	0.7149 P
(2)	Angle from Equilibrium to RAZero or Flood	> 20.00 deg	59.52 P
(3)	Angle from Equilibrium to 50% Dk Imm Angle	> 0.00 deg	6.14 P

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PONT00NCondition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

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GHS 17.90

PONTOON

Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.016f TCG = 0.000 VCG = 4.436
Free Surface Adjustment: 0.334
Adjusted CG: LCG = 25.015f TCG = 0.000 VCG = 4.770

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.178	0.01f	0.00	974.07	0.000	0.000	0.0000
1.163	0.01f	5.00s	974.00	0.000	1.747	0.0762
1.090	0.01f	10.00s	974.07	0.000	3.216	0.2948
0.877	0.01f	15.00s	974.23	0.000	3.763	0.6060
0.794	0.01f	16.76s	974.07	0.000	3.785	0.7222
0.639	0.01f	20.00s	974.07	0.000	3.710	0.9352
0.397	0.01f	25.00s	974.07	0.000	3.431	1.2493
0.151	0.02f	30.00s	974.07	0.000	3.047	1.5327
-0.095	0.02f	35.00s	974.07	0.000	2.602	1.7796
-0.341	0.02f	40.00s	974.07	0.000	2.118	1.9859
-0.584	0.03f	45.00s	974.07	0.000	1.605	2.1485
-0.823	0.03f	50.00s	974.07	0.000	1.073	2.2655
-1.056	0.03f	55.00s	974.06	0.000	0.528	2.3355
-1.271	0.03f	59.78s	974.07	0.000	0.000	2.3576
-1.281	0.03f	60.00s	974.07	0.000	-0.024	2.3575

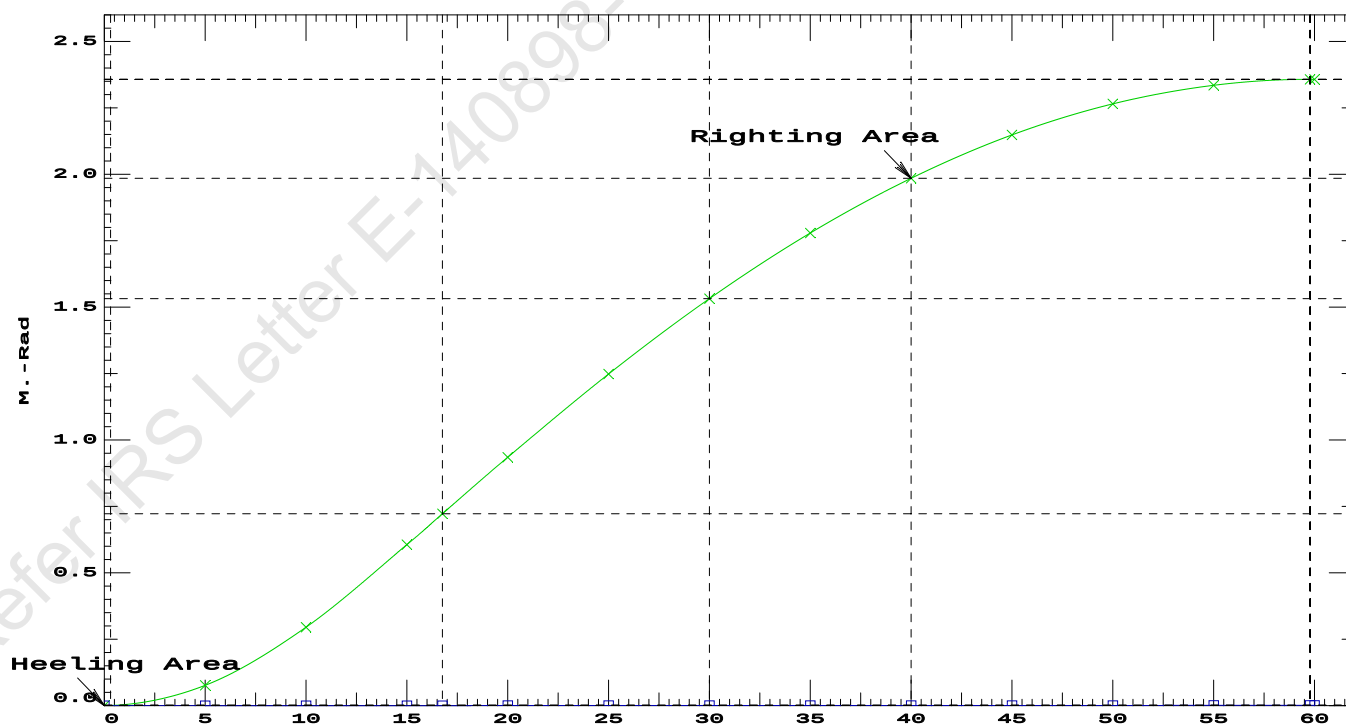
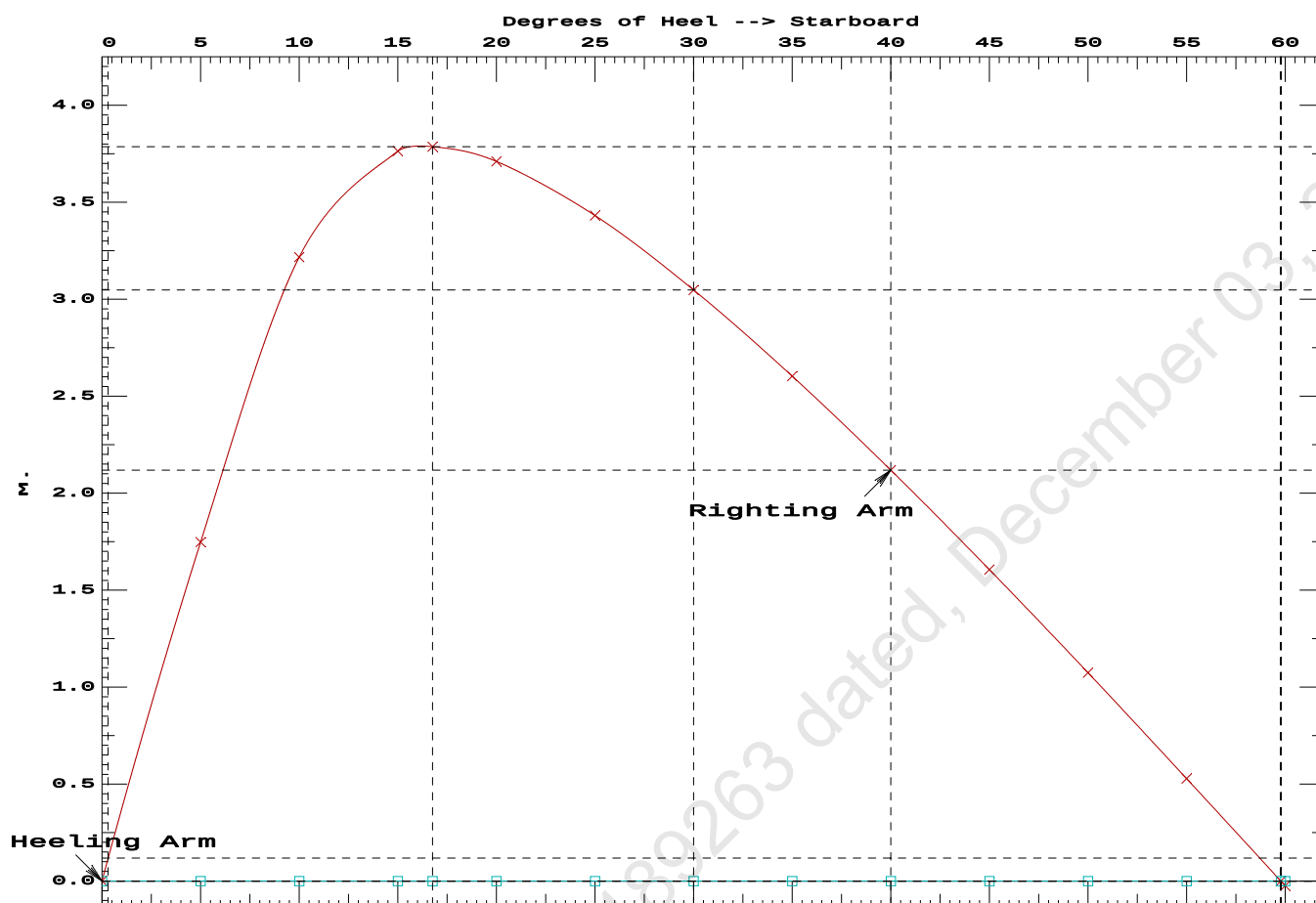
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM	STABILITY CRITERIA FOR CRANE OPERAT	Min/Max	Attained
(1)	Absolute Angle at Equilibrium	< 15.00 deg	0.00 P
(2)	Rise in Abs. RA from Equilibrium to MaxRA	> 66.7 %	368729 P
(3)	Abs Ratio from Equilibrium to RAZero or Flood	> 1.667	2202.03 P

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GHS 17.90Cybermarine Technologies PTE. Ltd.
PONT00NCondition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

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GHS 17.90 PONTON

Condition 15 : Operating Condition of 150T Crane**Crane Lifting 51.3T @ 12m Radius Fwd**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.195 @ 50.00f, 1.192 @ 25.00f, 1.190 @ 0.00

Trim: Fwd 0.005/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP	449.57	25.030f	0.000	3.000	
Crane	150.00	7.500f	0.000	5.200	
Crane Load	51.30	19.500f	0.000	33.218	
Total Fixed----->	650.87	20.554f	0.000	5.889	
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s
Total Tanks----->			323.20	34.000f	0.000
Total Weight----->			974.07	25.016f	0.000
			Displ(MT)----	LCB-----	TCB-----
HULL	1.025		974.06	25.016f	0.001s

Righting Arms:			0.000	0.001s	
External Arms:			0.000	0.001s	
Residual Righting Arms:			0.000	0.000	
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----
Total Waterplane----->	1.025	843.7	25.002f	0.027s	162.23
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----
		8.65	30.86	158.39	19.742
Distances in METERS.-----					
Moments in m.-MT.					

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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GHS 17.90 PONTON

After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.181 @ 50.00f, 1.133 @ 25.00f, 1.085 @ 0.00
Trim: Fwd 0.095/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----			
LIGHT SHIP	449.57	25.030f	0.000	3.000			
Crane	150.00	7.500f	0.000	5.200			
Total Fixed----->	599.57	20.644f	0.000	3.550			
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks----->			323.20	34.000f	0.000	1.509	325.46
Total Weight----->			922.77	25.322f	0.000	2.836	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL	1.025		922.77	25.326f	0.001s	0.569	

Righting Arms:			0.000	0.001s			
External Arms:			0.000	0.001s			
Residual Righting Arms:			0.000	0.000			
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----	
Total Waterplane---->	1.025	841.1	25.057f	0.017s	169.66	24.822*	
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----		
		8.62	30.89	167.39	22.556		
Distances in METERS.-----							
-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30; t: 15.00
limmarg: INFINITY; t3: 15; t: 7.50
limmarg: INFINITY; t3: 8; t: 3.75
limmarg: INFINITY; t3: 4; t: 1.88
limmarg: INFINITY; t3: 2; t: 0.94
limmarg: INFINITY; t3: 1; t: 0.47
limmarg: INFINITY; t3: 1; t: 0.23
limmarg: INFINITY; t3: 1; t: 0.12
limmarg: INFINITY; t3: 1; t: 0.06
limmarg: INFINITY; t3: 1; t: 0.03
limmarg: INFINITY; t3: 1; t: 0.01

Theta3 is 0.97 degrees, as demonstrated below.

01-12-21 09:42:46 Cybermarine Technologies PTE. Ltd.
GHS 17.90 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 20.644f TCG = 0.000 VCG = 3.550

Origin Depth	Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.073	0.11f	0.00	922.77	0.000	0.000	0.0000
1.073	0.11f	0.97s	922.77	0.000	0.395	0.0033
1.059	0.11f	5.00s	922.76	0.000	2.016	0.0881
0.969	0.13f	10.00s	922.76	0.000	3.673	0.3390
0.720	0.16f	15.00s	922.77	0.000	4.382	0.6973
0.470	0.21f	19.34s	922.77	0.000	4.507	1.0365
0.431	0.22f	20.00s	922.77	0.000	4.505	1.0888
0.138	0.27f	25.00s	922.77	0.000	4.385	1.4793
-0.158	0.33f	30.00s	922.77	0.000	4.149	1.8526

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 599.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 1 > 1.000 LARGE

01-12-21 09:42:46 Cybermarine Technologies PTE. Ltd.
GHS 17.90 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 20.644f TCG = 0.000 VCG = 3.550

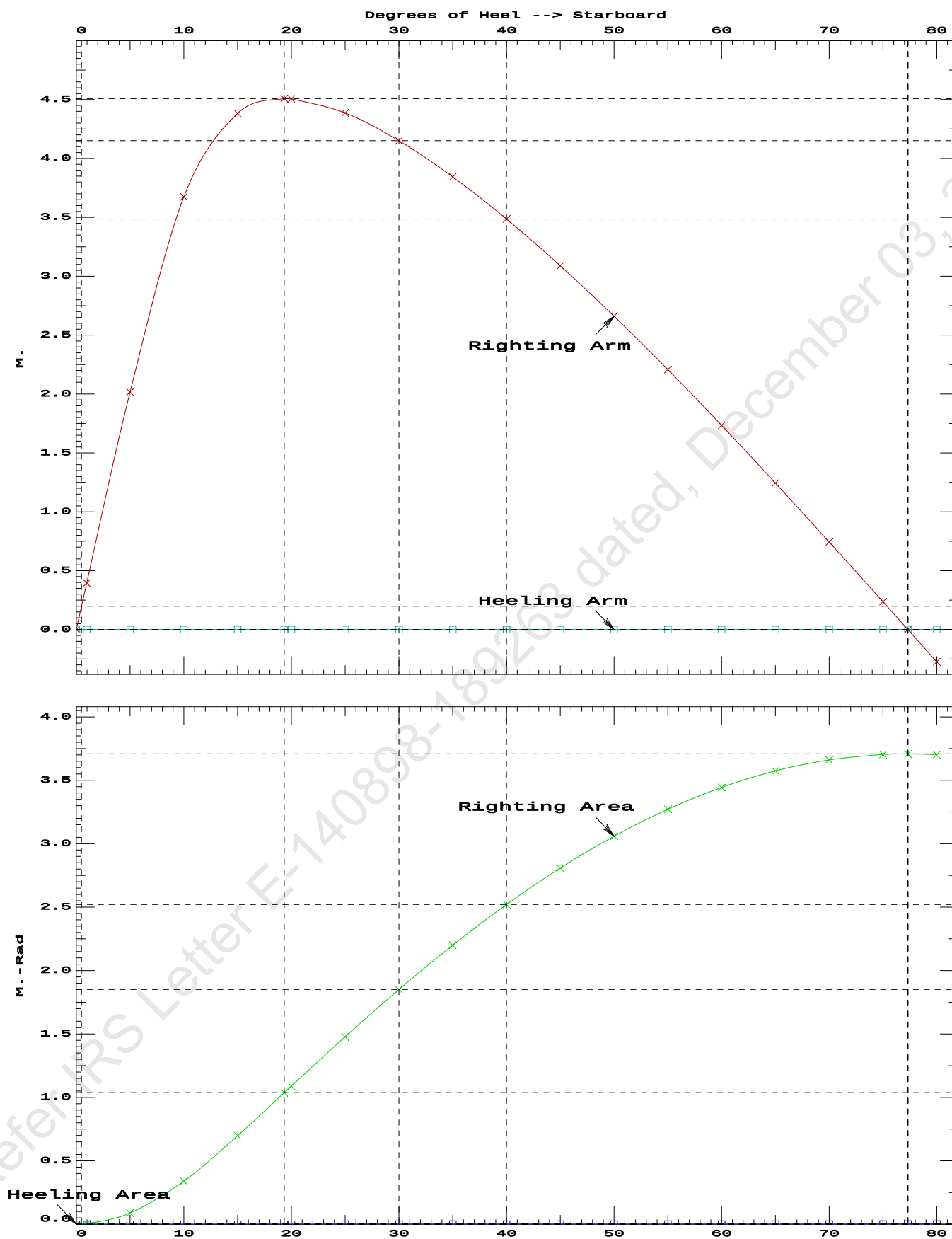
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.073	0.11f	0.00	922.77	0.000	0.000	0.0000
1.073	0.11f	0.97s	922.77	0.000	0.395	0.0033
1.059	0.11f	5.00s	922.76	0.000	2.016	0.0881
0.969	0.13f	10.00s	922.76	0.000	3.673	0.3390
0.720	0.16f	15.00s	922.77	0.000	4.382	0.6973
0.470	0.21f	19.34s	922.77	0.000	4.507	1.0365
0.431	0.22f	20.00s	922.77	0.000	4.505	1.0888
0.138	0.27f	25.00s	922.77	0.000	4.385	1.4793
-0.158	0.33f	30.00s	922.77	0.000	4.149	1.8526
-0.453	0.38f	35.00s	922.77	0.000	3.842	2.2018
-0.745	0.44f	40.00s	922.77	0.000	3.485	2.5219
-1.032	0.49f	45.00s	922.77	0.000	3.088	2.8090
-1.313	0.54f	50.00s	922.77	0.000	2.660	3.0600
-1.584	0.59f	55.00s	922.77	0.000	2.207	3.2725
-1.843	0.63f	60.00s	922.77	0.000	1.733	3.4446
-2.088	0.67f	65.00s	922.77	0.000	1.244	3.5746
-2.316	0.70f	70.00s	922.77	0.000	0.743	3.6614
-2.526	0.73f	75.00s	922.77	0.000	0.236	3.7042
-2.617	0.74f	77.32s	922.77	0.000	0.000	3.7089
-2.716	0.74f	80.00s	922.77	0.000	-0.274	3.7025

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 599.57 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 1 deg to RAZero > 20.00 deg 76.35 P

01-12-21 09:42:46
GHS 17.90Cybermarine Technologies PTE. Ltd.
PONT00N

LIGHTSHIP PARTICULARS

Lightship Weight	=	449.57	T
VCG	=	3.000	M above Baseline
LCG	=	25.030	M fwd of AP (Fr.0)
TCG	=	0.000	M off centerline

(Reference: Lightship particulars are taken from approved draft survey report.)

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GHS 17.22 PONT00N

USK draft refers to the line:

0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

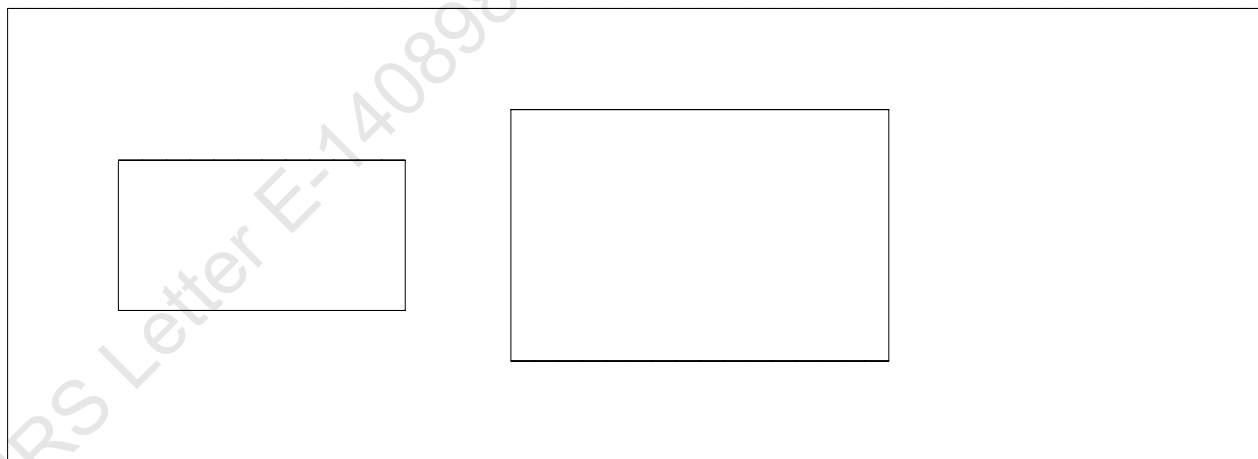
WINDAGE AREA CALCULATION AT DIFFERENT DRAFTS

Condition Graphic

Profile View



Plan View



01-08-20 21:20:15 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

LATERAL PLANE STATUS

USK draft: 0.512 @ 50.00f, 0.512 @ 25.00f, 0.512 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	22.3	25.000f	-0.253	121.4	25.000f	1.281
CRANE				57.0	10.100f	5.000
DECKCARGO.C				29.4	27.500f	3.500
Total Lateral Plane->	22.3	25.000f	-0.253	207.8	21.267f	2.615
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 0.612 @ 50.00f, 0.612 @ 25.00f, 0.612 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	26.8	25.000f	-0.302	116.9	25.000f	1.229
CRANE				57.0	10.100f	4.900
DECKCARGO.C				29.4	27.500f	3.400
Total Lateral Plane->	26.8	25.000f	-0.302	203.3	21.184f	2.572
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 0.712 @ 50.00f, 0.712 @ 25.00f, 0.712 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	31.3	25.000f	-0.351	112.4	25.000f	1.176
CRANE				57.0	10.100f	4.800
DECKCARGO.C				29.4	27.500f	3.300
Total Lateral Plane->	31.3	25.000f	-0.351	198.8	21.097f	2.529
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 0.812 @ 50.00f, 0.812 @ 25.00f, 0.812 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	35.9	25.000f	-0.400	107.8	25.000f	1.123
CRANE				57.0	10.100f	4.700
DECKCARGO.C				29.4	27.500f	3.200
Total Lateral Plane->	35.9	25.000f	-0.400	194.2	21.006f	2.487
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 0.912 @ 50.00f, 0.912 @ 25.00f, 0.912 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	40.4	25.000f	-0.450	103.3	25.000f	1.070
CRANE				57.0	10.100f	4.600
DECKCARGO.C				29.4	27.500f	3.100
Total Lateral Plane->	40.4	25.000f	-0.450	189.7	20.911f	2.445
Distances in METERS.-----						

01-08-20 21:20:15 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

LATERAL PLANE STATUS

USK draft: 1.012 @ 50.00f, 1.012 @ 25.00f, 1.012 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	45.0	25.000f	-0.497	98.7	25.000f	1.018
CRANE				57.0	10.100f	4.500
DECKCARGO.C				29.4	27.500f	3.000
Total Lateral Plane->	45.0	25.000f	-0.497	185.1	20.809f	2.405
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.112 @ 50.00f, 1.112 @ 25.00f, 1.112 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	49.7	25.000f	-0.545	94.0	25.000f	0.966
CRANE				57.0	10.100f	4.400
DECKCARGO.C				29.4	27.500f	2.900
Total Lateral Plane->	49.7	25.000f	-0.545	180.4	20.699f	2.367
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.212 @ 50.00f, 1.212 @ 25.00f, 1.212 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	54.4	25.000f	-0.594	89.3	25.000f	0.914
CRANE				57.0	10.100f	4.300
DECKCARGO.C				29.4	27.500f	2.800
Total Lateral Plane->	54.4	25.000f	-0.594	175.7	20.584f	2.329
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.312 @ 50.00f, 1.312 @ 25.00f, 1.312 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	59.1	25.000f	-0.642	84.5	25.000f	0.863
CRANE				57.0	10.100f	4.200
DECKCARGO.C				29.4	27.500f	2.700
Total Lateral Plane->	59.1	25.000f	-0.642	170.9	20.462f	2.291
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.412 @ 50.00f, 1.412 @ 25.00f, 1.412 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	63.9	25.000f	-0.691	79.8	25.000f	0.810
CRANE				57.0	10.100f	4.100
DECKCARGO.C				29.4	27.500f	2.600
Total Lateral Plane->	63.9	25.000f	-0.691	166.2	20.333f	2.255
Distances in METERS.-----						

01-08-20 21:20:15 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONTON

LATERAL PLANE STATUS

USK draft: 1.512 @ 50.00f, 1.512 @ 25.00f, 1.512 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	68.7	25.000f	-0.739	75.0	25.000f	0.759
CRANE				57.0	10.100f	4.000
DECKCARGO.C				29.4	27.500f	2.500
Total Lateral Plane->	68.7	25.000f	-0.739	161.4	20.194f	2.220
Distances in METERS.						

LATERAL PLANE STATUS

USK draft: 1.612 @ 50.00f, 1.612 @ 25.00f, 1.612 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	73.6	25.000f	-0.786	70.1	25.000f	0.708
CRANE				57.0	10.100f	3.900
DECKCARGO.C				29.4	27.500f	2.400
Total Lateral Plane->	73.6	25.000f	-0.786	156.5	20.044f	2.188
Distances in METERS.						

LATERAL PLANE STATUS

USK draft: 1.712 @ 50.00f, 1.712 @ 25.00f, 1.712 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	78.5	25.000f	-0.834	65.2	25.000f	0.657
CRANE				57.0	10.100f	3.800
DECKCARGO.C				29.4	27.500f	2.300
Total Lateral Plane->	78.5	25.000f	-0.834	151.6	19.883f	2.157
Distances in METERS.						

LATERAL PLANE STATUS

USK draft: 1.812 @ 50.00f, 1.812 @ 25.00f, 1.812 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	83.4	25.000f	-0.882	60.3	25.000f	0.607
CRANE				57.0	10.100f	3.700
DECKCARGO.C				29.4	27.500f	2.200
Total Lateral Plane->	83.4	25.000f	-0.882	146.7	19.712f	2.128
Distances in METERS.						

LATERAL PLANE STATUS

USK draft: 1.912 @ 50.00f, 1.912 @ 25.00f, 1.912 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	88.3	25.000f	-0.930	55.4	25.000f	0.556
CRANE				57.0	10.100f	3.600
DECKCARGO.C				29.4	27.500f	2.100
Total Lateral Plane->	88.3	25.000f	-0.930	141.8	19.529f	2.100
Distances in METERS.						

01-08-20 21:20:15 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

LATERAL PLANE STATUS

USK draft: 2.012 @ 50.00f, 2.012 @ 25.00f, 2.012 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	93.2	25.000f	-0.978	50.5	25.000f	0.505
CRANE				57.0	10.100f	3.500
DECKCARGO.C				29.4	27.500f	2.000
Total Lateral Plane->	93.2	25.000f	-0.978	136.9	19.333f	2.073
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 2.112 @ 50.00f, 2.112 @ 25.00f, 2.112 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	98.2	25.000f	-1.025	45.5	25.000f	0.455
CRANE				57.0	10.100f	3.400
DECKCARGO.C				29.4	27.500f	1.900
Total Lateral Plane->	98.2	25.000f	-1.025	131.9	19.118f	2.050
Distances in METERS.-----						

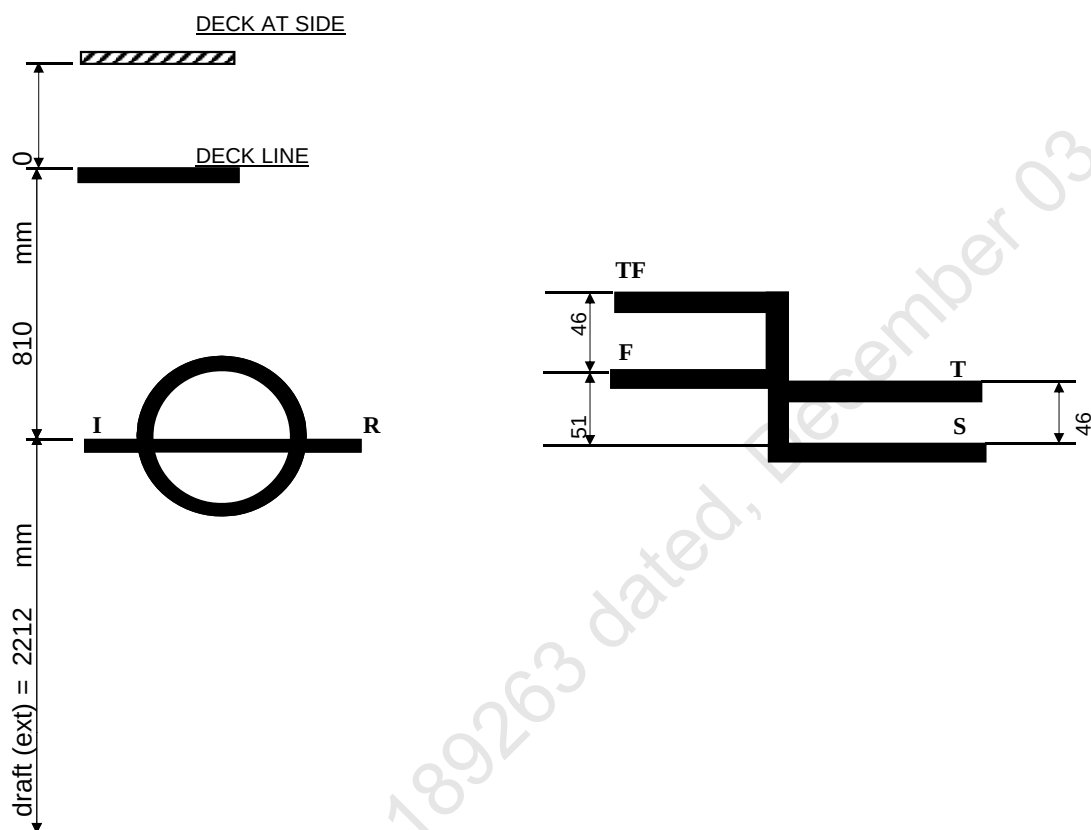
LATERAL PLANE STATUS

USK draft: 2.212 @ 50.00f, 2.212 @ 25.00f, 2.212 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	103.2	25.000f	-1.073	40.5	25.000f	0.405
CRANE				57.0	10.100f	3.300
DECKCARGO.C				29.4	27.500f	1.800
Total Lateral Plane->	103.2	25.000f	-1.073	126.9	18.887f	2.029
Distances in METERS.-----						

TABLE OF FREEBOARDS



LOADLINE	FREEBOARD (mm)	DRAFT (ext) KL (m)	DISPLACEMENT (ext) T	DEADWEIGHT T
SUMMER	810	2.212	1893.06	1443.49
FRESH WATER	759	2.263	1892.81	1443.24
TROPICAL	764	2.258	1935.52	1485.95
TROPICAL FRESH WATER	713	2.309	1934.23	1484.66

Deck Plate Thickness = 10 mm

Keel Plate Thickness = 12 mm



MEMORANDUM OF FREEBOARDS

29 November 2021

Chisti Marine Engineering Works.

Type B freeboard

Yard No. 019

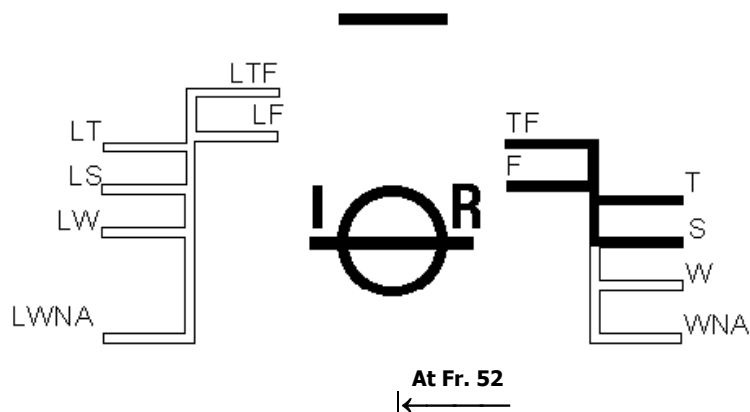
Freeboard Length : 48.000 m

Summer WL Extreme Draught : 2.212 m
- load line amidships

The upper edge of the deckline from which these freeboards are measured is 0 mm below top of
* ~~Wood~~ / * steel Deck * at side / * ~~continued to side~~. (*Delete words not applicable).

Freeboards from Deck Line		Seasonal, Fresh Water, or Timber allowances		
Tropical Fresh Water	713	mm.	97	mm. above summer
Fresh Water	759	mm.	51	mm. above summer
Tropical	764	mm.	46	mm. above summer
Summer	810	mm.	Upper edge of line through centre	
Winter	Not	mm.	Assigned	mm. below summer
Winter North Atlantic	Not	mm.	Assigned	mm. below summer
Timber Tropical fresh water	—	mm.	—	mm. above timber summer
Timber fresh water	—	mm.	—	mm. above timber summer
Timber Tropical	—	mm.	—	mm. above timber summer
Timber Summer	—	mm.	—	mm. above centre of ring
Timber Winter	—	mm.	—	mm. below timber summer
Timber Winter North Atlantic	—	mm.	—	mm. below timber summer

Valid only for unmanned operation while under tow along the Indian Coast, during the course of which the vessel does not go more than 20 nautical miles from the nearest land, and may cross gulfs or similar features recognised by the Administration.





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Indian Register of Shipping
CIN: U61100MH1975NPL018244

52 A, Adi Shankaracharya Marg, Opp. Powai Lake,
Powai, Mumbai - 400 072, India
Phone : +91 22 30519400 / +91 22 7119 9400
Fax : +91 22 2570 3611
E-mail : ho@irclass.org ★ Website : www.irclass.org

Our ref: E-140763-189090

01-December-2021

Company : Cybermarine Knowledge System Pvt. Ltd.

Dear Sir/Madam,

Your document (detailed below) has been reviewed.

Project Details	
Project number	NC002662
Project name	Chisti Marine Engineering Works - CHISTI/018 and CHISTI/019
Specific Yard No(s):	CHISTI/019
Project Manager	Rishi Doshi
Project In-Charge	Veerendra P

Document Details	
Plan No.	CM21-V-102-0209
Plan Title	Draft Survey Report_KB-33
Rev.No.	0
Upload.No.	1
Submitted Date	30-November-2021
Review Status	Approved
Parent Plan Title	

Notes

1. The review status Approved indicates:

The following Lightship particulars are approved based on the Draft Survey conducted on 30 Nov 2021:

Lightship Weight : 449.570 T

Lightship LCG : 25.030 m Forward of A.P (Frame 0)

Lightship VCG : 3.000 m Above Baseline.

Lightship TCG : 0.000 m from Centerline.

Final Stability Booklet prepared based on the approved Lightship particulars should be submitted for approval.

Copy of the approved Draft Survey report should form part of the Final Stability Booklet.

Thanking you,
Yours Sincerely,

For INDIAN REGISTER OF SHIPPING

Vimal R
Surveyor

This is an electronically published document and does not require signature.